DELOCALIZATION OF NON-MEAN-FIELD RANDOM MATRICES IN DIMENSIONS $d \geq 3$

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Abstract. We study $N \times N$ random band matrices $H = (H_{xy})$ with independent mean-zero Gaussian entries, where x, y lie on the discrete torus $(\mathbb{Z}/\sqrt[d]{N}\mathbb{Z})^d$ in dimensions $d \geq 3$. The variance profile satisfies $\mathbb{E}|H_{xy}|^2 = S_{xy}$, with $S_{xy} = 0$ whenever the distance between x and y exceeds a bandwidth parameter W. We prove that if $W \geq N^{\mathfrak{c}}$ for some constant $\mathfrak{c} > 0$, then in the large-N limit, bulk eigenvectors are delocalized, quantum unique ergodicity (QUE) holds, and the local bulk eigenvalue statistics are universal. Our proof is based on the tree approximation of the loop hierarchy [69] and diagrammatic techniques developed in earlier works [67, 65, 66, 27, 28].

The random band matrix model above is the classical Wegner orbital model; alongside it we also study a second classical non-mean-field model, the block Anderson model. Specifically, we consider Hermitian matrices $H = V + g\Psi$ on the same discrete torus $(\mathbb{Z}/\sqrt[d]{N}\mathbb{Z})^d$, where V is a random block potential consisting of i.i.d. complex Gaussian diagonal blocks of size $W^d \times W^d$, and Ψ encodes the interactions between neighboring blocks—random in the Wegner orbital model and deterministic in the block Anderson model. The parameter $g > 0$ represents the coupling strength between blocks. Assuming again that $W \geq N^{\mathfrak{c}}$, we establish delocalization of bulk eigenvectors, QUE, and bulk universality under the condition $W^{-d/2+\varepsilon} \leq g \leq \varepsilon^{-1}$ for any small constant $\varepsilon > 0$. Combined with the localization results of [52] for $g \ll W^{-d/2}$, this identifies a localization–delocalization transition at the scale $g = W^{-d/2}$ in dimensions $d \geq 3$.

Contents

1. Introduction 1
2. The model and main results 6
3. Steps 1 and 2: A priori G-loop estimates 19
4. Steps 3 and 4: Sharp maximum estimates for G-loops 31
5. Step 5: Pointwise estimate for $(\mathcal{L} - \mathcal{K})$-loops 45
6. Step 6: Expected 2-loop estimates 52
7. Estimation of the light-weight term 54
8. Extension to the block Anderson model 71
Appendix A. Proofs of some deterministic estimates 79
Appendix B. Proofs of auxiliary graphical lemmas 93
References 104

1. Introduction

The tight-binding model introduced by Anderson [9] describes electron transport in disordered semiconductors. It is formulated as a discrete random Schrödinger operator on $\mathbb{Z}^d$ of the form

$$H = -\Delta + \lambda V, \tag{1.1}$$

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where Δ denotes the graph Laplacian on $\mathbb{Z}^d$, V is a random potential with i.i.d. entries, and $\lambda > 0$ is a coupling parameter representing the strength of disorder. In his seminal work [9], Anderson predicted a transition from delocalized to localized behavior as the disorder strength λ increases.

Localization in the one-dimensional (1D) Anderson model, valid for all $\lambda > 0$, is well established; see, for example, [43, 47, 17, 23]. In dimensions $d \geq 2$, Anderson localization was first rigorously proved by Fröhlich and Spencer [40] using multi-scale analysis. An alternative approach was later developed by Aizenman and Molchanov [3] based on the fractional moment method. Since then, a substantial body of work has deepened our understanding of Anderson localization; see, for example, [39, 18, 55, 2, 4, 16, 42, 24, 50]. In contrast, the existence of delocalized states in the finite-dimensional Anderson model remains unproven—both for arbitrary disorder strengths and in all finite dimensions. In infinite dimensions, however, Aizenman and Warzel [6, 5] rigorously established the presence of a delocalized phase for the Anderson model on the Bethe lattice, an infinite regular tree. More recently, the mobility edge phenomenon has also been proved in this setting [1]. The tree geometry differs fundamentally from Euclidean lattices, as tree graphs are loop-free.

To bridge the understanding between random Schrödinger operators and random matrix theory, the random band matrix model was introduced [20, 19, 38]. A d-dimensional random band matrix $H = (H_{xy})$ is an $N \times N$ Hermitian random matrix defined on the discrete torus $(\mathbb{Z}/\sqrt[d]{N}\mathbb{Z})^d$ (assuming $\sqrt[d]{N} \in \mathbb{N}$ for simplicity). Subject to Hermitian symmetry, the entries H_{xy} are independent real or complex Gaussian variables with mean zero. The variance profile $S_{xy} = \mathbb{E}|H_{xy}|^2$ decays to zero when the distance between x and y exceeds a bandwidth parameter W, and satisfies the normalization $\sum_y S_{xy} \equiv 1$. It was conjectured in [20, 19, 38, 41] that this model exhibits a localization–delocalization transition, accompanied by a change in spectral statistics from Poisson to GOE/GUE (Gaussian Orthogonal/Unitary Ensemble) statistics, at a critical bandwidth $W_c(N)$. For eigenvalues in the bulk of the spectrum, i.e., $|E| \leq 2 - \kappa$ with some fixed $\kappa > 0$, localization and Poisson statistics are expected when $W \ll W_c$, while delocalization and GOE/GUE statistics are expected when $W \gg W_c$, where

$$W_c = \begin{cases} \sqrt{N}, & d = 1, \\ \sqrt{\log N}, & d = 2, \\ \mathrm{O}(1), & d \geq 3. \end{cases} \tag{1.2}$$

See [12, 56, 57, 58] for further discussion of these conjectures.

The density of states for random band matrices in dimensions $d \geq 1$ was established in [25]. Delocalization of 1D band matrices under the assumption $W \geq N^a$, for various exponents $a > 1/2$, was proved in a series of works [29, 30, 33, 10, 45, 13, 14, 15, 67, 27]. The full conjecture was recently resolved in [69], which established both delocalization and universality of bulk eigenvalue statistics under the optimal condition $W \gg \sqrt{N}$. These results were later extended to non-Gaussian random band matrices in [35]. On the other hand, localization has been proved under the condition $W \ll N^a$ for various exponents $a < 1/2$ in [54, 52, 22, 21], and under the sharp condition $W \ll \sqrt{N}$ in [26]. Therefore, the localization-delocalization transition at the critical bandwidth $W = \sqrt{N}$ is now rigorously established for 1D random band matrices. In two dimensions, delocalization and bulk universality under the condition $W \geq N^\varepsilon$ (for arbitrarily small constant $\varepsilon > 0$) was proved in [28].

In this paper, we study band matrices H defined on the d-dimensional torus $\mathbb{Z}^d_{WL} = \mathbb{Z}^d/((WL)\cdot\mathbb{Z}^d)$ with $N = (WL)^d$ and $d \geq 3$. For simplicity, we assume that the torus $\mathbb{Z}^d_{WL}$ is partitioned into d-dimensional cubes of side length W, indexed by the smaller torus $\mathbb{Z}^d_L$. For $x, y \in \mathbb{Z}^d_{WL}$ belonging to blocks labeled by $a, b \in \mathbb{Z}^d_L$, the variance profile is given by

$$S_{xy} = c_{a-b} W^{-d} \mathbf{1}(\|a - b\|_1 \leq 1), \tag{1.3}$$

where $(c_{a-b}\mathbf{1}(\|a-b\|_1 \leq 1) : a, b \in \mathbb{Z}^d_L)$ is a symmetric doubly stochastic matrix. Our main results establish delocalization of bulk eigenvectors and universality of bulk eigenvalue statistics for the band matrix H in dimensions $d \geq 3$, under the assumption $W \geq N^\varepsilon$ for any fixed $\varepsilon > 0$. In higher dimensions $d \geq 7$, weaker forms of delocalization and bulk universality were previously obtained in [65, 66, 64].

Beyond standard band matrices, the block Anderson model provides an even closer analogue to the original Anderson model (1.1). This model, inspired by Wegner's orbital model [62, 53, 51], describes quantum particles with multiple internal degrees of freedom (e.g., orbitals or spin) moving in a disordered medium. Concretely, we consider the matrix $H = \Psi + \lambda V$, where $V = \mathrm{diag}(V_1, \ldots, V_{L^d})$ and the blocks

V_i are i.i.d. $W^d \times W^d$ Gaussian random matrices. The matrix Ψ encodes the hopping between neighboring blocks, and in the block Anderson model, we take Ψ to be the block Laplacian with the diagonal removed.

It was shown in [52] that, in all dimensions $d \geq 1$, the block Anderson model exhibits localization with localization length of order $\mathrm{O}(W)$ whenever $\lambda \gg W^{d/2}$. In this paper, we prove that all results obtained for random band matrices extend to the block Anderson model, provided $c \leq \lambda \leq W^{d/2-c}$ for a small constant $c > 0$. Combined with the localization results of [52], this establishes the localization–delocalization transition for the block Anderson model in all dimensions $d \geq 3$. In dimensions $d \in \{1, 2\}$, delocalization and bulk universality were previously proved in [59], while partial delocalization results for dimensions $d \geq 7$ were obtained in [68]. However, for the physically most relevant case $d = 3$, no rigorous results on delocalization were available for either random band matrices or the block Anderson model prior to this work. One reason why $d = 3$ is the final dimension in which these conjectures are resolved lies in certain critical difficulties that are intrinsic to three dimensions, as we now explain.

Denote by $G(z) = (H - z)^{-1}$ the resolvent (Green's function) of the random band and block Anderson models, for $z \in \mathbb{C}$. It is believed that the size of the resolvent entries $|G_{0x}(z)|^2$ is approximated by the resolvent of a certain random walk on $\mathbb{Z}^d_{WL}$, see e.g., [32, 65]. In dimensions $d \in \{1, 2\}$, the random-walk resolvent remains roughly constant (in magnitude, up to a logarithmic correction when $d = 2$) below a certain cutoff scale. Hence, an L^∞-bound on $G(z)$, together with control of this cutoff scale, suffices to estimate $G(z)$. Starting from $d = 3$, however, $|G_{0x}|^2 \asymp |x|^{-d+2}$. This forces us to track the *pointwise estimate* of G_{0x}. At the other extreme, in high dimensions, the lattice $\mathbb{Z}^d$ increasingly resembles a Bethe lattice. Although this analogy is difficult to exploit rigorously, we note that $(|x|^{-d+2})^\alpha$ is integrable whenever $\alpha > d/(d-2) \to 1$ as $d \to \infty$. In other words, $|x|^{-d+2}$ becomes "almost integrable" for large d. This observation was crucially used in the proofs for $d \geq 7$ in [65, 66, 64]. In dimension $d = 3$, we need to combine the *tree-approximation* method for $d \in \{1, 2\}$ [69, 28] with the *diagrammatic expansion* approach for $d \geq 7$ [65, 66, 64].

To illustrate the above points, we briefly explain the main ideas developed in this paper. For simplicity, we focus on the random band matrix model with variance profile (1.3); the analysis for the block Anderson model is analogous, up to some additional technical details. We begin by recalling the matrix Brownian motion used in [27, 69, 28]:

$$\mathrm{d}(H_t)_{xy} = \sqrt{S_{xy}}\,\mathrm{d}(\boldsymbol{B}_t)_{xy}, \quad \text{with} \quad H_0 = 0,$$

where $(\boldsymbol{B}_t)_{xy}$ are standard independent complex Brownian motions for $x, y \in \mathbb{Z}^d_{WL}$, subject to the Hermitian symmetry condition $(\boldsymbol{B}_t)_{xy} = \overline{(\boldsymbol{B}_t)_{yx}}$. Following [61, 60, 27], we consider the Green's function of H_t with a carefully chosen time-dependent spectral parameter z_t (see (2.37) below), defined by

$$G_t := (H_t - z_t)^{-1},$$

whose dynamics are naturally renormalized at leading order. We then define the n-G-loop observable $\mathcal{L}^{(n)}$ as an n-tensor:

$$\mathcal{L}^{(n)}_{a_1,\ldots,a_n} \equiv \mathcal{L}^{(n)}_{a_1,\ldots,a_n}(t) := \mathrm{Tr} \prod_{i=1}^{n} (G_t \cdot E_{a_i}), \quad \text{for} \quad a_i \in \mathbb{Z}^d_L, \quad 1 \leq i \leq n, \tag{1.4}$$

where each E_a is a block-averaging matrix, defined by $(E_a)_{xy} = W^{-d}\delta_{xy}$ when x, y belong to the a-th block, and $(E_a)_{xy} = 0$ otherwise. In the proof, the G-loops considered in this paper involve combinations of G_t and its Hermitian conjugate G_t^*; however, we omit this detail in the following heuristic discussion for simplicity.

The G-loops $\mathcal{L}^{(n)}(t)$ defined above satisfy a system of evolution equations known as the *loop hierarchy*; see equation (2.47) below for its precise formulation. The dynamics of n-loops depend on $(n+1)$-loops and a martingale term, whose quadratic variation involves $(2n+2)$-loops. Schematically, the loop hierarchy takes the form

$$\mathrm{d}\mathcal{L}^{(n)} = (\mathcal{L} * \mathcal{L})^{(n)}\mathrm{d}t + \mathcal{E}^{\mathring{G},(n)}\mathrm{d}t + \mathrm{d}\mathcal{E}^{M,(n)}, \tag{1.5}$$

where each term on the right-hand side has the following interpretation.

- The *light-weight* term $\mathcal{E}^{\mathring{G},(n)}$ depends on $(n+1)$-loops, typically of the form

$$W^d \sum_{u,v \in \mathbb{Z}^d_L} S^{(\mathsf{B})}_{uv} \cdot \mathrm{Tr}[\mathring{G}_t E_u] \cdot \mathcal{L}^{(n+1)}_{a_1,\ldots,a_n,v}, \quad \text{with} \quad \mathring{G}_t := G_t - mI_N. \tag{1.6}$$

  Here, $\mathrm{Tr}[\mathring{G}_t E_u]$ is referred to as a *light-weight*, $S^{(\mathsf{B})}_{uv} = c_{u-v}\mathbf{1}(\|u - v\|_1 \leq 1)$ denotes the block variance profile corresponding to (1.3), and $m \equiv m_{\mathrm{sc}}$ is the Stieltjes transform of the semicircle law (see (2.7)).

- The *martingale* term $\mathrm{d}\mathcal{E}^{M,(n)}$ has a quadratic variation depending on $(2n+2)$-loops.
- The *convolution* term $(\mathcal{L} * \mathcal{L})^{(n)}$ consists of sums of the form $\mathcal{L}^{(p)} \diamond \mathcal{L}^{(q)}$ with $p+q = n+2$ and $p, q \geq 2$, where

$$(\mathcal{L}^{(p)} \diamond \mathcal{L}^{(q)})_{a_1,\ldots,a_n} := W^d \sum_{u,v} S^{(\mathsf{B})}_{uv} \cdot \mathcal{L}^{(p)}_{a_1,\ldots,a_{p-1},u} \cdot \mathcal{L}^{(q)}_{a_p,\ldots,a_n,v}. \tag{1.7}$$

Although the formulation above is heuristic, it captures the essential structure of the loop hierarchy.

The loop equation (1.5) can be generalized to random band matrices without block structure. This issue has already been addressed in earlier works; see [35] for $d = 1$ and [37] for dimensions $d \in \{1, 2\}$. In fact, our method can be automatically extended to the case where the variance matrix takes the following form:

$$S_{xy} = \sum_{u,v \in \mathbb{Z}^d_{WL}} \chi(x-u) \widetilde{S}_{uv} \chi(v-y),$$

where $\widetilde{S}$ is a symmetric doubly stochastic matrix and $\chi : \mathbb{Z}^d_{WL} \to [0, \infty)$ is a mollifier at scale $W^{1-\varepsilon}$. In this case, let F_u denote the diagonal mollifier matrices with entries $(F_u)_{xy} = \delta_{xy}\chi(x-u)$ for $u, x, y \in \mathbb{Z}^d_{WL}$. Then, for $(x_1, \ldots, x_n) \in (\mathbb{Z}^d_{WL})^n$, we define the n-G-loops $\mathcal{L}^{(n)}_{x_1,\ldots,x_n}(t)$ in analogy with (1.4), replacing the matrices E_{a_i} with the mollifier matrices F_{x_i}. The dynamics of these loops satisfy equations of the same form as (1.5), with $S^{(\mathsf{B})}$ in (1.6) and (1.7) replaced by $\widetilde{S}$. Consequently, all our arguments extend to this setting with only minor notational modifications. To streamline the presentation, we continue to work with the simpler block structure in (1.3), which allows us to focus on the core technical challenges in dimensions $d \geq 3$.

A key observation from [69, 28] is that both the light-weight and martingale terms are small errors. Consequently, $\mathcal{L}^{(n)}$ can be approximated by $\mathcal{K}^{(n)}$, the solution of the deterministic *convolution tree equation*:

$$\mathrm{d}\mathcal{K}^{(n)} = (\mathcal{K} * \mathcal{K})^{(n)} \mathrm{d}t. \tag{1.8}$$

For any fixed n, the right-hand side of equation (1.8) depends only on $\mathcal{K}^{(k)}$ with $k \leq n$, forming a *closed system* of equations for $(\mathcal{K}^{(2)}, \ldots, \mathcal{K}^{(n)})$ that can be solved inductively and admits explicit solutions, as discovered in [69]. Since these solutions have an explicit tree representation, we refer to $\mathcal{K}^{(n)}$ as the *tree approximation* to $\mathcal{L}^{(n)}$. In particular, for $n = 2$, equation (1.8) has the solution $\mathcal{K}^{(2)}(t) = W^{-d}\Theta_t$, where Θ_t denotes the propagator defined in Definition 2.17. This propagator can carry different charges. For simplicity, in the discussion below, we use the notation Θ_t exclusively for the most relevant $(+,-)$-charged propagator, denoted $\Theta^{(+,-)}_t$. It can be viewed as the resolvent of a Laplacian on the torus $\mathbb{Z}^d_L$, describing the classical diffusion of a random walk. This propagator serves as the basis for establishing quantum diffusion, delocalization, and the universality of eigenvalue statistics—provided the approximation $\mathcal{L}^{(2)} \approx \mathcal{K}^{(2)}$ can be justified in a sufficiently strong sense.

To rigorously justify the tree approximation, three main ingredients are required:

(1) A strategy to manage the dependence of $\mathcal{E}^{\mathring{G},(n)}$ and $\mathrm{d}\mathcal{E}^{M,(n)}$ on higher-order loops $\mathcal{L}^{(k)}$ with $k > n$.
(2) A stability theory for the perturbed equation $\mathrm{d}\mathcal{L} = (\mathcal{L} * \mathcal{L})\mathrm{d}t + \text{errors}$.
(3) Explicit quantitative bounds showing that $\mathcal{E}^{\mathring{G},(n)}$ and $\mathrm{d}\mathcal{E}^{M,(n)}$ are indeed small.

In the absence of additional structure in the hierarchy, issue (1) reflects a classical obstacle in many-body dynamics, reminiscent of the well-known BBGKY hierarchy. In our setting, however, a key observation is that higher-order loops appear only in the error terms. This feature enables the development of a general inductive and self-improving framework, first introduced in [69], based on a bootstrap argument that applies across all dimensions. It should be noted that the smallness of the martingale and light-weight terms is a highly nontrivial fact. Without the correct perspective, these terms may seem much larger than the leading contribution. Indeed, in the early development of the theory of random band matrices, researchers often encountered the seemingly impenetrable issue that the error terms in the so-called *T-equation* diverge. While we do not elaborate on the historical developments here, we will demonstrate below that the light-weight term is, in fact, a genuinely small error.

For both issues (2) and (3), we need a suitable norm to control the error terms and develop a stability theory for the loop hierarchy. In dimensions $d \in \{1, 2\}$, the natural choice is the L^∞-norm, justified by the fact that Θ_t is approximately constant up to a cutoff. Surprisingly, the loop hierarchy remains stable in dimensions $d \geq 3$ under this norm. In dimensions $d \geq 3$, the propagator satisfies

$$\Theta_{t,ab} \lesssim (|a-b|+1)^{-d+2} e^{-c|a-b|/\ell_t}$$

for some constant $c > 0$, where $\ell_t = \min(|1-t|^{-1/2}, L)$. The tree approximation $\mathcal{K}^{(n)}_{a_1,\dots,a_n}$ is a complicated function of the propagator entries. Hence, the L^∞-norm is a very crude norm for these functions, and one would not a priori expect stability in this norm. For clarity, in the following heuristic discussion, we assume $\operatorname{Im} z_t \geq L^{-2}$, so that $\ell_t^{-2} = |1-t| \asymp \operatorname{Im} z_t$. The key new ingredient enabling L^∞-stability of the loop hierarchy in dimensions $d \geq 3$ is a class of *loop-contraction inequalities* (Lemma 4.1), which extend the classical Ward's identity:

$$\sum_y |G_{xy}(z)|^2 = \sum_y |G_{yx}(z)|^2 = \operatorname{Im} G_{xx} / \operatorname{Im} z, \tag{1.9}$$

and apply to partial sums involving the absolute values of the n-G-loops.

However, the L^∞-stability of the tree approximation remains incomplete without a sufficiently precise *pointwise estimate* $\mathcal{L}^{(2)}_{ab} \approx \mathcal{K}^{(2)}_{ab}$ for the 2-G-loops, that is,

$$\left|(\mathcal{L}-\mathcal{K})^{(2)}_{ab}\right| \ll W^{-d}\Theta_{t,ab}, \quad \forall a, b \in \mathbb{Z}^d_L. \tag{1.10}$$

Such a strong pointwise control is essential because of the leading term $\mathcal{L}^{(2)} \diamond \mathcal{L}^{(n)}$ in the loop hierarchy (1.5). Subtracting the corresponding term $\mathcal{K}^{(2)} \diamond \mathcal{K}^{(n)}$ from the convolution tree equation (1.8) produces three terms $\mathcal{K}^{(2)} \diamond (\mathcal{L}-\mathcal{K})^{(n)}$, $(\mathcal{L}-\mathcal{K})^{(2)} \diamond \mathcal{K}^{(n)}$, and $(\mathcal{L}-\mathcal{K})^{(2)} \diamond (\mathcal{L}-\mathcal{K})^{(n)}$. While it is intuitively clear that the last two terms are negligible compared to the first, a rigorous justification requires establishing the stability estimate (1.10). This requires a significantly more delicate analysis of the equation (1.5) for $n = 2$. One main difficulty is that this equation involves $\mathcal{L}^{(n)}$ with $n \geq 3$, for which only L^∞-bounds are available. Therefore, we need to develop a method to estimate the error terms in the 2-loop equation using only the L^∞-bounds on the higher-order loops. We will discuss the details of the L^∞-stability and loop-contraction inequalities in Section 4, and the proof of the pointwise stability estimate (1.10) in Section 3.3.

Issue (3) involves bounding the martingale term $\mathrm{d}\mathcal{E}^{M,(n)}$ and the light-weight term $\mathcal{E}^{\mathring{G},(n)}$. In dimensions $d \geq 3$, estimating these terms presents a serious challenge. We will discuss the difficulties related to bounding the martingale term in Section 3.5. For now, we focus on estimating the light-weight term for $n = 2$:

$$\mathcal{E}^{\mathring{G},(2)}_{ab} = W^d \sum_{u,v} S^{(\mathsf{B})}_{uv} \cdot \operatorname{Tr}[\mathring{G}_t E_u] \cdot \mathcal{L}^{(3)}_{a,b,v}, \quad \text{with} \quad \mathcal{L}^{(3)}_{a,b,v} = W^{-3d} \sum_{x\in[a],y\in[b],z\in[v]} (G_t)_{xy}(G_t)_{yz}(G_t)_{zx},$$

where $x \in [a]$ means that the lattice point x belongs to the block labeled by a. We expect an averaged local law to hold in the form $|\operatorname{Tr}[\mathring{G}_t E_u]| \leq W^{-c}$ for some constant $c > 0$. Next, assuming the *sharp entrywise local law* for G_t:

$$|(G_t)_{xy} - m\delta_{xy}| \lesssim W^{-d/2}\big(\Theta_{t,ab}\big)^{1/2}, \quad \forall x \in [a],\ y \in [b], \tag{1.11}$$

we obtain the estimate

$$\begin{aligned} |\mathcal{E}^{\mathring{G},(2)}_{ab}| &\lesssim W^{-c-d/2} \sum_{|v-a|\vee|v-b|\lesssim \ell_t} (\Theta_{t,ab}\Theta_{t,bv}\Theta_{t,va})^{1/2} \\ &\lesssim W^{-c-d/2}\big(\Theta_{t,ab}\big)^{1/2}\ell_t^2 = W^{-c-d/2}\big(\Theta_{t,ab}\big)^{1/2} \cdot (1-t)^{-1}. \end{aligned} \tag{1.12}$$

Upon integrating in time, the singularity $(1-t)^{-1}$ produces only a harmless logarithmic factor. However, the resulting bound contains only $(\Theta_{t,ab})^{1/2}$ rather than the optimal $\Theta_{t,ab}$. This gap cannot be compensated by the W^{-c} factor from the averaged local law, as c is at best $c = d$. Without the extra factor $(\Theta_{t,ab})^{1/2}$, the light-weight term would dominate the leading term and violate the approximation in (1.10). In the actual proof, the situation is even more delicate, since the estimate (1.11) is not available a priori.

To overcome this difficulty, we employ the *diagrammatic methods* developed in [65, 66, 67]. While this method is technically involved and was previously applied only in high dimensions $d \geq 7$, our objective here is more modest: we aim to generate an additional "*long edge*"—that is, a resolvent entry $G_{x'y'}$ with $|x'-y'| \gtrsim W|a-b|$—in order to recover the missing factor $(\Theta_{t,ab})^{1/2}$. For this purpose, the diagrammatic expansions can be extended to all dimensions $d \geq 3$.

Specifically, we apply Gaussian integration by parts to express the expectation of high moments $\mathbb{E}|\mathcal{E}^{\mathring{G},(2)}_{ab}|^{2p}$ as a sum of graphs, where each graph either contains $2p$ *additional long edges* or contributes to a sufficiently small error. Roughly speaking, each application of Gaussian integration by parts produces a $W^{-d/2}$ factor, unless it generates a long edge. To ensure that these $W^{-d/2}$ factors compensate for the missing $(\Theta_{t,ab})^{1/2}$ when $|a-b| \asymp L$, we must expand the graphs to a sufficiently high order k, chosen so that $W^{-kd/2} \leq (\Theta_{t,ab})^p$. As a result, the contributing graphs are extremely complicated. To bound them, we need to sum over their

internal vertices in such a way that the final contribution retains at least the $2p$ additional long edges. We will use Ward's identity (1.9) as a key tool to control the summation over internal vertices, provided the order of the summations follows a carefully designed "nested order". This nested ordering is a crucial component in our estimates of the graphs arising from the complicated diagrammatic expansions. See Section 7.2 for a detailed discussion.

In addition to the above considerations, two key ingredients continue to play a central role in our analysis: the *sum-zero property* [65, 66, 69], and the CLT mechanism for the fluctuation cancellation of resolvents, which was previously employed in the $d = 2$ analysis [28]. Our final objective is to incorporate all these components—diagrammatic tools, the sum-zero property, fluctuation cancellation, and the stability of the tree approximation—into the flow-based framework.

In summary, our proof relies on a flow-based analysis of the tree approximation to the G-loop hierarchy, combining *sharp max-norm estimates* for higher-order $(\mathcal{L} - \mathcal{K})^{(n)}$-loops with $n \geq 3$ and a *precise pointwise estimate* for the $(\mathcal{L} - \mathcal{K})^{(2)}$-loops. It is perhaps surprising that this seemingly "simple" strategy succeeds without tracking the precise pointwise decay of higher-order G-loops. This is made possible by several key technical inputs, including *loop-contraction inequalities* and *diagrammatic techniques*. The main advantage of our approach is that it closes the analysis using only a minimal set of technical ingredients, thereby avoiding an unnecessarily complicated treatment in dimensions $d \geq 3$. A complete implementation of this strategy will be presented in the subsequent sections after we state the main results in Section 2.

Notations. To facilitate the presentation, we introduce some necessary notations that will be used throughout this paper. We will use the set of natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$ and the upper half complex plane $\mathbb{C}_+ := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. In this paper, we are interested in the asymptotic regime with $N \to \infty$. When we refer to a constant, it will not depend on N or W. Unless otherwise noted, we will use C, D etc. to denote large positive constants, whose values may change from line to line. Similarly, we will use ε, δ, τ, c, $\mathfrak{c}$, $\mathfrak{d}$ etc. to denote small positive constants. For any two (possibly complex) sequences ξ_N and ζ_N depending on N, $\xi_N = \mathrm{O}(\zeta_N)$, $\zeta_N = \Omega(\xi_N)$, or $\xi_N \lesssim \zeta_N$ means that $|\xi_N| \leq C|\zeta_N|$ for some constant $C > 0$, whereas $\xi_N = \mathrm{o}(\zeta_N)$ or $|\xi_N| \ll |\zeta_N|$ means that $|\xi_N|/|\zeta_N| \to 0$ as $N \to \infty$. We say that $\xi_N \asymp \zeta_N$ if $\xi_N = \mathrm{O}(\zeta_N)$ and $\zeta_N = \mathrm{O}(\xi_N)$. For any $\alpha, \beta \in \mathbb{R}$, we denote $[\![\alpha, \beta]\!] := [\alpha, \beta] \cap \mathbb{Z}$, $[\![\alpha]\!] := [\![1, \alpha]\!]$, $\alpha \vee \beta := \max\{\alpha, \beta\}$, and $\alpha \wedge \beta := \min\{\alpha, \beta\}$. Given a vector $\mathbf{v}$, $|\mathbf{v}| \equiv \|\mathbf{v}\|_2$ denotes the Euclidean norm and $\|\mathbf{v}\|_p$ denotes the L^p-norm. Given a matrix $\mathcal{A} = (\mathcal{A}_{ij})$, $\|\mathcal{A}\|$, $\|\mathcal{A}\|_{p\to p}$, and $\|\mathcal{A}\|_\infty \equiv \|\mathcal{A}\|_{\max} := \max_{i,j} |\mathcal{A}_{ij}|$ denote the operator (i.e., $L^2 \to L^2$) norm, $L^p \to L^p$ norm (where we allow $p = \infty$), and maximum (i.e., L^∞) norm, respectively. We will use $\mathcal{A}_{ij}$ and $\mathcal{A}(i, j)$ interchangeably in this paper. We will use I_n to denote an $n \times n$ identity matrix.

Given an event Ξ, let $\mathbf{1}_\Xi$ or $\mathbf{1}(\Xi)$ denote its indicator function. We will say an event Ξ holds with high probability (w.h.p.) if for any constant $D > 0$, $\mathbb{P}(\Xi) \geq 1 - N^{-D}$ for large enough N. More generally, we say an event Ω holds *w.h.p.* in Ξ if for any constant $D > 0$, $\mathbb{P}(\Xi \setminus \Omega) \leq N^{-D}$ for large enough N. For clarity of presentation, we will use the following notion of stochastic domination introduced in [31]. Let

$$\xi = \left(\xi^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)}\right), \quad \zeta = \left(\zeta^{(N)}(u) : N \in \mathbb{N}, u \in U^{(N)}\right),$$

be two families of non-negative random variables, where $U^{(N)}$ is a possibly N-dependent parameter set. We say ξ is stochastically dominated by ζ, uniformly in u, if for any fixed (small) $\tau > 0$ and (large) $D > 0$,

$$\mathbb{P}\left(\bigcup_{u \in U^{(N)}} \left\{ \xi^{(N)}(u) > N^\tau \zeta^{(N)}(u) \right\} \right) \leq N^{-D} \tag{1.13}$$

for large enough $N \geq N_0(\tau, D)$, and we will use the notation $\xi \prec \zeta$. If for some complex family ξ we have $|\xi| \prec \zeta$, then we will also write $\xi \prec \zeta$ or $\xi = \mathrm{O}_\prec(\zeta)$. As a convention, for two *deterministic* non-negative quantities ξ and ζ, we will write $\xi \prec \zeta$ if and only if $\xi \leq N^\tau \zeta$ for any constant $\tau > 0$.

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2. The model and main results

We focus in this paper on the classical block random band matrix model (i.e., the *Wegner orbital model*) and the *block Anderson model* with identity interactions. As noted in the introduction, these are the models studied in [52], where localization of bulk eigenvectors was established under the condition $\lambda \gg W^{d/2}$. In contrast, the present work establishes a complementary result describing the delocalized regime of the Wegner orbital and block Anderson models when $\lambda \ll W^{d/2}$. Our approach can be readily extended to a much

broader class of random band matrices and block Anderson models with general (non-identity) interactions—by combining the techniques developed here with those from [59] and the block reduction method of [37]. However, the models considered in this paper allow us to avoid inessential technical complications and to focus on the core challenges that arise in dimensions $d \geq 3$. Extensions to more general settings will be addressed in future work. For clarity of presentation, we rescale our models by

$$g = \lambda^{-1}, \tag{2.1}$$

and consider matrices of the form $H = V + g\Psi$. This rescaling has the advantage that, in the regime $g \ll 1$ (i.e., $\lambda \gg 1$), the density of states of H may be viewed as a small perturbation of the semicircle law.

Our models are defined on a d-dimensional discrete torus $\mathbb{Z}^d_{WL} \subset \mathbb{Z}^d$ with $d \geq 3$, consisting of N lattice points and side length WL:

$$\mathbb{Z}^d_{WL} := [\![-(WL)/2 + 1, (WL)/2]\!]^d \quad \text{with} \quad N = (W \cdot L)^d.$$

We partition $\mathbb{Z}^d_{WL}$ into L^d disjoint blocks, indexed by $\mathbb{Z}^d_L := [\![-L/2 + 1, L/2]\!]^d$, which we refer to as the *block lattice*. Each index $a = (a(1), \dots, a(d)) \in \mathbb{Z}^d_L$ corresponds to a block

$$[a] := \prod_{i=1}^{d} [\![(a(i) - 1)W + 1,\ a(i)W]\!], \tag{2.2}$$

where $a(i)$ denotes the i-th coordinate of a for $i \in [\![d]\!]$.[1] We view both $\mathbb{Z}^d_{WL}$ and $\mathbb{Z}^d_L$ as discrete d-dimensional tori. We will denote the vertices of $\mathbb{Z}^d_{WL}$ by $x, y, \dots$, and denote those of $\mathbb{Z}^d_L$ by $a, b, \dots$ Given $x, y \in \mathbb{Z}^d_{WL}$ and $a, b \in \mathbb{Z}^d_L$, we denote the periodic representatives of $x - y$ and $a - b$ by $(x - y)_{WL}$ and $(a - b)_L$, respectively:

$$(x - y)_{WL} := \big((x - y) + (WL)\mathbb{Z}^d\big) \cap \mathbb{Z}^d_{WL}, \quad (a - b)_L := \big((a - b) + L\mathbb{Z}^d\big) \cap \mathbb{Z}^d_L. \tag{2.3}$$

For definiteness, we use the L^∞-metric to define (periodic) distances on $\mathbb{Z}^d_{WL}$ and $\mathbb{Z}^d_L$:

$$|x - y| \equiv \|(x - y)_{WL}\|_\infty, \quad \forall x, y \in \mathbb{Z}^d_{WL}, \quad \text{and} \quad |a - b| \equiv \|(a - b)_L\|_\infty, \quad \forall a, b \in \mathbb{Z}^d_L.$$

We write $x \sim y$ if x and y are neighbors in $\mathbb{Z}^d_{WL}$, and similarly $a \sim b$ if a and b are neighbors in $\mathbb{Z}^d_L$.

2.1. **Main results for the random band matrix model.** Our *random band matrix* (or referred to as the *Wegner orbital model* following [52]) is defined by a complex Hermitian random block Hamiltonian $H = (H_{xy} : x, y \in \mathbb{Z}^d_{WL})$, where the entries H_{xy} are independent (up to the Hermitian symmetry $H_{xy} = \overline{H}_{yx}$) Gaussian random variables. More precisely, given a symmetric doubly stochastic variance matrix $S = (S_{xy} : x, y \in \mathbb{Z}^d_{WL})$, the diagonal entries of H are real Gaussian random variables, and the off-diagonal entries are complex Gaussian random variables, distributed as follows:

$$H_{xy} \sim \mathcal{N}_{\mathbb{R}}(0, S_{xy}) \cdot \mathbf{1}_{x=y} + \mathcal{N}_{\mathbb{C}}(0, S_{xy}) \cdot \mathbf{1}_{x \neq y}. \tag{2.4}$$

For $x \in [a]$ and $y \in [b]$, the variance matrix $S \equiv S(g)$ is given by

$$S_{xy} \equiv \mathrm{Var}(H_{xy}) := W^{-d} S^{(\mathsf{B})}_{ab}(g), \quad \text{with} \quad S^{(\mathsf{B})}_{ab}(g) := \frac{1}{1 + 2dg^2} \mathbf{1}_{a=b} + \frac{g^2}{1 + 2dg^2} \mathbf{1}_{a \sim b}, \tag{2.5}$$

where $g > 0$ (recall (2.1)) is a coupling parameter that quantifies the interaction strength between neighboring blocks, and $S^{(\mathsf{B})}(g)$ denotes an $L^d \times L^d$ matrix. Informally, the matrix H consists of i.i.d. GUE blocks on the diagonal and i.i.d. Ginibre blocks (up to Hermitian symmetry and scaling) on the off-diagonal. Note that when $g = 1$, the present model reduces to the standard block random band matrices studied in [69, 28].

Let $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$ denote the eigenvalues of H. The corresponding normalized eigenvectors of H are denoted by $(\boldsymbol{\psi}_k)_{k=1}^N$. It is well-known that the empirical spectral measure $N^{-1} \sum_{k=1}^N \delta_{\lambda_k}$ converges almost surely to the Wigner semicircle law [63] with density $\rho_{\mathrm{sc}}(x) = \sqrt{(4 - x^2)_+}/2\pi$. Define the Green's function (or resolvent) of the Hamiltonian H as

$$G(z) := (H - z)^{-1}, \quad z \in \mathbb{C}_+. \tag{2.6}$$

[1] In the above definitions, we implicitly assume that L is even; the case of odd L can be treated similarly by defining the block lattice as $\mathbb{Z}^d_L := [\![-(L-1)/2, (L-1)/2]\!]^d$.

It is also known that as $N \to \infty$, $G(z)$ converges to the scalar matrix $m(z)I_N$ entrywise, where $m(z)$ denotes the Stieltjes transform of ρ_{sc}, defined by

$$m(z) \equiv m_{\mathrm{sc}}(z) := \frac{-z + \sqrt{z^2 - 4}}{2} = \int_{\mathbb{R}} \frac{\rho_{\mathrm{sc}}(x)}{x - z} \mathrm{d}x. \tag{2.7}$$

Moreover, we define the matrix[2]

$$M(z) \equiv M_N(z) := m(z)I_N. \tag{2.8}$$

We now state the main results for the random band matrix model. Our first main result establishes the delocalization of the bulk eigenvectors of H in dimensions $d \geq 3$.

Theorem 2.1 (Delocalization)**.** *Fix any dimension $d \geq 3$, and consider the random band matrix model defined above. Assume there exist constants $\mathfrak{c}, \mathfrak{d} > 0$ such that*

$$W \geq N^{\mathfrak{c}}, \tag{2.9}$$

and that g satisfies the condition

$$W^{-d/2+\mathfrak{d}} \leq g \leq \mathfrak{d}^{-1}. \tag{2.10}$$

Then, for any small constants $\kappa, \tau > 0$ and large constant $D > 0$, the following delocalization estimate holds, provided N is sufficiently large:

$$\mathbb{P}\left(\max_{k:|\lambda_k| \leq 2-\kappa} \|\boldsymbol{\psi}_k\|_\infty^2 \leq N^{-1+\tau} \right) \geq 1 - N^{-D}. \tag{2.11}$$

Since the eigenvectors $\boldsymbol{\psi}_k$ are L^2-normalized, the L^∞-bound in (2.11) implies that bulk eigenvectors have localization length at least $\Omega((WL)^{1-\varepsilon})$, for any small constant $\varepsilon > 0$. The upper bound in condition (2.10) is not essential; we include it only for clarity of presentation, as our main interest is the small g regime.[3] The key aspect of condition (2.10) is the lower bound $g \geq W^{-d/2+\mathfrak{d}}$. Indeed, by the fractional moment method [3], it was shown in [52] that when $g \ll W^{-d/2}$, the eigenvectors of both the random band matrix model and the block Anderson model (defined in Section 2.2 below) are localized with localization length of order $\mathrm{O}(W)$. Combined with this result, Theorem 2.1 demonstrates a localization–delocalization transition for the random band matrix model in the bulk spectrum as g crosses the critical threshold $W^{-d/2}$.

Theorem 2.1 follows immediately from the following sharp local law for the Green's function of H, defined as in (2.6). For simplicity of notation, given constants $\kappa, \varepsilon > 0$, define the spectral domain

$$\mathbf{D}_{\kappa,\varepsilon} := \{z = \hat{E} + \mathrm{i}\eta \in \mathbb{C}_+ : |\hat{E}| \leq 2 - \kappa, N^{-1+\varepsilon} \leq \eta \leq 1\}, \tag{2.12}$$

and the simplified notation

$$\mathcal{B}_{\eta,K} := \frac{(g^2 + \eta)^{-1}}{W^2(K + W)^{d-2}} + \frac{1}{N\eta}, \quad \forall K \geq 0. \tag{2.13}$$

Theorem 2.2 (Local semicircle law)**.** *In the setting of Theorem 2.1, for any (small) constants $\kappa, \varepsilon, \tau > 0$ and (large) constant $D > 0$, the following events hold with probability $\geq 1 - N^{-D}$ for large enough N:*

$$\bigcap_{z=\hat{E}+\mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}} \bigcap_{x,y \in \mathbb{Z}^d_{WL}} \left\{ |G_{xy}(z) - M_{xy}(z)|^2 \leq W^\tau \mathcal{B}_{\eta,|x-y|} \right\}, \tag{2.14}$$

$$\bigcap_{z=\hat{E}+\mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}} \left\{ \max_{a \in \mathbb{Z}^d_L} \left| W^{-d} \sum_{x \in [a]} G_{xx}(z) - m(z) \right| \leq W^\tau \mathcal{B}_{\eta,0} \right\}, \tag{2.15}$$

where G is defined in (2.6), $m(z)$ and $M(z)$ are defined in (2.7) and (2.8), and $[a]$ is defined in (2.2).

Proof of Theorem 2.1. Theorem 2.1 follows directly from the entrywise local law (2.14) via the bound

$$|\boldsymbol{\psi}_k(x)|^2 \leq \eta \operatorname{Im} G_{xx}(\lambda_k + \mathrm{i}\eta), \quad \forall \eta > 0\,. \tag{2.16}$$

Applying the local law (2.14) to $G_{xx}(\lambda_k + \mathrm{i}\eta)$ with $\eta = N^{-1+\tau}$, we deduce that $\operatorname{Im} G_{xx}(\lambda_k + \mathrm{i}\eta) = \mathrm{O}(1)$ with high probability. Combined with (2.16), this completes the proof. □

[2] We introduce this notation to maintain consistency with the block Anderson model, where $M(z)$ is no longer a scalar matrix proportional to $m(z)$; see (2.32) and (2.33).

[3] All results below remain valid for $g \gg 1$, provided g in the relevant equations is replaced by $g \wedge 1$.

Under the assumptions of Theorem 2.1, we can further establish a stronger *quantum unique ergodicity* (QUE) estimate for the bulk eigenvectors of H, albeit at the cost of a slightly weaker probability bound. Roughly speaking, the QUE estimates (2.18) and (2.19) below indicate that every bulk eigenvector of H is asymptotically uniformly distributed (in the sense of L^2-mass) across all scales larger than W. In particular, this implies that the localization length of every bulk eigenvector is of order $\Omega(WL)$ with probability $1-o(1)$.

Theorem 2.3 (Quantum unique ergodicity)**.** *In the setting of Theorem 2.1, given $E \in [-2+\kappa, 2-\kappa]$ and a constant $\varepsilon_0 \in (0, \mathfrak{d}/2)$, define the interval*

$$\mathcal{I}_E \equiv \mathcal{I}_E(\varepsilon_0) := \left\{ x : |x-E| \le W^{-\varepsilon_0}\left(gW^{d/2}/N\right) \right\}. \tag{2.17}$$

For each $d \ge 3$ and constant $0 < c < \varepsilon_0 \wedge (\mathfrak{d}/5)$, the following estimate holds for any small constant $\tau > 0$:

$$\sup_{E:|E|\le 2-\kappa} \max_{a\in\mathbb{Z}_L^d} \mathbb{P}\left(\max_{i,j:\lambda_i,\lambda_j\in\mathcal{I}_E} \left| \sum_{x\in[a]} \overline{\boldsymbol{\psi}}_i(x)\boldsymbol{\psi}_j(x) - \frac{W^d}{N}\delta_{ij} \right| \ge \frac{W^{d-c}}{N} \right) \le W^{-(2\varepsilon_0)\wedge(2\mathfrak{d}/5)+2c+\tau} \tag{2.18}$$

provided N is large enough. More generally, for any subset $A \subset \mathbb{Z}_L^d$, we have

$$\sup_{E:|E|\le 2-\kappa} \mathbb{P}\left(\max_{k:\lambda_k\in\mathcal{I}_E} \left| \sum_{a\in A}\sum_{x\in[a]} |\boldsymbol{\psi}_k(x)|^2 - \frac{W^d}{N}|A| \right| \ge \frac{W^{d-c}}{N}|A| \right) \le W^{-(2\varepsilon_0)\wedge(2\mathfrak{d}/5)+2c+\tau}. \tag{2.19}$$

As an important consequence of the above QUE estimates, and by employing the Green's function comparison argument developed in [64], we can derive the following *universality of the local bulk eigenvalue statistics* for our random band matrices. Let $p_H(\lambda_1, \dots, \lambda_N)$ denote the joint symmetrized probability density of the (unordered) eigenvalues of H. For any $1 \le n \le N$, define the n-point correlation function as

$$p_H^{(n)}(\lambda_1, \cdots, \lambda_n) := \int_{\mathbb{R}^{N-n}} p_H(\lambda_1, \cdots, \lambda_N)\, \mathrm{d}\lambda_{n+1} \cdots \mathrm{d}\lambda_N.$$

Moreover, let $p_{\mathrm{GUE}}^{(n)}$ denote the corresponding n-point correlation function for an $N \times N$ GUE matrix.

Theorem 2.4 (Bulk universality)**.** *In the setting of Theorem 2.1, let $O \in C_c^\infty(\mathbb{R}^n)$ be an arbitrary smooth, compactly supported function. Then, for any $|E| \le 2-\kappa$ and fixed $n \in \mathbb{N}$, we have*

$$\lim_{N\to\infty} \int_{\mathbb{R}^n} \mathrm{d}\boldsymbol{\alpha}\, O(\boldsymbol{\alpha}) \left[p_H^{(n)} - p_{\mathrm{GUE}}^{(n)} \right] \left(E + \frac{\alpha_1}{N}, \dots, E + \frac{\alpha_n}{N} \right) = 0, \tag{2.20}$$

where $\boldsymbol{\alpha}$ denotes $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n)$.

This bulk universality result shows that, for our random band matrices, the local eigenvalue gap statistics near any fixed bulk energy level E asymptotically coincide with those of Wigner matrices. However, we note that the distribution of an individual bulk eigenvalue λ_k of H may differ significantly from that of Wigner matrices, as λ_k can exhibit fluctuations that are much larger than N^{-1}.

Similar to the one- and two-dimensional cases [69, 28, 35, 59], our random band matrix model in dimensions $d \ge 3$ also satisfies the *quantum diffusion conjecture*, which serves as a key input for establishing the QUE estimates in Theorem 2.3. To state it, we define the following Θ-matrices:

$$\Theta^{(+,-)}(z) = \Theta^{(-,+)}(z) := \frac{1}{1-|m(z)|^2 S^{(\mathsf{B})}}, \quad \Theta^{(+,+)}(z) = (\Theta^{(-,-)}(z))^* := \frac{1}{1-m(z)^2 S^{(\mathsf{B})}}, \tag{2.21}$$

where we recall the matrix $S^{(\mathsf{B})}$ defined in (2.5) and $m(z)$ defined in (2.7).

Theorem 2.5 (Quantum diffusion)**.** *In the setting of Theorem 2.1, for any (small) constants $\kappa, \varepsilon, \tau > 0$ and (large) constant $D > 0$, the following events hold with probability $\ge 1 - N^{-D}$ for all $a, b \in \mathbb{Z}_L^d$ and for large enough N:*

$$\bigcap_{z=\hat{E}+\mathrm{i}\eta\in\mathbf{D}_{\kappa,\varepsilon}} \left\{ \left| \frac{1}{W^{2d}} \sum_{x\in[a],y\in[b]} |G_{xy}(z)|^2 - \frac{|m|^2\Theta_{ab}^{(+,-)}(z)}{W^d} \right| \le W^\tau \left[\left((\mathcal{B}_{\eta,0})^{\frac{1}{5}} \mathcal{B}_{\eta,W|a-b|} \right) \wedge (\mathcal{B}_{\eta,0})^2 \right] \right\}, \tag{2.22}$$

$$\bigcap_{z=\hat{E}+\mathrm{i}\eta\in\mathbf{D}_{\kappa,\varepsilon}} \left\{ \left| \frac{1}{W^{2d}} \sum_{x\in[a],y\in[b]} (G_{xy}G_{yx})(z) - \frac{m^2\Theta_{ab}^{(+,+)}(z)}{W^d} \right| \le W^\tau \left[\left((\mathcal{B}_{\eta,0})^{\frac{1}{5}} \mathcal{B}_{\eta,W|a-b|} \right) \wedge (\mathcal{B}_{\eta,0})^2 \right] \right\}. \tag{2.23}$$

Moreover, for each $z = \hat{E}+\mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$, the expectations $\mathbb{E}|G_{xy}(z)|^2$ and $\mathbb{E}\left(G_{xy}(z)G_{yx}(z)\right)$ satisfy the following bounds for any small constant $\tau > 0$ and large enough N:

$$\max_{a,b}\left|\frac{1}{W^{2d}}\sum_{x\in[a],y\in[b]}\mathbb{E}|G_{xy}(z)|^2 - \frac{|m|^2\Theta_{ab}^{(+,-)}(z)}{W^d}\right| \le W^{\tau}(\mathcal{B}_{\eta,0})^2\left(\left(g^2W^d\right)^{-1/5}+\mathcal{B}_{\eta,0}\right), \tag{2.24}$$

$$\max_{a,b}\left|\frac{1}{W^{2d}}\sum_{x\in[a],y\in[b]}\mathbb{E}(G_{xy}G_{yx})(z) - \frac{m^2\Theta_{ab}^{(+,+)}(z)}{W^d}\right| \le W^{\tau}(\mathcal{B}_{\eta,0})^2\left(\left(g^2W^d\right)^{-1/5}+\mathcal{B}_{\eta,0}\right). \tag{2.25}$$

Note that when $\eta \le g^2/L^d$, we have

$$\mathcal{B}_{\eta,K} \asymp (N\eta)^{-1} \ge (g^2W^d)^{-1}, \quad \forall K \ge 0. \tag{2.26}$$

In this case, the expected estimates in (2.24) and (2.25) provide improvements over those in (2.22) and (2.23), which in turn lead to the QUE estimates in Theorem 2.3.

Proof of Theorem 2.3. With the quantum diffusion estimates (2.24) and (2.25), the proof of Theorem 2.3 is similar to those for [69, Theorem 2.5], [28, Theorem 2.4], and [59, Theorem 2.2]. We now briefly outline the proof without giving all details.

Take $z = E+\mathrm{i}\eta$ with $|E| \le 2-\kappa$ and $\eta = W^{-\varepsilon_0}(gW^{d/2}/N) \ge W^{\mathfrak{d}-\varepsilon_0}/N$. With the spectral decomposition of $G(z)$ and the definition of $\mathcal{I}_E$, we find that

$$\begin{aligned}\mathbb{E}\sum_{i,j:\lambda_i,\lambda_j\in\mathcal{I}_E}\left|\boldsymbol{\psi}_i^*\left(E_a-N^{-1}\right)\boldsymbol{\psi}_j\right|^2 &\lesssim \eta^2\mathbb{E}\operatorname{Tr}\left[\operatorname{Im}G(z)\left(E_a-N^{-1}\right)\operatorname{Im}G(z)\left(E_a-N^{-1}\right)\right]\\ &= \frac{\eta^2}{L^{2d}}\sum_{b,b'\in\mathbb{Z}_L^d}\mathbb{E}\operatorname{Tr}\left[\operatorname{Im}G(z)\left(E_a-E_b\right)\operatorname{Im}G(z)\left(E_a-E_{b'}\right)\right],\end{aligned} \tag{2.27}$$

where we recall that E_a is the block-averaging matrix restricted to the a-th block: $(E_a)_{xy} = W^{-d}\mathbf{1}(x=y\in[a])$. Expanding $\operatorname{Im}G$ as $\operatorname{Im}G = (G-G^*)/(2\mathrm{i})$ and applying the QUE estimates (2.24) and (2.25), we can bound the right-hand side of (2.27) by

$$W^{\tau}\eta^2\left(\frac{1}{N\eta}\right)^2\left[\frac{1}{N\eta}+\frac{1}{(g^2W^d)^{1/5}}\right]+\frac{C\eta^2}{W^d}\max_{\sigma,\sigma'\in\{+,-\}}\max_{a,b,b'}\left|\Theta_{ab}^{(\sigma,\sigma')}-\Theta_{ab'}^{(\sigma,\sigma')}\right| \tag{2.28}$$

for any small constant $\tau > 0$ and a large constant $C > 0$, where the first term comes from the application of (2.24)–(2.26). Then, with the estimate (2.70) below for the Θ-matrices (note that $\Theta_{ab}^{(\sigma,\sigma')}-\Theta_{ab'}^{(\sigma,\sigma')} = \mathring{\Theta}_{ab}^{(\sigma,\sigma')}-\mathring{\Theta}_{ab'}^{(\sigma,\sigma')}$ by (2.61)), we get

$$\max_{\sigma,\sigma'\in\{+,-\}}\max_{a,b,b'}\left|\Theta_{ab}^{(\sigma,\sigma')}-\Theta_{ab'}^{(\sigma,\sigma')}\right| \prec g^{-2}.$$

Plugging it into (2.28) and further into (2.27) yields that

$$\mathbb{E}\sum_{i,j:\lambda_i,\lambda_j\in\mathcal{I}_E}\left|\boldsymbol{\psi}_i^*\left(E_a-N^{-1}\right)\boldsymbol{\psi}_j\right|^2 \lesssim W^{\tau}N^{-2}\cdot\left(W^{-\mathfrak{d}+\varepsilon_0}+W^{-2\mathfrak{d}/5}+W^{-2\varepsilon_0}\right).$$

Finally, applying Markov's inequality concludes (2.18). The proof of (2.19) follows a similar argument, where we simply replace E_a in (2.27) with $|A|^{-1}\sum_{a\in A}E_a$, after which all subsequent arguments remain valid. □

Proof of Theorem 2.4. Using Theorem 2.1, Theorem 2.2, and the QUE estimate (2.18) as inputs, the proof of Theorem 2.4 follows from the Green's function comparison method developed in [64]. The argument is essentially identical to that used for one-dimensional [69, Theorem 2.6] and two-dimensional [28, Theorem 2.6] random band matrices. For instance, adapting the proof of [28, Theorem 2.6], we only need to adjust certain parameters in the proofs of equations (2.22) and (2.23) therein. In that setting, given $y \in \mathbb{Z}_{WL}^d$, the "bad" event $\mathcal{B} \equiv \mathcal{B}(y)$ was defined by the existence of an index α such that $|\lambda_\alpha - E| \le N^{-1+\mathfrak{c}/6}$ and $|M_{y,\alpha}| \ge N^{-\mathfrak{c}/18}$ ($\mathfrak{c}$ is the constant in the assumption $W \ge N^{\mathfrak{c}}$ and $M_{y,\alpha}$ is defined below equation (2.24) of

[28]).[4] In our setting, we modify the definition of $\mathcal{B}$ to $\mathcal{B}(y) := \{\exists \alpha : |\lambda_\alpha - E| \le N^{-1}W^{\mathfrak{d}/3}, |M_{y,\alpha}| \ge W^{-\mathfrak{d}/6}\}$. Applying (2.18) with $\varepsilon_0 = \mathfrak{d}/3$ and $c = \mathfrak{d}/6$, we can deduce that $\mathbb{P}(\mathcal{B}) \le W^{-\mathfrak{d}/15+\tau}$. With these modified parameters and the new definition of $\mathcal{B}$, the remainder of the proof of (2.20) proceeds exactly as in the proof of [28, Theorem 2.6]. Hence, we omit further details. □

2.2. **Main results for the block Anderson model.** We next define the block Anderson model. We begin by introducing the *random block potential* V, an $N \times N$ complex Hermitian random block matrix whose diagonal blocks are i.i.d. GUE matrices. In other words, V can be regarded as a random band matrix whose entries are distributed according to (2.4), with variance matrix $S = (S_{xy})$ given by:

$$S_{xy} \equiv \mathrm{Var}(V_{xy}) := W^{-d} S^{(\mathsf{B})}_{ab}(0) = W^{-d}\mathbf{1}(a = b), \quad \text{for } x \in [a],\ y \in [b], \tag{2.29}$$

where $S^{(\mathsf{B})}(0) = I_{L^d}$ denotes the matrix defined in (2.5) with $g = 0$. Next, we define the *block Anderson model* as a random block Schrödinger operator of the form

$$H \equiv H(g) = V + g\Psi, \tag{2.30}$$

where $g > 0$ (recall (2.1)) is a coupling parameter, and Ψ represents the interaction Hamiltonian that introduces hopping between neighboring blocks. For definiteness, we focus on the classical block Anderson model in which each block is the identity matrix:

$$\Psi = \Psi^{(\mathsf{B})} \otimes I_{W^d}, \quad \text{where} \quad \Psi^{(\mathsf{B})}_{ab} = \mathbf{1}(a \sim b). \tag{2.31}$$

For the block Anderson model, we again denote its eigenvalues by $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$, and the corresponding normalized eigenvectors by $(\boldsymbol{\psi}_k)_{k=1}^N$. The Green's function $G(z)$ is defined analogously to (2.6).

Remark 2.6. For simplicity, we have slightly abused notation by using the same symbols (such as H, S, λ_i, $\boldsymbol{\psi}_i$, and $G(z)$ defined above, as well as $m(z)$ and $M(z)$ defined below) for both the random band matrix model and the block Anderson model. When a distinction is needed, we will add a superscript BA to denote quantities associated with the block Anderson model.

It is well-known that as $N \to \infty$, the empirical spectral measure $N^{-1}\sum_{k=1}^N \delta_{\lambda_k}$ is well approximated by a deterministic probability measure μ_N, known as the free convolution of the semicircle law and the empirical measure of $g\Psi$. This measure has a continuous probability density $\rho_N(x)$ on $\mathbb{R}$ [11], whose support is contained in $[-2-2dg, 2+2dg]$. (For finite L, this support need not be symmetric about 0, e.g., when L is odd, and it may have gaps.) The Stieltjes transform $m(z) \equiv m(z,g)$ of the measure μ_N is defined as the unique solution to the following self-consistent equation

$$\frac{1}{N}\,\mathrm{Tr}\,\frac{1}{g\Psi - z - m(z)} = m(z) \tag{2.32}$$

such that $\mathrm{Im}\, m(z) > 0$ for $z \in \mathbb{C}_+$. In addition, we define the $N \times N$ matrix $M(z) \equiv M_N(z,g)$ and the $L^d \times L^d$ matrix $M^{(\mathsf{B})}(z) \equiv M^{(\mathsf{B})}_L(z,g)$ as

$$M(z) := \frac{1}{g\Psi - z - m(z)} = M^{(\mathsf{B})}(z) \otimes I_{W^d}, \quad M^{(\mathsf{B})}(z) := \frac{1}{g\Psi^{(\mathsf{B})} - z - m(z)}. \tag{2.33}$$

Note M and $M^{(\mathsf{B})}$ are both (complex) symmetric matrices. In the random matrix theory literature (see e.g., [49, 46, 44, 8, 34] for various settings of deformed Wigner-type matrices), the Green's function $G(z)$ is known to converge to $M(z)$ in the sense of local laws. (However, existing convergence estimates in the literature are generally non-optimal in the non-mean-field setting $W^d \ll N$.)

Theorem 2.7 (Main results for the block Anderson model)**.** *Fix any dimension $d \ge 3$, and assume there exist constants $\mathfrak{c}, \mathfrak{d} > 0$ such that (2.9) and (2.10) hold. Then, for the block Anderson model defined above, the following results remain valid if, throughout their statements, we replace the energy condition $|E| \le 2-\kappa$*

[4] In dimension $d \ge 3$, $M_{y,\alpha}$ is defined with the weights of the variance matrix S: $M_{y,\alpha} := N\sum_{x \in \mathbb{Z}^d_{WL}} S_{xy}|\boldsymbol{\psi}_\alpha(x)|^2 - 1 = \sum_{b \in \mathbb{Z}^d_L} S^{(\mathsf{B})}_{a_y b}(NW^{-d}\sum_{x \in [b]}|\boldsymbol{\psi}_\alpha(x)|^2 - 1)$, where $a_y \in \mathbb{Z}^d_L$ denotes the block containing y, i.e., $y \in [a_y]$. This is an average over at most $2d+1$ blocks (for the block Anderson model, where S is given by (2.29), only the block $[a_y]$ appears). Hence, if $|M_{y,\alpha}| \ge W^{-\mathfrak{d}/6}$, then the event in (2.18) with $i = j = \alpha$ and $c = \mathfrak{d}/6$ occurs for at least one of these blocks, and $\mathbb{P}(\mathcal{B}(y))$ is bounded by $2d+1$ times the right-hand side of (2.18).

(for $E = \lambda_k$ in (2.11), for $E = \hat{E}$ in (2.12), and for the energy E in Theorem 2.3 and Theorem 2.4) with the bulk condition

$$E \in \mathbf{B}_\kappa := \{E \in \mathbb{R} : \ \rho_N(E) \geq \kappa\}, \tag{2.34}$$

where, by the Stieltjes inversion formula, $\rho_N(E) = \pi^{-1} \operatorname{Im} m(E + \mathrm{i}0_+, g)$:

- *The delocalization estimate (2.11) holds.*
- *The local laws (2.14) and (2.15) hold for large enough N, where G is defined in (2.6), and $m(z)$ and $M(z)$ are defined in (2.32) and (2.33), respectively.*
- *The QUE estimates (2.18) and (2.19) hold, and the bulk universality holds in the following form, normalized by the local densities: for any fixed $n \in \mathbb{N}$, $O \in C_c^\infty(\mathbb{R}^n)$, $E \in \mathbf{B}_\kappa$ and $E' \in (-2, 2)$,*

$$\lim_{N\to\infty} \int_{\mathbb{R}^n} \mathrm{d}\boldsymbol{\alpha}\, O(\boldsymbol{\alpha}) \left[\frac{p_H^{(n)}\big(E + \frac{\alpha_1}{N\rho_N(E)}, \dots, E + \frac{\alpha_n}{N\rho_N(E)}\big)}{\rho_N(E)^n} - \frac{p_{\mathrm{GUE}}^{(n)}\big(E' + \frac{\alpha_1}{N\rho_{\mathrm{sc}}(E')}, \dots, E' + \frac{\alpha_n}{N\rho_{\mathrm{sc}}(E')}\big)}{\rho_{\mathrm{sc}}(E')^n} \right] = 0.$$

(For the random band matrix model, $\rho_N = \rho_{\mathrm{sc}}$, and the choice $E' = E$ gives back (2.20) after the change of variables $\boldsymbol{\alpha} \mapsto \rho_{\mathrm{sc}}(E)\boldsymbol{\alpha}$. For the block Anderson model $\rho_N(E) \neq \rho_{\mathrm{sc}}(E)$ in general, so the comparison with the GUE has to be made at the local density.)

- *Recall the variance matrix S from (2.29) and the matrix M from (2.33). Define the Θ-matrices as*

$$\Theta^{(+,-)}(z) := \big(1 - M^{(+,-)}(z) S^{(\mathsf{B})}\big)^{-1}, \quad \Theta^{(+,+)}(z) = (\Theta^{(-,-)}(z))^* := \big(1 - M^{(+,+)}(z) S^{(\mathsf{B})}\big)^{-1}, \tag{2.35}$$

where $S^{(\mathsf{B})} = I_{L^d}$, and the matrices $M^{(+,-)} = M^{(-,+)}$ and $M^{(+,+)} = (M^{(-,-)})^*$ are defined by[5]

$$M_{ab}^{(+,-)} = \frac{1}{W^d} \sum_{x\in[a], y\in[b]} |M_{xy}|^2 = |M_{ab}^{(\mathsf{B})}|^2, \quad M_{ab}^{(+,+)} = \frac{1}{W^d} \sum_{x\in[a], y\in[b]} M_{xy} M_{yx} = (M_{ab}^{(\mathsf{B})})^2.$$

Then, the quantum diffusion estimates (2.22)–(2.25) hold if we replace $|m|^2 \Theta_{ab}^{(+,-)}$ and $m^2 \Theta_{ab}^{(+,+)}$ with $(\Theta^{(+,-)} M^{(+,-)})_{ab}$ and $(\Theta^{(+,+)} M^{(+,+)})_{ab}$, respectively.

The lower bound $g \gg W^{-d/2}$ in (2.10) is sharp, as explained below Theorem 2.1: when $g \ll W^{-d/2}$, the block Anderson model is localized, as proven in [52]. To establish Theorem 2.7, it is sufficient to prove the local laws (2.14) and (2.15), together with the quantum diffusion estimates (2.22)–(2.25), while the delocalization, QUE, and bulk universality can be derived as corollaries, as shown in Section 2.1.

2.3. **Stochastic flow and loop hierarchy.** The remainder of this paper is devoted to proving the local laws in Theorem 2.2 and the quantum diffusion estimates in Theorem 2.5 for both models. Before proceeding with the proofs, we first introduce some key tools and convenient notations for the remainder of this section, which will be utilized throughout the subsequent arguments. First, we introduce the flow framework, which is the same as that employed for 1D and 2D random band matrices [69, 28] except for a dimension-dependent scaling W^{-d}. Consider the following matrix Brownian motion:

$$\mathrm{d}(H_t)_{xy} = \sqrt{S_{xy}}\mathrm{d}(\boldsymbol{B}_t)_{xy}, \quad \forall x, y \in \mathbb{Z}_{WL}^d, \quad \text{where} \quad H_0 = \begin{cases} 0, & \text{for band matrix} \\ g\Psi, & \text{for block Anderson} \end{cases}. \tag{2.36}$$

Here, $(\boldsymbol{B}_t)_{xy}$ are independent complex Brownian motions up to the Hermitian symmetry $(\boldsymbol{B}_t)_{xy} = \overline{(\boldsymbol{B}_t)_{yx}}$, i.e., $t^{-1/2}\boldsymbol{B}_t$ is an $N \times N$ GUE whose entries have zero mean and unit variance; $S = (S_{xy})$ is the variance matrix defined in (2.5) or (2.29). Following [61, 60, 27], we consider the Green's function of H_t with a carefully chosen time-dependent spectral parameter z_t, whose dynamics are naturally renormalized at leading order.

Definition 2.8 (Flow framework)**.** *For any $E \in \mathbb{R}$ and $g > 0$, we denote $m(E, g) \equiv m(E + \mathrm{i}0_+, g)$, as defined in (2.7) and (2.32) for the random band matrix and block Anderson model, respectively. Based on this, we define the* spectral parameter flow *z_t by*

$$z_t(E, g) = E + (1 - t)m(E, g), \quad \text{for} \ \ t \in [0, 1]. \tag{2.37}$$

We refer to E and g as **flow parameters***, which remain fixed throughout the flow. Let $z_t = E_t + \mathrm{i}\eta_t$ with*

$$E_t \equiv E_t(E, g) = E + (1 - t) \operatorname{Re} m(E, g), \quad \eta_t \equiv \eta_t(E, g) = (1 - t) \operatorname{Im} m(E, g). \tag{2.38}$$

[5] Note the definition (2.35) is consistent with (2.21), where $M^{(+,-)} = |m|^2 I$ and $M^{(+,+)} = m^2 I$.

Furthermore, we define the matrix $M(E,g)$, as in (2.8) and (2.33) for the random band matrix and block Anderson model, respectively, with z and $m(z)$ replaced by E and $m(E,g)$. Then, we denote the Green's function of H_t as $G_t(z,g) := (H_t(g)-z)^{-1}$, and define the resolvent flow as

$$G_{t;E,g} \equiv G_t(z_t(E,g),g) := (H_t(g)-z_t(E,g))^{-1}. \tag{2.39}$$

Remark 2.9. To explain the choice of the flow z_t in (2.37), let $M_t(z,g) \in \mathcal{M}_N(\mathbb{C})$ (where $\mathcal{M}_N(\mathbb{C})$ denotes the set of $N \times N$ complex matrices) be the unique solution to the matrix Dyson equation

$$M_t(z,g) := (H_0 - z - t\mathcal{S}[M_t(z,g)])^{-1}, \tag{2.40}$$

such that $\operatorname{Im} M_t$ is positive definite whenever $z \in \mathbb{C}_+$. Here, the linear operator $\mathcal{S}$ is defined by

$$\mathcal{S}[X]_{xy} := \delta_{xy} \sum_{w=1}^{N} S_{xw} X_{ww}, \quad \text{for} \quad X \in \mathcal{M}_N(\mathbb{C}), \tag{2.41}$$

where S_{xy} is given by (2.5) and (2.29) for the respective models. It is known that M_t describes the deterministic limit of the Green's function $(H_t - z)^{-1}$. Moreover, since $\mathcal{S}[M(E,g)] = m(E,g)I_N$, one can check that $M_t(z_t(E,g),g) \equiv M(E,g)$. In other words, the deterministic limit of G_t remains invariant under the evolution.

Given any target spectral parameter z, we are interested in the original resolvent $G(z) = (H-z)^{-1}$. For random band matrices, this can be achieved through the stochastic flow by carefully choosing the spectral parameter E. The corresponding flow for the block Anderson model will be presented later in Section 8.

Lemma 2.10 (Lemma 2.8 of [69])**.** *Fix any $z \in \mathbb{C}_+$ with $\operatorname{Im} z \in (0,1]$ and $|\operatorname{Re} z| \le 2-\kappa$. We choose*

$$t_0 \equiv t_0(z) = |m(z)|^2 = \frac{\operatorname{Im} m(z)}{\operatorname{Im} m(z) + \operatorname{Im} z}, \quad E \equiv E(z) = -2\frac{\operatorname{Re} m(z)}{|m(z)|}. \tag{2.42}$$

Then, for the random band matrix model, we have

$$\sqrt{t_0} m(E) = m(z), \quad z_{t_0}(E) = \sqrt{t_0} z, \quad G(z) \overset{d}{=} \sqrt{t_0} G_{t_0;E}, \tag{2.43}$$

where "$\overset{d}{=}$" means equality in distribution.

In the main proofs for random band matrices, we will fix a target spectral parameter $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$ for an arbitrarily small constant $\varepsilon > 0$ (recall (2.12)). Accordingly, we choose the parameters t_0 and E as specified in (2.42). The second identity in (2.38) implies that $1 - t \asymp \eta_t$ uniformly in $t \in [0,t_0]$, i.e., during the flow from $t=0$ to t_0, the imaginary part η_t decreases from $\eta_0 \asymp 1$ to $\eta_{t_0} \asymp 1 - t_0 \asymp \eta \ge N^{-1+\varepsilon}$. For clarity, unless we want to emphasize their dependence on E (or g), we will often omit this variable from various notations, such as $z_t(E,g)$, $E_t(E,g)$, $\eta_t(E,g)$, $m(E,g)$, $M(E,g)$, and most importantly, $G_{t;E,g} \equiv G_t$. Our focus will be on the dynamics of G_t and the corresponding G-loops defined below.

Definition 2.11 (G-loop)**.** *For $\sigma \in \{+,-\}$, we denote*

$$G_t(\sigma) := \begin{cases} (H_t - z_t)^{-1}, & \text{if } \sigma = +, \\ (H_t - \overline{z}_t)^{-1}, & \text{if } \sigma = -. \end{cases}$$

*In other words, we let $G_t(+) \equiv G_t$ and $G_t(-) \equiv G_t^*$. For any $n \in \mathbb{N}$, fix indices $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n) \in \{+,-\}^n$ and $\mathbf{a} = (a_1, \dots, a_n) \in (\mathbb{Z}_L^d)^n$. We define the corresponding n-G-**loop** by*

$$\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \operatorname{Tr}\left(\prod_{i=1}^{n} (G_t(\sigma_i) E_{a_i}) \right), \quad \text{where} \quad (E_{a_i})_{xy} = W^{-d} \mathbf{1}(x = y \in [a_i]). \tag{2.44}$$

Sometimes, we will also refer to G-loops as $\mathcal{L}$-loops. Furthermore, we denote

$$m(\sigma) := \begin{cases} m(E,g), & \text{if } \sigma = + \\ \overline{m}(E,g), & \text{if } \sigma = - \end{cases}, \quad M(\sigma) := \begin{cases} M(E,g), & \text{if } \sigma = + \\ M(E,g)^*, & \text{if } \sigma = - \end{cases}. \tag{2.45}$$

Finally, we define the centered resolvent $\mathring{G}$ *as*

$$\mathring{G}_t(\sigma) := G_t(\sigma) - M(\sigma), \quad \forall t \in [0,1],\ \sigma \in \{+,-\}. \tag{2.46}$$

(Note that $\mathring{G}_t(\sigma)$ was denoted by $\widetilde{G}_t(\sigma)$ in the preceding papers [69, 28]; here we adopt a different notation for clarity.) For any $a \in \mathbb{Z}_L^d$, we refer to $\mathrm{Tr}[\mathring{G}_t(\sigma)E_a]$ *as a* light-weight.

To describe the loop hierarchy for the G-loops, we introduce the following operations from [69]. (For a graphical illustration of these operations, see also the diagrams in [69, Definition 2.10].)

Definition 2.12. *For any fixed $n \in \mathbb{N}$, take an n-loop of the form (2.44).*

1. *For $k \in [\![n]\!]$ and $a \in \mathbb{Z}_L^d$, we define a "cut-and-glue" operator $\mathcal{G}_k^{(a)}$ as follows: $\mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}$ is defined to be the loop obtained by replacing $G_t(\sigma_k)$ with $G_t(\sigma_k)E_aG_t(\sigma_k)$. In other words, it cuts the k-th edge $G_t(\sigma_k)$ and glues the two new ends with E_a to get a new loop that is one unit longer. This operator can also be considered as an operator on $(\boldsymbol{\sigma}, \mathbf{a})$, that is,*

$$\mathcal{G}_k^{(a)}(\boldsymbol{\sigma}, \mathbf{a}) = \big((\sigma_1, \dots, \sigma_{k-1}, \sigma_k, \sigma_k, \sigma_{k+1}, \dots, \sigma_n),\ (a_1, \dots, a_{k-1}, a, a_k, a_{k+1}, \dots, a_n)\big).$$

Hence, we will sometimes write $\mathcal{G}_k^{(a)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} \equiv \mathcal{L}_{t,\,\mathcal{G}_k^{(a)}(\boldsymbol{\sigma},\mathbf{a})}^{(n+1)}$.

2. *For $k < l \in [\![n]\!]$, we define two other types of "cut-and-glue" operators—$(\mathcal{G}_L)_{k,l}^{(a)}$ from the left ("L") of k, and $(\mathcal{G}_R)_{k,l}^{(a)}$ from the right ("R") of k—as follows. In other words, these operators cut the k-th and l-th edges $G_t(\sigma_k)$ and $G_t(\sigma_l)$, and create two chains: the left chain to the vertex a_k is of length $(n+k-l+1)$ and contains the vertex a_n, while the right chain to the vertex a_k is of length $(l-k+1)$ and does not contain the vertex a_n. Then, $(\mathcal{G}_L)_{k,l}^{(a)}$ (resp. $(\mathcal{G}_R)_{k,l}^{(a)}$) gives an $(n+k-l+1)$-loop (resp. $(l-k+1)$-loop) obtained by gluing the left chain (resp. right chain) at the new vertex a. Again, we can also consider the two operators to be defined on the indices $(\boldsymbol{\sigma}, \mathbf{a})$:*

$$(\mathcal{G}_L)_{k,l}^{(a)}(\boldsymbol{\sigma}, \mathbf{a}) = ((\sigma_1, \dots, \sigma_k, \sigma_l, \dots, \sigma_n), (a_1, \dots, a_{k-1}, a, a_l, \dots, a_n)),$$
$$(\mathcal{G}_R)_{k,l}^{(a)}(\boldsymbol{\sigma}, \mathbf{a}) = ((\sigma_k, \dots, \sigma_l), (a_k, \dots, a_{l-1}, a)).$$

Hence, we will sometimes write $(\mathcal{G}_L)_{k,l}^{(a)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} \equiv \mathcal{L}_{t,(\mathcal{G}_L)_{k,l}^{(a)}(\boldsymbol{\sigma},\mathbf{a})}^{(n+k-l+1)}$ and $(\mathcal{G}_R)_{k,l}^{(a)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} \equiv \mathcal{L}_{t,(\mathcal{G}_R)_{k,l}^{(a)}(\boldsymbol{\sigma},\mathbf{a})}^{(l-k+1)}$.

For $x, y \in \mathbb{Z}_{WL}^d$, we abbreviate $\partial_{xy} := \partial_{(H_t)_{xy}}$. By Itô's formula, we can derive the following SDE satisfied by the G-loops, called *loop hierarchy*; see Lemma 2.11 of [69].

Lemma 2.13 (Loop hierarchy)**.** *An n-G-loop satisfies the following SDE, called the* loop hierarchy*:*

$$\mathrm{d}\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} = \mathrm{d}\mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{M,(n)} + \mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{\mathring{G},(n)}\mathrm{d}t + W^d \sum_{1\le k<l\le n} \sum_{a,b} \Big((\mathcal{G}_L)_{k,l}^{(a)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big) S_{ab}^{(\mathsf{B})} \Big((\mathcal{G}_R)_{k,l}^{(b)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big)\,\mathrm{d}t, \tag{2.47}$$

where $S^{(\mathsf{B})}$ denotes $S^{(\mathsf{B})}(g)$ in (2.5) (for the random band matrix model) or $S^{(\mathsf{B})}(0) = I_{L^d}$ (for the block Anderson model). Moreover, the martingale term $\mathrm{d}\mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{M,(n)}$ and the light-wight term $\mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{\mathring{G},(n)}$ are defined by

$$\mathrm{d}\mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{M,(n)} := \sum_{x,y\in\mathbb{Z}_{WL}^d} \Big(\partial_{xy}\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big) \cdot \sqrt{S_{xy}}\,(\mathrm{d}\boldsymbol{B}_t)_{xy}, \tag{2.48}$$

$$\mathcal{E}_{t,\boldsymbol{\sigma},\mathbf{a}}^{\mathring{G},(n)} := W^d \sum_{k=1}^{n} \sum_{a,b\in\mathbb{Z}_L^d} \mathrm{Tr}\Big(\mathring{G}_t(\sigma_k)E_a\Big) S_{ab}^{(\mathsf{B})} \Big(\mathcal{G}_k^{(b)} \circ \mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big). \tag{2.49}$$

We emphasize that the superscript (n) indicates the length of the G-loop on the left-hand side of the equation. For clarity and conciseness, we may omit this superscript when its value is clear from the context.

Note that the right-hand side (RHS) of equation (2.47) involves G-loops of length larger than n, and hence represents a "hierarchy" rather than a "self-consistent equation" for the G-loops. This loop hierarchy is well-approximated by the $\mathcal{K}$-loops, defined as follows.

Definition 2.14 (Tree approximation)**.** *We define the $\mathcal{K}$-loop of length 1 as:*

$$\mathcal{K}_{t,\sigma,a}^{(1)} = m(\sigma), \quad \forall t \in [0,1),\ \ \sigma \in \{+,-\},\ \ a \in \mathbb{Z}_L^d.$$

For $n \ge 2$, we define the function $\mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}$ (of $t \in [0,1)$, $\boldsymbol{\sigma} \in \{+,-\}^n$, and $\mathbf{a} \in (\mathbb{Z}_L^d)^n$) to be the unique solution to the following system of equations, referred to as the **convolution tree equations***:*

$$\partial_t \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} = W^d \sum_{1\le k<l\le n} \sum_{a,b} \Big((\mathcal{G}_L)_{k,l}^{(a)} \circ \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big) S_{ab}^{(\mathsf{B})} \Big((\mathcal{G}_R)_{k,l}^{(b)} \circ \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\Big), \tag{2.50}$$

where the operators $(\mathcal{G}_L)$ and $(\mathcal{G}_R)$ act on $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ through the actions on indices:

$$(\mathcal{G}_L)^{(a)}_{k,l} \circ \mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} := \mathcal{K}^{(n+k-l+1)}_{t,(\mathcal{G}_L)^{(a)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})}, \quad (\mathcal{G}_R)^{(b)}_{k,l} \circ \mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} := \mathcal{K}^{(l-k+1)}_{t,(\mathcal{G}_R)^{(b)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})}. \tag{2.51}$$

We impose the following initial condition at $t = 0$:

$$\mathcal{K}^{(k)}_{0,\boldsymbol{\sigma},\mathbf{a}} = \mathcal{M}^{(k)}_{\boldsymbol{\sigma},\mathbf{a}}, \quad \forall k \in \mathbb{N},\ \boldsymbol{\sigma} \in \{+,-\}^k,\ \mathbf{a} \in (\mathbb{Z}^d_L)^k, \tag{2.52}$$

where $\mathcal{M}^{(k)}_{\boldsymbol{\sigma},\mathbf{a}}$ is a k-M-loop defined as

$$\mathcal{M}^{(k)}_{\boldsymbol{\sigma},\mathbf{a}} := \mathrm{Tr}\left(\prod_{i=1}^{k} (M(\sigma_i)E_{a_i})\right). \tag{2.53}$$

(Note that this M-loop can be simplified as $\mathcal{M}^{(k)}_{\boldsymbol{\sigma},\mathbf{a}} = W^{-(k-1)d}\prod_{i=1}^{k} m(\sigma_i)\mathbf{1}(a_1 = \cdots = a_k)$ for the random band matrix model.) We call $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ an n-$\mathcal{K}$-loop. Moreover, we refer to $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ as providing a **tree approximation** *of the G-loop $\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$, since it admits an explicit tree representation, as shown in [69, 59] (see also Appendix A.5 for the detailed construction).*

Given any Hermitian matrix $\mathcal{A}$, define its resolvent as $R(z) := (\mathcal{A} - z)^{-1}$ for $z = E + \mathrm{i}\eta \in \mathbb{C}_+$. Then, with the algebraic identity $R - R^* = 2\mathrm{i}\eta RR^* = 2\mathrm{i}\eta R^*R$, we get the well-known Ward's identity:

$$\sum_x \overline{R_{xy'}}R_{xy} = \frac{1}{2\mathrm{i}\eta}\left(R_{y'y} - \overline{R_{yy'}}\right), \quad \sum_x \overline{R_{y'x}}R_{yx} = \frac{1}{2\mathrm{i}\eta}\left(R_{yy'} - \overline{R_{y'y}}\right). \tag{2.54}$$

As a special case, if $y = y'$, we have

$$\sum_x |R_{xy}(z)|^2 = \sum_x |R_{yx}(z)|^2 = \mathrm{Im}\, R_{yy}(z)/\eta. \tag{2.55}$$

Applying (2.54) to G_t, we can show that the G-loops satisfy the following identity (2.56), which we also refer to as a "Ward's identity". In [69, 59], it is shown that a similar Ward's identity (2.57) holds for the $\mathcal{K}$-loops.

Lemma 2.15 (Ward's identity for $\mathcal{L}$-loops and $\mathcal{K}$-loops)**.** *Given $\boldsymbol{\sigma} \in \{+,-\}^n$ with $n \geq 2$ and $\sigma_1 = -\sigma_n$, we have the following identities, which are called Ward's identities at the vertex a_n:*

$$\sum_{a_n} \mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \frac{1}{2\mathrm{i}W^d\eta_t}\left(\mathcal{L}^{(n-1)}_{t,\widehat{\boldsymbol{\sigma}}^{(+,n)},\widehat{\mathbf{a}}^{(n)}} - \mathcal{L}^{(n-1)}_{t,\widehat{\boldsymbol{\sigma}}^{(-,n)},\widehat{\mathbf{a}}^{(n)}}\right), \tag{2.56}$$

$$\sum_{a_n} \mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \frac{1}{2\mathrm{i}W^d\eta_t}\left(\mathcal{K}^{(n-1)}_{t,\widehat{\boldsymbol{\sigma}}^{(+,n)},\widehat{\mathbf{a}}^{(n)}} - \mathcal{K}^{(n-1)}_{t,\widehat{\boldsymbol{\sigma}}^{(-,n)},\widehat{\mathbf{a}}^{(n)}}\right), \tag{2.57}$$

where η_t is defined in (2.38), $\widehat{\boldsymbol{\sigma}}^{(\pm,n)}$ is obtained by removing σ_n from $\boldsymbol{\sigma}$ and replacing σ_1 with $\pm$, i.e., $\widehat{\boldsymbol{\sigma}}^{(\pm,n)} := (\pm, \sigma_2, \cdots \sigma_{n-1})$, and $\widehat{\mathbf{a}}^{(n)}$ is obtained by removing a_n from $\mathbf{a}$, i.e., $\widehat{\mathbf{a}}^{(n)} := (a_1, a_2, \cdots, a_{n-1})$.

Proof. For the random band matrix, this corresponds to Lemma 3.6 in [69], while for the block Anderson model, it corresponds to Lemma 3.17 in [59]. □

In Appendix A.5, we will present a tree representation formula for the $\mathcal{K}$-loops, originally discovered in [69] for the random band matrix model and in [59] for the block Anderson model. Using this tree representation, we can establish the following upper bound (2.58) for $\mathcal{K}$-loops. The proof, being similar to those in [69, 59], is deferred to Appendix A.5.

Lemma 2.16 (Upper bounds on $\mathcal{K}$-loops)**.** *Fix any dimension $d \geq 3$. Then, for every $n \in \mathbb{N}$ and $t \in [0,1)$, the $\mathcal{K}$-loops satisfy the upper bound*

$$\max_{\boldsymbol{\sigma}\in\{+,-\}^n} \max_{\mathbf{a}\in(\mathbb{Z}^d_L)^n} \left|\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(W^{-d}B_{t,0}\right)^{n-1}, \tag{2.58}$$

where $B_{t,0} = (g^2 + |1-t|)^{-1} + (L^d|1-t|)^{-1}$ is defined precisely in Definition 2.18 below.

2.4. **Propagators.** The quantum diffusion behavior of the resolvents is governed by the Θ-propagators, defined as follows.

Definition 2.17. *Given $\sigma_1, \sigma_2 \in \{+,-\}$, define an $L^d \times L^d$ matrix $M^{(\sigma_1,\sigma_2)}$ as a rescaled 2-M-loop:*

$$M_{ab}^{(\sigma_1,\sigma_2)} := W^d \operatorname{Tr}\left(M(\sigma_1)E_a M(\sigma_2)E_b\right), \quad \forall a,b \in \mathbb{Z}_L^d. \tag{2.59}$$

For the random band matrix model, we have $M^{(\sigma_1,\sigma_2)} = m(\sigma_1)m(\sigma_2)I_{L^d}$; for the block Anderson model, we have $M_{ab}^{(\sigma_1,\sigma_2)} = M_{ba}^{(\mathsf{B})}(\sigma_1)M_{ab}^{(\mathsf{B})}(\sigma_2)$, where $M^{(\mathsf{B})}$ is defined in (2.33), with $M^{(\mathsf{B})}(+) \equiv M^{(\mathsf{B})}$ and $M^{(\mathsf{B})}(-) \equiv (M^{(\mathsf{B})})^*$. For $t \in [0,1)$ and $\sigma_1, \sigma_2 \in \{+,-\}$, the Θ-propagator $\Theta_t^{(\sigma_1,\sigma_2)}$ is an $L^d \times L^d$ matrix defined as:

$$\Theta_t^{(\sigma_1,\sigma_2)} := \left(1 - tM^{(\sigma_1,\sigma_2)}S^{(\mathsf{B})}\right)^{-1}, \tag{2.60}$$

where $S^{(\mathsf{B})}$ is given by $S^{(\mathsf{B})}(g)$ in (2.5) or $S^{(\mathsf{B})}(0) = I_{L^d}$, depending on whether we consider the random band matrix model or the block Anderson model. We denote the entries of $\Theta_t^{(\sigma_1,\sigma_2)}$ as $\Theta_t^{(\sigma_1,\sigma_2)}(a,b)$ or $\Theta_{t,ab}^{(\sigma_1,\sigma_2)}$. We further define the zero-mode-removed propagator $\mathring{\Theta}_t^{(\sigma_1,\sigma_2)}$ by

$$\mathring{\Theta}_t^{(\sigma_1,\sigma_2)}(a,b) := \Theta_t^{(\sigma_1,\sigma_2)}(a,b) - L^{-2d}\sum_{a',b'}\Theta_t^{(\sigma_1,\sigma_2)}(a',b'). \tag{2.61}$$

Note that for the random band matrix model, the Θ-propagators $\Theta_{t_0}^{(\sigma,\sigma')}$ for $\sigma,\sigma' \in \{+,-\}$ reduce to the Θ-matrices defined in (2.21) under the relation (2.43). Similarly, for the block Anderson model, there is a corresponding reduction to the definition in (2.35) at $t = t_0$ under the relation (8.2) below.

To capture the decay profile of the Θ-propagators, we introduce the following control parameter $B_{t,K}$ in (2.62). It is an order parameter that will appear in many estimates throughout the proof.

Definition 2.18 (Definition of B)**.** *For any $t \in [0,1)$, define*

$$B_{t,K} := \frac{(g^2 + |1-t|)^{-1}}{(K+1)^{d-2}} + \frac{1}{L^d|1-t|}, \quad \forall K \geq 0. \tag{2.62}$$

Since $\eta_t \asymp 1-t$ by (2.38), the parameter $\mathcal{B}_{\eta_t,K}$ defined in (2.13) satisfies

$$\mathcal{B}_{\eta_t,K} \asymp W^{-d}B_{t,(K/W)}. \tag{2.63}$$

We now summarize some fundamental properties of the Θ-propagators that will be used extensively in the main proof. These properties have essentially been established in previous works [28, 68, 59]. For the reader's convenience, we briefly outline the proof of Lemma 2.19 in Appendix A.1.

Lemma 2.19. *For any $t \in [0,1)$, define*

$$\ell_t := \min\left(\max\left(g|1-t|^{-\frac{1}{2}}, 1\right), L\right). \tag{2.64}$$

For any $\sigma_1, \sigma_2 \in \{+,-\}$, $\Theta_t^{(\sigma_1,\sigma_2)}$ defined in Definition 2.17 satisfies the following properties:

(1) **Symmetry**: $\Theta_t^{(\sigma_2,\sigma_1)} = (\Theta_t^{(\sigma_1,\sigma_2)})^\top = \Theta_t^{(\sigma_1,\sigma_2)}$.

(2) **Translation invariance**: *For any $a,b,r \in \mathbb{Z}_L^d$, we have $\Theta_t^{(\sigma_1,\sigma_2)}(a+r,b+r) = \Theta_t^{(\sigma_1,\sigma_2)}(a,b)$.*

(3) **Commutativity**: *We have $\left[S^{(\mathsf{B})}, \Theta_t^{(\sigma_1,\sigma_2)}\right] = \left[\Theta_t^{(\sigma_1,\sigma_2)}, \Theta_{t'}^{(\sigma_1,\sigma_2)}\right] = 0$ for all $t \neq t'$.*

(4) **$(\infty \to \infty)$-norm**: *For any $a,b \in \mathbb{Z}_L^d$, we have $|\Theta_{t,ab}^{(\sigma_1,\sigma_2)}| \leq \Theta_{t,ab}^{(+,-)}$. Moreover,*

$$\|\Theta_t^{(\sigma_1,\sigma_2)}\|_{\infty\to\infty} = \max_a \sum_b \left|\Theta_{t,ab}^{(\sigma_1,\sigma_2)}\right| \leq \max_a \sum_b \Theta_{t,ab}^{(+,-)} = \frac{1}{1-t}. \tag{2.65}$$

(5) **Polynomial and exponential decay**: *For all $\sigma_1,\sigma_2 \in \{+,-\}$, there exist constants $c_d, C_d > 0$ (depending on d) such that the following bound holds:*

$$\left|\Theta_t^{(\sigma_1,\sigma_2)}(0,a)\right| \leq C_d B_{t,|a|} \cdot e^{-c_d|a|/\ell_t}, \quad \forall a \in \mathbb{Z}_L^d. \tag{2.66}$$

Furthermore, when $\sigma_1 = \sigma_2$, we have a much stronger exponential decay: there exist constants $c_\kappa, C_\kappa > 0$ (depending on d and κ) such that

$$\left|\Theta_t^{(\sigma_1,\sigma_2)}(0,a)\right| \leq C_\kappa\left(1_{a=0} + g^2 e^{-c_\kappa|a|}\right), \quad \forall a \in \mathbb{Z}_L^d. \tag{2.67}$$

In the setting of the block Anderson model, these constants, including c_d and C_d in (2.66), may also depend on κ and on $\mathfrak{d}$ in (2.10).

(6) **First-order difference**: *Fix any constant $c_0 \in (0,1)$. The following estimate holds for all $a, r \in \mathbb{Z}_L^d$ satisfying $|r| \le c_0|a|$ (the implicit constants in $\prec$ may depend on c_0):*

$$\left|\Theta_t^{(\sigma_1,\sigma_2)}(0,a+r) - \Theta_t^{(\sigma_1,\sigma_2)}(0,a)\right| \prec \frac{1}{g^2+|1-t|}\frac{|r|}{(|a|+1)^{d-1}}. \tag{2.68}$$

(7) **Second-order difference**: *Fix any constant $c_0 \in (0,1)$. The following estimate holds for all $a, r \in \mathbb{Z}_L^d$ satisfying $|r| \le c_0|a|$ (the implicit constants in $\prec$ may depend on c_0):*

$$\left|\Theta_t^{(\sigma_1,\sigma_2)}(0,a+r) + \Theta_t^{(\sigma_1,\sigma_2)}(0,a-r) - 2\Theta_t^{(\sigma_1,\sigma_2)}(0,a)\right| \prec \frac{1}{g^2+|1-t|}\frac{|r|^2}{(|a|+1)^{d}}. \tag{2.69}$$

(8) **Propagator without zero mode.** *The following estimate holds for all $a \in \mathbb{Z}_L^d$:*

$$\left|\mathring{\Theta}_t^{(\sigma_1,\sigma_2)}(0,a)\right| \prec \frac{1}{g^2+|1-t|}\frac{1}{(|a|+1)^{d-2}}. \tag{2.70}$$

Example 2.20. As shown in [69, 59], the 2-$\mathcal{K}$ and 3-$\mathcal{K}$ loops are given by

$$\mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(2)} = \sum_b \Theta_t^{(\sigma_1,\sigma_2)}(a_1,b)\mathcal{M}_{\boldsymbol{\sigma},(b,a_2)}^{(2)} = W^{-d}\left(\Theta_t^{(\sigma_1,\sigma_2)} M^{(\sigma_1,\sigma_2)}\right)_{a_1 a_2}, \tag{2.71}$$

$$\mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(3)} = \sum_{b_1,b_2,b_3} \Theta_t^{(\sigma_1,\sigma_2)}(a_1,b_1)\Theta_t^{(\sigma_2,\sigma_3)}(a_2,b_2)\Theta_t^{(\sigma_3,\sigma_1)}(a_3,b_3)\mathcal{M}_{\boldsymbol{\sigma},\mathbf{b}}^{(3)}, \tag{2.72}$$

where $\boldsymbol{\sigma} = (\sigma_1,\dots,\sigma_n)$ and $\mathbf{a} = (a_1,\dots,a_n)$ for $n \in \{2,3\}$, and the M-loops are defined in (2.53).

2.5. **Proof of the main results.** The main results, Theorems 2.2 and 2.5, for random band matrices follow directly from the following key lemmas on the G-loop estimates, while the proof of Theorem 2.7 for the block Anderson model will be given separately in Section 8. Recall the notions of stochastic domination, $B_{t,K}$, and ℓ_t defined in (1.13), (2.62), and (2.64), respectively.

Lemma 2.21 (G-loop estimates)**.** *In the setting of Theorem 2.2, fix any $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$ and consider the flow framework in Lemma 2.10. For each fixed $n \in \mathbb{N}$, the following estimate holds uniformly in $t \in [0,t_0]$:*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)} - \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\right| \prec (W^{-d}B_{t,0})^n, \tag{2.73}$$

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}\right| \prec (W^{-d}B_{t,0})^{n-1}. \tag{2.74}$$

Lemma 2.22 (2-loop estimates)**.** *In the setting of Lemma 2.21, for $\boldsymbol{\sigma} \in \{+,-\}^2$ and $\mathbf{a} = (a_1,a_2) \in (\mathbb{Z}_L^d)^2$, the following pointwise estimate holds uniformly in $t \in [0,t_0]$ for any large constant $D > 0$:*

$$\left|\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(2)} - \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(2)}\right| \prec \left(W^{-d}B_{t,0}\right)^{1/5} \cdot \left(W^{-d}B_{t,|a_1-a_2|}\right) e^{-(|a_1-a_2|/\ell_t)^{1/2}} + W^{-D}. \tag{2.75}$$

Moreover, the expectation of a 2-G-loop satisfies the following L^∞-estimate uniformly in $t \in [0,t_0]$:

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathbb{E}\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(2)} - \mathcal{K}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(2)}\right| \prec \left(W^{-d}B_{t,0}\right)^2 \left((g^2W^d)^{-1/5} + W^{-d}B_{t,0}\right). \tag{2.76}$$

In the above estimates, the exponent $1/5$ can be replaced by any other positive constant less than $1/4$.

Lemma 2.23 (Local law for G_t)**.** *In the setting of Lemma 2.21, the following entrywise local law holds uniformly in $t \in [0,t_0]$:*

$$|(G_t - M)_{xy}|^2 \prec W^{-d}B_{t,(|x-y|/W)}. \tag{2.77}$$

With these lemmas, we readily conclude the proofs of our main results, Theorems 2.2 and 2.5.

Proof of Theorems 2.2 and 2.5. For a fixed $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$, we can choose the flow framework as in Lemma 2.10. Then, with (2.43), (2.63), and (2.71), we observe that Lemma 2.23, Lemma 2.21 (with $n = 1$), and Lemma 2.22 respectively give the entrywise local law (2.14), the averaged local law (2.15), and the quantum diffusion estimates in Theorem 2.5 at each *fixed* z. To extend these estimates uniformly to all z, we can use a standard N^{-C}-net and perturbation argument, whose details we omit. □

Before concluding this section, we now outline the proofs of Lemmas 2.21 to 2.23. At $t = 0$, we have $G_0(\sigma) = M(\sigma)$ for $\sigma \in \{+,-\}$. Together with Definitions 2.11 and 2.14, it implies that for any fixed $n \in \mathbb{N}$:

$$\mathcal{L}^{(n)}_{0,\boldsymbol{\sigma},\mathbf{a}} = \mathcal{K}^{(n)}_{0,\boldsymbol{\sigma},\mathbf{a}}, \quad \forall\, \boldsymbol{\sigma} \in \{+,-\}^n, \quad \mathbf{a} \in (\mathbb{Z}^d_L)^n.$$

Now, for $t > 0$, we will establish the following theorem, which provides an induction result that extends the G-loop estimate progressively along the stochastic flow.

Theorem 2.24. *In the setting of Theorem 2.2, fix any $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$ and consider the flow framework in Lemma 2.10. Suppose the following estimates hold at a fixed $s \in [0, t_0]$:*

(a) **G-loop estimate**: *For each fixed $n \geq 1$, we have*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left| \mathcal{L}^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}} \right| \prec (W^{-d} B_{s,0})^n. \tag{2.78}$$

(b) **2-loop estimate**: *For $\boldsymbol{\sigma} \in \{+,-\}^2$, $\mathbf{a} = (a_1, a_2) \in (\mathbb{Z}^d_L)^2$, and any large constant $D > 0$, we have*

$$\left| \mathcal{L}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} \right| \prec \left(W^{-d} B_{s,0}\right)^{1/5} \cdot \left(W^{-d} B_{s,|a_1-a_2|}\right) e^{-(|a_1-a_2|/\ell_s)^{1/2}} + W^{-D}. \tag{2.79}$$

Furthermore, if $1 - s \geq g^2$ (where $\ell_s = 1$ by (2.64)), we assume a stronger estimate:

$$\left| \mathcal{L}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} \right| \prec \left(W^{-d} B_{s,0}\right)^2 e^{-|a_1-a_2|^{1/2}} + W^{-D}. \tag{2.80}$$

(c) **Local law**: *We have the (maximum) entrywise local law*

$$\|G_s - M\|_{\max} \prec (W^{-d} B_{s,0})^{1/2}. \tag{2.81}$$

(d) **Expected 2-loop estimate**: *For all $\boldsymbol{\sigma} \in \{+,-\}^2$ and $\mathbf{a} = (a_1, a_2) \in (\mathbb{Z}^d_L)^2$, we have*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left| \mathbb{E}\mathcal{L}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} \right| \prec \left(W^{-d} B_{s,0}\right)^2 \left((g^2 W^d)^{-1/5} + W^{-d} B_{s,0} \right). \tag{2.82}$$

Then, there exists a constant $0 < \mathfrak{c}_d \leq 10^{-2}$ (depending on d, κ, ε, and $\mathfrak{d}$ in (2.10)) such that for any $s < t < 1$ satisfying

$$\left(W^{-d} B_{t,0}\right)^{\mathfrak{c}_d} \leq \frac{1-t}{1-s} < 1, \tag{2.83}$$

the estimates (2.73)–(2.77) hold. In the proof, we do not track the exact value of $\mathfrak{c}_d$ (although it can be done by keeping track of the constants carefully in our proof). In addition, if $1 - t \geq g^2$, we have

$$\left| \mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \right| \prec \left(W^{-d} B_{t,0}\right)^2 e^{-|a_1-a_2|^{1/2}} + W^{-D}, \quad \forall \boldsymbol{\sigma} \in \{+,-\}^2,\ \mathbf{a} \in (\mathbb{Z}^d_L)^2. \tag{2.84}$$

With Theorem 2.24 in hand, we can establish Lemmas 2.21 to 2.23 via induction on t.

Proof of Lemmas 2.21 to 2.23. We first perform induction from $t = 0$ up to $t = 1 - g^2$ (when $g^2 \leq 1/2$) or $t = 1/2$ (when $g^2 > 1/2$), establishing (2.73)–(2.77) at $t_1 := (1 - g^2) \vee (1/2)$. In the case $g^2 \leq 1/2$, we additionally maintain the stronger 2-loop estimate (2.84) throughout this step. Next, starting from t_1, we continue the induction using Theorem 2.24 up to $t = t_0$, thereby completing the proof of the estimates stated in Lemmas 2.21 to 2.23. □

The proof of Theorem 2.24 is divided into six steps, which comprise the remainder of this paper. Throughout these steps, we assume the hypotheses of Theorem 2.24 hold. Moreover, each step builds upon the results obtained in the preceding ones.

Step 1 (A priori G-loop bound): The n-G-loops satisfy the a priori bound:

$$\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec \left(\frac{1-s}{1-u}\right)^{n-1} (W^{-d} B_{s,0})^{n-1}, \quad \forall u \in [s,t]. \tag{2.85}$$

Furthermore, a weak local law holds in the sense

$$\|G_u - M\|_{\max} \prec (W^{-d} B_{u,0})^{1/4}, \quad \forall u \in [s,t]. \tag{2.86}$$

Step 2 (A priori 2-G-loop decay): The following sharp local laws hold uniformly for all $u \in [s,t]$:

$$|(G_u - M)_{xy}|^2 \prec W^{-d} B_{u,(|x-y|/W)}, \quad \forall x, y \in \mathbb{Z}^d_{WL}, \tag{2.87}$$

$$\max_{a \in \mathbb{Z}^d_L} |\mathrm{Tr}\left((G_u - M) E_a\right)| \prec W^{-d} B_{u,0}. \tag{2.88}$$

In particular, the entrywise local law (2.77) and the 1-loop estimate in (2.73) hold at time t. In addition, there exists a constant $C_d > 0$ (independent of $\mathfrak{c}_d$ in (2.83)) such that, for any $\boldsymbol{\sigma} \in \{+,-\}^2$, $\mathbf{a} = (a_1, a_2) \in (\mathbb{Z}_L^d)^2$, and $u \in [s,t]$, the following estimate holds for arbitrarily large constant $D > 0$:

$$\left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(\frac{1-s}{1-u}\right)^{C_d} \left(W^{-d}B_{u,0}\right)^{1/5} \cdot \left(W^{-d}B_{u,|a_1-a_2|}\right) e^{-(|a_1-a_2|/\ell_u)^{1/2}} + W^{-D}. \tag{2.89}$$

Hence, at this step, the estimate (2.79) deteriorates at time t by a factor of $(|1-s|/|1-t|)^{C_d}$. We note that the exponent $1/5$—as an arbitrarily chosen positive constant less than $1/4$—in (2.89) is not optimal. Achieving the optimal decay rate for a 2-loop estimate would require this exponent to be 1, which is beyond the scope of the current paper (see also Remark 3.14 below).

Step 3 (Sharp n-loop bound): The following bound on n-G-loops holds for any fixed $n \in \mathbb{N}$:

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec (W^{-d}B_{u,0})^{n-1}, \quad \forall u \in [s,t]. \tag{2.90}$$

In particular, this gives the estimate (2.74) at time t.

Step 4 (Sharp $(\mathcal{L}-\mathcal{K})$-loop estimate): The following estimate on $(\mathcal{L}-\mathcal{K})$-loops holds for any fixed $n \in \mathbb{N}$:

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec (W^{-d}B_{u,0})^{n}, \quad \forall u \in [s,t]. \tag{2.91}$$

In particular, this gives the estimate (2.73) at time t.

Step 5 (Pointwise estimate for 2-loops): For any $\boldsymbol{\sigma} \in \{+,-\}^2$, $\mathbf{a} = (a_1, a_2) \in (\mathbb{Z}_L^d)^2$, and $u \in [s,t]$, the following estimate holds for any large constant $D > 0$:

$$\left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(W^{-d}B_{u,0}\right)^{1/5} \cdot \left(W^{-d}B_{u,|a_1-a_2|}\right) e^{-(|a_1-a_2|/\ell_u)^{1/2}} + W^{-D}. \tag{2.92}$$

Hence, the pointwise decay estimate (2.75) holds at time t. Furthermore, if $1 - t \geq g^2$, then we have

$$\left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(W^{-d}B_{u,0}\right)^2 e^{-|a_1-a_2|^{1/2}} + W^{-D}, \quad \forall u \in [s,t]. \tag{2.93}$$

Step 6 (Expected 2-G-loop estimate): The following estimate holds:

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathbb{E}\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} - \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(W^{-d}B_{u,0}\right)^2 \left((g^2W^d)^{-1/5} + W^{-d}B_{u,0}\right), \quad \forall u \in [s,t]. \tag{2.94}$$

Hence, the estimate (2.76) holds at time t.

We remark that the estimates established in each step hold uniformly in $u \in [s,t]$ (recall (1.13)) due to a standard N^{-C}-net and perturbation argument. For simplicity of presentation, we will not emphasize this uniformity at every step of the proof.

3. Steps 1 and 2: A priori G-loop estimates

The remainder of the paper is devoted to proving Theorem 2.24, according to the six steps outlined following its statement. Most of the arguments extend directly to the block Anderson model, with technical differences arising in Step 1 and in the treatment of the light-weight term $\mathcal{E}^{\mathring{G},(2)}$ (defined in (2.49)) when controlling the 2-G-loops—specifically, in the proofs of Lemmas 3.10 and 3.11 below. We will describe the necessary modifications to adapt the argument to the block Anderson model in Section 8.

3.1. Proof of Step 1.

Our proof depends crucially on the following lemma, which provides estimates on the resolvent entries via bounds on 2-G-loops. Specifically, (3.2) and (3.5) establish bounds on both the entrywise and averaged differences between G and M in the max-norm sense, while (3.3) provides a finer estimate on the off-diagonal resolvent entries, which allows us to derive the decay of the resolvent entries from the decay of the 2-G-loops.

Lemma 3.1. *Consider the setting of the random band matrix model. For any $t \in [0, t_0]$, define the event*

$$\Omega(t, \varepsilon_0) := \left\{\|G_t - M\|_{\max} \leq W^{-\varepsilon_0}\right\} \tag{3.1}$$

for a small constant $\varepsilon_0 > 0$. Then, the following entrywise local law *holds for any constants $\tau, D > 0$:*

$$\mathbb{P}\left(\mathbf{1}(\Omega(t,\varepsilon_0)) \cdot \|G_t - M\|_{\max}^2 \leq W^{\tau} \max_{a,b \in \mathbb{Z}_L^d} \mathcal{L}^{(2)}_{t,(-,+),(a,b)}\right) \geq 1 - W^{-D}. \tag{3.2}$$

Furthermore, we have the following pointwise decay estimate of G_t for all $a, b \in \mathbb{Z}_L^d$:

$$\mathbf{1}(\Omega(t,\varepsilon_0)) \cdot \max_{x\in[a],y\in[b],x\neq y} |(G_t)_{xy}|^2 \prec \sum_{\substack{|a'-a|\le 1,|b'-b|\le 1,\\ \boldsymbol{\sigma}\in\{(+,-),(-,+)\}}} \mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},(a',b')} + W^{-d}\mathbf{1}_{|a-b|\le 1}\,. \tag{3.3}$$

Finally, suppose the following estimates hold for a deterministic control parameter $W^{-d/2} \le \Psi_t \le W^{-\varepsilon_0}$:[6]

$$\|G_t - M\|_{\max} \prec W^{-\varepsilon_0}, \quad \max_{a,b\in\mathbb{Z}_L^d} \mathcal{L}^{(2)}_{t,(-,+),(a,b)} \prec \Psi_t^2\,. \tag{3.4}$$

Then, we have the averaged local law*:*

$$\max_a |\mathrm{Tr}\left((G_t - M)\,E_a\right)| \prec \Psi_t^2\,. \tag{3.5}$$

Proof. These estimates have been proven as Lemma 4.1 in [69] for 1D random band matrices. However, their proofs are dimension-independent and use standard arguments based on resolvent identities and large deviation estimates as in [36, 32]. □

Step 1 of the proof of Theorem 2.24 is similar to that in [69, Section 5.1], where the core is to establish the following continuity estimate for the G-loops.

Lemma 3.2. *Fix any $\varepsilon \le s \le t \le 1$ for a constant $\varepsilon > 0$. In the flow setting given by Definition 2.8 and Lemma 2.10, assume that the following bound holds at time s for any fixed $n \in \mathbb{N}$:*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}}\right| \prec (W^{-d}B_{s,0})^{n-1}. \tag{3.6}$$

Then, on the event $\Omega_t = \{\|G_t\|_{\max} \le C_0\}$ for a constant $C_0 > 0$, the following estimate holds for any fixed $n \in \mathbb{N}$ with $n \ge 2$:

$$\mathbf{1}(\Omega_t) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left((W^{-d}B_{s,0}) \cdot \frac{\eta_s}{\eta_t}\right)^{n-1} \le \left((W^{-d}B_{t,0}) \cdot \frac{\eta_s}{\eta_t}\right)^{n-1}. \tag{3.7}$$

Proof. The proof of this lemma is exactly the same as that for Lemma 5.1 in [69], except for some minor changes in notation. Hence, we omit the details. □

With Lemmas 3.1 and 3.2, Step 1 of the proof of Theorem 2.24 for the random band matrix model (i.e., the proof of (2.85) and (2.86)) is the same as that in [69, Section 5.1]. Hence, we omit the details.

3.2. **Dynamics of $(\mathcal{L}-\mathcal{K})$-loops.** We begin by presenting some representations of the G-loop dynamics, formulated using the loop hierarchy (2.47), Duhamel's principle, and certain evolution kernels, which we introduce below. This dynamics has already been derived for random band matrices in [69] and for the block Anderson model in [59], and will be the basis for Step 2 and subsequent steps of the proof of Theorem 2.24. For any fixed $n \in \mathbb{N}$, combining the loop hierarchy (2.47) and the convolution tree equation (2.50), we obtain the following SDE for the $(\mathcal{L}-\mathcal{K})$-loops as shown in equation (5.12) of [69]:

$$\begin{aligned}\mathrm{d}(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = &\left[\mathcal{K}^{(2)} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t + \sum_{l_\mathcal{K}=3}^{n} \left[\mathcal{K}^{(l_\mathcal{K})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t \\ &+ \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}t + \mathcal{E}^{\mathring{G},(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}t + \mathrm{d}\mathcal{E}^{M,(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}.\end{aligned} \tag{3.8}$$

Here, every $\mathcal{K}^{(l_\mathcal{K})}$, $2 \le l_\mathcal{K} \le n$, is regarded as a linear operator acting on the $(\mathcal{L}-\mathcal{K})$-loops, defined as:

$$\begin{aligned}\left[\mathcal{K}^{(l_\mathcal{K})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} := &\, W^d \sum_{1\le k<l\le n: l-k=l_\mathcal{K}-1} \sum_{a,b} (\mathcal{L}-\mathcal{K})^{(n-l_\mathcal{K}+2)}_{t,(\mathcal{G}_L)^{(a)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})} S^{(\mathsf{B})}_{ab} \mathcal{K}^{(l_\mathcal{K})}_{t,(\mathcal{G}_R)^{(b)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})} \\ &+ W^d \sum_{1\le k<l\le n: l-k=n-l_\mathcal{K}+1} \sum_{a,b} \mathcal{K}^{(l_\mathcal{K})}_{t,(\mathcal{G}_L)^{(a)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})} S^{(\mathsf{B})}_{ab} (\mathcal{L}-\mathcal{K})^{(n-l_\mathcal{K}+2)}_{t,(\mathcal{G}_R)^{(b)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})},\end{aligned} \tag{3.9}$$

$\mathcal{E}^{\mathring{G}}$ and $\mathrm{d}\mathcal{E}^M$ are defined in (2.49) and (2.48), respectively, and the term $\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}$ is defined by

$$\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} := W^d \sum_{1\le k<l\le n} \sum_{a,b} (\mathcal{L}-\mathcal{K})^{(n+k-l+1)}_{t,(\mathcal{G}_L)^{(a)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})} S^{(\mathsf{B})}_{ab} (\mathcal{L}-\mathcal{K})^{(l-k+1)}_{t,(\mathcal{G}_R)^{(b)}_{k,l}(\boldsymbol{\sigma},\mathbf{a})}. \tag{3.10}$$

[6] This parameter Ψ_t should be distinguished from the matrix Ψ appearing in the block Anderson Hamiltonian (2.30).

In (3.8), the superscript "(n)" indicates the length of the $(\mathcal{L}-\mathcal{K})$-loop on the LHS of the equation, and we will sometimes omit it from our notations when the value of n is clear from the context.

We now rewrite (3.8) into an integral equation using Duhamel's principle. First, we introduce the evolution kernel associated with this integral equation.

Definition 3.3 (Evolution kernel)**.** *For each $t \in [0,1)$, fixed $n \geq 2$ and $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n) \in \{+,-\}^n$, we define the linear operator $\Theta^{(n)}_{t,\boldsymbol{\sigma}}$ acting on n-dimensional tensors $\mathcal{A} : (\mathbb{Z}^d_L)^n \to \mathbb{C}$ as follows (recall (2.59)):*

$$\left(\Theta^{(n)}_{t,\boldsymbol{\sigma}} \circ \mathcal{A}\right)_{\mathbf{a}} = \sum_{i=1}^{n} \sum_{b_i \in \mathbb{Z}^d_L} \left(\frac{M^{(\sigma_i,\sigma_{i+1})} S^{(\mathsf{B})}}{1 - tM^{(\sigma_i,\sigma_{i+1})} S^{(\mathsf{B})}} \right)_{a_i b_i} \mathcal{A}_{\mathbf{a}^{(i)}(b_i)}, \tag{3.11}$$

where $\mathbf{a} = (a_1, \dots, a_n) \in (\mathbb{Z}^d_L)^n$, $\mathbf{a}^{(i)}(b_i) := (a_1, \dots, a_{i-1}, b_i, a_{i+1}, \dots, a_n)$, and we adopt the cyclic convention that $\sigma_{n+1} = \sigma_1$. The evolution kernel corresponding to $\Theta^{(n)}_{t,\boldsymbol{\sigma}}$ is given by

$$\left(\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A}\right)_{\mathbf{a}} = \sum_{\mathbf{b}=(b_1,\dots,b_n)} \prod_{i=1}^{n} \left(\frac{1 - s \cdot M^{(\sigma_i,\sigma_{i+1})} S^{(\mathsf{B})}}{1 - t \cdot M^{(\sigma_i,\sigma_{i+1})} S^{(\mathsf{B})}} \right)_{a_i b_i} \cdot \mathcal{A}_{\mathbf{b}}. \tag{3.12}$$

By the definition of the 2-$\mathcal{K}$-loop in (2.71), we observe that the first term on the RHS of (3.8) can be rewritten as $\Theta^{(n)}_{t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma}}$. With this fact and using Duhamel's principle, we obtain the following lemma.

Lemma 3.4 (Integrated loop hierarchy, Lemma 5.3 of [69])**.** *First, (3.8) is equivalent to the integral equation: for any stopping time $\tau \geq s$ with respect to the matrix Brownian motion $\{H_t\}$,*

$$\begin{aligned}(\mathcal{L}-\mathcal{K})^{(n)}_{\tau,\boldsymbol{\sigma},\mathbf{a}} = (\mathcal{L}-\mathcal{K})^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}} + \int_s^\tau \left(\Theta^{(n)}_{u,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u + \sum_{l_{\mathcal{K}}=3}^{n} \int_s^\tau \left[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}u \\ + \int_s^\tau \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}u + \int_s^\tau \mathcal{E}^{\mathring{G},(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}u + \int_s^\tau \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}.\end{aligned} \tag{3.13}$$

Second, applying Duhamel's principle to (3.8), we obtain the following integrated loop hierarchy*:*

$$\begin{aligned}(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \left(\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \sum_{l_{\mathcal{K}}=3}^{n} \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \left[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \\ + \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u + \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{E}^{\mathring{G},(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u + \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}.\end{aligned} \tag{3.14}$$

Before returning to the proof of Theorem 2.24, we introduce the following notation, which controls the quadratic variation of the martingale term in (2.48). More precisely, $\mathrm{d}\mathcal{E}^{M,(n)}$ is a sum over the n edges of the loop at which the loop is cut, and the tensor defined below keeps only the terms with the same cut on both sides. Hence, by the Cauchy–Schwarz inequality over the n cuts, the quadratic variation of any linear combination $\sum_{\mathbf{a}} \varphi_{\mathbf{a}} \, \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}$ is bounded by $n \sum_{\mathbf{a},\mathbf{a}'} \varphi_{\mathbf{a}} \overline{\varphi_{\mathbf{a}'}} \, (\mathcal{E} \otimes \mathcal{E})^{M,(n)}_{u,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} \, \mathrm{d}u$; in general, the two are not equal because of the cross terms between different cuts. The factor n is absorbed into the constant $C_{n,p}$ in Lemma 3.6 below.

Definition 3.5 (Quadratic variation loop)**.** *For $t \in [0,1]$ and $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n) \in \{+,-\}^n$, we introduce the $(2n)$-dimensional* quadratic variation tensor *for any $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{a}' = (a'_1, \dots, a'_n)$:*

$$(\mathcal{E} \otimes \mathcal{E})^{M,(n)}_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} := \sum_{k=1}^{n} (\mathcal{E} \otimes \mathcal{E})^{M,(n;k)}_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'}, \quad \text{where} \quad (\mathcal{E} \otimes \mathcal{E})^{M,(n;k)}_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} := W^d \sum_{b,b'} S^{(\mathsf{B})}_{bb'} \mathcal{L}^{(2n+2)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(k)},(\mathbf{a}\otimes\mathbf{a}')^{(k)}(b,b')}. \tag{3.15}$$

Here, $\mathcal{L}^{(2n+2)}$ denotes a $(2n+2)$-loop obtained by cutting the k-th edge of $\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ and then gluing it (with indices $\mathbf{a}$) with its conjugate loop (with indices $\mathbf{a}'$) along the newly introduced vertices b and b'. Formally:

$$\begin{aligned}\mathcal{L}^{(2n+2)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(k)},(\mathbf{a}\otimes\mathbf{a}')^{(k)}(b,b')} := \mathrm{Tr}\Bigg\{ \prod_{i=k}^{n} (G_t(\sigma_i)E_{a_i}) \cdot \prod_{i=1}^{k-1} (G_t(\sigma_i)E_{a_i}) \cdot G_t(\sigma_k) E_b G_t(-\sigma_k) \\ \times \prod_{i=1}^{k-1} \left(E_{a'_{k-i}} G_t(-\sigma_{k-i})\right) \cdot \prod_{i=k}^{n} \left(E_{a'_{n+k-i}} G_t(-\sigma_{n+k-i})\right) E_{b'} \Bigg\},\end{aligned}$$

where the notations $(\mathbf{a}\otimes\mathbf{a}')^{(k)}$ *and* $(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(k)}$ *(with* $\overline{\boldsymbol{\sigma}}$ *denoting* $(-\sigma_1,\dots,-\sigma_n)$*) represent respectively*

$$(\mathbf{a}\otimes\mathbf{a}')^{(k)}(b,b') = (a_k,\dots,a_n,a_1,\dots,a_{k-1},b,a'_{k-1},\dots,a'_1,a'_n,\dots,a'_k,b'),$$
$$(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(k)} = (\sigma_k,\dots,\sigma_n,\sigma_1,\dots,\sigma_k,-\sigma_k,\dots,-\sigma_1,-\sigma_n,\dots,-\sigma_k). \tag{3.16}$$

Under the above notations, applying the Burkholder-Davis-Gundy inequality, we obtain the following lemma that provides high-moment bounds on the martingale term.

Lemma 3.6 (Lemma 5.5 of [69])**.** *Let* τ *be a stopping time with respect to the matrix Brownian motion* $\{H_t\}$*. Then, for any fixed* $p\in\mathbb{N}$*, there exists a constant* $C_{n,p}$ *such that*

$$\mathbb{E}\left[\int_s^\tau \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right]^{2p} \le C_{n,p}\mathbb{E}\left(\int_s^\tau \left((\mathcal{E}\otimes\mathcal{E})^{M,(n)}_{u,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}}\right)\mathrm{d}u\right)^p, \tag{3.17}$$

$$\mathbb{E}\left[\int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\right]^{2p} \le C_{n,p}\mathbb{E}\left(\int_s^t \left(\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\otimes\mathcal{U}^{(n)}_{u,t,\overline{\boldsymbol{\sigma}}}\right)\circ(\mathcal{E}\otimes\mathcal{E})^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a},\mathbf{a}}\mathrm{d}u\right)^p. \tag{3.18}$$

Here, $\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\otimes\mathcal{U}^{(n)}_{u,t,\overline{\boldsymbol{\sigma}}}$ *denotes the tensor product of the evolution kernel defined in (3.12):*

$$\left[\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\otimes\mathcal{U}^{(n)}_{u,t,\overline{\boldsymbol{\sigma}}}\right)\circ\mathcal{A}\right]_{\mathbf{a},\mathbf{a}'} = \sum_{\mathbf{b},\mathbf{b}'\in(\mathbb{Z}^d_L)^n}\prod_{i=1}^n\left(\frac{1-u\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}\right)_{a_ib_i}\prod_{i=1}^n\left(\frac{1-u\cdot M^{(-\sigma_i,-\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(-\sigma_i,-\sigma_{i+1})}S^{(\mathsf{B})}}\right)_{a'_ib'_i}\mathcal{A}_{\mathbf{b},\mathbf{b}'}$$

for any $(2n)$*-dimensional tensor* $\mathcal{A}:(\mathbb{Z}^d_L)^{2n}\to\mathbb{C}$ *and* $\mathbf{b}=(b_1,\dots,b_n)$*,* $\mathbf{b}'=(b'_1,\dots,b'_n)$*.*

3.3. **Proof of Step 2.** In this subsection, we focus on the proof of (2.89). The local laws (2.87) and (2.88) then follow directly from (2.89) together with Lemma 3.1. Because of the hierarchical structure of (2.47), establishing the pointwise stability estimate (2.89) would, in principle, require *pointwise control* of all higher-order G-loops with length $n\ge 3$. Truncating this "pointwise" loop hierarchy (even if possible) would be substantially more involved than handling the "maximum" loop hierarchy used in our current approach. Fortunately, by combining two key technical ingredients—the diagrammatic techniques developed for the light-weight estimate and the loop-contraction inequality (see Lemma 3.16) used in the martingale estimate—we can overcome this difficulty and effectively truncate the pointwise loop hierarchy at order $n=2$. This is one of the pivotal components of our proof: without it, both Step 2 and the L^∞-stability of the tree approximation of the loop hierarchy, established in Steps 3 and 4 below, would break down.

We begin by defining the following tail functions to quantify the pointwise decay.

Definition 3.7 (Tail functions)**.** *For any* $t\in[0,1)$*, we define a tail function* $\mathcal{T}_t:[0,\infty)\to[0,\infty)$ *as*

$$\mathcal{T}_t(r) := B_{t,r}\cdot e^{-(r/\ell_t)^{1/2}},\quad \forall r\ge 0, \tag{3.19}$$

which corresponds to the Θ*-propagator bound in (2.66). Given any* $0\le\ell\le L$ *and large constant* $D>0$*, we will also use a truncated tail function* $\widetilde{\mathcal{T}}$ *defined as:*

$$\widetilde{\mathcal{T}}^\ell_{t,D}(r) := \max\left(\mathcal{T}_t(r\wedge\ell),\ W^{-D}\right). \tag{3.20}$$

Note that $\mathcal{T}_t$ is a decreasing function of $r\ge 0$. Moreover, for $0\le r\le L$, when $1-t\ge g^2/L^2$, the term $(L^d|1-t|)^{-1}$ is always dominated by the term $(g^2+|1-t|)^{-1}/(r+1)^{d-2}$ in $B_{t,r}$, while when $1-t\le g^2/L^2$, $\ell_t=L$ and the exponential factor $\exp(-(r/\ell_t)^{1/2})$ in (3.19) is of constant order. The tail function $\mathcal{T}_t$ satisfies the following elementary estimate, whose proof is provided in Appendix A.3.

Lemma 3.8 (Property of $\mathcal{T}$)**.** *There exists a constant* $C_d>0$ *(depending only on* d*) such that the following bound holds for any* $0\le u\le t<1$ *satisfying (i)* $1-u\ge 1-t\ge g^2/L^2$*, or (ii)* $1-t\le 1-u\le g^2/L^2$*:*

$$\sum_{c\in\mathbb{Z}^d_L}\mathcal{T}_u(|a-c|)\cdot\mathcal{T}_t(|c-b|)\le\frac{C_d}{1-u}\cdot\mathcal{T}_t(|a-b|),\quad\forall a,b\in\mathbb{Z}^d_L. \tag{3.21}$$

Our core strategy for the proof of Step 2 involves iterating the following self-improving estimates: for any large constant $D>0$, we have that

$$\sup_{u\in[s,t]}\max_{\mathbf{a}=(a,b)}\frac{\left|\mathcal{L}^{(2)}_{u,(+,-),\mathbf{a}}\right|}{W^{-d}\widetilde{\mathcal{T}}^\ell_{u,D}(|a-b|)}\prec 1\implies\sup_{u\in[s,t]}\max_{\mathbf{a}=(a,b)}\max_{\boldsymbol{\sigma}}\frac{\left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right|}{W^{-d}\widetilde{\mathcal{T}}^\ell_{u,D}(|a-b|)}\prec\left(\frac{1-s}{1-t}\right)^{C_d}(W^{-d}B_{t,0})^{1/5}$$

$$\Longrightarrow \sup_{u\in[s,t]}\max_{\mathbf{a}=(a,b)}\frac{\left|\mathcal{L}^{(2)}_{u,(+,-),\mathbf{a}}\right|}{W^{-d}\widetilde{\mathcal{T}}^{\ell'}_{u,D}(|a-b|)}\prec 1, \tag{3.22}$$

where the two scales $\ell'>\ell$ are roughly connected through the relation $\mathcal{T}_t(\ell')\asymp(W^{-d}B_{t,0})^{1/6}\cdot\mathcal{T}_t(\ell)$. Here, the decrease from 1/5 to 1/6 accounts for the prefactor $(|1-s|/|1-t|)^{C_d}$. In other words, (3.22) shows that once we have obtained a sharp 2-G-loop bound up to the scale ℓ, then after one iteration, we can push this bound to a slightly larger scale ℓ'. After O(1) iterations, we can improve the 2-G-loop bound up to a scale $\widehat{\ell}$ with $\mathcal{T}_t(\widehat{\ell})=\mathrm{O}(W^{-D})$. At this scale, we have

$$(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\prec W^{-d}\widetilde{\mathcal{T}}^{\widehat{\ell}}_{u,D}(|a-b|)\asymp W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a-b|).$$

Together with the middle step of (3.22), it implies the desired estimate (2.89).

To establish (3.22), we use equation (3.13) with $n=2$:

$$(\mathcal{L}-\mathcal{K})^{(2)}_{\tau,\boldsymbol{\sigma},\mathbf{a}}=(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}}+\int_s^\tau\left(\Theta^{(2)}_{u,\boldsymbol{\sigma}}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}+\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}_{u,\boldsymbol{\sigma}}+\mathcal{E}^{\mathring{G}}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\mathrm{d}u+\int_s^\tau\mathrm{d}\mathcal{E}^{M}_{u,\boldsymbol{\sigma},\mathbf{a}}, \tag{3.23}$$

where we omit "(2)" from some superscripts. Define the following (random) control parameter:

$$\widehat{\mathcal{J}}^{\ell}_{u,D}:=\max_{\mathbf{a}=(a,b)}\max_{\boldsymbol{\sigma}}\left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right|/\left[W^{-d}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|)\right]. \tag{3.24}$$

The first two terms in the integrand on the RHS of (3.23) can be bounded as in the following lemma, whose proof is postponed to Section 3.4.

Lemma 3.9. *In the setting of Theorem 2.24, suppose the weak local law (2.86) holds. Then, there exists a constant $C>0$ (depending only on c_d and C_d in (2.66)) such that the following estimates hold with high probability for all $0\le\ell\le L$:*

$$\max_{\mathbf{a}=(a,b)}\max_{\boldsymbol{\sigma}}\left|\left(\Theta^{(2)}_{u,\boldsymbol{\sigma}}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\right|\Big/\left[W^{-d}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|)\right]\le\frac{C}{1-u}\cdot\widehat{\mathcal{J}}^{\ell}_{u,D}, \tag{3.25}$$

$$\max_{\mathbf{a}=(a,b)}\max_{\boldsymbol{\sigma}}\left|\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}_{u,\boldsymbol{\sigma},\mathbf{a}}\right|\Big/\left[W^{-d}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|)\right]\le\frac{C}{1-u}\cdot\left(\widehat{\mathcal{J}}^{\ell}_{u,D}+\left(\widehat{\mathcal{J}}^{\ell}_{u,D}\right)^2\cdot\mathbf{1}_{\ell\ge1}\right). \tag{3.26}$$

Next, we bound the light-weight term $\mathcal{E}^{\mathring{G}}$ using the following two lemmas. Lemma 3.10 is applied to recover the polynomial decay $B_{t,\cdot}$ in (2.62), while Lemma 3.11 is used to establish the tail behavior described by the tail function in (3.20).

Lemma 3.10 (Light-weight estimate: B-bound)**.** *In the setting of Theorem 2.24, assume that the bounds in (3.4) hold for some constant $\varepsilon_0>0$ and deterministic control parameter $W^{-d/2}\le\Psi_t\le W^{-\varepsilon_0}$. Suppose we have the estimate*

$$\mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},(a,b)}\prec\Psi_t^2(|a-b|),\quad\forall\,\boldsymbol{\sigma}\in\{(+,-),(-,+)\},\ a,b\in\mathbb{Z}^d_L, \tag{3.27}$$

for a class of deterministic parameters $0<\Psi_t(|a-b|)\le W^{-\varepsilon_0}$. Without loss of generality, assume that $\Psi_t(\ell)$ is monotonically decreasing for $\ell\ge0$ (otherwise, one may replace it by the function $\sup_{\ell'\ge\ell}\Psi_t(\ell')$). Suppose there exist constants $C_1,C_2>1$ such that the following relations hold for any constant $C>1$:

$$W^{-d/2}\lesssim\Psi_t(0)\asymp\Psi_t(\ell)\ \ \forall\,0\le\ell\le C,\quad\text{and}\quad\Psi_t(\ell_1)/\Psi_t(\ell_2)\le C_1(\ell_2/\ell_1)^{C_2}\ \ \forall\ell_2\ge\ell_1\ge1. \tag{3.28}$$

Then, the following estimate holds:

$$\mathcal{E}^{\mathring{G},(2)}_{t,\boldsymbol{\sigma},(a,b)}\prec\frac{1}{\eta_t}\Psi_t(0)\cdot\Psi_t^2(|a-b|),\quad\forall\,\boldsymbol{\sigma}\in\{+,-\}^2,\ a,b\in\mathbb{Z}^d_L. \tag{3.29}$$

In particular, if we take $\Psi_t(|a-b|)=(W^{-c_0}B_{t,|a-b|\wedge K})^{1/2}$ for a constant $c_0>0$ and some $0\le K\le L$, then

$$\mathcal{E}^{\mathring{G},(2)}_{t,\boldsymbol{\sigma},(a,b)}\prec\frac{1}{\eta_t}\left(W^{-c_0}B_{t,0}\right)^{1/2}\cdot W^{-c_0}B_{t,|a-b|\wedge K}. \tag{3.30}$$

Lemma 3.11 (Light-weight estimate: $\mathcal{T}$-bound)**.** *Given any $t\in[0,1)$, assume that the bounds in (3.4) hold for some constant $\varepsilon_0>0$ and deterministic control parameter $W^{-d/2}\le\Psi_t\le W^{-\varepsilon_0}$. Suppose for some $0\le\ell\le(\log W)^{10}\ell_t$, the following estimate holds for any large constant $D>0$:*

$$\mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},(a,b)}\prec W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|),\quad\forall\,\boldsymbol{\sigma}\in\{(+,-),(-,+)\},\ a,b\in\mathbb{Z}^d_L. \tag{3.31}$$

Then, the following estimate holds for any large constant $D > 0$:

$$\mathcal{E}^{\mathring{G},(2)}_{t,\boldsymbol{\sigma},(a,b)} \prec \frac{1}{\eta_t}(W^{-d}B_{t,0})^{1/2} \cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|), \quad \forall\, \boldsymbol{\sigma} \in \{+,-\}^2,\ a,b \in \mathbb{Z}^d_L. \tag{3.32}$$

Remark 3.12. The second condition in (3.28) implies that the parameter $\Psi_t(r)$ decays polynomially in r. In particular, the parameter $W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(r)$ in (3.31) does not satisfy this condition because of the exponential factor $\exp\left(-(r/\ell_t)^{1/2}\right)$ in (3.19). Consequently, Lemma 3.11 cannot be deduced directly from Lemma 3.10. The corresponding proof is technically more delicate, since factors of the form $\Psi_t(c|a-b|)$, for some constant $c \in (0,1)$, can no longer be replaced by $\Psi_t(|a-b|)$ as in the polynomial-decay setting. In Lemma 3.11, the assumption $\ell \le (\log W)^{10}\ell_t$ is made without loss of generality, since $\mathcal{T}_t(\ell) \le W^{-D}$ whenever $\ell > (\log W)^{10}\ell_t$.

The proofs of Lemmas 3.10 and 3.11 are based on bounding the high moments using Gaussian integration by parts and some diagrammatic tools developed in [65, 67]. We will present the details in Section 7. Finally, the martingale term is bounded as in the following lemma, whose proof will be postponed to Section 3.5.

Lemma 3.13 (Martingale estimate)**.** *Given any $t \in [0,1)$, suppose the setting of Lemma 3.10 holds. Consider the term $(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} \equiv (\mathcal{E}\otimes\mathcal{E})^{M,(2;k)}_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}}$ defined in (3.15) for any $\mathbf{a} = (a,b) \in (\mathbb{Z}^d_L)^2$, $\boldsymbol{\sigma} = (\sigma_1,\sigma_2) \in \{+,-\}^2$, and $k \in \{1,2\}$. First, the following estimate holds:*

$$(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} \prec \frac{1}{\eta_t}\Psi_t(0)\cdot\Psi_t^4\left(|a-b|\right). \tag{3.33}$$

In particular, if we take $\Psi_t(|a-b|) = (W^{-c_0}B_{t,|a-b|\wedge K})^{1/2}$ for a constant $c_0 > 0$ and some $0 \le K \le L$, then

$$(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} \prec \frac{1}{\eta_t}\left(W^{-c_0}B_{t,0}\right)^{1/2}\cdot\left(W^{-c_0}B_{t,|a-b|\wedge K}\right)^2. \tag{3.34}$$

Second, suppose for some $0 \le \ell \le (\log W)^{10}\ell_t$, the estimate (3.31) holds for any large constant $D > 0$. Then, the following estimate holds for any large constant $D > 0$:

$$(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} \prec \frac{1}{\eta_t}\left[\left(W^{-d}B_{t,0}\right)^{1/2} + \left(\widehat{\mathcal{T}}^{\ell}_{t,D}\right)^3\right]\cdot\left(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|)\right)^2. \tag{3.35}$$

Remark 3.14. We remark that the martingale estimate is the main bottleneck of our proof, preventing us from improving the exponent 1/5 in (2.92) to the optimal value 1. This might be overcome by incorporating certain (not necessarily sharp) spatial decay properties of higher-order G-loops into the loop hierarchy analysis. However, such a refinement is far from a straightforward extension of the current argument. Even if achievable, it would entail substantial additional technical complexity while yielding only a "marginal" improvement in the quantum diffusion estimates. Since this refinement does not affect the main results—delocalization (Theorem 2.1), local laws (Theorem 2.2), QUE (Theorem 2.3), and bulk universality (Theorem 2.4)—we leave this direction for future investigation.

With the above Lemmas 3.9–3.13 as inputs, we are ready to control the terms in equation (3.23) and complete Step 2 of the proof of Theorem 2.24.

Step 2: Proof of (2.87)–(2.89). Suppose the estimate (2.89) has been established. Then, using (2.71) and (2.66), we obtain

$$\mathcal{K}^{(2)}_{u,(-,+),(a_1,a_2)} \prec W^{-d}B_{u,|a_1-a_2|}, \quad \forall a_1,a_2 \in \mathbb{Z}^d_L. \tag{3.36}$$

Together with (2.89), this implies—provided $\mathfrak{c}_d$ in (2.83) is chosen sufficiently small—the 2-G-loop bound

$$\mathcal{L}^{(2)}_{u,(-,+),(a_1,a_2)} \prec W^{-d}B_{u,|a_1-a_2|}, \quad \forall a_1,a_2 \in \mathbb{Z}^d_L. \tag{3.37}$$

Applying the weak local law (2.86) from Step 1 to verify the first condition in (3.4), and then invoking Lemma 3.1 with the control parameter $\Psi_u := \max\{W^{-d/2}, (W^{-d}B_{u,0})^{1/2}\}$, we obtain the entrywise local law (2.87) and the averaged local law (2.88). Here, the maximum in the definition of Ψ_u only ensures the lower bound $\Psi_u \ge W^{-d/2}$ required in Lemma 3.1: since $B_{u,0} \ge (g^2+1)^{-1} \ge (\mathfrak{d}^{-2}+1)^{-1}$ by (2.10), Ψ_u is comparable to $(W^{-d}B_{u,0})^{1/2}$, so the second condition in (3.4) follows from (3.37), and the resulting bounds are the same as with $\Psi_u = (W^{-d}B_{u,0})^{1/2}$.

It remains to establish the estimate (2.89). From the bound (2.85) proved in Step 1, we know

$$\max_{\boldsymbol{\sigma},\mathbf{a}}\left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \frac{1-s}{1-t}(W^{-d}B_{t,0}) =: W^{-c_0}, \quad \forall u \in [s,t], \tag{3.38}$$

where $c_0 \equiv c_0(W) \gtrsim 1$ under the condition (2.83). We first prove the following maximum bound:

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec W^{-d} B_{u,0}, \quad \forall u \in [s,t]. \tag{3.39}$$

Using Lemma 3.9 (for the case $\ell = 0$), Lemma 3.10, and the inductive hypothesis (2.78) with $n = 2$, we obtain from the flow (3.23) (with $\tau = t$) that

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \le \int_s^t \frac{C_0}{1-u} \max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \mathrm{d}u + \mathrm{O}_{\prec}\left((W^{-d}B_{s,0})^2 + W^{-\frac{3}{2}c_0}\right) + \left|\int_s^t \mathrm{d}\mathcal{E}^M_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \tag{3.40}$$

for a constant $C_0 > 0$ depending only on c_d and C_d in (2.66). Combining the estimate (3.33) in Lemma 3.13 with Lemma 3.6, and applying Markov's inequality, we obtain that

$$\left|\int_s^t \mathrm{d}\mathcal{E}^M_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(\int_s^t \frac{W^{-5c_0/2}}{\eta_u}\mathrm{d}u\right)^{1/2} \prec W^{-\frac{5}{4}c_0}.$$

Together with (3.40), it gives

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \le \int_s^t \frac{C_0}{1-u} \max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \mathrm{d}u + \mathrm{O}_{\prec}\left((W^{-d}B_{s,0})^2 + W^{-\frac{5}{4}c_0}\right). \tag{3.41}$$

Recall the following classical Grönwall's inequality, where β is non-negative and α is non-decreasing:

$$f(t) \le \alpha(t) + \int_s^t \beta(u) f(u)\mathrm{d}u \implies f(t) \le \alpha(t)\exp\left(\int_s^t \beta(u)\mathrm{d}u\right). \tag{3.42}$$

Applying it to (3.41) yields that

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(\frac{1-s}{1-t}\right)^{C_0} \left((W^{-d}B_{t,0})^2 + W^{-\frac{5}{4}c_0}\right) \le W^{-d}B_{t,0}, \tag{3.43}$$

where, in the last step, we use that

$$\left(\frac{1-s}{1-t}\right)^{C_0} W^{-\frac{5}{4}c_0} \le W^{-d}B_{t,0} \cdot \left(\frac{1-s}{1-t}\right)^{C_0+5/4} \left(W^{-d}B_{t,0}\right)^{1/4} \ll W^{-d}B_{t,0}$$

as long as we choose $\mathfrak{c}_d$ in (2.83) sufficiently small depending on C_0. The last bound in (3.43) is precisely (3.39) at $u = t$. The same argument applies at each fixed $u \in [s,t]$, and a standard N^{-C}-net and perturbation argument extends it uniformly to all $u \in [s,t]$, which concludes (3.39).

We can obtain from (3.39) that for $\ell^{(0)} = 0$ and any large constant $D > 0$,

$$\max_{\mathbf{a}=(a,b)} \max_{\boldsymbol{\sigma}} \left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \Big/ \left(W^{-d}\widetilde{\mathcal{T}}^{\ell^{(0)}}_{u,D}(|a-b|)\right) \prec 1, \quad \forall u \in [s,t], \tag{3.44}$$

where we also use that

$$\mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a-b|) \tag{3.45}$$

by (2.66) and the definition of 2-$\mathcal{K}$-loop in (2.71). Now, fix a large constant $D > 0$. Suppose we have the following bound for a collection of length scales $\{K_u \ge 0 : u \in [s,t]\}$:

$$\max_{\mathbf{a}=(a,b)} \max_{\boldsymbol{\sigma}} \left|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \Big/ \left(W^{-d}\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\right) \prec 1, \quad \forall u \in [s,t]. \tag{3.46}$$

Moreover, assume that $\mathcal{T}_u(K_u)$ is non-decreasing in u, i.e.,

$$\mathcal{T}_u(K_u) \le \mathcal{T}_v(K_v), \quad \widetilde{\mathcal{T}}^{K_u}_{u,D}(r) \le \widetilde{\mathcal{T}}^{K_v}_{v,D}(r), \quad \forall s \le u \le v \le t,\ r \ge 0. \tag{3.47}$$

Our goal is to establish the following estimate for a constant $C_0 > 0$ depending only on c_d and C_d in (2.66):

$$\widehat{\mathcal{J}}^{K_u}_{u,D} \prec (|1-s|/|1-u|)^{C_0} (W^{-d}B_{t,0})^{1/5}, \quad \forall u \in [s,\tau], \tag{3.48}$$

where τ is a stopping time defined as

$$\tau := t \wedge T, \quad \text{with} \quad T := \inf\left\{u \ge s : \widehat{\mathcal{J}}^{K_u}_{u,D} \ge (W^{-d}B_{u,0})^{1/6}\right\}. \tag{3.49}$$

For the proof of (3.48), using Lemma 3.11, we obtain that

$$\int_s^\tau \mathring{\mathcal{E}}^{G}_{u,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}u \prec \int_s^\tau \frac{1}{\eta_u}(W^{-d}B_{u,0})^{1/2} \cdot \left(W^{-d}\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\right)\mathrm{d}u \prec (W^{-d}B_{\tau,0})^{1/2} \cdot \left(W^{-d}\widetilde{\mathcal{T}}^{K_\tau}_{\tau,D}(|a-b|)\right), \tag{3.50}$$

where, in the second step, we also use the fact that $B_{u,0}$ and $\widetilde{\mathcal{T}}^{K_u}_{u,D}(r)$ are monotonically increasing in u. Similarly, using Lemma 3.13 along with Lemma 3.6 and the definition of the stopping time τ, we obtain that

$$\begin{aligned}\int_s^\tau \mathrm{d}\mathcal{E}^M_{u,\boldsymbol{\sigma},\mathbf{a}} &\prec \Big[\big(W^{-d}B_{\tau,0}\big)^{1/4} + \sup_{u\in[s,\tau]} \Big(\widehat{\mathcal{J}}^{K_u}_{u,D}\Big)^{3/2} \Big] \cdot \Big(W^{-d}\widetilde{\mathcal{T}}^{K_\tau}_{\tau,D}(|a-b|)\Big) \\ &\lesssim \big(W^{-d}B_{\tau,0}\big)^{1/4} \cdot \Big(W^{-d}\widetilde{\mathcal{T}}^{K_\tau}_{\tau,D}(|a-b|)\Big).\end{aligned} \tag{3.51}$$

Applying Lemma 3.9, (3.50), and (3.51) to equation (3.23) yields that

$$\begin{aligned}\Big|(\mathcal{L}-\mathcal{K})^{(2)}_{\tau,\boldsymbol{\sigma},\mathbf{a}}\Big| \le \Big|(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}}\Big| + \int_s^\tau \frac{C_0}{1-u}\widehat{\mathcal{J}}^{K_u}_{u,D}\cdot W^{-d}\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\,\mathrm{d}u \\ + \mathrm{O}_\prec\Big(\big(W^{-d}B_{\tau,0}\big)^{1/4}\cdot W^{-d}\widetilde{\mathcal{T}}^{K_\tau}_{\tau,D}(|a-b|)\Big).\end{aligned} \tag{3.52}$$

From this equation, using the induction hypothesis (2.79) at time s and the definition (3.24), we obtain that

$$\widehat{\mathcal{J}}^{K_\tau}_{\tau,D} \le \int_s^\tau \frac{C_0}{1-u}\widehat{\mathcal{J}}^{K_u}_{u,D}\mathrm{d}u + \mathrm{O}_\prec\Big(\big(W^{-d}B_{\tau,0}\big)^{1/5}\Big).$$

Applying Grönwall's inequality (3.42) again to this equation, we derive (3.48).

By the induction hypothesis (2.79) at time s, the stopping time T defined in (3.49) satisfies $T>s$ with high probability. In (3.48), using again (2.83) with $\mathfrak{c}_d$ chosen sufficiently small depending on C_0, we can ensure that with high probability, $\widehat{\mathcal{J}}^{K_u}_{u,D} \ll (W^{-d}B_{u,0})^{1/6}$ for all $u\in[s,\tau]$. This implies that $T\ge t$ with high probability, so that (3.48) holds for all $u\in[s,t]$. Next, we define a new collection of parameters $\{K'_u\ge 0 : u\in[s,t]\}$ as follows: for each $u\in[s,t]$, let K'_u be the unique positive solution to

$$\mathcal{T}_u(K'_u) = \mathcal{T}_u(K_u)\cdot(W^{-d}B_{u,0})^{1/6}. \tag{3.53}$$

Under this definition, the monotonicity relation (3.47) continues to hold for the family $\{K'_u\}_{u\in[s,t]}$. Furthermore, from (3.45) and (3.48), we obtain

$$\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec W^{-d}\Big(\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\cdot(W^{-d}B_{u,0})^{1/6} + \widetilde{\mathcal{T}}^{L}_{u,D}(|a-b|)\Big) \lesssim W^{-d}\widetilde{\mathcal{T}}^{K'_u}_{u,D}(|a-b|),$$

where, in the second step, we use (3.53) along with the relation $\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\asymp\widetilde{\mathcal{T}}^{L}_{u,D}(|a-b|)$ for $K_u\ge L$. We now use this as the input (with K_u replaced by K'_u) in (3.46) for the next iteration of the above argument, and show that (3.46) continues to hold at an even larger scale K''_u. From (3.53), it follows that, for any fixed $D>0$, after performing at most $\mathrm{O}(1)$ iterations (where the number of iterations depends on D), we reach scales $\{K_u : u\in[s,t]\}$ such that $\mathcal{T}_u(K_u)\le W^{-D}$ for all $u\in[s,t]$. Consequently, we have

$$\widetilde{\mathcal{T}}^{K_u}_{u,D}(|a-b|)\asymp\widetilde{\mathcal{T}}^{L}_{u,D}(|a-b|),\quad \forall u\in[s,t],\ a,b\in\mathbb{Z}^d_L.$$

Finally, applying (3.48) once more yields (2.89). □

Remark 3.15. We briefly explain why the stability estimate (2.89) *cannot* be established continuously along the flow using Duhamel's formula, as was done in the $d\in\{1,2\}$ case [69, 28, 35]. For a large constant $D>0$, these works consider the random control parameter $J_{u,D}\equiv\widehat{\mathcal{J}}^{L}_{u,D}$ (defined in (3.24)) at the full-system scale $\ell=L$. Roughly speaking, the previous approach begins with an initial high-probability bound $J_{s,D}\le W^{-c}$ (for some constant $c>0$) at $u=s$ and then propagates this bound along a W^{-C}-net of $[s,t]$. The essential idea is that if $J_{u,D}\le W^{-c}$ holds, a perturbative argument yields a slightly weaker bound $J_{u',D}\le W^{-c+\varepsilon}$ at $u'=u+W^{-C}$. This can then be bootstrapped back to $J_{u',D}\le W^{-c}$ using the loop hierarchy (3.14), at the cost of a probability loss W^{-D}, which remains acceptable since the total accumulation over the net is only W^{-D+C}. However, this mechanism breaks down once we attempt to incorporate the light-weight estimates from Lemmas 3.10 and 3.11. They rely on a moment method that degrades the probability bound much more severely—from W^{-D} in the original assumptions (3.27) (resp. (3.31)) to $W^{-\delta D}$ in (3.29) (resp. (3.32)) for some constant $\delta<1$. As a result, Lemmas 3.10 and 3.11 can each be applied only $\mathrm{O}(1)$ times, and thus the flow-based propagation strategy is no longer applicable. Instead, we must prove the light-weight estimate simultaneously for all $u\in[s,t]$, starting from the weak local law (2.86) established in Step 1. This necessitates the self-improving argument developed in (3.22).

Moreover, since our initial control of $J_{u,D}$ is too weak (it can be as large as W^D), the quadratic term $\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}$ cannot be effectively bounded by $J^2_{u,D}$, unlike in [69, 28, 35]. We must therefore rely on the

bound (3.26). However, this estimate invalidates the Duhamel-based argument used in dimensions $d \in \{1,2\}$: when inserted into Duhamel's formula, it contributes a term of size $(|1-s|/|1-t|)\cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|)$, which already breaks the first step of (3.22). As a result, we must treat $\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}$ as a main term rather than an error term, and reorganize the proof around a new Grönwall-type argument.

3.4. **Proof of Lemma 3.9.** Without loss of generality, it suffices to consider the case where

$$\mathcal{T}_u(\ell) \geq W^{-D}. \tag{3.54}$$

Otherwise, we can find $K \leq \ell$ such that $\mathcal{T}_u(K) = W^{-D}$, for which we always have $\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|) = \widetilde{\mathcal{T}}^{K}_{u,D}(|a-b|)$ by definition (3.20). For the proof of (3.25), by the definitions (3.11), (3.20), and (3.24), we have that

$$\begin{aligned} W^d\left|\left(\Theta^{(2)}_{u,\boldsymbol{\sigma}}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\right| &\leq C\widehat{\mathcal{J}}^{\ell}_{u,D}\sum_c\left|\left(\Theta^{(\sigma_1,\sigma_2)}_u M^{(\sigma_1,\sigma_2)}\right)_{ac}\right|\cdot\widetilde{\mathcal{T}}^{\ell}_{u,D}(|c-b|) \\ &\leq C\widehat{\mathcal{J}}^{\ell}_{u,D}\sum_{c:|c-b|\geq\ell}\left|\Theta^{(\sigma_1,\sigma_2)}_{u,ac}\right|\cdot\mathcal{T}_u(\ell) + C\widehat{\mathcal{J}}^{\ell}_{u,D}\sum_{c:|c-b|<\ell}\mathcal{T}_u(|a-c|)\mathcal{T}_u(|c-b|) \\ &\leq \frac{C}{1-u}\widehat{\mathcal{J}}^{\ell}_{u,D}\left[\mathcal{T}_u(\ell)+\mathcal{T}_u(|a-b|)\right] \leq \frac{C}{1-u}\widehat{\mathcal{J}}^{\ell}_{u,D}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-b|), \end{aligned} \tag{3.55}$$

where, in the second step, we use the estimate (2.66), and the facts that under the condition (3.54),

$$\widetilde{\mathcal{T}}^{\ell}_{u,D}(|c-b|) \leq \mathcal{T}_u(\ell) \quad \text{for} \quad |c-b|\geq\ell, \quad \widetilde{\mathcal{T}}^{\ell}_{u,D}(|c-b|) \leq \mathcal{T}_u(|c-b|) \quad \text{for} \quad |c-b|<\ell, \tag{3.56}$$

and in the third step, we use (2.65) and (3.21). This concludes (3.25).

The proof of (3.26) follows a similar argument. We have that for any $\boldsymbol{\sigma}_1,\boldsymbol{\sigma}_2 \in \{+,-\}^2$,

$$\begin{aligned} & W^d\sum_{c_1,c_2}\left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}_1,(a,c_1)}S^{(\mathsf{B})}_{c_1c_2}(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}_2,(c_2,b)}\right| \\ \leq & \sum_c\left(|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma}_1,(a,c)}|+|\mathcal{K}^{(2)}_{u,\boldsymbol{\sigma}_1,(a,c)}|\right)\cdot\max_{c':|c'-b|\geq(\ell-1)_+}\widehat{\mathcal{J}}^{\ell}_{u,D}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|c'-b|) \\ + & \sum_c\left(|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma}_2,(c,b)}|+|\mathcal{K}^{(2)}_{u,\boldsymbol{\sigma}_2,(c,b)}|\right)\cdot\max_{c':|c'-a|\geq(\ell-1)_+}\widehat{\mathcal{J}}^{\ell}_{u,D}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-c'|) \\ + & \mathbf{1}_{\ell\geq1}\cdot W^{-d}\left(\widehat{\mathcal{J}}^{\ell}_{u,D}\right)^2\sum_{c:|c-a|\vee|c-b|<\ell}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|a-c|)\,\widetilde{\mathcal{T}}^{\ell}_{u,D}(|c-b|). \end{aligned} \tag{3.57}$$

For the first term on the RHS of (3.57), applying the Cauchy–Schwarz inequality and Ward's identities (2.56) and (2.57), we get that with high probability,

$$\begin{aligned} \sum_c\left(|\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma}_1,(a,c)}|+|\mathcal{K}^{(2)}_{u,\boldsymbol{\sigma}_1,(a,c)}|\right) &\leq \sum_c\left(\mathcal{L}^{(2)}_{u,(-,+),(a,c)}+\mathcal{K}^{(2)}_{u,(-,+),(a,c)}\right) \\ &= \frac{\operatorname{Im}m+\operatorname{Im}\operatorname{Tr}(G_uE_a)}{W^d\eta_u} = \frac{1+\mathrm{o}(1)}{W^d(1-u)}, \end{aligned}$$

where we use the weak local law (2.86) and the relation (2.38) in the last step. With this estimate, we can bound the first term on the RHS of (3.57) as follows with high probability:

$$\frac{1+\mathrm{o}(1)}{W^d(1-u)}\max_{c':|c'-b|\geq(\ell-1)_+}\widehat{\mathcal{J}}^{\ell}_{u,D}\widetilde{\mathcal{T}}^{\ell}_{u,D}(|c'-b|) \leq \frac{C}{W^d(1-u)}\widehat{\mathcal{J}}^{\ell}_{u,D}\mathcal{T}_u(\ell),$$

where we have used (3.56) and $\mathcal{T}_u(K+1) \asymp \mathcal{T}_u(K)$ for any $K \geq 0$. The second term on the RHS of (3.57) can be bounded in the same way. For the last term on the RHS of (3.57), using (3.56), we bound it by

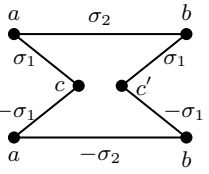
$$\mathbf{1}_{\ell\geq1}\cdot CW^{-d}\left(\widehat{\mathcal{J}}^{\ell}_{u,D}\right)^2\sum_c\mathcal{T}_u(|a-c|)\,\mathcal{T}_u(|c-b|) \leq \mathbf{1}_{\ell\geq1}\cdot\frac{C}{W^d(1-u)}\left(\widehat{\mathcal{J}}^{\ell}_{u,D}\right)^2\mathcal{T}_u(|a-b|), \tag{3.58}$$

where we use (3.21) in the second step. Combining the above two bounds, we conclude (3.26).

3.5. **Proof of Lemma 3.13.** We only control the term $(\mathcal{E}\otimes\mathcal{E})^{M}_{t,\boldsymbol{\sigma},\boldsymbol{a},\boldsymbol{a}} \equiv (\mathcal{E}\otimes\mathcal{E})^{M,(2;1)}_{t,\boldsymbol{\sigma},\boldsymbol{a},\boldsymbol{a}}$ without loss of generality. By definition (3.15), we can write it as follows with $\mathbf{a}(c,c') = (a,b,c',b,a,c)$ and $(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)} = (\sigma_1,\sigma_2,\sigma_1,-\sigma_1,-\sigma_2,-\sigma_1)$:

$$(\mathcal{E}\otimes\mathcal{E})^{M}_{t,\boldsymbol{\sigma},\boldsymbol{a},\boldsymbol{a}} = W^d \sum_{c,c'} S^{(\mathsf{B})}_{cc'} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')}. \tag{3.59}$$

We can represent the RHS using the following graph:

$W^d \sum_{c\sim c'} S^{(\mathsf{B})}_{cc'}$ 

(3.60)

Each solid edge between two block vertices (e.g., a and b) with charge $\sigma\in\{+,-\}$ denotes a resolvent entry $(G_t(\sigma))_{xy}$ or $(G_t(\sigma))_{yx}$ with $x\in[a]$ and $y\in[b]$. The orientation of the graph is assumed to be clockwise. To illustrate the idea for the proof of (3.33), assume without loss of generality that the vertex c is closer to b than to a. Then, the graph in (3.60) already contains four "long legs": in addition to the two edges connecting a and b, the two edges connecting a and c contribute a factor $\Psi_t^2(|a-c|)\le\Psi_t^2(|a-b|/2)\lesssim\Psi_t^2(|a-b|)$. Summing over the two remaining edges connected to c' and applying Cauchy–Schwarz together with Ward's identity yields an additional factor of η_t^{-1}. However, compared with the target estimate (3.33), we still miss the factor $\Psi_t(0)$. To recover it, we require a sharper treatment of the 6-G-loop that avoids applying Cauchy–Schwarz directly to the two edges connected to c'. Indeed, if we can directly perform the global summation over these two edges without taking absolute values, Ward's identity (2.54) produces a solid edge, which contributes the desired small factor $\Psi_t(0)$. This is precisely achieved via the following *loop-contraction inequality* for the 6-G-loop in (3.60), which can be regarded as a pointwise version of the more general loop-contraction inequality (4.2) below. For simplicity, we abbreviate $G\equiv G_t$ in the following proof.

Lemma 3.16 (Loop-contraction inequality)**.** *For any subset $\mathcal{A}\subset\mathbb{Z}_L^d$, we have that*

$$\sum_{c'\in\mathcal{A}}\sum_{c\sim c'}\mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')} \lesssim \frac{1}{W^d\eta_t}\max_{c'\in\mathcal{A}}\left(\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c',b,c',b)}\right)^{1/2}\cdot\max_{\sigma\in\{+,-\}}\left|\mathcal{L}^{(3)}_{t,(\sigma,\sigma_2,-\sigma_2),(a,b,a)}\right|, \tag{3.61}$$

where $\boldsymbol{\sigma}^{(\mathrm{alt})} = (\sigma_1,-\sigma_1,\sigma_1,-\sigma_1)$ is an alternating loop. By symmetry, a similar inequality holds:

$$\sum_{c\in\mathcal{A}}\sum_{c'\sim c}\mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')} \lesssim \frac{1}{W^d\eta_t}\max_{c\in\mathcal{A}}\left(\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(a,c,a,c)}\right)^{1/2}\cdot\max_{\sigma\in\{+,-\}}\left|\mathcal{L}^{(3)}_{t,(\sigma,-\sigma_2,\sigma_2),(b,a,b)}\right|. \tag{3.62}$$

Proof. We only prove the inequality (3.61), while the proof of (3.62) is similar by switching the roles of (a,c) and (b,c'). We view the symmetric 6-loop as a quadratic form. For any $c,c'\in\mathbb{Z}_L^d$ with $|c-c'|\le 1$ and any $z\in[c]$, we define the column vector $\psi^z\in\mathbb{C}^{W^d}$ and the $W^d\times W^d$ matrix $A^{(c')}$ as

$$\psi^z_y = (G(\sigma_1)E_aG(\sigma_2))_{zy},\quad\forall y\in[b],\quad\text{and}\quad A^{(c')}_{y_1y_2} = (G(\sigma_1)E_{c'}G(-\sigma_1))_{y_1y_2},\quad\forall y_1,y_2\in[b].$$

Then, each summand on the left-hand side (LHS) of (3.61) can be written as

$$\mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')} = W^{-3d}\sum_{z\in[c]}(\psi^z)^*A^{(c')}\psi^z \le W^{-3d}\|A^{(c')}\|\sum_{z\in[c]}\|\psi^z\|_2^2. \tag{3.63}$$

Since the operator norm of $A^{(c')}$ is bounded above by its Hilbert-Schmidt norm, we have

$$\|A^{(c')}\|^2 \le \sum_{y_1,y_2\in[b]}(G(\sigma_1)E_{c'}G(-\sigma_1))_{y_1y_2}(G(\sigma_1)E_{c'}G(-\sigma_1))_{y_2y_1} =: W^{2d}\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c',b,c',b)}. \tag{3.64}$$

On the other hand, the sum of the squared L^2-norms of ψ^z is equal to

$$\sum_{z\in[c]}\|\psi^z\|_2^2 = \sum_{z\in[c],y\in[b]}(G(\sigma_1)E_aG(\sigma_2))_{zy}(G(-\sigma_2)E_aG(-\sigma_1))_{yz} = W^{2d}\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}',(a,b,a,c)},$$

where $\boldsymbol{\sigma}' = (\sigma_1,\sigma_2,-\sigma_2,-\sigma_1)$. Plugging the previous two displays into (3.63) yields

$$\mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')} \le \left(\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c',b,c',b)}\right)^{1/2}\cdot\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}',(a,b,a,c)}. \tag{3.65}$$

Summing this inequality over $c' \in \mathcal{A}$ and $c \sim c'$, we obtain

$$\sum_{c' \in \mathcal{A}} \sum_{c \sim c'} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma} \otimes \overline{\boldsymbol{\sigma}})^{(1)}, \mathbf{a}(c,c')} \lesssim \max_{c' \in \mathcal{A}} \left(\mathcal{L}^{(4)}_{t, \boldsymbol{\sigma}^{(\mathrm{alt})}, (c', b, c', b)} \right)^{1/2} \cdot \sum_{c} \mathcal{L}^{(4)}_{t, \boldsymbol{\sigma}', (a,b,a,c)},$$

where the summation over c has been extended to all of $\mathbb{Z}^d_L$, at the cost of a dimensional constant, since each c has at most 3^d neighbours c'. Finally, applying Ward's identity (2.56) to the sum over c yields the desired bound (3.61). □

Now, we are ready to complete the proof of Lemma 3.13. We first prove the bound (3.33). We split the summation over c into two regions according to the distances from c to the fixed indices a and b:

$$(\mathcal{E} \otimes \mathcal{E})^M_{t, \boldsymbol{\sigma}, \boldsymbol{a}, \boldsymbol{a}} \lesssim W^d \sum_{\substack{c' \sim c \\ |c-b| > |c-a|}} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma} \otimes \overline{\boldsymbol{\sigma}})^{(1)}, \mathbf{a}(c,c')} + W^d \sum_{\substack{c' \sim c \\ |c-b| \le |c-a|}} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma} \otimes \overline{\boldsymbol{\sigma}})^{(1)}, \mathbf{a}(c,c')} =: \mathcal{S}_1 + \mathcal{S}_2. \tag{3.66}$$

By symmetry, it suffices to prove the bound (3.33) for the sum $\mathcal{S}_1$ using (3.61). (To control the sum $\mathcal{S}_2$, we only need to utilize the inequality (3.62) in place of (3.61) in the following proof.) Under the conditions $|c - c'| \le 1$ and $|c - b| \ge |a - b|/2$, we bound each off-diagonal G-entry of the 4-loop $\mathcal{L}^{(4)}_{t, \boldsymbol{\sigma}^{(\mathrm{alt})}, (c', b, c', b)}$ using (3.3), and each diagonal G-entry using (3.2). This gives

$$\left(\mathcal{L}^{(4)}_{t, \boldsymbol{\sigma}^{(\mathrm{alt})}, (c', b, c', b)} \right)^{1/2} \prec \max_{\substack{|b'-b| \le 1, |c''-c'| \le 1, \\ \boldsymbol{\sigma} \in \{(+,-),(-,+)\}}} \mathcal{L}^{(2)}_{t, \boldsymbol{\sigma}, (b', c'')} + W^{-d} \mathbf{1}_{|b-c'| \le 1} \prec \Psi_t^2(|b - c'|) \lesssim \Psi_t^2(|a - b|). \tag{3.67}$$

Here, the second step uses the assumption (3.27) on the 2-loops together with condition (3.28), while the third step follows from $|c - c'| \le 1$, $|c - b| \ge |a - b|/2$, and another application of (3.28). Plugging this bound into the loop-contraction inequality (3.61), we get that

$$\mathcal{S}_1 \prec \frac{1}{\eta_t} \Psi_t^2 \left(|a - b| \right) \max_{\sigma \in \{+,-\}} \left| \mathcal{L}^{(3)}_{t, (\sigma, \sigma_2, -\sigma_2), (a,b,a)} \right|. \tag{3.68}$$

To bound the 3-G-loop on the RHS, we write

$$\mathcal{L}^{(3)}_{t, (\sigma, \sigma_2, -\sigma_2), (a,b,a)} = W^{-3d} \sum_{x, x' \in [a]} \sum_{y \in [b]} G_{x x'}(\sigma) G_{x' y}(\sigma_2) G_{y x}(-\sigma_2),$$

and apply an argument analogous to that in (3.67), obtaining

$$\left| \mathcal{L}^{(3)}_{t, (\sigma, \sigma_2, -\sigma_2), (a,b,a)} \right| \prec \Psi_t(0) \cdot \Psi_t^2(|a - b|). \tag{3.69}$$

Roughly speaking, the factor $\Psi_t^2(|a - b|)$ arises from the two long legs—namely, the entries $G_{x' y}$ and $G_{y x}$ between $[a]$ and $[b]$. The short leg $G_{x x'}$, with both indices in $[a]$, contributes a factor of $\Psi_t(0) + W^{-d}$ by (3.2). (More precisely, each off-diagonal entry of G contributes a factor of $\Psi_t(0)$, while diagonal entries contribute W^{-d}.) Substituting (3.69) into (3.68) gives the desired bound (3.33).

In order to show the bound (3.35), we define two cutoff scales

$$\ell_t^* = (\log W)^{3/2} \ell_t, \quad \ell_t^\dagger = (\log W)^{7/4} \ell_t. \tag{3.70}$$

Then, we divide the proof into two cases. First, suppose $|a - b| \le \ell_t^\dagger$. In this case, the function $\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a - b|)$ exhibits no exponential decay. Hence, it suffices to apply the polynomial decay bound (3.34) with $c_0 = d$ and $K = \ell$, i.e. with the *truncated* profile function $\Psi_t(r) = (W^{-d} B_{t, r \wedge \ell})^{1/2}$. The conclusion (3.35) then follows directly from (3.34) and the facts

$$[\Psi_t(r)]^2 \prec W^{-d} \widetilde{\mathcal{T}}^{\ell}_{t,D}(r), \quad W^{-d} \widetilde{\mathcal{T}}^{\ell}_{t,D}(r) \prec [\Psi_t(r)]^2, \quad \forall\, 0 \le r \le \ell_t^\dagger,$$

both of which hold by the definition (3.20) of $\widetilde{\mathcal{T}}^{\ell}_{t,D}$, since $\mathcal{T}_t(\rho) \asymp B_{t,\rho} \exp(-(\rho/\ell_t)^{1/2})$ and $\exp(-(\rho/\ell_t)^{1/2}) \ge W^{-o(1)}$ for $0 \le \rho \le \ell \wedge \ell_t^\dagger$. We stress that the truncation is needed here: for the untruncated profile $(W^{-d} B_{t,r})^{1/2}$ the second bound fails in the range $\ell < r \le \ell_t^\dagger$, where $\widetilde{\mathcal{T}}^{\ell}_{t,D}(r) = \mathcal{T}_t(\ell)$ while $B_{t,r} \ll B_{t,\ell}$. It remains to deal with the case

$$|a - b| > \ell_t^\dagger = (\log W)^{7/4} \ell_t. \tag{3.71}$$

In this case, we split the summation (3.59) into three parts as follows:

$$(\mathcal{E}\otimes\mathcal{E})^{M}_{t,\boldsymbol{\sigma},\boldsymbol{a},\boldsymbol{a}} \lesssim W^{d}\Bigg(\sum_{|c-a|\le \ell_t^* \text{ or } |c'-b|>\ell}^{\star} + \sum_{|c'-b|\le \ell_t^* \text{ or } |c-a|>\ell}^{\star} + \sum_{\ell_t^*<|c'-b|,|c-a|\le \ell}^{\star}\Bigg)\mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\overline{\boldsymbol{\sigma}})^{(1)},\mathbf{a}(c,c')}$$
$$=: \widetilde{\mathcal{S}}_1+\widetilde{\mathcal{S}}_2+\widetilde{\mathcal{S}}_3. \tag{3.72}$$

where $\sum^{\star}$ refers to the summation subject to the constraint $|c-c'|\le 1$. To estimate $\widetilde{\mathcal{S}}_1$, we combine (3.3) with (3.31) and proceed analogously to (3.67). This yields

$$\left(\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c',b,c',b)}\right)^{1/2} \prec W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|b-c'|) \prec W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|), \tag{3.73}$$

where the second inequality is justified as follows: (1) if $|c'-b|>\ell$, then $\widetilde{\mathcal{T}}^{\ell}_{t,D}(|b-c'|)=\widetilde{\mathcal{T}}^{\ell}_{t,D}(\ell)\le\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|)$; (2) if $|c-a|\le\ell_t^*$, then by (3.71), we have $|b-c'|\ge|a-b|-(\ell_t^*+1)=(1-\mathrm{o}(1))|a-b|$, which implies

$$\widetilde{\mathcal{T}}^{\ell}_{t,D}(|b-c'|)\le\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|-(\ell_t^*+1)\right)\prec\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|).$$

Substituting (3.73) into the loop-contraction inequality (3.61), we obtain

$$\widetilde{\mathcal{S}}_1 \prec \frac{1}{\eta_t}\left[W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\right]\max_{\sigma\in\{+,-\}}\left|\mathcal{L}^{(3)}_{t,(\sigma,\sigma_2,-\sigma_2),(a,b,a)}\right|. \tag{3.74}$$

Next, using (3.3) and (3.31), we obtain a similar bound on the 3-G-loop as in (3.69):

$$\left|\mathcal{L}^{(3)}_{t,(\sigma,\sigma_2,-\sigma_2),(a,b,a)}\right| \prec (W^{-d}B_{t,0})^{1/2}\cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right).$$

Plugging this into (3.74) gives

$$\widetilde{\mathcal{S}}_1 \prec \frac{1}{\eta_t}\left(W^{-d}B_{t,0}\right)^{1/2}\left[W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\right]^2, \tag{3.75}$$

which is controlled by the RHS of (3.35). By symmetry, an identical bound holds for $\widetilde{\mathcal{S}}_2$.

It remains to control the sum $\widetilde{\mathcal{S}}_3$. In this case, we bound all legs of the original 6-loop directly using (3.3). Here, all legs of this loop have lengths (i.e., $|a-b|$, $|a-c|$, and $|c'-b|$) at least ℓ_t^*. Then, instead of using the assumption (3.31), we will bound the 2-loops using the parameter defined in (3.24) as follows: for any $a,b\in\mathbb{Z}_L^d$ satisfying $|a-b|\gtrsim\ell_t^*$,

$$\mathcal{L}^{(2)}_{t,(-,+),(a,b)} \prec \widehat{\mathcal{J}}^{\ell}_{t,D}\cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)+\mathcal{K}^{(2)}_{t,(-,+),(a,b)} \lesssim \left(\widehat{\mathcal{J}}^{\ell}_{t,D}+W^{-D}\right)\cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right), \tag{3.76}$$

where the second step is due to the exponential decay of the 2-$\mathcal{K}$-loop given by (2.66). Then, using (3.3) to bound the 6 legs of the 6-loop, we obtain that

$$\widetilde{\mathcal{S}}_3 \prec W^{-2d}\left(\widehat{\mathcal{J}}^{\ell}_{t,D}+W^{-D}\right)^3\sum_{\ell_t^*<|c'-b|,|c-a|\le\ell}^{\star}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-c|\right)\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|c-b|\right), \tag{3.77}$$

where we also use that $\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|c'-b|\right)\asymp\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|c-b|\right)$ for $|c-c'|\le1$. We can bound the RHS of (3.77) by

$$\left(\widehat{\mathcal{J}}^{\ell}_{t,D}+W^{-D}\right)^3\cdot W^{-2d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\cdot\sum_{c}\left[\mathcal{T}_t\left(|a-c|\right)+W^{-D}\right]\left[\mathcal{T}_t\left(|c-b|\right)+W^{-D}\right]$$
$$\lesssim\frac{(\widehat{\mathcal{J}}^{\ell}_{t,D}+W^{-D})^3}{1-t}\cdot W^{-2d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\cdot\left[\mathcal{T}_t\left(|a-b|\right)+W^{-D}\right]\lesssim\frac{(\widehat{\mathcal{J}}^{\ell}_{t,D}+W^{-D})^3}{\eta_t}\cdot\left(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}\left(|a-b|\right)\right)^2,$$

where we use (3.21) and the fact $\sum_a\mathcal{T}_t\left(|a|\right)\lesssim(1-t)^{-1}$. This finishes the estimation of $\widetilde{\mathcal{S}}_3$, and hence completes the proof of the bound (3.35).

4. Steps 3 and 4: Sharp maximum estimates for G-loops

In this section, we apply the integrated loop hierarchy (3.14) to complete the proofs of Steps 3 and 4. The key task is to show that the second through fifth terms in (3.14) are errors in the max-norm sense.

The integrands in the second to fourth terms on the RHS of (3.14) each involve a global sum over an index of an $\mathcal{L}$- or $(\mathcal{L}-\mathcal{K})$-loop (see (3.9), (3.10), and (2.49)). In low dimensions $d \in \{1,2\}$, these terms can be controlled directly using their max-norms, together with a factor ℓ_t^d that captures the range of exponential decay. For $d \in \{1,2\}$, the factor ℓ_t^d is bounded by $|1-t|^{-1}$, which incurs only a harmless logarithmic contribution upon integration in time. However, in higher dimensions $d \geq 3$, an additional factor ℓ_t^{d-2} arises on top of the $(1-t)^{-1}$, creating a substantial term that cannot be canceled. To obtain sufficiently sharp bounds without relying on the precise pointwise decay of higher-order G-loops, we develop a new *loop-contraction inequality* (4.1), derived from the Cauchy–Schwarz inequality and the Ward's identities (2.56) and (2.57). Although technically simple, this inequality provides a powerful tool: it allows us to control the absolute sum of a higher-order G-loop in terms of lower-order ones in a sharp max-norm sense, and crucially produces the desired $(1-t)^{-1}$ factor without introducing any excess powers of ℓ_t. Among all these terms, one remains that cannot be effectively controlled using the loop-contraction inequalities—namely, the term of the form $(\mathcal{L}-\mathcal{K})^{(2)} \diamond (\mathcal{L}-\mathcal{K})^{(n)}$ (recall the notation in (1.7)) in (3.10). This term is instead shown to be an error by applying the pointwise $(\mathcal{L}-\mathcal{K})^{(2)}$-estimate (2.89).

Bounding the martingale term requires controlling a $(2n+2)$-loop (recall Definition 3.5). The corresponding loop structure is symmetric and involves two summation indices along the loop (see e.g., (3.60)). To handle it, we derive another loop-contraction inequality (4.2), obtained by extending the argument in the proof of Lemma 3.16. This inequality yields a non-sharp but sufficient bound for the martingale term, introducing an additional power of W, i.e., the factor $(W^{-d}B_{t,0})^{-1/(2p)}$ in (4.15). However, this factor remains small enough for our proof for large enough p.

Our overall strategy for Steps 3 and 4 can now be summarized as follows. Using the loop-contraction inequalities together with a good stability estimate (2.89) for $(\mathcal{L}-\mathcal{K})^{(2)}$-loops, we can bound each term in (3.14) either by lower-order $(\mathcal{L}-\mathcal{K})$-loops or by higher-order $(\mathcal{L}-\mathcal{K})$-loops with an extra small factor $W^{-\varepsilon}$. Then, with the integrated hierarchy (3.14), we can bootstrap from a sequence of weaker bounds to tighter ones, ultimately yielding improved control on the entire hierarchy of $(\mathcal{L}-\mathcal{K})$-loops in each induction. As in [69, 28, 35], after O(1) iterations, this process yields (almost) sharp max-norm bounds for all $(\mathcal{L}-\mathcal{K})$-loops.

Lemma 4.1 (Loop-contraction inequality)**.** *For each $1 \leq k \leq n-1$, we have the bound*

$$\max_{\boldsymbol{\sigma}} \sum_{a_n} |\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \leq \frac{1}{W^d \eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \cdot \max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2n-2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \right)^{\frac{1}{2}}. \tag{4.1}$$

Furthermore, for any $n \geq 4$, $1 \leq k < j < l \leq n-1$, and subsets $\mathcal{A}(a_n) \subset \mathbb{Z}_L^d$ (which may depend on a_n) of cardinality $|\mathcal{A}(a_n)| \leq C$ for a constant $C > 0$, the following bound holds for any fixed $p \geq 1$:

$$\max_{\boldsymbol{\sigma}} \sum_{a_n} \sum_{a_j \in \mathcal{A}(a_n)} |\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \leq \frac{C}{W^d \eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \cdot \max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2n-2l-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \right)^{\frac{1}{2}} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2(l-k)p)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \right)^{\frac{1}{2p}}. \tag{4.2}$$

Proof. We first prove (4.1). Applying the Cauchy-Schwarz inequality with respect to the averages over $[a_k]$ and $[a_n]$ in the loop $\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$, we obtain that for $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_n)$ and $\mathbf{a} = (a_1, \ldots, a_n)$,

$$|\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \leq \left(\mathcal{L}^{(2k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1} \cdot \mathcal{L}^{(2n-2k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2} \right)^{1/2}, \tag{4.3}$$

where $\mathbf{a}_1$, $\mathbf{a}_2$, $\boldsymbol{\sigma}_1$, and $\boldsymbol{\sigma}_2$ are defined as

$$\begin{aligned} \mathbf{a}_1 &= (a_1, \cdots, a_{k-1}, a_k, a_{k-1}, \cdots, a_1, a_n), \quad \boldsymbol{\sigma}_1 = (\sigma_1, \cdots, \sigma_k, -\sigma_k, \cdots, -\sigma_1), \\ \mathbf{a}_2 &= (a_{n-1}, \cdots, a_{k+1}, a_k, a_{k+1}, \cdots, a_{n-1}, a_n), \quad \boldsymbol{\sigma}_2 = (-\sigma_n, \cdots, -\sigma_{k+1}, \sigma_{k+1}, \cdots, \sigma_n). \end{aligned}$$

Note that the loops $\mathcal{L}^{(2k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1}$ and $\mathcal{L}^{(2n-2k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2}$ are both non-negative since $\boldsymbol{\sigma}_1$ and $\boldsymbol{\sigma}_2$ are symmetric. Then, applying the Cauchy-Schwarz inequality to (4.3) again, we obtain that

$$\sum_{a_n} |\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \leq \Big(\sum_{a_n} \mathcal{L}^{(2k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1} \Big)^{1/2} \Big(\sum_{a_n} \mathcal{L}^{(2n-2k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2} \Big)^{1/2}. \tag{4.4}$$

Using Ward's identity (2.56), we can express the G-loops on the RHS as

$$\sum_{a_n} \mathcal{L}^{(2k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1} = \frac{\mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma}_1^+,\mathbf{a}_1'} - \mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma}_1^-,\mathbf{a}_1'}}{2\mathrm{i}W^d\eta_t}, \quad \sum_{a_n} \mathcal{L}^{(2n-2k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2} = \frac{\mathcal{L}^{(2n-2k-1)}_{t,\boldsymbol{\sigma}_2^+,\mathbf{a}_2'} - \mathcal{L}^{(2n-2k-1)}_{t,\boldsymbol{\sigma}_2^-,\mathbf{a}_2'}}{2\mathrm{i}W^d\eta_t}, \tag{4.5}$$

where $\mathbf{a}_1'$, $\mathbf{a}_2'$, $\boldsymbol{\sigma}_1^\pm$, and $\boldsymbol{\sigma}_2^\pm$ are defined as

$$\mathbf{a}_1' = (a_1, \cdots, a_{k-1}, a_k, a_{k-1} \cdots, a_1), \quad \boldsymbol{\sigma}_1^\pm = (\pm, \sigma_2, \cdots, \sigma_k, -\sigma_k, \cdots, -\sigma_2),$$
$$\mathbf{a}_2' = (a_{n-1} \cdots, a_{k+1}, a_k, a_{k+1}, \cdots, a_{n-1}), \quad \boldsymbol{\sigma}_2^\pm = (\pm, -\sigma_{n-1}, \cdots, -\sigma_{k+1}, \sigma_{k+1}, \cdots \sigma_{n-1}).$$

Plugging (4.5) into (4.4), we conclude (4.1).

For $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_n) \in \{+,-\}^n$ and $\mathbf{a} = (a_1, \ldots, a_{n-1}) \in (\mathbb{Z}_L^d)^{n-1}$, denote a G-chain of length n by

$$\mathcal{C}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \prod_{i=1}^{n-1} (G_t(\sigma_i)E_{a_i}) \cdot G_t(\sigma_n). \tag{4.6}$$

To show (4.2), we split the loop $\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ at a_k, a_l, and a_n, and write it as

$$\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = W^{-3d} \sum_{x_k \in [a_k]} \sum_{x_l \in [a_l]} \sum_{x_n \in [a_n]} \left(\mathcal{C}^{(k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1}\right)_{x_n x_k} \left(\mathcal{C}^{(l-k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2}\right)_{x_k x_l} \left(\mathcal{C}^{(n-l)}_{t,\boldsymbol{\sigma}_3,\mathbf{a}_3}\right)_{x_l x_n},$$

where $\mathbf{a}_1 = (a_1, \ldots, a_{k-1})$, $\boldsymbol{\sigma}_1 = (\sigma_1, \ldots, \sigma_k)$, $\mathbf{a}_2 = (a_{k+1}, \ldots, a_{l-1})$, $\boldsymbol{\sigma}_2 = (\sigma_{k+1}, \ldots, \sigma_l)$, $\mathbf{a}_3 = (a_{l+1}, \ldots, a_{n-1})$, and $\boldsymbol{\sigma}_3 = (\sigma_{l+1}, \ldots, \sigma_n)$. We can bound the RHS by using the operator norm of the $W^d \times W^d$ matrix $A = (A_{xy} : x \in [a_k], y \in [a_l])$ with $A_{xy} = (\mathcal{C}^{(l-k)}_{t,\boldsymbol{\sigma}_2,\mathbf{a}_2})_{xy}$:

$$\left|\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \le \frac{1}{W^{2d}} \sum_{x_n \in [a_n]} \left(\sum_{x_k \in [a_k]} \left| \left(\mathcal{C}^{(k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1}\right)_{x_n x_k} \right|^2 \right)^{1/2} \left(\sum_{x_l \in [a_l]} \left| \left(\mathcal{C}^{(n-l)}_{t,\boldsymbol{\sigma}_3,\mathbf{a}_3}\right)_{x_l x_n} \right|^2 \right)^{1/2} \cdot \frac{1}{W^d}\|A\|. \tag{4.7}$$

To control the operator norm $\|A\|$, we use the simple linear algebra fact $\|A\| \le \{\mathrm{Tr}[(AA^*)^p]\}^{\frac{1}{2p}}$ for any $p \in \mathbb{N}$, which gives that

$$\frac{1}{W^d}\|A\| \le \left\{ \frac{1}{W^{2pd}} \mathrm{Tr}\left[(AA^*)^p\right] \right\}^{\frac{1}{2p}} \le \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left| \mathcal{L}^{(2(l-k)p)}_{t,\boldsymbol{\sigma},\mathbf{b}} \right| \right)^{\frac{1}{2p}} \tag{4.8}$$

Plugging (4.8) into (4.7) and applying the Cauchy–Schwarz inequality, we obtain that

$$\begin{aligned}
\sum_{a_n} \sum_{a_j \in \mathcal{A}(a_n)} \left|\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| &\le C \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left| \mathcal{L}^{(2(l-k)p)}_{t,\boldsymbol{\sigma},\mathbf{b}} \right| \right)^{\frac{1}{2p}} \left(\frac{1}{W^{2d}} \sum_{a_n} \sum_{x_n \in [a_n]} \sum_{x_k \in [a_k]} \left| \left(\mathcal{C}^{(k)}_{t,\boldsymbol{\sigma}_1,\mathbf{a}_1}\right)_{x_n x_k} \right|^2 \right)^{1/2} \\
&\quad \times \left(\frac{1}{W^{2d}} \sum_{a_n} \sum_{x_n \in [a_n]} \sum_{x_l \in [a_l]} \left| \left(\mathcal{C}^{(n-l)}_{t,\boldsymbol{\sigma}_3,\mathbf{a}_3}\right)_{x_l x_n} \right|^2 \right)^{1/2} \\
&\le \frac{C}{W^d\eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left| \mathcal{L}^{(2(l-k)p)}_{t,\boldsymbol{\sigma},\mathbf{b}} \right| \right)^{\frac{1}{2p}} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left| \mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}} \right| \cdot \max_{\boldsymbol{\sigma},\mathbf{b}} \left| \mathcal{L}^{(2n-2l-1)}_{t,\boldsymbol{\sigma},\mathbf{b}} \right| \right)^{1/2},
\end{aligned}$$

where we applied Ward's identity (2.56) in the second step. This concludes (4.2). □

In the course of proving Lemma 2.16, we will also establish the following bound, whose proof is deferred to Appendix A.5.

Lemma 4.2. *For any $n \ge 2$ and $t \in [0,1)$, we have that*

$$\max_{\boldsymbol{\sigma} \in \{+,-\}^n} \sum_{a_n} |\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \prec \frac{1}{W^d\eta_t} (W^{-d}B_{t,0})^{n-2}. \tag{4.9}$$

For any $n \ge 1$, let $\Xi^{(\mathcal{L})}_{t,n} \ge 1$ and $\Xi^{(\mathcal{L}-\mathcal{K})}_{t,n} \ge 1$ be *deterministic* control parameters for $\mathcal{L}$-loops and $(\mathcal{L}-\mathcal{K})$-loops of length n such that the following bounds hold:

$$\widehat{\Xi}^{(\mathcal{L})}_{t,n} := 1 + \max_{\boldsymbol{\sigma} \in \{+,-\}^n} \max_{\mathbf{a} \in (\mathbb{Z}_L^d)^n} \left| \mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \right| \Big/ \left(W^{-d}B_{t,0}\right)^{n-1} \prec \Xi^{(\mathcal{L})}_{t,n}, \tag{4.10}$$

$$\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{t,n} := 1 + \max_{\boldsymbol{\sigma}\in\{+,-\}^n} \max_{\mathbf{a}\in(\mathbb{Z}^d_L)^n} \left|(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \Big/ \left(W^{-d}B_{t,0}\right)^n \prec \Xi^{(\mathcal{L}-\mathcal{K})}_{t,n}. \tag{4.11}$$

Using these control parameters and the loop-contraction inequalities in Lemmas 4.1 and 4.2, we can control the terms on the RHS of equation (3.14) as follows.

Lemma 4.3 (Estimates of $\mathcal{E}$ terms)**.** *In the setting of Theorem 2.24, suppose the local laws (2.87) and (2.88) and the 2-G-loop estimate (2.89) hold. Then, the following estimates hold for any fixed $n \geq 2$:*

(1) *The light-weight term defined in (2.49) satisfies that*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{E}^{\mathring{G},(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \frac{\left(W^{-d}B_{t,0}\right)^n}{\eta_t} \cdot \left(\widehat{\Xi}^{(\mathcal{L})}_{t,n_1} \cdot \widehat{\Xi}^{(\mathcal{L})}_{t,n_2}\right)^{\frac{1}{2}}, \tag{4.12}$$

where $n_1 = n-1$ and $n_2 = n+1$ if n is even, and $n_1 = n_2 = n$ if n is odd.

(2) *For $3 \leq l_{\mathcal{K}} \leq n$, the $[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})]^{(n)}$ term defined in (3.9) satisfies that*

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \frac{\left(W^{-d}B_{t,0}\right)^n}{\eta_t} \cdot \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{t,n-l_{\mathcal{K}}+2}. \tag{4.13}$$

(3) *The $\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}$ term defined in (3.10) satisfies that*

$$\begin{aligned}\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec{}& \frac{\left(W^{-d}B_{t,0}\right)^n}{\eta_t} \sum_{n'=\lceil n/2\rceil+1}^{n-1} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{t,n+2-n'} \cdot \left(\widehat{\Xi}^{(\mathcal{L})}_{t,n'_1} \cdot \widehat{\Xi}^{(\mathcal{L})}_{t,n'_2}\right)^{1/2} \\ &+ \frac{\left(W^{-d}B_{t,0}\right)^n}{\eta_t} \cdot \left(W^{-d}B_{t,0}\right)^{\frac{1}{6}} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{t,n},\end{aligned} \tag{4.14}$$

where $n'_1 = n'_2 = n'-1$ if n' is even, and $n'_1 = n'-2$ and $n'_2 = n'$ if n' is odd. Note that the first term on the RHS is zero when $n \in \{2,3\}$.

(4) *The term $(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} \equiv (\mathcal{E}\otimes\mathcal{E})^{M,(n;k)}_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'}$ defined in (3.15) satisfies the following estimate for each $k \in [\![n]\!]$ and any fixed $p \in \mathbb{N}$:*

$$\max_{\boldsymbol{\sigma}} \max_{\mathbf{a},\mathbf{a}'} (\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} \prec \frac{\left(W^{-d}B_{t,0}\right)^{2n-\frac{1}{2p}}}{\eta_t} \cdot \widehat{\Xi}^{(\mathcal{L})}_{t,2n-1} \left(\widehat{\Xi}^{(\mathcal{L})}_{t,4p}\right)^{\frac{1}{2p}}. \tag{4.15}$$

Proof. Using the inequality (4.1) (with $k = \lceil n/2\rceil$) and the averaged local law (2.88) established in Step 2, we can bound that

$$\begin{aligned}\left|\mathcal{E}^{\mathring{G},(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| &\leq W^d \sum_{k=1}^{n} \sum_{a,b\in\mathbb{Z}^d_L} \left|\mathrm{Tr}\left(\mathring{G}_t(\sigma_k)E_a\right)\right| S^{(\mathsf{B})}_{ab} \left|\left(\mathcal{G}^{(b)}_k \circ \mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right)\right| \\ &\prec W^d \cdot \left(W^{-d}B_{t,0}\right) \cdot \frac{1}{W^d\eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(n_1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \cdot \max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(n_2)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right|\right)^{\frac{1}{2}},\end{aligned}$$

which concludes (4.12) together with the definition (4.10). For (4.13), with the $\mathcal{K}$-loop bounds (2.58) and (4.9), we obtain that

$$\begin{aligned}\max_{\boldsymbol{\sigma},\mathbf{a}} \left|[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| &\lesssim W^d \left(\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n-l_{\mathcal{K}}+2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} \sum_{a_{l_{\mathcal{K}}}} \left|\mathcal{K}^{(l_{\mathcal{K}})}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \\ &\lesssim \frac{1}{\eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n-l_{\mathcal{K}}+2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \cdot \left(W^{-d}B_{t,0}\right)^{l_{\mathcal{K}}-2},\end{aligned}$$

which concludes (4.13) together with the definition (4.11).

For the estimate (4.14), we first consider the case $n \in \{2,3\}$. In this case, we have that

$$\max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \lesssim \left(\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} W^d \sum_{a_2} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \frac{\left(W^{-d}B_{t,0}\right)^{\frac{1}{6}}}{\eta_t} \max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|,$$

which concludes (4.14) for $n \in \{2,3\}$ by using the definition (4.11). Above, in the second step, we use the 2-G-loop estimate (2.89) to get that

$$W^d \sum_{a_2} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(\frac{1-s}{1-t}\right)^{C_d} (W^{-d}B_{t,0})^{1/5} \cdot \sum_{a_2} \mathcal{T}_t(|a_1-a_2|) + W^{-D} \prec \frac{(W^{-d}B_{t,0})^{1/6}}{\eta_t} \tag{4.16}$$

under the condition (2.83) as long as $\mathfrak{c}_d$ is chosen sufficiently small depending on C_d. For general $n \geq 4$, we need to bound that

$$\begin{aligned} \max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \lesssim & \sum_{n'=\lceil n/2\rceil+1}^{n-1} \left(\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n+2-n')}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} W^d \sum_{a_{n'}} \left|(\mathcal{L}-\mathcal{K})^{(n')}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| \\ & + \left(\max_{\boldsymbol{\sigma},\mathbf{a}} \left|(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} W^d \sum_{a_2} \left|(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|. \end{aligned} \tag{4.17}$$

Note that the second term on the RHS can be controlled using (4.16) again, while the first term on the RHS can be handled with the loop-contraction inequality (4.1) for $\mathcal{L}$-loops and the bound (4.9) for $\mathcal{K}$-loops:

$$\begin{aligned} W^d \sum_{a_{n'}} \left|(\mathcal{L}-\mathcal{K})^{(n')}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| &\leq W^d \sum_{a_{n'}} \left(\left|\mathcal{L}^{(n')}_{t,\boldsymbol{\sigma},\mathbf{a}}\right| + \left|\mathcal{K}^{(n')}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|\right) \\ &\leq \frac{1}{\eta_t} \min_{k=1}^{n'-1} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right| \cdot \max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2n'-2k-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right|\right)^{1/2} + \frac{1}{\eta_t}(W^{-d}B_{t,0})^{n'-2}. \end{aligned}$$

Then, using (4.10) and (4.11), and setting $k = \lceil n'/2\rceil$, we can bound the RHS of (4.17) by that of (4.14).

Finally, to establish the estimate (4.15), we apply the inequality (4.2)—with n replaced by $2n+2$, and choosing $k = n$, $j = n+1$, and $l = n+2$. This yields that for any $p \geq 1$,

$$\max_{\boldsymbol{\sigma}} \max_{\mathbf{a},\mathbf{a}'} \left|(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'}\right| \lesssim \max_{\boldsymbol{\sigma},\mathbf{a}} W^d \sum_{a_{n+1}\sim a_{2n+2}} |\mathcal{L}^{(2n+2)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \lesssim \frac{1}{\eta_t} \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(2n-1)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right|\right) \cdot \left(\max_{\boldsymbol{\sigma},\mathbf{b}} \left|\mathcal{L}^{(4p)}_{t,\boldsymbol{\sigma},\mathbf{b}}\right|\right)^{\frac{1}{2p}}.$$

Combined with the definition (4.10), this completes the proof of (4.15). □

With the estimates from Lemma 4.3, we now proceed to analyze the integrated loop hierarchy in (3.14). Without loss of generality, the proof can be divided into two cases, depending on whether (i) $1-t \geq g^2/L^2$, or (ii) $1-s \leq g^2/L^2$.[7] In case (i), we employ the sum-zero operator introduced in [69]. In contrast, case (ii) requires a new approach based on the removal of zero modes from the $\mathcal{L}$-loops.

4.1. **Proof of Step 3: The case $1-t \geq g^2/L^2$.** Throughout this subsection, we always assume that $1-t \geq g^2/L^2$, in which case we have

$$B_{u,0} \asymp (g^2+|1-u|)^{-1}, \quad \ell_u \asymp g\eta_u^{-1/2}+1, \quad \forall u \in [s,t]. \tag{4.18}$$

In Step 2 of the proof of Theorem 2.24, we have established an exponential decay of the 2-G-loops beyond the scale ℓ_u as shown in (2.89). With (3.3), we can easily extend this decay to general G-loops.

Definition 4.4 (Fast decay property)**.** *Let $\mathcal{A} : (\mathbb{Z}_L^d)^n \to \mathbb{C}$ be an n-dimensional tensor for a fixed $n \geq 2$. Given $u \in [s,t]$ and constants $\varepsilon, D > 0$, we say $\mathcal{A}$ satisfies the (u,ε,D)-decay property if*

$$\max_{i,j\in[\![n]\!]} |a_i - a_j| \geq W^{\varepsilon}\ell_u \implies \mathcal{A}_{\mathbf{a}} = \mathrm{O}(W^{-D}) \quad \text{for} \quad \mathbf{a} = (a_1, a_2, \ldots, a_n). \tag{4.19}$$

It is easy to see that the G-loops satisfy the (u,ε,D)-decay property for any constants $\varepsilon, D > 0$ under the estimate (2.89) by using Lemma 3.1. Moreover, in Appendix A.5, we will present a tree representation formula for the $\mathcal{K}$-loops, which is formed with the Θ-propagators. Thus, the $\mathcal{K}$-loops also satisfy the (u,ε,D)-decay property for any constants $\varepsilon, D > 0$ by using (2.66).

[7] If $1-t < g^2/L^2 < 1-s$, then we can add a middle time $u = 1-g^2/L^2$ and perform the proofs for case (i) from s to u, and for case (ii) from u to t.

Claim 4.5. *Suppose the estimates (2.87) and (2.89) hold. For any $n \geq 2$, $\boldsymbol{\sigma} \in \{+,-\}^n$, $u \in [s,t]$, and constants $\varepsilon, D > 0$, the loops $\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}$ and $\mathcal{K}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}$ satisfy the (u,ε,D)-decay property with probability $1-\mathrm{O}(W^{-D'})$ for any large constant $D' > 0$. In other words, we have that*

$$\mathbb{P}\left(\max_{\boldsymbol{\sigma}}\left(\left|\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| + \left|\mathcal{K}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right|\right)\cdot \mathbf{1}\left(\max_{i,j\in[\![n]\!]}|a_i - a_j| \geq W^{\varepsilon}\ell_u\right) \geq W^{-D}\right) \leq W^{-D'}. \tag{4.20}$$

Due to the fast decay property of the G-loops and $\mathcal{K}$-loops, when the evolution kernels act on them, we can apply the evolution kernel estimates in Lemma 4.16. In particular, for non-alternating loops $\boldsymbol{\sigma}$, using the estimates in Lemma 4.3 and the estimate (4.54) in Lemma 4.16, we readily establish the following lemma.

Lemma 4.6 (Non-alternating loops)**.** *Under the assumptions of Theorem 2.24, suppose $1-t \geq g^2/L^2$ and the estimates (2.87)–(2.89) hold uniformly in $u \in [s,t]$. Fix any $n \geq 2$ and $\boldsymbol{\sigma} \in \{+,-\}^n$ satisfying*

$$\sigma_k = \sigma_{k+1} \quad \text{for some} \quad k \in [\![n]\!]. \tag{4.21}$$

(Recall that $\sigma_{n+1} = \sigma_1$ as a convention.) Then, we have the following estimate for any fixed $p \geq 1$:

$$\begin{aligned}\max_{\mathbf{a}\in(\mathbb{Z}^d_L)^n}\left|(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\right|/(W^{-d}B_{t,0})^n &\prec \sup_{u\in[s,t]}\left((W^{-d}B_{t,0})^{\frac{1}{6}}\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} + \max_{n'=2}^{n-1}\Xi^{(\mathcal{L}-\mathcal{K})}_{u,n'} + \max_{n'=n-1}^{n+1}\Xi^{(\mathcal{L})}_{u,n'}\right) \\ &+ \sup_{u\in[s,t]}\left(\max_{n'=\lceil n/2\rceil+1}^{n-1}\Xi^{(\mathcal{L}-\mathcal{K})}_{u,n+2-n'}\left(\Xi^{(\mathcal{L})}_{u,n'_1}\Xi^{(\mathcal{L})}_{u,n'_2}\right)^{\frac{1}{2}} + (W^{-d}B_{s,0})^{-\frac{1}{4p}}\left(\Xi^{(\mathcal{L})}_{u,2n-1}\right)^{\frac{1}{2}}\left(\Xi^{(\mathcal{L})}_{u,4p}\right)^{\frac{1}{4p}}\right),\end{aligned} \tag{4.22}$$

where, as defined below (4.14), $n'_1 = n'_2 = n'-1$ if n' is even, and $n'_1 = n'-2$ and $n'_2 = n'$ if n' is odd.

Proof. By Claim 4.5, the light-weight term $\mathring{\mathcal{E}}^{G,(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ satisfies condition (4.52) below for any constants $\varepsilon, D > 0$. Then, under the assumption (4.21), using the bound (4.12) together with the evolution kernel estimate (4.54) below, we can control the fourth term on the RHS of (3.14) as

$$\begin{aligned}\int_s^t\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\mathring{\mathcal{E}}^{G,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\mathrm{d}u &\prec \int_s^t\left(\frac{g^2+|1-u|}{g^2+|1-t|}\right)^{n-1}\frac{\left(W^{-d}B_{u,0}\right)^n}{\eta_u}\left(\widehat{\Xi}^{(\mathcal{L})}_{u,n_1}\cdot\widehat{\Xi}^{(\mathcal{L})}_{u,n_2}\right)^{\frac{1}{2}}\mathrm{d}u \\ &\prec \left(W^{-d}B_{t,0}\right)^n\int_s^t\frac{1}{\eta_u}\max_{n'=n-1}^{n+1}\Xi^{(\mathcal{L})}_{u,n'}\mathrm{d}u,\end{aligned}$$

where in the second step, we use $\frac{g^2+|1-u|}{g^2+|1-t|}B_{u,0} \leq B_{t,0}$ and the control parameter in (4.10). Using the bounds (4.13) and (4.14), together with the assumption (2.78) on $(\mathcal{L}-\mathcal{K})^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}}$ and the evolution kernel estimate (4.54), the first three terms on the RHS of (3.14) can be bounded in the same manner. This yields that

$$\begin{aligned}(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}/(W^{-d}B_{t,0})^n &\prec 1 + (W^{-d}B_{t,0})^{-n}\int_s^t\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \int_s^t\frac{1}{\eta_u}\max_{n'=2}^{n-1}\Xi^{(\mathcal{L}-\mathcal{K})}_{u,n'}\mathrm{d}u \\ &+ \int_s^t\frac{1}{\eta_u}\max_{n'=n-1}^{n+1}\Xi^{(\mathcal{L})}_{u,n'}\mathrm{d}u + \int_s^t\frac{1}{\eta_u}\max_{n'=\lceil n/2\rceil+1}^{n-1}\Xi^{(\mathcal{L}-\mathcal{K})}_{u,n+2-n'}\left(\Xi^{(\mathcal{L})}_{u,n'_1}\Xi^{(\mathcal{L})}_{u,n'_2}\right)^{\frac{1}{2}}\mathrm{d}u + (W^{-d}B_{t,0})^{\frac{1}{6}}\int_s^t\frac{1}{\eta_u}\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n}\mathrm{d}u.\end{aligned} \tag{4.23}$$

For the martingale term, by Lemma 3.6, and using (4.15) together with (4.54), we obtain that

$$\begin{aligned}(W^{-d}B_{t,0})^{-n}\int_s^t\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} &\prec (W^{-d}B_{t,0})^{-n}\left\{\int_s^t\left(\left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\otimes\mathcal{U}^{(n)}_{u,t,\overline{\boldsymbol{\sigma}}}\right)\circ(\mathcal{E}\otimes\mathcal{E})^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a},\mathbf{a}}\mathrm{d}u\right\}^{1/2} \\ &\prec (W^{-d}B_{s,0})^{-\frac{1}{4p}}\left\{\int_s^t\frac{1}{\eta_u}\Xi^{(\mathcal{L})}_{u,2n-1}\left(\Xi^{(\mathcal{L})}_{u,4p}\right)^{\frac{1}{2p}}\mathrm{d}u\right\}^{1/2}\end{aligned}$$

for any fixed $p \geq 1$. Here, the bound (4.15) produces the factor $(W^{-d}B_{u,0})^{2n-\frac{1}{2p}}$ in the integrand. Together with the factors produced by (4.54), its positive powers $B^{2n}_{u,0}$ are bounded by $B^{2n}_{t,0}$ using $\frac{g^2+|1-u|}{g^2+|1-t|}B_{u,0} \leq B_{t,0}$ and $B_{u,0} \leq B_{t,0}$, while for the remaining negative power we can only use $B^{-\frac{1}{2p}}_{u,0} \leq B^{-\frac{1}{2p}}_{s,0}$, since $B_{s,0} \leq B_{u,0} \leq B_{t,0}$. Plugging it into (4.23) and performing the integral over u, we conclude (4.22). □

Alternating cases: It remains to deal with the case with alternating signs, where (4.21) does not occur:

$$\sigma_k = -\sigma_{k+1}, \quad \forall\, k \in [\![n]\!]. \tag{4.24}$$

For this purpose, we introduce another key tool—a sum-zero operator $\mathcal{Q}_t$.

Definition 4.7 (Partial sum and sum-zero operator)**.** *Let $\mathcal{A} : (\mathbb{Z}_L^d)^n \to \mathbb{C}$ be an n-dimensional tensor for a fixed $n \in \mathbb{N}$ with $n \geq 2$. Define the partial sum operator $\mathcal{P}$ as*

$$(\mathcal{P} \circ \mathcal{A})_{a_1} := \sum_{a_i : i \in [\![2,n]\!]} \mathcal{A}_{\mathbf{a}}, \quad \mathbf{a} = (a_1, \dots, a_n).$$

We say a tensor $\mathcal{A}$ satisfies the sum-zero property *if $\mathcal{P} \circ \mathcal{A} \equiv 0$. For $t \in [0,1)$, we define the operator $\mathcal{Q}_t$ as*

$$(\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} := \mathcal{A}_{\mathbf{a}} - (\mathcal{P} \circ \mathcal{A})_{a_1} \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}, \tag{4.25}$$

where the tensor $\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$ is a mollifier satisfying

$$\sum_{a_2,\dots,a_n} \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)} \equiv 1, \quad \forall a_1 \in \mathbb{Z}_L^d, \tag{4.26}$$

along with the following estimates for a constant $c > 0$:

$$\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)} \prec (\ell_t^d)^{-(n-1)} \exp\left(-c \sum_{i=2}^n |a_i - a_1|/\ell_t\right), \quad \left|\partial_t \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}\right| \prec |1-t|^{-1} (\ell_t^d)^{-(n-1)} \exp\left(-c \sum_{i=2}^n |a_i - a_1|/\ell_t\right). \tag{4.27}$$

The detailed form of $\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$ is not important to us, and we will give an example in Example 4.8 below. With equation (4.26), we see that $\mathcal{P} \circ \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)} \equiv 1$, $\mathcal{P} \circ \mathcal{Q}_t = 0$, and that for any tensor $\mathcal{A}$,

$$\mathcal{P} \circ \mathcal{A} \equiv 0 \quad \Longrightarrow \quad \mathcal{P} \circ \left(\Theta_{t,\boldsymbol{\sigma}}^{(n)} \circ \mathcal{A}\right) \equiv 0, \tag{4.28}$$

where we recall that $\Theta_{t,\boldsymbol{\sigma}}^{(n)}$ is the operator defined in Definition 3.3. In other words, if $\mathcal{A}$ satisfies the sum-zero property, then so does $\Theta_{t,\boldsymbol{\sigma}}^{(n)} \circ \mathcal{A}$.

Example 4.8. To construct the mollifier tensor $\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$, we first choose a compactly supported, smooth, non-negative function $f \in C_c^\infty(\mathbb{R}^d)$ that is not identically equal to zero, and rescale it as $f_t(a) := \ell_t^{-d} f(|a|/\ell_t)$ for $a \in \mathbb{Z}_L^d$. We then define, with an appropriate normalization constant $C_{f,t}$,

$$\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)} := C_{f,t} \cdot \prod_{i=2}^n f_t(a_i - a_1), \quad \forall \mathbf{a} \in (\mathbb{Z}_L^d)^n. \tag{4.29}$$

It is easy to see that this function satisfies the required properties in (4.27); in particular, both $\boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$ and $\partial_t \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$ vanish unless $\max_i |a_i - a_1| \lesssim \ell_t$.

We will use the sum-zero operator to get improved estimates on the terms on the RHS of (3.14) when (4.24) holds. Roughly speaking, we will decompose a tensor $\mathcal{A}$ as $\mathcal{A}_{\mathbf{a}} = (\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} + (\mathcal{P} \circ \mathcal{A})_{a_1} \boldsymbol{\chi}_{t,\mathbf{a}}^{(n)}$ using (4.25). For the first part, we can get an improvement by using the evolution kernel estimate (4.56), while for the second part, we can apply Ward's identity to $\mathcal{P} \circ \mathcal{A}$.

We first claim that when $\boldsymbol{\sigma}$ satisfies (4.24), the following estimate holds uniformly in $u \in [s,t]$:

$$\left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\boldsymbol{\sigma}}^{(n)}\right]_{a_1} \boldsymbol{\chi}_{u,\mathbf{a}}^{(n)} \prec (W^{-d} B_{u,0})^n \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}. \tag{4.30}$$

To see this, we apply Ward's identities in Lemma 2.15 at the vertex a_n and get

$$\left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\boldsymbol{\sigma}}^{(n)}\right]_{a_1} = \frac{1}{2\mathrm{i} W^d \eta_u} \left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\widehat{\boldsymbol{\sigma}}^{(+,n)}}^{(n-1)} - \mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\widehat{\boldsymbol{\sigma}}^{(-,n)}}^{(n-1)}\right]_{a_1}.$$

By (4.11), the two $(\mathcal{L} - \mathcal{K})^{(n-1)}$-loops on the RHS are controlled by

$$(\mathcal{L} - \mathcal{K})_{u,\widehat{\boldsymbol{\sigma}}^{(\pm,n)},\widehat{\mathbf{a}}^{(n)}}^{(n-1)} \prec (W^{-d} B_{u,0})^{n-1} \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}.$$

Moreover, due to the fast decay property of the $(\mathcal{L} - \mathcal{K})$-loops, the partial sums over the remaining $(n-2)$ vertices lead to an additional $\ell_u^{d(n-2)}$ factor up to a negligible error $\mathrm{O}(W^{-D})$. This leads to

$$\left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\boldsymbol{\sigma}}^{(n)}\right]_{a_1} \prec (W^d \eta_u)^{-1} (\ell_u^d)^{n-2} \cdot (W^{-d} B_{u,0})^{n-1} \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}. \tag{4.31}$$

Together with (4.27), it implies that

$$\left[\mathcal{P} \circ (\mathcal{L} - \mathcal{K})_{u,\boldsymbol{\sigma}}^{(n)}\right]_{a_1} \boldsymbol{\chi}_{u,\mathbf{a}}^{(n)} \prec (W^d \ell_u^d \eta_u)^{-1} \cdot (W^{-d} B_{u,0})^{n-1} \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})} \lesssim (W^{-d} B_{u,0})^n \Xi_{u,n-1}^{(\mathcal{L}-\mathcal{K})}$$

uniformly in $u \in [s,t]$, where, in the second step, we use $(\ell_u^d \eta_u)^{-1} \lesssim B_{u,0}$ by (4.18).

It remains to control $\mathcal{Q}_t \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$. For this purpose, we need the following claim on the $(\infty \to \infty)$-norm of the sum-zero operator, which follows easily from Definition 4.7 and the estimate (4.27).

Claim 4.9. *Let $\mathcal{A} : (\mathbb{Z}_L^d)^n \to \mathbb{C}$ be an n-dimensional tensor for a fixed $n \geq 2$. If $\mathcal{A}$ satisfies the (t,ε,D)-decay property, then we have that*

$$\|\mathcal{Q}_t \circ \mathcal{A}\|_\infty \leq W^{C_n \varepsilon} \|\mathcal{A}\|_\infty + W^{-D+C_n} \tag{4.32}$$

for a constant C_n that does not depend on ε or D. Furthermore, if $\|\mathcal{A}\|_\infty \leq W^C$ for a constant $C > 0$, then $\mathcal{A}_{\mathbf{a}} - (\mathcal{Q}_t \circ \mathcal{A})_{\mathbf{a}} = (\mathcal{P} \circ \mathcal{A})_{a_1} \boldsymbol{\chi}^{(n)}_{t,\mathbf{a}}$ satisfies the (t,ε',D')-decay property for any constants $\varepsilon', D' > 0$.

We derive from equation (3.8) that

$$\begin{aligned} \mathrm{d}\mathcal{Q}_t \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} &= \mathcal{Q}_t \circ \left[\mathcal{K}^{(2)} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t + \sum_{l_{\mathcal{K}}=3}^{n} \mathcal{Q}_t \circ \left[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t + \mathcal{Q}_t \circ \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t \\ &\quad + \mathcal{Q}_t \circ \mathcal{E}^{\mathring{G},(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \mathrm{d}t + \mathcal{Q}_t \circ \mathrm{d}\mathcal{E}^{M,(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} - \left[\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma}}\right]_{a_1} \left(\partial_t \boldsymbol{\chi}^{(n)}_{t,\mathbf{a}}\right) \mathrm{d}t. \end{aligned} \tag{4.33}$$

Recalling Definition 3.3, we can rewrite the first term on the RHS as

$$\mathcal{Q}_t \circ \left[\mathcal{K}^{(2)} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \Theta^{(n)}_{t,\boldsymbol{\sigma}} \circ \left[\mathcal{Q}_t \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma}}\right]_{\mathbf{a}} + \left[\mathcal{Q}_t, \Theta^{(n)}_{t,\boldsymbol{\sigma}}\right] \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}, \tag{4.34}$$

where $[\mathcal{Q}_t, \Theta^{(n)}_{t,\boldsymbol{\sigma}}] = \mathcal{Q}_t \circ \Theta^{(n)}_{t,\boldsymbol{\sigma}} - \Theta^{(n)}_{t,\boldsymbol{\sigma}} \circ \mathcal{Q}_t$ denotes the commutator between $\mathcal{Q}_t$ and $\Theta^{(n)}_{t,\boldsymbol{\sigma}}$. Since $\mathcal{P} \circ \mathcal{Q}_t = 0$, we notice that the first 5 terms on the RHS of (4.33) satisfy the sum zero property. Since $\mathcal{P} \circ \boldsymbol{\chi}^{(n)}_{t,\mathbf{a}} \equiv 1$, we have $\mathcal{P} \circ \left(\partial_t \boldsymbol{\chi}^{(n)}_{t,\mathbf{a}}\right) = 0$, so the last term on the RHS of (4.33) also satisfies the sum zero property. Next, due to (4.28), the first term on the RHS of (4.34) also satisfies the sum-zero property. Finally, since the LHS of (4.34) has the sum-zero property, the second term on the RHS of (4.34) also satisfies the sum-zero property.

With Duhamel's principle, we can derive from (4.33) and (4.34) the following counterpart of (3.14):

$$\begin{aligned} \mathcal{Q}_t \circ (\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} &= \left(\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_s \circ \mathcal{B}_0(s)\right)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_u \circ \sum_{k=1}^{5} \mathcal{B}_k(u)\right)_{\mathbf{a}} \mathrm{d}u \\ &\quad + \int_s^t \left(\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_u \circ \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}, \end{aligned} \tag{4.35}$$

where the tensors $\mathcal{B}_i$ for $i \in [\![0,5]\!]$ are defined as follows:

$$\begin{aligned} &\mathcal{B}_0(s) := (\mathcal{L}-\mathcal{K})^{(n)}_{s,\boldsymbol{\sigma}}, \quad \mathcal{B}_1(u) := \sum_{l_{\mathcal{K}}=3}^{n} \left[\mathcal{K}^{(l_{\mathcal{K}})} \sim (\mathcal{L}-\mathcal{K})\right]^{(n)}_{u,\boldsymbol{\sigma}}, \quad \mathcal{B}_2(u) := \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{u,\boldsymbol{\sigma}}, \\ &\mathcal{B}_3(u) := \mathcal{E}^{\mathring{G},(n)}_{u,\boldsymbol{\sigma}}, \quad \mathcal{B}_4(u) := \left[\mathcal{Q}_u, \Theta^{(n)}_{u,\boldsymbol{\sigma}}\right] \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}}, \quad \mathcal{B}_5(u) := -\left[\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}}\right] \cdot \partial_u \boldsymbol{\chi}^{(n)}_u. \end{aligned}$$

We can control the terms on the RHS of (4.35) by applying the improved evolution kernel estimate (4.56), which exploits both the sum-zero and fast-decay properties of the terms on the RHS of (4.33). Combining this with (4.30) and Lemma 4.6 for the non-alternating case, we obtain the following inductive bootstrap bounds for the Ξ-parameters. The proof proceeds by estimating the RHS of (4.35) in a manner analogous to the proof of Lemma 4.6, using the bounds established in Lemma 4.3 together with straightforward controls of the $\mathcal{B}_4$ and $\mathcal{B}_5$ terms, based on the properties in (4.27). Hence, we defer the detailed proof to Section 4.5.

Lemma 4.10 (Inductive bootstrap bound for Ξ-parameters)**.** *Under the assumptions of Theorem 2.24, suppose $1-t \geq g^2/L^2$ and the estimates (2.87)–(2.89) hold uniformly in $u \in [s,t]$. Then, for any fixed $n \geq 2$ and $p \geq 1$, the following bound holds uniformly in $u \in [s,t]$:*

$$\begin{aligned} \sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} &\prec \sup_{v\in[s,u]} \left(\left(W^{-d} B_{s,0}\right)^{-\frac{1}{4p}} \left(\Xi^{(\mathcal{L})}_{v,2n-1}\right)^{\frac{1}{2}} \left(\Xi^{(\mathcal{L})}_{v,4p}\right)^{\frac{1}{4p}} \right) \\ &\quad + \sup_{v\in[s,u]} \left(\max_{n'=1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{v,n'} + \max_{n'=\lceil n/2\rceil+1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n+2-n'} \left(\Xi^{(\mathcal{L})}_{v,n'_1} \Xi^{(\mathcal{L})}_{v,n'_2}\right)^{\frac{1}{2}} \right). \end{aligned} \tag{4.36}$$

Now, similar to the argument in Section 5.6 of [69], we will iterate the bootstrap bound (4.36) to obtain the sharp L^∞-bound (2.90) on the G-loops in the regime $1-t \geq g^2/L^2$. That is, for any fixed $n \in \mathbb{N}$,

$$\sup_{u\in[s,t]} \widehat{\Xi}^{(\mathcal{L})}_{u,n} \prec 1. \tag{4.37}$$

Observe that when $1-s > g^2$, we have $B_{u,0} \asymp |1-u|^{-1}$ for all $u \in [s, 1-g^2]$. Consequently, the G-loop bound (2.90) follows directly from (2.85). Hence, in the remainder of the proof, it suffices to consider the case $1-t \leq 1-s \leq g^2$, where $W^{-d}B_{u,0} \asymp (g^2W^d)^{-1}$ for all $u \in [s,t]$. First, the averaged local law (2.88) gives that $\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,1} \prec 1 =: \Xi^{(\mathcal{L}-\mathcal{K})}_{u,1}$ uniformly for $u \in [s,t]$. Second, by the $\mathcal{K}$-loop bound (2.58), we have

$$\widehat{\Xi}^{(\mathcal{L})}_{u,n} \prec 1 + (g^2W^d)^{-1}\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n}, \qquad \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} \prec (g^2W^d)\left(\widehat{\Xi}^{(\mathcal{L})}_{u,n}+1\right), \quad \text{for} \quad n \geq 2. \tag{4.38}$$

Moreover, the a priori G-loop bound (2.85) provides the following initial estimates, uniform in $u \in [s,t]$:

$$\widehat{\Xi}^{(\mathcal{L})}_{u,n} \prec (\eta_s/\eta_u)^{n-1}, \quad \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} \prec (\eta_s/\eta_u)^{n-1}\cdot(g^2W^d). \tag{4.39}$$

We then introduce the control parameter

$$\Psi(n,k;s,t) := (g^2W^d)^{3/4} + (\eta_s/\eta_t)^{n-1}\,(g^2W^d)^{1-k/8}. \tag{4.40}$$

The iteration will proceed simultaneously in the indices n and k. The outcome of each step is summarized in the following lemma, whose proof—being analogous to that of (5.109) in [69]—is deferred to Section 4.5.

Lemma 4.11. *In the setting of Theorem 2.24, suppose (2.85)–(2.89) and (4.36) hold uniformly for all $u \in [s,t]$ with $1-s \leq g^2$. Fix any $(n,k) \in \mathbb{N}^2$ with $n \geq 2$ and $k \geq 1$. Assume that, uniformly for $u \in [s,t]$,*

$$\sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,r} \prec \Psi(r,l;s,u) \tag{4.41}$$

holds for all index pairs $(r,l) \in \{(r,k) : 2 \leq r \leq n-1\} \cup \{(r,k-1) : 2 \leq r \leq n+2\}$. Then, (4.41) also holds for $(r,l) = (n,k)$.

Roughly speaking, this lemma states that if we have already established a "good" bound for all shorter G-loops of length $r \leq n-1$, along with a "weaker" bound for G-loops of length $r \leq n+2$, then we can derive the "good" bound for all G-loops of length n. Using Lemma 4.11, we apply a simple iterative argument to complete the proof of (2.90) in Step 3 for the case $1-t \geq g^2/L^2$.

Proof of (2.90) when $1-t \geq g^2/L^2$. As discussed above, it remains to deal with the case $1-s \leq g^2$. By (4.39), we initially have a weak bound for G-loops of arbitrarily large lengths, meaning that (4.41) holds with $l=0$ for every fixed $r \in \mathbb{N}$. Applying Lemma 4.11 once, we obtain a slightly improved bound (4.41) for $r=2$ and $l=1$. Then, continuing the iteration in r while keeping $l=1$ fixed, we establish the bound (4.41) for each fixed $r \in \mathbb{N}$ with $l=1$. Next, applying the iteration in Lemma 4.11 again yields an even stronger bound (4.41) with $r=2$ and $l=2$. Repeating the iteration in r with $l=2$ fixed, we further establish the bound (4.41) for every fixed $r \in \mathbb{N}$ with $l=2$. This process continues, progressively improving the bound to (4.41) for each fixed $r \in \mathbb{N}$ with $l=3$, and so forth.

For any given $(n,k) \in \mathbb{N}^2$, by repeating the above procedure for $\mathrm{O}(1)$ times, we conclude that the estimate (4.41) holds for $(r,l) = (n,k)$. In particular, if we choose $\mathfrak{c}_d$ in condition (2.83) sufficiently small and take k large enough so that $(\eta_s/\eta_t)^{n-1}\cdot(g^2W^d)^{1-k/8} \ll (g^2W^d)^{3/4}$, then we get from (4.41) that

$$\sup_{u\in[s,t]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} \prec \Psi(n,k;s,t) \lesssim (g^2W^d)^{3/4}. \tag{4.42}$$

Together with (4.38), this implies the estimate (4.37), which completes the proof of the bound (2.90). □

4.2. **Proof of Step 3: The case $1-s \leq g^2/L^2$.** In this setting, we have $\ell_u \equiv L$ for all $u \in [s,t]$. Here, a key difference from the $d=2$ case in [28] arises in the intermediate regime $g^2/L^d \leq 1-t \leq g^2/L^2$, where the polynomial decay mode $g^{-2}/(|a-b|+1)^{d-2}$ and the zero mode $(L^d|1-t|)^{-1}$ mix in the quantity $B_{t,|a-b|}$ (recall (2.62)). In this regime, we lose the fast decay property for G-loops that is required by Lemma 4.16 below, and instead the weaker evolution kernel estimate (4.51) becomes relevant. This estimate introduces a factor of $(1-s)/(1-t)$, which transforms the zero mode $(L^d|1-s|)^{-1}$ at time s into the zero mode $(L^d|1-t|)^{-1}$ at time t. However, this transformation breaks the polynomial decay mode, potentially destabilizing the tree approximation at time t. To address this difficulty, we observe that the two modes actually propagate independently along the flow. Motivated by this, we introduce a zero-mode-removing

operator, which decomposes the $(\mathcal{L}-\mathcal{K})$-loops into two parts: (1) the zero-mode components, which can be controlled using Ward's identities, and (2) the zero-mode-free components, whose evolution kernel satisfies sharper estimates thanks to the bound (2.70) (see (4.57) below).

Definition 4.12 (Zero-mode-removing operators)**.** *Let $\mathcal{A}:(\mathbb{Z}_L^d)^n\to\mathbb{C}$ be an arbitrary n-dimensional tensor for a fixed $n\geq 1$. Define $P^{(i)}$ as the* partial averaging operator *with respect to the i-th index of the loop:*

$$(P^{(i)}\circ\mathcal{A})_{(a_1,\dots,a_{i-1},b,a_{i+1},\dots,a_n)}=L^{-d}\sum_{a_i}\mathcal{A}_{(a_1,\dots,a_{i-1},a_i,a_{i+1},\dots,a_n)},\quad \forall i\in[\![n]\!],\ b\in\mathbb{Z}_L^d.$$

Correspondingly, we define the zero-mode-removing *operator as $Q^{(i)}:=I-P^{(i)}$. Furthermore, for any subset $A\subset[\![n]\!]$, we denote*

$$P^{(A)}:=\prod_{i\in A}P^{(i)},\quad \text{and}\quad Q^{(A)}:=\prod_{i\in A}Q^{(i)}.$$

As a convention, when $A=\emptyset$, we define $P^{(\emptyset)}$ and $Q^{(\emptyset)}$ as the identity operator. Note that for any subsets A and A', the operators $P^{(A)}$, $P^{(A')}$, $Q^{(A)}$, and $Q^{(A')}$ all commute with each other.

By definition, the $(\infty\to\infty)$-operator norm of $Q^{(i)}$ is trivially bounded:

$$\big\|Q^{(i)}\circ\mathcal{A}\big\|_\infty\leq 2\|\mathcal{A}\|_\infty. \tag{4.43}$$

Given any $\boldsymbol{\sigma}\in\{+,-\}^n$, we denote $I_{\mathrm{diff}}(\boldsymbol{\sigma}):=\{i\in[\![n]\!]:\sigma_i\neq\sigma_{i+1}\}$, where we again adopt the cyclic convention $\sigma_{n+1}=\sigma_1$. Given any $\mathcal{L}$- or $\mathcal{K}$-loop, we can express it as a linear combination of $(Q^{(A)}\circ\mathcal{L})_{t,\boldsymbol{\sigma},\mathbf{a}}$ or $(Q^{(A)}\circ\mathcal{K})_{t,\boldsymbol{\sigma},\mathbf{a}}$ loops with $A\supset I_{\mathrm{diff}}(\boldsymbol{\sigma})$, by repeatedly applying (2.56) and (2.57). This is summarized in Lemma 4.13, whose proof is deferred to Section 4.5 below.

Lemma 4.13. *For any fixed $n\in\mathbb{N}$, $\boldsymbol{\sigma}\in\{+,-\}^n$, $\mathbf{a}\in(\mathbb{Z}_L^d)^n$, and subset $A\subset[\![n]\!]$, we have the expansion:*

$$\Big(Q^{(A)}\circ\mathcal{L}^{(n)}\Big)_{t,\boldsymbol{\sigma},\mathbf{a}}=\Big(Q^{(A_n)}\circ\mathcal{L}^{(n)}\Big)_{t,\boldsymbol{\sigma},\mathbf{a}}+\sum_\alpha\frac{\xi_\alpha}{(2\mathrm{i}N\eta_t)^{n-k_\alpha}}\Big(Q^{(A_\alpha)}\circ\mathcal{L}^{(k_\alpha)}\Big)_{t,\boldsymbol{\sigma}_\alpha,\mathbf{a}_\alpha}, \tag{4.44}$$

where $A_n:=A\cup I_{\mathrm{diff}}(\boldsymbol{\sigma})$, $\{\alpha\}$ denotes a collection of $\mathrm{O}(1)$ many labels, $1\leq k_\alpha\leq n-1$, $\mathbf{a}_\alpha\in(\mathbb{Z}_L^d)^{k_\alpha}$ consists of a subset of indices in $\mathbf{a}$, $\boldsymbol{\sigma}_\alpha\in\{+,-\}^{k_\alpha}$, $[\![k_\alpha]\!]\supset A_\alpha\supset I_{\mathrm{diff}}(\boldsymbol{\sigma}_\alpha)$, and ξ_α denote some deterministic and integer-valued coefficients of order $\mathrm{O}(1)$. The same expansion also holds for the $\mathcal{K}$-loop:

$$\Big(Q^{(A)}\circ\mathcal{K}^{(n)}\Big)_{t,\boldsymbol{\sigma},\mathbf{a}}=\Big(Q^{(A_n)}\circ\mathcal{K}^{(n)}\Big)_{t,\boldsymbol{\sigma},\mathbf{a}}+\sum_\alpha\frac{\xi_\alpha}{(2\mathrm{i}N\eta_t)^{n-k_\alpha}}\Big(Q^{(A_\alpha)}\circ\mathcal{K}^{(k_\alpha)}\Big)_{t,\boldsymbol{\sigma}_\alpha,\mathbf{a}_\alpha}. \tag{4.45}$$

As a special case that is of particular importance to us, the above expansions hold for $A=\emptyset$.

Given any $n\geq 2$ and $A\subset[\![n]\!]$, we derive from equation (3.8) that

$$\begin{aligned}\mathrm{d}Q^{(A)}\circ(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}=Q^{(A)}\circ\Big[\Theta^{(n)}_{t,\boldsymbol{\sigma}}\circ(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma}}\Big]_{\mathbf{a}}\mathrm{d}t+\sum_{l_{\mathcal{K}}=3}^{n}Q^{(A)}\circ\Big[\mathcal{K}^{(l_{\mathcal{K}})}\sim(\mathcal{L}-\mathcal{K})\Big]^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}t\\+Q^{(A)}\circ\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}t+Q^{(A)}\circ\mathcal{E}^{\mathring{G},(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}\mathrm{d}t+Q^{(A)}\circ\mathrm{d}\mathcal{E}^{M,(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}.\end{aligned} \tag{4.46}$$

Due to the translation invariance of $S^{(\mathsf{B})}$ and $M^{(\sigma_i,\sigma_{i+1})}$, they both have an eigenvector $\mathbf{e}$ with $\mathbf{e}(a)\equiv L^{-d/2}$ for all $a\in\mathbb{Z}_L^d$. As a consequence, the operators $\Theta^{(n)}$ and $\mathcal{U}^{(n)}$ defined in (3.11) and (3.12) commute with the operators $P^{(i)}$, and hence also commute with the operators $Q^{(A)}$ for all $A\subset[\![n]\!]$. Hence, the first term on the RHS of (4.46) can be written as

$$\Big[\Theta^{(n)}_{t,\boldsymbol{\sigma}}\circ Q^{(A)}\circ(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma}}\Big]_{\mathbf{a}}.$$

Then, applying Duhamel's principle to the equation (4.46), we derive the following integral equation:

$$\begin{aligned}Q^{(A)}\circ(\mathcal{L}-\mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}=\Big(Q^{(A)}\circ\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}\circ\mathcal{B}_0(s)\Big)_{\mathbf{a}}+\int_s^t\bigg(Q^{(A)}\circ\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\sum_{k=1}^{3}\mathcal{B}_k(u)\bigg)_{\mathbf{a}}\mathrm{d}u\\+\int_s^t\Big(Q^{(A)}\circ\mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}}\circ\mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\Big)_{\mathbf{a}},\end{aligned} \tag{4.47}$$

where we use the notations in (4.35). If $A \supset I_{\mathrm{diff}}(\boldsymbol{\sigma})$, then the new kernel $Q^{(A)} \circ \mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ has a better $(\infty \to \infty)$-norm estimate (as stated in Lemma 4.17 below) than the original kernel $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ when $1 - s \le g^2/L^2$. In fact, corresponding to each $i \in I_{\mathrm{diff}}(\boldsymbol{\sigma})$, the $(\infty \to \infty)$-norm estimate of $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ will be weaker by an η_s/η_t factor.

Combining the expansions (4.44) and (4.45) (for the case $A = \emptyset$), the equation (4.47), the evolution kernel estimate in Lemma 4.17, and the estimates in Lemma 4.3, we can establish a result similar to that in Lemma 4.10. We postpone the proof of Lemma 4.14 to Section 4.5 below.

Lemma 4.14. *Under the assumptions of Theorem 2.24, suppose $1 - s \le g^2/L^2$ and the estimates (2.87), (2.88), and (2.89) hold uniformly in $u \in [s,t]$. Then, for any fixed $n \ge 2$ and $p \ge 1$, the bound (4.36) holds uniformly in $u \in [s,t]$.*

Now, we are ready to complete the proof of Step 3 for the case $1 - s \le g^2/L^2$.

Proof of (2.90) when $1 - s \le g^2/L^2$. Given Lemma 4.14, we can prove a similar iterative result to Lemma 4.11, but with a different control parameter defined as follows:

$$\Psi(n,k;s,t) := (W^{-d}B_{s,0})^{-3/4} + (\eta_s/\eta_t)^{n-1}(W^{-d}B_{s,0})^{k/8-1}. \tag{4.48}$$

More precisely, suppose the estimate (4.41) holds uniformly in $u \in [s,t]$ for all index pairs $(r,l) \in \{(r,k) : 2 \le r \le n-1\} \cup \{(r,k-1) : 2 \le r \le n+2\}$. Then, we have the following estimate uniformly in $u \in [s,t]$:

$$\sup_{v \in [s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} \prec \Psi(n,k;s,u). \tag{4.49}$$

Since the proof of (4.49) is very similar to that for Lemma 4.11 by using Lemma 4.14, we omit the details. With this result, performing exactly the same iterative argument as in the case $1 - t \ge g^2/L^2$ (i.e., the argument around (4.42)), we can conclude (2.90) for the case $1 - s \le g^2/L^2$. □

4.3. **Proof of Step 4.** We again divide the proof of (2.91) into two cases according to whether $1-t \ge g^2/L^2$ or $1 - s \le g^2/L^2$. In the former case, we apply Lemma 4.10 established in Step 3, while in the latter we apply Lemma 4.14. At this step, using the sharp G-loop bound (2.90) together with the averaged local law (2.88), we may choose the optimal parameters $\Xi^{(\mathcal{L})}_{v,n'} = 1$ for all G-loops and $\Xi^{(\mathcal{L}-\mathcal{K})}_{v,1} = 1$. Then, from (4.36), we get the following bound for any fixed $n \ge 2$ and $p \ge 1$:

$$\sup_{v \in [s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} \prec (W^{-d}B_{s,0})^{-\frac{1}{4p}} + \sup_{v \in [s,u]} \left(\max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} \right). \tag{4.50}$$

First, taking $n = 2$ in (4.50), the second term on the RHS vanishes. Since p can be chosen arbitrarily large, we conclude that $\sup_{v \in [s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,2} \prec 1$. Starting from this base case, we can derive that $\sup_{v \in [s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} \prec 1$ for any fixed $n \in \mathbb{N}$ by applying (4.50) inductively in n. This completes the proof of (2.91) in Step 4.

4.4. **Evolution kernel estimates.** In the above proof, we have used the following estimates on the evolution kernel defined in Definition 3.3. Their proofs are postponed to Appendix A.2. We first have an easy bound on the $(\infty \to \infty)$-norm of $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$.

Lemma 4.15. *Let $\mathcal{A} : (\mathbb{Z}^d_L)^n \to \mathbb{C}$ be an n-dimensional tensor for a fixed $n \ge 2$. Then, for any $0 \le s \le t < 1$, we have that*

$$\|\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A}\|_\infty \le \left(\frac{1-s}{1-t}\right)^n \cdot \|\mathcal{A}\|_\infty, \tag{4.51}$$

where the L^∞-norm of $\mathcal{A}$ is defined as $\|\mathcal{A}\|_\infty = \max_{\mathbf{a} \in (\mathbb{Z}^d_L)^n} |\mathcal{A}_{\mathbf{a}}|$.

If the tensor $\mathcal{A}$ exhibits faster-than-polynomial decay on scales larger than ℓ_s, we show that the $(\infty \to \infty)$-norm of the evolution kernel $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ satisfies a better bound. This bound can be further improved when $\boldsymbol{\sigma}$ is non-alternating or when $\mathcal{A}$ satisfies a sum-zero property.

Lemma 4.16. *Let $\mathcal{A} : (\mathbb{Z}^d_L)^n \to \mathbb{C}$ be an n-dimensional tensor for a fixed $n \ge 2$. Suppose it satisfies the following fast-decay property for some small constant $\varepsilon \in (0,1)$ and large constant $D > 1$:*

$$|\mathcal{A}_{\mathbf{a}}| \le W^{-D}, \quad \forall\, \mathbf{a} = (a_1, \dots, a_n) \in (\mathbb{Z}^d_L)^n \quad \text{with} \quad \max_{i,j \in [\![n]\!]} |a_i - a_j| \ge W^{\varepsilon} \ell_s\,. \tag{4.52}$$

Fix any $0 \le s \le t \le 1 - g^2/L^2$ such that $(1-t)/(1-s) \ge W^{-1}$. Then, there exists a constant $C_n > 0$ that does not depend on ε or D such that the following bound holds:

$$\left\| \mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A} \right\|_\infty \le W^{C_n\varepsilon} \frac{\ell_t^2}{\ell_s^2} \left(\frac{g^2 + |1-s|}{g^2 + |1-t|} \right)^n \|\mathcal{A}\|_\infty + W^{-D+C_n}. \tag{4.53}$$

This estimate can be further improved in the following cases:

(I) *If $\boldsymbol{\sigma}$ is non-alternating, i.e., $\sigma_k = \sigma_{k+1}$ for some $k \in [\![n]\!]$, then we have*

$$\left\| \mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A} \right\|_\infty \le W^{C_n\varepsilon} \left(\frac{g^2 + |1-s|}{g^2 + |1-t|} \right)^{n-1} \|\mathcal{A}\|_\infty + W^{-D+C_n} \tag{4.54}$$

for a constant $C_n > 0$ that does not depend on ε or D.

(II) *If $\mathcal{A}$ satisfies the following sum-zero property:*

$$\sum_{a_2,\dots,a_n \in \mathbb{Z}_L^d} \mathcal{A}_{\mathbf{a}} = 0, \quad \forall a_1 \in \mathbb{Z}_L^d, \tag{4.55}$$

then there exists a constant $C_n > 0$ that does not depend on ε or D such that

$$\left\| \mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A} \right\|_\infty \le W^{C_n\varepsilon} \left(\frac{g^2 + |1-s|}{g^2 + |1-t|} \right)^{n} \|\mathcal{A}\|_\infty + W^{-D+C_n}. \tag{4.56}$$

If $1 - s \le g^2/L^2$, we obtain the following estimates for the evolution kernels with zero modes removed.

Lemma 4.17. *Fix any $1 - g^2/L^2 \le s \le t < 1$ and $\boldsymbol{\sigma} \in \{+,-\}^n$. Let $\mathcal{A} : (\mathbb{Z}_L^d)^n \to \mathbb{C}$ be an arbitrary n-dimensional tensor for a fixed $n \ge 2$. Then, for any subset $A \subset [\![n]\!]$ satisfying $A \supset I_{\mathrm{diff}}(\boldsymbol{\sigma})$, we have*

$$\| Q^{(A)} \circ \mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A} \|_\infty \prec \|\mathcal{A}\|_\infty. \tag{4.57}$$

4.5. **Proofs of supporting lemmas.** This subsection presents the proofs of several supporting lemmas.

Proof of Lemma 4.10. Lemma 4.6 already gives a good enough bound for non-alternating $(\mathcal{L}-\mathcal{K})$-loops. It remains to control $\max_{\mathbf{a}} |(\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}| / (W^{-d} B_{u,0})^n$ for alternating $\boldsymbol{\sigma}$ using (4.30) and equation (4.35).

First, the partial sum term $[\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}}]_{a_1} \boldsymbol{\chi}^{(n)}_{u,\mathbf{a}}$ has been bounded in (4.30). Second, using Claim 4.9, the induction hypothesis (2.78) at time s, the estimates established in (4.12)–(4.15), and the evolution kernel estimate (4.56), and adopting an argument similar to that in the proof of Lemma 4.6, we can control the terms involving $\mathcal{B}_i$ for $i \in \{0,1,2,3\}$ and the martingale term on the RHS of (4.35) as follows:

$$\begin{aligned}
&[\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}}]_{a_1} \boldsymbol{\chi}^{(n)}_{u,\mathbf{a}} + \left(\mathcal{U}^{(n)}_{s,u,\boldsymbol{\sigma}} \circ \mathcal{Q}_s \circ \mathcal{B}_0(s) \right)_{\mathbf{a}} + \int_s^u \left(\mathcal{U}^{(n)}_{v,u,\boldsymbol{\sigma}} \circ \mathcal{Q}_v \circ \sum_{k=1}^3 \mathcal{B}_k(v) \right)_{\mathbf{a}} \mathrm{d}v + \int_s^u \left(\mathcal{U}^{(n)}_{v,u,\boldsymbol{\sigma}} \circ \mathcal{Q}_v \circ \mathrm{d}\mathcal{E}^{M,(n)}_{v,\boldsymbol{\sigma}} \right)_{\mathbf{a}} \\
&\prec (W^{-d} B_{u,0})^n \sup_{v \in [s,u]} \left((W^{-d} B_{u,0})^{\frac{1}{6}} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} + \max_{n'=1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{v,n'} + \max_{n'=\lceil n/2 \rceil + 1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n+2-n'} \left(\Xi^{(\mathcal{L})}_{v,n'_1} \Xi^{(\mathcal{L})}_{v,n'_2} \right)^{\frac{1}{2}} \right) \\
&\quad + (W^{-d} B_{u,0})^n \sup_{v \in [s,u]} \left((W^{-d} B_{s,0})^{-\frac{1}{4p}} (\Xi^{(\mathcal{L})}_{v,2n-1})^{\frac{1}{2}} (\Xi^{(\mathcal{L})}_{v,4p})^{\frac{1}{4p}} \right).
\end{aligned} \tag{4.58}$$

It remains to address the terms involving $\mathcal{B}_4$ and $\mathcal{B}_5$ on the RHS of (4.35). We claim the following bounds:

$$\|\mathcal{B}_4(u)\|_\infty + \|\mathcal{B}_5(u)\|_\infty \prec \eta_u^{-1} (W^{-d} B_{u,0})^n \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n-1}. \tag{4.59}$$

First, combining (4.31) with (4.18) and the second estimate in (4.27), we obtain that

$$\left\| \left[\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}} \right] \partial_u \boldsymbol{\chi}^{(n)}_u \right\|_\infty \prec \frac{1}{\eta_u} \frac{g^2 + |1-u|}{|1-u| \ell_u^d} \cdot (W^{-d} B_{u,0})^n \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n-1} \lesssim \eta_u^{-1} (W^{-d} B_{u,0})^n \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n-1}. \tag{4.60}$$

Second, using the definition of $\mathcal{Q}_t$ in (4.25) and the definition of $\Theta^{(n)}_{u,\boldsymbol{\sigma}}$ in (3.11), we can bound that

$$\begin{aligned}
\left[\mathcal{Q}_u, \Theta^{(n)}_{u,\boldsymbol{\sigma}} \right] \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}} &= \Theta^{(n)}_{u,\boldsymbol{\sigma}} \circ \left[\left(\mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}} \right) \cdot \boldsymbol{\chi}^{(n)}_u \right]_{\mathbf{a}} - \left[\left(\mathcal{P} \circ \Theta^{(n)}_{u,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}} \right) \cdot \boldsymbol{\chi}^{(n)}_u \right]_{\mathbf{a}} \\
&\prec \eta_u^{-1} \left\| \boldsymbol{\chi}^{(n)}_u \right\|_\infty \cdot \left\| \mathcal{P} \circ (\mathcal{L}-\mathcal{K})^{(n)}_{u,\boldsymbol{\sigma}} \right\|_\infty \prec \eta_u^{-1} (W^{-d} B_{u,0})^n \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n-1},
\end{aligned} \tag{4.61}$$

where in the second step, we use the simple fact that $\|\Theta^{(n)}_{u,\boldsymbol{\sigma}}\|_{\infty \to \infty} \lesssim (1-u)^{-1}$ for any $\boldsymbol{\sigma} \in \{+,-\}^n$ due to (2.65), and in the third step, we use (4.31) and the first bound in (4.27). Combining (4.60) and (4.61)

yields (4.59). To apply the evolution kernel estimate (4.56), we also need the fast-decay property (4.52) (with s replaced by v) for $\mathcal{Q}_v \circ \mathcal{B}_4(v)$ and $\mathcal{Q}_v \circ \mathcal{B}_5(v)$. We claim that this property holds for any constants $\varepsilon, D > 0$. Note that $\|\mathcal{B}_4(v)\|_\infty + \|\mathcal{B}_5(v)\|_\infty \le W^C$ for a constant $C > 0$, since $\|G_v\| \le \eta_v^{-1}$, the $\mathcal{K}$-loops are bounded by (2.58), $L^d \le W^C$, and $(1-v)^{-1} \le L^2/g^2 \le W^C$. Hence, by Claim 4.9, it suffices to check the fast-decay property for $\mathcal{B}_4(v)$ and $\mathcal{B}_5(v)$ themselves. For $\mathcal{B}_5(v)$, it follows from the decay of $\partial_v \boldsymbol{\chi}_v^{(n)}$ given by the second estimate in (4.27). For $\mathcal{B}_4(v)$, we use the expression in the first line of (4.61). Its second term decays by the first estimate in (4.27). For its first term, the tensor $\mathcal{X} := (\mathcal{P} \circ (\mathcal{L} - \mathcal{K})^{(n)}_{v,\boldsymbol{\sigma}}) \cdot \boldsymbol{\chi}_v^{(n)}$ satisfies the (v, ε, D)-decay property for any $\varepsilon, D > 0$ by Claim 4.9, and the kernel of $\Theta^{(n)}_{v,\boldsymbol{\sigma}}$ in (3.11) is bounded by $C(1-v)^{-1} e^{-c|a_i - b_i|/\ell_v}$ by (2.66) (together with (8.6) or (8.7) in the block Anderson model). Thus, if $\max_{i,j} |a_i - a_j| \ge W^{2\varepsilon}\ell_v$, then in the i-th term of (3.11), either $|a_i - b_i| \ge W^{2\varepsilon}\ell_v/2$, in which case the kernel is $\mathrm{O}(W^{-D})$, or $\max_{j,k} |(\mathbf{a}^{(i)}(b_i))_j - (\mathbf{a}^{(i)}(b_i))_k| \ge W^{\varepsilon}\ell_v$, in which case $\mathcal{X}_{\mathbf{a}^{(i)}(b_i)} = \mathrm{O}(W^{-D})$. Summing over $b_i \in \mathbb{Z}^d_L$, we see that $\Theta^{(n)}_{v,\boldsymbol{\sigma}} \circ \mathcal{X}$ satisfies the $(v, 2\varepsilon, D - C)$-decay property. Since ε and D are arbitrary, this proves the claim. Now, applying the evolution kernel estimate (4.56) and the estimates in (4.59), and performing the integral over v, we can obtain that

$$\int_s^u \left(\mathcal{U}^{(n)}_{v,u,\boldsymbol{\sigma}} \circ \mathcal{Q}_v \circ \sum_{k=4}^{5} \mathcal{B}_k(v) \right)_{\mathbf{a}} \mathrm{d}v \prec (W^{-d} B_{u,0})^n \sup_{v \in [s,u]} \left(\Xi^{(\mathcal{L}-\mathcal{K})}_{v,n-1} \right). \tag{4.62}$$

Finally, combining Lemma 4.6 with (4.58) and (4.62), we obtain that

$$\begin{aligned} \sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} &\prec (W^{-d}B_{u,0})^{\frac{1}{6}} \sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} + \sup_{v\in[s,u]} \left((W^{-d}B_{s,0})^{-\frac{1}{4p}} (\Xi^{(\mathcal{L})}_{v,2n-1})^{\frac{1}{2}} (\Xi^{(\mathcal{L})}_{v,4p})^{\frac{1}{4p}} \right) \\ &+ \sup_{v\in[s,u]} \left(\max_{n'=1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{v,n'} + \max_{n'=\lceil n/2\rceil+1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n+2-n'} \left(\Xi^{(\mathcal{L})}_{v,n'_1} \Xi^{(\mathcal{L})}_{v,n'_2} \right)^{\frac{1}{2}} \right), \end{aligned}$$

which yields (4.36) at each fixed $u \in [s,t]$ upon solving for $\sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n}$. Then, applying a standard N^{-C}-net argument, together with the continuity estimate of Lemma 3.2 which controls the increments of the G-loops between two nearby times, extends it uniformly to all $u \in [s,t]$. □

Proof of Lemma 4.11. By the averaged local law (2.88), we have $\widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,1} \prec 1 =: \Xi^{(\mathcal{L}-\mathcal{K})}_{u,1}$ uniformly for all $u \in [s,t]$. Substituting this into (4.36), we obtain the following bound, also uniformly in $u \in [s,t]$ under the assumption $1 - s \le g^2$:

$$\begin{aligned} \sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} &\prec \sup_{v\in[s,u]} \left(\left(W^{-d}B_{s,0}\right)^{-\frac{1}{4p}} (\Xi^{(\mathcal{L})}_{v,2n-1})^{\frac{1}{2}} (\Xi^{(\mathcal{L})}_{v,4p})^{\frac{1}{4p}} \right) \\ &+ \sup_{v\in[s,u]} \left(\max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{v,n'} + \max_{n'=3}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n+2-n'} \left(\Xi^{(\mathcal{L})}_{v,n'_1} \Xi^{(\mathcal{L})}_{v,n'_2} \right)^{\frac{1}{2}} \right), \end{aligned} \tag{4.63}$$

where we also use that $\lceil n/2 \rceil + 1 \le n - 1$ only when $n \ge 4$, in which case we have $\lceil n/2 \rceil + 1 \ge 3$.

To show Lemma 4.11 using the bootstrap bound (4.63), we first claim that under the assumption (4.41), the following bound holds uniformly in $u \in [s,t]$ as long as we choose $p \ge 4$:

$$\sup_{v\in[s,u]} \left((g^2 W^d)^{\frac{1}{4p}} (\widehat{\Xi}^{(\mathcal{L})}_{v,2n-1})^{\frac{1}{2}} (\widehat{\Xi}^{(\mathcal{L})}_{v,4p})^{\frac{1}{4p}} \right) \prec \Psi(n,k;s,u) \,. \tag{4.64}$$

Given the bound (4.64), we obtain from (4.63) that

$$\sup_{v\in[s,u]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{v,n} \prec \Psi(n,k;s,u) + \sup_{v\in[s,u]} \left(\max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{v,n'} + \max_{n'=3}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{v,n+2-n'} \left(\Xi^{(\mathcal{L})}_{v,n'_1} \Xi^{(\mathcal{L})}_{v,n'_2} \right)^{\frac{1}{2}} \right) \tag{4.65}$$

uniformly in $u \in [s,t]$. By the induction hypothesis (4.41), we may choose the $\Xi^{(\mathcal{L}-\mathcal{K})}$-parameters as

$$\Xi^{(\mathcal{L}-\mathcal{K})}_{v,n'} = \Psi(n',k;s,u) \le \Psi(n,k;s,u) \quad \text{for} \quad 2 \le n' \le n-1,$$

and, by (4.38) and (4.41), we can choose the $\Xi^{(\mathcal{L})}$-parameters as

$$\Xi^{(\mathcal{L})}_{v,n'} = \begin{cases} 1 + (g^2 W^d)^{-1} \Psi(n',k;s,u), & \text{for} \quad n' \le n-1, \\ 1 + (g^2 W^d)^{-1} \Psi(n',k-1;s,u) & \text{for} \quad n \le n' \le n+2. \end{cases}$$

With these choices, we readily verify that

$$\max_{n'=n-1}^{n+1} \Xi_{v,n'}^{(\mathcal{L})} \le \Psi(n,k;s,u), \quad \text{and} \quad \left(\Xi_{v,n_1'}^{(\mathcal{L})}\Xi_{v,n_2'}^{(\mathcal{L})}\right)^{\frac{1}{2}} \lesssim 1+(\eta_s/\eta_u)^{n'}\,(g^2W^d)^{-\frac{k}{8}} \quad \text{for} \quad 3\le n'\le n-1.$$

Substituting these bounds into (4.65), we obtain

$$\sup_{v\in[s,u]} \widehat{\Xi}_{v,n}^{(\mathcal{L}-\mathcal{K})} \prec \Psi(n,k;s,u) + \max_{n'=3}^{n-1} \Psi(n+2-n',k;s,u)\left[1+\left(\frac{\eta_s}{\eta_u}\right)^{n'}(g^2W^d)^{-\frac{k}{8}}\right] \lesssim \Psi(n,k;s,u),$$

which concludes the proof of Lemma 4.11.

It remains to prove (4.64). For $n\in\{2,3\}$, using the a priori bound (4.39) for $\widehat{\Xi}_{v,2n-1}^{(\mathcal{L})}$ and $\widehat{\Xi}_{v,4p}^{(\mathcal{L})}$, along with the relation (4.38), we obtain for $p\ge 4$ that,

$$\sup_{v\in[s,u]}\left((g^2W^d)^{\frac{1}{4p}}(\widehat{\Xi}_{v,2n-1}^{(\mathcal{L})})^{\frac{1}{2}}(\widehat{\Xi}_{v,4p}^{(\mathcal{L})})^{\frac{1}{4p}}\right) \prec (g^2W^d)^{\frac{1}{16}}\,(\eta_s/\eta_u)^n \le \Psi(n,k;s,u).$$

We now consider the $n\ge 4$ case. Recall the G-chain defined in (4.6). Given an arbitrary $(2n-1)$-G-loop $\mathcal{L}_{v,\boldsymbol{\sigma},\mathbf{a}}^{(2n-1)}$ with

$$\boldsymbol{\sigma}=(\sigma,\sigma_1,\dots,\sigma_{n-1},\sigma',\sigma_1',\dots,\sigma_{n-2}'), \quad \mathbf{a}=(b,a_1,\dots,a_{n-1},b',a_1',\dots,a_{n-2}'),$$

we can write it as

$$\mathcal{L}_{v,\boldsymbol{\sigma},\mathbf{a}}^{(2n-1)} = W^{-2d}\sum_{x\in[b],x'\in[b']}\left(\mathcal{C}_{v,\boldsymbol{\sigma}_1,\mathbf{a}_1}^{(n)}\right)_{xx'}\left(\mathcal{C}_{v,\boldsymbol{\sigma}_2,\mathbf{a}_2}^{(n-1)}\right)_{x'x},$$

where $\boldsymbol{\sigma}_1=(\sigma_1,\dots,\sigma_{n-1},\sigma')$, $\boldsymbol{\sigma}_2=(\sigma_1',\dots,\sigma_{n-2}',\sigma)$, $\mathbf{a}_1=([a_1],\dots,[a_{n-1}])$, and $\mathbf{a}_2=([a_1'],\dots,[a_{n-2}'])$. Applying the Cauchy-Schwarz inequality gives

$$\left|\mathcal{L}_{v,\boldsymbol{\sigma},\mathbf{a}}^{(2n-1)}\right| \le \left(W^{-2d}\sum_{x\in[b],x'\in[b']}\left|\left(\mathcal{C}_{v,\boldsymbol{\sigma}_1,\mathbf{a}_1}^{(n)}\right)_{xx'}\right|^2\right)^{1/2}\left(W^{-2d}\sum_{x\in[b],x'\in[b']}\left|\left(\mathcal{C}_{v,\boldsymbol{\sigma}_2,\mathbf{a}_2}^{(n-1)}\right)_{x'x}\right|^2\right)^{1/2}. \tag{4.66}$$

Following the argument for equation (5.118) in [69], we obtain that

$$\begin{aligned} W^{-2d}\sum_{x\in[b],x'\in[b']}\left|\left(\mathcal{C}_{v,\boldsymbol{\sigma}_1,\mathbf{a}_1}^{(n)}\right)_{xx'}\right|^2 &\le (g^2W^d)\Xi_{v,2l_1}^{(\mathcal{L})}\Xi_{v,2l_2}^{(\mathcal{L})},\\ W^{-2d}\sum_{x\in[b],x'\in[b']}\left|\left(\mathcal{C}_{v,\boldsymbol{\sigma}_2,\mathbf{a}_2}^{(n-1)}\right)_{x'x}\right|^2 &\le (g^2W^d)\Xi_{v,2l_3}^{(\mathcal{L})}\Xi_{v,2l_4}^{(\mathcal{L})}, \end{aligned} \tag{4.67}$$

where $l_1=\lceil n/2\rceil$, $l_2=\lfloor n/2\rfloor$, $l_3=\lceil (n-1)/2\rceil$, and $l_4=\lfloor (n-1)/2\rfloor$. By the induction hypothesis (4.41) (noting $\max_i(2l_i)\le n+1$) and the relation (4.38), we have

$$\widehat{\Xi}_{v,2l_i}^{(\mathcal{L})} \prec 1+(g^2W^d)^{-1}\Psi(2l_i,k-1;s,u) \prec 1+(\eta_s/\eta_u)^{2l_i-1}\,(g^2W^d)^{-(k-1)/8}. \tag{4.68}$$

Plugging (4.68) into (4.67) and then applying (4.66), we obtain for $k\ge 1$ that,

$$\begin{aligned} \sup_{v\in[s,u]}\left(\widehat{\Xi}_{v,2n-1}^{(\mathcal{L})}\right)^{\frac{1}{2}} &\prec \sup_{v\in[s,u]}(g^2W^d)^{1/2}\left[\prod_{i=1}^4\left(1+(\eta_s/\eta_u)^{2l_i-1}\,(g^2W^d)^{-(k-1)/8}\right)\right]^{1/4}\\ &\lesssim (g^2W^d)^{1/2}\left(1+(\eta_s/\eta_u)^n\,(g^2W^d)^{-(k-1)/8}\right). \end{aligned} \tag{4.69}$$

On the other hand, by (4.39), we have $\sup_{v\in[s,u]}(\widehat{\Xi}_{v,4p}^{(\mathcal{L})})^{\frac{1}{4p}} \prec \eta_s/\eta_u$. Combining it with (4.69), and using the condition (2.83), we conclude (4.64) when $p\ge 4$. □

Proof of Lemma 4.13. We only prove the expansion (4.44) using Ward's identity (2.56), while (4.45) follows from the same argument by using the identity (2.57) instead. We first record the following consequence of (2.56): for any $\boldsymbol{\sigma}\in\{+,-\}^n$ with $\sigma_i\ne\sigma_{i+1}$ and any subset $A\subset[\![n]\!]\setminus\{i\}$,

$$\left(Q^{(A)}\circ P^{(i)}\circ\mathcal{L}^{(n)}\right)_{t,\boldsymbol{\sigma},\mathbf{a}} = \frac{1}{2\mathrm{i}N\eta_t}\left(\left(Q^{(A_{(i)})}\circ\mathcal{L}^{(n-1)}\right)_{t,\boldsymbol{\sigma}_+,\mathbf{a}_+} - \left(Q^{(A_{(i)})}\circ\mathcal{L}^{(n-1)}\right)_{t,\boldsymbol{\sigma}_-,\mathbf{a}_-}\right), \tag{4.70}$$

where $A_{(i)}:=\{1\le j<i: j\in A\}\cup\{i\le j\le n-1: j+1\in A\}$, $\mathbf{a}_\pm=(a_1,\dots,a_{i-1},a_{i+1},\dots,a_n)$, and $\boldsymbol{\sigma}_\pm=(\sigma_1\cdots\sigma_{i-1},\pm,\sigma_{i+2},\cdots\sigma_n)$. We now prove (4.44) by induction in n and using (4.70).

First, it is easy to see that (4.44) trivially holds for $n = 1$ and $|A| \in \{0, 1\}$. Next, suppose we have shown that there exists a decomposition (4.44) for any $n \le l - 1$. We prove (4.44) for $n = l$. Given any $A \subset [\![n]\!]$, if $A \supset I_{\mathrm{diff}}(\boldsymbol{\sigma})$, then (4.44) holds trivially with only one term on the RHS (i.e., the term $Q^{(A)} \circ \mathcal{L}^{(n)}$ itself). Otherwise, suppose there exists $i \in I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus A$, then we write

$$\left(Q^{(A)} \circ \mathcal{L}^{(n)}\right)_{t,\boldsymbol{\sigma},\mathbf{a}} = \left(Q^{(A\cup\{i\})} \circ \mathcal{L}^{(n)}\right)_{t,\mathbf{a},\boldsymbol{\sigma}} + \left(Q^{(A)} \circ P^{(i)} \circ \mathcal{L}^{(n)}\right)_{t,\boldsymbol{\sigma},\mathbf{a}}. \tag{4.71}$$

Applying (4.70) to the second term on the RHS and using the induction hypothesis for $Q^{(A_{(i)})} \circ \mathcal{L}^{(n-1)}$, we can expand $(Q^{(A)} \circ P^{(i)} \circ \mathcal{L}^{(n)})_{t,\boldsymbol{\sigma},\mathbf{a}}$ into an expression that matches the form of a summand on the RHS of (4.44). On the other hand, in the first term of (4.71), the number of elements in $I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus (A \cup \{i\})$ is reduced by 1 relative to $I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus A$. If $I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus (A \cup \{i\}) = \emptyset$, then the induction is completed. Otherwise, we choose the next element $j \in I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus (A \cup \{i\})$ and repeat the same argument with A and i replaced by $A \cup \{i\}$ and j, respectively. Iterating this procedure at most $|I_{\mathrm{diff}}(\boldsymbol{\sigma}) \setminus A|$ times, we finally express $Q^{(A)} \circ \mathcal{L}^{(n)}$ in the form (4.44). This completes the induction, thereby concluding the proof of Lemma 4.13. □

Proof of Lemma 4.14. Given any $\boldsymbol{\sigma} \in \{+, -\}^n$ and $\mathbf{a} \in (\mathbb{Z}_L^d)^n$, we expand $\mathcal{L}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ and $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ as in (4.44) and (4.45) with $A_n = I_{\mathrm{diff}}(\boldsymbol{\sigma})$. This yields

$$(\mathcal{L} - \mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = Q^{(A_n)} \circ (\mathcal{L} - \mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} + \sum_{\alpha} \frac{\xi_\alpha}{(2\mathrm{i}N\eta_t)^{n-k_\alpha}} Q^{(A_\alpha)} \circ (\mathcal{L} - \mathcal{K})^{(k_\alpha)}_{t,\boldsymbol{\sigma}_\alpha,\mathbf{a}_\alpha}. \tag{4.72}$$

For the $k_\alpha = 1$ terms, using (4.43) and the averaged local law (2.88), we bound

$$\frac{\xi_\alpha}{(2\mathrm{i}N\eta_t)^{n-k_\alpha}} Q^{(A_\alpha)} \circ (\mathcal{L} - \mathcal{K})^{(k_\alpha)}_{t,\boldsymbol{\sigma}_\alpha,\mathbf{a}_\alpha} \prec (N\eta_t)^{-(n-1)} \cdot W^{-d} B_{t,0} \lesssim (W^{-d} B_{t,0})^n. \tag{4.73}$$

All other terms on the RHS of (4.72) involve loops of length ≥ 2, and therefore satisfy equation (4.47). Take the first term as an example. Using the evolution kernel estimate in Lemma 4.17, the bounds (4.12)–(4.15), and the assumption (2.78) on $(\mathcal{L} - \mathcal{K})^{(n)}_{s,\boldsymbol{\sigma},\mathbf{a}}$, we obtain that

$$\begin{aligned}
\frac{Q^{(A_n)} \circ (\mathcal{L} - \mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}}{(W^{-d} B_{t,0})^n} &\prec 1 + (W^{-d} B_{t,0})^{\frac{1}{6}} \int_s^t \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} \frac{\mathrm{d}u}{\eta_u} + \int_s^t \max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n'} \frac{\mathrm{d}u}{\eta_u} + \int_s^t \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{u,n'} \frac{\mathrm{d}u}{\eta_u} \\
&+ \int_s^t \max_{n'=\lceil n/2\rceil+1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n+2-n'} \left(\Xi^{(\mathcal{L})}_{u,n'_1} \Xi^{(\mathcal{L})}_{u,n'_2}\right)^{\frac{1}{2}} \frac{\mathrm{d}u}{\eta_u} + \frac{1}{(W^{-d} B_{t,0})^n} \int_s^t \left(Q^{(A_n)} \circ \mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}.
\end{aligned} \tag{4.74}$$

By Lemma 3.6, together with (4.15) and Lemma 4.17, we bound the martingale term as

$$(W^{-d} B_{t,0})^{-n} \int_s^t \left(Q^{(A_n)} \circ \mathcal{U}^{(n)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(n)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \prec \left(W^{-d} B_{s,0}\right)^{-\frac{1}{4p}} \left\{\int_s^t \eta_u^{-1} \Xi^{(\mathcal{L})}_{u,2n-1} \left(\Xi^{(\mathcal{L})}_{u,4p}\right)^{\frac{1}{2p}} \mathrm{d}u\right\}^{1/2} \tag{4.75}$$

for any fixed $p \ge 1$. Plugging it into (4.74) and performing the integral over u, we obtain

$$\begin{aligned}
\frac{Q^{(A_n)} \circ (\mathcal{L} - \mathcal{K})^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}}{(W^{-d} B_{t,0})^n} &\prec \sup_{u\in[s,t]} \left((W^{-d} B_{t,0})^{\frac{1}{6}} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} + \max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{u,n'}\right) \\
&+ \sup_{u\in[s,t]} \left(\max_{n'=\lceil n/2\rceil+1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n+2-n'} \left(\Xi^{(\mathcal{L})}_{u,n'_1} \Xi^{(\mathcal{L})}_{u,n'_2}\right)^{\frac{1}{2}} + \left(W^{-d} B_{s,0}\right)^{-\frac{1}{4p}} \left(\Xi^{(\mathcal{L})}_{u,2n-1}\right)^{\frac{1}{2}} \left(\Xi^{(\mathcal{L})}_{u,4p}\right)^{\frac{1}{4p}}\right).
\end{aligned} \tag{4.76}$$

This bound also applies to each term $Q^{(A_\alpha)} \circ (\mathcal{L} - \mathcal{K})^{(k_\alpha)}_{t,\boldsymbol{\sigma}_\alpha,\mathbf{a}_\alpha}$ in (4.72), with n replaced by k_α. Using the fact $(N\eta_t)^{-1} \le W^{-d} B_{t,0}$, every summand of (4.72), after being rescaled by $(W^{-d} B_{t,0})^{-n}$, is controlled by the RHS of (4.76). Now, taking the maximum of both sides of (4.72) over $(\boldsymbol{\sigma}, \mathbf{a})$ gives

$$\begin{aligned}
\sup_{u\in[s,t]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} &\prec (W^{-d} B_{t,0})^{\frac{1}{6}} \sup_{u\in[s,t]} \widehat{\Xi}^{(\mathcal{L}-\mathcal{K})}_{u,n} + \sup_{u\in[s,t]} \left(\left(W^{-d} B_{s,0}\right)^{-\frac{1}{4p}} \left(\Xi^{(\mathcal{L})}_{u,2n-1}\right)^{\frac{1}{2}} \left(\Xi^{(\mathcal{L})}_{u,4p}\right)^{\frac{1}{4p}}\right) \\
&+ \sup_{u\in[s,t]} \left(\max_{n'=2}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n'} + \max_{n'=n-1}^{n+1} \Xi^{(\mathcal{L})}_{u,n'} + \max_{n'=\lceil n/2\rceil+1}^{n-1} \Xi^{(\mathcal{L}-\mathcal{K})}_{u,n+2-n'} \left(\Xi^{(\mathcal{L})}_{u,n'_1} \Xi^{(\mathcal{L})}_{u,n'_2}\right)^{\frac{1}{2}}\right),
\end{aligned}$$

solving which yields (4.36) at $u = t$. Obviously, the same argument applies to each fixed $u \in [s, t]$. Finally, a standard N^{-C}-net argument, together with the continuity estimate of Lemma 3.2, extends (4.36) uniformly to the entire interval $u \in [s, t]$. □

5. Step 5: Pointwise estimate for $(\mathcal{L}-\mathcal{K})$-loops

As in the proof for Step 3, by adding intermediate times if necessary, we can divide the proof into the following four cases: (i) $g^2/L^2 \le 1-t \le 1-s \le g^2$, (ii) $g^2/L^d \le 1-t \le 1-s \le g^2/L^2$, (iii) $1-s \ge 1-t \ge g^2$, and (iv) $1-t \le 1-s \le g^2/L^d$. In case (iv), $\ell_t = L$ and $B_{t,0}$ is dominated by the $(L^d|1-t|)^{-1}$ term, so the desired bound already follows from the conclusion (2.91) in Step 4. Hence, we only need to focus on the first three cases in the following proof. Our goal is to remove the $(|1-s|/|1-u|)^{C_d}$ factor in (2.89). Our proofs below are all based on the integrated loop hierarchy in equation (3.14) with $n=2$:

$$\begin{aligned}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \\ + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{E}^{\mathring{G},(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}}. \end{aligned} \tag{5.1}$$

With the stability estimate (2.89) in hand, we can now show that the quadratic term $\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}$ yields a negligible contribution. Using the light-weight estimate (3.32) and the martingale estimate (3.35) established in Step 2, we can also obtain sufficient control of the third and fourth terms on the RHS of (5.1). The remaining technical challenge arises from the contribution of the initial condition, i.e., the first term on the RHS of (5.1). In case (i), bounding this term requires leveraging a CLT-type cancellation mechanism, similar to that developed in [28].[8] In case (ii), we apply the zero-mode-removing operator introduced in Definition 4.12. Finally, case (iii) can be regarded as a special instance of the lower-dimensional proofs for $d \in \{1,2\}$, as explained in Section 5.3.

If $|1-s|/|1-t| \le (\log W)^{10}$, then the factor $(|1-s|/|1-u|)^{C_d} \le (\log W)^{10C_d}$ can be absorbed into the stochastic domination notation "$\prec$". Hence, throughout the following proof, we always assume that

$$|1-s|/|1-t| > (\log W)^{10} \implies t \ge 1-(\log W)^{-10}. \tag{5.2}$$

5.1. **The case $g^2/L^2 \le 1-t \le 1-s \le g^2$.** In this case, we have $B_{u,0} \asymp g^{-2}$ and $\ell_u \asymp g|1-u|^{-1/2}$ for all $u \in [s,t]$. In Step 2, we have established the following estimates for any large constant $D>0$:

$$\mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},(a_1,a_2)} \prec W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a_1-a_2|), \quad (\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},(a_1,a_2)} \prec (g^2W^d)^{-\frac{1}{6}} \cdot W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a_1-a_2|), \quad \forall u \in [s,t]. \tag{5.3}$$

We now apply (5.3) to (3.57) with $\ell = L$. In this case, the index sets $\{c' : |c'-b| \ge L-1\}$ and $\{c' : |c'-a| \ge L-1\}$ appearing in the first two terms on the RHS are empty, because $\max_{a,b\in\mathbb{Z}^d_L}|a-b| \le L/2$ for the periodic L^∞-distance on $\mathbb{Z}^d_L$. Hence only the third term remains, and it can be bounded as in (3.58). Consequently, we obtain that

$$\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec \frac{(g^2W^d)^{-\frac{1}{3}}}{\eta_u}\left[W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a_1-a_2|)\right]. \tag{5.4}$$

Moreover, using Lemmas 3.11 and 3.13, we get that for any $\boldsymbol{\sigma} \in \{+,-\}^2$,

$$\begin{aligned}\mathcal{E}^{\mathring{G},(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} &\prec \frac{(g^2W^d)^{-\frac{1}{2}}}{\eta_u}\left[W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a_1-a_2|)\right], \\ (\mathcal{E}\otimes\mathcal{E})^{M,(2)}_{u,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} &\prec \frac{(g^2W^d)^{-\frac{1}{2}}}{\eta_u}\left[W^{-d}\widetilde{\mathcal{T}}^{L}_{u,D}(|a_1-a_2|)\right]^2. \end{aligned} \tag{5.5}$$

We recall the definition of $\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}}$ in (3.12) and notice the following identity (recall that $t \gtrsim 1$ under (5.2)):

$$\frac{1-s\cdot M^{(\sigma_1,\sigma_2)}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_1,\sigma_2)}S^{(\mathsf{B})}} = \frac{s}{t} + \frac{t-s}{t}\Theta^{(\sigma_1,\sigma_2)}_t. \tag{5.6}$$

If $\sigma_1 = \sigma_2$, then $\Theta^{(\sigma_1,\sigma_2)}_t$ decays exponentially by (2.67). Using the estimates (5.4) and (5.5), the assumption (2.79), the decomposition (5.6), and the estimate (2.67), we obtain from (5.1) that

$$(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \prec (g^2W^d)^{-\frac{1}{5}} \cdot \left[W^{-d}\widetilde{\mathcal{T}}^{L}_{t,D}(|a_1-a_2|)\right].$$

[8] Interestingly, in two dimensions [28], the CLT cancellation was used in Step 3 but not in Step 5, whereas in our case the situation is reversed. In lower dimensions, Step 5 becomes almost trivial once Steps 2 and 4 are established; however, in dimensions $d \ge 3$, the additional complexity in Step 5 arises once again due to the polynomial decay of the 2-G-loops.

Since the proof is considerably simpler than in the case $\sigma_1 \neq \sigma_2$ discussed below, we omit the details.

We now focus on the key challenge with $\sigma_1 \neq \sigma_2$. When $\sigma_1 \neq \sigma_2$, with the estimates in (5.5) and using the decomposition (5.6), we can bound the second and third terms on the RHS of (5.1) as

$$\int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \prec (g^2W^d)^{-\frac{1}{3}} \int_s^t \frac{\mathcal{A}_{u,t,\mathbf{a}}}{\eta_u}\mathrm{d}u \prec (g^2W^d)^{-\frac{1}{3}} \sup_{u\in[s,t]} \mathcal{A}_{u,t,\mathbf{a}}, \tag{5.7}$$

$$\int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathring{\mathcal{E}}^{\mathring{G},(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \prec (g^2W^d)^{-\frac{1}{2}} \int_s^t \frac{\mathcal{A}_{u,t,\mathbf{a}}}{\eta_u}\mathrm{d}u \prec (g^2W^d)^{-\frac{1}{2}} \sup_{u\in[s,t]} \mathcal{A}_{u,t,\mathbf{a}}, \tag{5.8}$$

where the two-dimensional tensor $\mathcal{A}$ is defined as

$$\begin{aligned}\mathcal{A}_{u,t,\mathbf{a}} &:= W^{-d}\widetilde{\mathcal{T}}^L_{u,D}\left(|a_1-a_2|\right) + (t-u)\sum_b \left[\Theta^{(+,-)}_{t,a_1b}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{u,D}\left(|b-a_2|\right) + W^{-d}\widetilde{\mathcal{T}}^L_{u,D}\left(|a_1-b|\right)\cdot\Theta^{(+,-)}_{t,ba_2}\right] \\ &\quad + (t-u)^2\sum_{b_1,b_2}\Theta^{(+,-)}_{t,a_1b_1}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{u,D}\left(|b_1-b_2|\right)\cdot\Theta^{(+,-)}_{t,b_2a_2}, \quad \forall \mathbf{a}=(a_1,a_2)\in(\mathbb{Z}^d_L)^2. \end{aligned}\tag{5.9}$$

Similarly, we write the martingale term as

$$\begin{aligned}\int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} &= \int_s^t \frac{u^2}{t^2}\mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} + \left(\Theta^{(+,-)}_t \cdot \left(\int_s^t \frac{t-u}{t}\mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)\right)_{\mathbf{a}} + \left(\left(\int_s^t \frac{t-u}{t}\mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)\cdot\Theta^{(+,-)}_t\right)_{\mathbf{a}} \\ &\quad + \left(\Theta^{(+,-)}_t \cdot \left(\int_s^t \frac{(t-u)^2}{t^2}\mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)\cdot\Theta^{(+,-)}_t\right)_{\mathbf{a}}. \end{aligned}\tag{5.10}$$

Combining the second bound in (5.5) with the Burkholder-Davis-Gundy inequality (recall (3.17)), we get

$$\int_s^t (t-u)^r \cdot \mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec (1-s)^r (g^2W^d)^{-\frac{1}{4}} \cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}\left(|a_1-a_2|\right), \quad \forall r\in\{0,1,2\}.$$

Plugging it into (5.10) and using the $(\infty\to\infty)$-norm of $\Theta^{(+,-)}_t$ given by (2.65), we obtain

$$\int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathrm{d}\mathcal{E}^{M,(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \prec \frac{(1-s)^2}{(1-t)^2}\cdot(g^2W^d)^{-\frac{1}{4}} \max_{u\in[s,t]} \mathcal{A}_{u,t,\mathbf{a}}. \tag{5.11}$$

Now, inserting the bounds (5.7), (5.8), and (5.11) into (5.1) gives that

$$(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \prec \left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \frac{(1-s)^2}{(1-t)^2}\cdot(g^2W^d)^{-\frac{1}{4}} \sup_{u\in[s,t]} \mathcal{A}_{u,t,\mathbf{a}}. \tag{5.12}$$

Applying the estimate (2.66) to $\Theta^{(+,-)}_t$, along with the bound (3.21), we obtain that

$$\begin{aligned}(1-u)\sum_b \Theta^{(+,-)}_{t,a_1b}\cdot\widetilde{\mathcal{T}}^L_{u,D}\left(|b-a_2|\right) &\prec (1-u)\sum_b \mathcal{T}_t\left(|a_1-b|\right)\left[\mathcal{T}_u\left(|b-a_2|\right)+W^{-D}\right] \\ &\prec \mathcal{T}_t\left(|a_1-a_2|\right) + \frac{1-u}{1-t}W^{-D} \lesssim \widetilde{\mathcal{T}}^L_{t,D-\theta}\left(|a_1-a_2|\right), \end{aligned}\tag{5.13}$$

where $\theta := 2/(d\mathfrak{c})$. In the last step, we used that $(1-u)/(1-t) \le (1-s)/(1-t) \le L^2 \le N^{2/d} \le W^\theta$ in the present regime, by (2.9). Inserting this estimate into the definition of $\mathcal{A}$ in (5.9), we can bound $\mathcal{A}_{u,t,\mathbf{a}}$ as

$$\max_{u\in[s,t]} \mathcal{A}_{u,t,\mathbf{a}} \prec \frac{1-s}{1-t}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D-2\theta}\left(|a_1-a_2|\right). \tag{5.14}$$

Combining this with (5.12), we obtain, for any large constant $D>0$,

$$(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \prec \left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + (g^2W^d)^{-\frac{1}{5}}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}\left(|a_1-a_2|\right). \tag{5.15}$$

For the first term on the RHS, we claim that

$$\left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \prec (g^2W^d)^{-\frac{1}{5}}\cdot W^{-d}\mathcal{T}_t\left(|a_1-a_2|\right) + W^{-D}. \tag{5.16}$$

Combining (5.15) and (5.16), we conclude (2.92) for the case $g^2/L^2 \le 1-t \le 1-s \le g^2$.

Proof of (5.16). Using the decomposition (5.6), we can expand the LHS of (5.16) as

$$\left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} = \frac{s^2}{t^2}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}} + \frac{t-s}{t}\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \frac{t-s}{t}\left((\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}} + \frac{(t-s)^2}{t^2}\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}}. \tag{5.17}$$

With the inductive hypothesis (2.79) on $(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}}$ and using (5.13) again, we get that

$$(t-s)\left|\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}}\right| + (t-s)\left|\left((\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}}\right| \prec (g^2W^d)^{-\frac{1}{5}} \cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}(|a_1-a_2|). \tag{5.18}$$

From (2.79), (5.17), and (5.18), we obtain that

$$\left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \prec (g^2W^d)^{-\frac{1}{5}} \cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}(|a_1-a_2|) + (1-s)^2\left|\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}}\right|. \tag{5.19}$$

To show (5.16), it remains to prove the following estimate for $\mathbf{a}=(a_1,a_2)$ and $\boldsymbol{\sigma} \in \{(+,-),(-,+)\}$:

$$(1-s)^2\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}} \prec (g^2W^d)^{-\frac{1}{5}} \cdot W^{-d}\mathcal{T}_t(|a_1-a_2|) + W^{-D}. \tag{5.20}$$

Recall that we assume (5.2), under which we have

$$\ell_t \geq \ell_s \cdot (\log W)^5. \tag{5.21}$$

Due to the exponential decay of $\Theta^{(+,-)}_t$ and $(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}$ given by (2.66) and (2.79), respectively, we only need to focus on the following regime:

$$|a_1-a_2| \leq \frac{1}{2}(\log W)^{3/2} \cdot \ell_t + (\log W)^{5/2} \cdot \ell_s \leq (\log W)^{3/2} \cdot \ell_t; \tag{5.22}$$

otherwise the LHS of (5.20) is an error of order W^{-D} for any large constant $D>0$. Under this condition, we have $\exp\left(\left(|a_1-a_2|/\ell_t\right)^{1/2}\right) \prec 1$, so proving (5.20) amounts to establishing the following bound:

$$(1-s)^2\left(\Theta^{(+,-)}_t \cdot (\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}} \cdot \Theta^{(+,-)}_t\right)_{\mathbf{a}} \prec \frac{(g^2W^d)^{-\frac{6}{5}}}{|a_1-a_2|^{d-2}+1}. \tag{5.23}$$

To show this estimate, we write the LHS of (5.23) as

$$\begin{aligned}
&(1-s)^2\sum_{b_1,b_2}\Theta^{(+,-)}_{t,a_1b_1}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},(b_1,b_2)}\left(\Theta^{(+,-)}_{t,b_2a_2}-\Theta^{(+,-)}_{t,b_1a_2}\right) + (1-s)^2\sum_{b_1,b_2}\left(\Theta^{(+,-)}_{t,a_1b_1}\Theta^{(+,-)}_{t,a_2b_1}\right)(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},(b_1,b_2)} \\
&= (1-s)^2\sum_{b_1,b_2}\Theta^{(+,-)}_{t,a_1b_1}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},(b_1,b_2)}\left(\Theta^{(+,-)}_{t,b_2a_2}-\Theta^{(+,-)}_{t,b_1a_2}\right) + (1-s)^2\sum_{b_1}\Theta^{(+,-)}_{t,a_1b_1}\Theta^{(+,-)}_{t,a_2b_1}\frac{\operatorname{Im}\operatorname{Tr}\left((G_s-M)E_{b_1}\right)}{W^d\eta_s} \\
&=: f_{\mathbf{a}} + g_{\mathbf{a}}.
\end{aligned}$$

Above, we have applied Ward's identities (2.56) and (2.57) to express $\sum_{b_2}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},(b_1,b_2)}$ as

$$\sum_{b_2}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},(b_1,b_2)} = \frac{\operatorname{Im}(\mathcal{L}-\mathcal{K})^{(1)}_{s,+,b_1}}{W^d\eta_s} = \frac{\operatorname{Im}\operatorname{Tr}\left((G_s-M)E_{b_1}\right)}{W^d\eta_s} \prec \frac{(g^2W^d)^{-1}}{W^d\eta_s}, \tag{5.24}$$

where we use the averaged local law (2.88) for $\operatorname{Tr}\left((G_s-M)E_{b_1}\right)$ in the last step. With (5.24), we can bound $g_{\mathbf{a}}$ as follows for any large constant $D>0$:

$$\begin{aligned}
g_{\mathbf{a}} &\prec \frac{1-s}{g^2W^{2d}}\sum_{b_1}\Theta^{(+,-)}_{t,a_1b_1}\Theta^{(+,-)}_{t,a_2b_1} \lesssim \frac{1-s}{g^2W^{2d}}\sum_{b_1}\mathcal{T}_t(|a_1-b_1|)\mathcal{T}_t(|a_2-b_1|) \\
&\lesssim \frac{1-s}{1-t}\left(g^2W^d\right)^{-1} \cdot W^{-d}\mathcal{T}_t(|a_1-a_2|) \leq \frac{(g^2W^d)^{-\frac{6}{5}}}{|a_1-a_2|^{d-2}+1},
\end{aligned} \tag{5.25}$$

where, in the second step, we apply the estimate (2.66), and in the third step, we use the bound (3.21).

To conclude (5.23), it remains to show that

$$f_{\mathbf{a}} \prec \frac{(g^2W^d)^{-\frac{6}{5}}}{|a_1-a_2|^{d-2}+1}. \tag{5.26}$$

To simplify notation, we abbreviate $\mathcal{B} := (\mathcal{L} - \mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}$. By the assumption (2.79), we have

$$\mathcal{B}_{b_1 b_2} \prec \frac{(g^2 W^d)^{-\frac{6}{5}}}{|b_1 - b_2|^{d-2} + 1}, \quad \text{and} \quad \mathcal{B}_{b_1 b_2} \prec W^{-D} \quad \text{whenever} \quad |b_1 - b_2| \geq (\log W)^3 \ell_s, \tag{5.27}$$

for any large constant $D > 0$. To show (5.26), we decompose $f_{\mathbf{a}}$ into the following two parts:

$$f_{\mathbf{a}}^{\text{near}} := (1-s)^2 \sum_{b_1 : |b_1 - a_1| \wedge |b_1 - a_2| \leq (\log W)^4 \ell_s} \sum_{b_2} \Theta^{(+,-)}_{t,a_1 b_1} \mathcal{B}_{b_1 b_2} \left(\Theta^{(+,-)}_{t,b_2 a_2} - \Theta^{(+,-)}_{t,b_1 a_2} \right),$$

$$f_{\mathbf{a}}^{\text{far}} := (1-s)^2 \sum_{b_1 : |b_1 - a_1| \wedge |b_1 - a_2| > (\log W)^4 \ell_s} \sum_{b_2} \Theta^{(+,-)}_{t,a_1 b_1} \mathcal{B}_{b_1 b_2} \left(\Theta^{(+,-)}_{t,b_2 a_2} - \Theta^{(+,-)}_{t,b_1 a_2} \right).$$

For the term $f_{\mathbf{a}}^{\text{near}}$, using (5.27) and (2.66), we get that for any constant $D > 0$,

$$\begin{aligned} f_{\mathbf{a}}^{\text{near}} &\prec (1-s)^2 \sum_{b_1}^{*} \sum_{b_2 : |b_2 - b_1| \leq (\log W)^3 \ell_s} \frac{g^{-2}}{|a_1 - b_1|^{d-2} + 1} \frac{(g^2 W^d)^{-\frac{6}{5}}}{|b_1 - b_2|^{d-2} + 1} \left(\frac{g^{-2}}{|b_2 - a_2|^{d-2} + 1} + \frac{g^{-2}}{|b_1 - a_2|^{d-2} + 1} \right) + W^{-D} \\ &\prec (1-s)^2 \sum_{b_1}^{*} \frac{(g^2 W^d)^{-\frac{6}{5}}}{|a_1 - b_1|^{d-2} + 1} \cdot \frac{g^{-4} \ell_s^2}{|b_1 - a_2|^{d-2} + 1} \prec (g^2 W^d)^{-\frac{6}{5}} \cdot \frac{(1-s)^2 g^{-4} \ell_s^4}{|a_1 - a_2|^{d-2} + 1} \lesssim \frac{(g^2 W^d)^{-\frac{6}{5}}}{|a_1 - a_2|^{d-2} + 1}, \end{aligned}$$

where $\sum_{b_1}^{*}$ refers to the summation over $\{b_1 : |b_1 - a_1| \wedge |b_1 - a_2| \leq (\log W)^4 \ell_s\}$. To control the term $f_{\mathbf{a}}^{\text{far}}$, we employ a CLT-type cancellation mechanism, extending the approach developed for 2D random band matrices in [28].

Lemma 5.1. *In the setting of Theorem 2.24, fix any $g^2/L^2 \leq 1 - t \leq 1 - s \leq g^2$ so that (2.83) holds. Then, $f_{\mathbf{a}}^{\text{far}}$ satisfies the following estimate under the conditions (5.21) and (5.22):*

$$f_{\mathbf{a}}^{\text{far}} \prec \frac{(g^2 W^d)^{-\frac{6}{5}}}{|a_1 - a_2|^{d-2} + 1}. \tag{5.28}$$

With this lemma, we conclude the estimate (5.26), which implies (5.20), and hence completes the proof of (5.16) for the case $g^2/L^2 \leq 1 - t \leq 1 - s \leq g^2$ together with (5.19). □

Proof of Lemma 5.1. We decompose $f_{\mathbf{a}}^{\text{far}}$ as its expectation $\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ plus the fluctuation part $\mathbb{I}\mathbb{E} f_{\mathbf{a}}^{\text{far}}$, where $\mathbb{I}\mathbb{E}$ denotes $\mathbb{I}\mathbb{E} := 1 - \mathbb{E}$, and $\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ and $\mathbb{I}\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ represent the expressions obtained by replacing $\mathcal{B}$ in $f_{\mathbf{a}}^{\text{far}}$ with $\mathbb{E}\mathcal{B}$ and $\mathbb{I}\mathbb{E}\mathcal{B}$, respectively. Due to the translation invariance and symmetry of $\mathbb{E}\mathcal{B}$, the expectation part $\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ involves a second-order difference of the $\Theta^{(+,-)}$-propagator, which satisfies the improved bound (2.69) that is summable in a. For the fluctuation part $\mathbb{I}\mathbb{E} f_{\mathbf{a}}^{\text{far}}$, the $\Theta^{(+,-)}$-propagator effectively transfers the estimate of $\mathbb{I}\mathbb{E}\mathcal{B}$ from shorter scales of order ℓ_s to the larger scale ℓ_t. Moreover, intuitively, the random variables $\mathbb{I}\mathbb{E}\mathcal{B}_{b_1 b_2}$ and $\mathbb{I}\mathbb{E}\mathcal{B}_{b_1' b_2'}$ become asymptotically independent when $|b_1 - b_1'|$ exceeds the typical decay scale ℓ_s of $\mathcal{B}$. Thus, the term $\mathbb{I}\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ can be viewed as a superposition of roughly ℓ_t^d / ℓ_s^d asymptotically independent, centered random variables, yielding an additional improvement through a CLT-type argument.

We now proceed to the formal proof. By the translation invariance and symmetry of our model on the block level, $\mathbb{E}\mathcal{B}$ is also translationally invariant and symmetric in the following sense:

$$\mathbb{E}\mathcal{B}(a + c, b + c) = \mathbb{E}\mathcal{B}(a, b), \quad \mathbb{E}\mathcal{B}(a, b) = \mathbb{E}\mathcal{B}(b, a), \quad \forall a, b, c \in \mathbb{Z}_L^d.$$

These imply that $\mathbb{E}\mathcal{B}(b_1, b_2) = \mathbb{E}\mathcal{B}(b_2, b_1) = \mathbb{E}\mathcal{B}(b_1, 2b_1 - b_2)$, with which we get

$$\sum_{b_2} \mathbb{E}\mathcal{B}_{b_1 b_2} \left(\Theta^{(+,-)}_{t,b_2 a_2} - \Theta^{(+,-)}_{t,b_1 a_2} \right) = \sum_{b_2} \mathbb{E}\mathcal{B}_{b_1, (2b_1 - b_2)} \left(\Theta^{(+,-)}_{t,b_2 a_2} - \Theta^{(+,-)}_{t,b_1 a_2} \right) = \sum_{b_2} \mathbb{E}\mathcal{B}_{b_1 b_2} \left(\Theta^{(+,-)}_{t,(2b_1 - b_2) a_2} - \Theta^{(+,-)}_{t,b_1 a_2} \right).$$

Thus, we can rewrite $\mathbb{E} f_{\mathbf{a}}^{\text{far}}$ as

$$\mathbb{E} f_{\mathbf{a}}^{\text{far}} = (1-s)^2 \sum_{b_1}^{\star} \sum_{b_2} \Theta^{(+,-)}_{t,a_1 b_1} \mathbb{E}\mathcal{B}_{b_1 b_2} \cdot \frac{1}{2} \left(\Theta^{(+,-)}_{t,b_2 a_2} + \Theta^{(+,-)}_{t,(2b_1 - b_2) a_2} - 2\Theta^{(+,-)}_{t,b_1 a_2} \right),$$

where $\sum_{b_1}^{\star}$ refers to the summation over $\{b_1 : |b_1 - a_1| \wedge |b_1 - a_2| > (\log W)^4 \ell_s\}$. Using (2.66) and (2.69), along with the properties of $\mathcal{B}$ in (5.27), we can bound the above expression by

$$\mathbb{E} f_{\mathbf{a}}^{\text{far}} \prec (1-s)^2 \sum_{b_1}^{\star} \frac{g^{-2}}{|a_1 - b_1|^{d-2} + 1} \frac{g^{-2} \ell_s^2}{|a_2 - b_1|^d + 1} \sum_{b_2 : |b_2 - b_1| \leq (\log W)^3 \ell_s} \frac{(g^2 W^d)^{-\frac{6}{5}}}{|b_1 - b_2|^{d-2} + 1} + W^{-D}$$

$$\prec \frac{(g^2W^d)^{-\frac{6}{5}}\cdot(1-s)^2g^{-4}\ell_s^4}{|a_1-a_2|^{d-2}+1} \lesssim \frac{(g^2W^d)^{-\frac{6}{5}}}{|a_1-a_2|^{d-2}+1} \tag{5.29}$$

for arbitrarily large constant $D>0$.

To deal with the term $\mathbb{I}\mathbb{E}f_{\mathbf{a}}^{\mathrm{far}}$, we need to explore the CLT cancellation within it as in [28, Section 7] and [59, Appendix A.10]. By Markov's inequality and using $g^4(1-s)^{-2}\asymp \ell_s^4$, it suffices to prove the following $(2p)$-th moment bound for any fixed $p\in\mathbb{N}$:

$$\mathbb{E}\left|g^4(1-s)^{-2}(g^2W^d)^{\frac{6}{5}}\cdot \mathbb{I}\mathbb{E}f_{\mathbf{a}}^{\mathrm{far}}\right|^{2p} \prec \left(\frac{\ell_s^4}{|a_1-a_2|^{d-2}+1}\right)^{2p}. \tag{5.30}$$

Abbreviating $\Xi_{a_2}(b_1,b_2):=g^2(\Theta^{(+,-)}_{t,b_2a_2}-\Theta^{(+,-)}_{t,b_1a_2})$ and $\mathsf{B}:=(g^2W^d)^{\frac{6}{5}}\mathcal{B}$, we can write the LHS of (5.30) as

$$\sum_{\mathbf{b}^{(1)},\dots,\mathbf{b}^{(2p)}}^{\star}\prod_{k=1}^{2p}\left(g^2\Theta_t^{(+,-)}(a_1,b_1^{(k)})\cdot\Xi_{a_2}(b_1^{(k)},b_2^{(k)})\right)\cdot\mathbb{E}\Big\{\prod_{k=1}^{p}\mathbb{I}\mathbb{E}\mathsf{B}_{b_1^{(k)}b_2^{(k)}}\cdot\prod_{k=p+1}^{2p}\mathbb{I}\mathbb{E}\overline{\mathsf{B}}_{b_1^{(k)}b_2^{(k)}}\Big\}+\mathrm{O}(W^{-D}) \tag{5.31}$$

for any large constant $D>0$, where $\sum^{\star}$ refers to the summation of $\mathbf{b}^{(k)}=(b_1^{(k)},b_2^{(k)})$ over the regions

$$\left\{(b_1^{(k)},b_2^{(k)}):|b_1^{(k)}-a_1|\wedge|b_1^{(k)}-a_2|>(\log W)^4\ell_s,\ |b_1^{(k)}-b_2^{(k)}|\le(\log W)^3\ell_s\right\}. \tag{5.32}$$

First, suppose the following pairing condition holds:

$$\text{for any}\ \ k\in[\![2p]\!],\ \ \text{there exists}\ \ l\neq k\ \ \text{such that}\ \ |b_1^{(k)}-b_1^{(l)}|\le 10(\log W)^3\ell_s. \tag{5.33}$$

We can divide $[\![2p]\!]$ into a disjoint union of r subsets $[\![2p]\!]=\sqcup_{i=1}^r A_i$, constructed such that for every $k\in A_i$, any $l\in[\![2p]\!]$ satisfying $|b_1^{(k)}-b_1^{(l)}|\le 10(\log W)^3\ell_s$ also belongs to A_i. According to the pairing condition, we have $r\le p$. For each $i\in[\![r]\!]$, let k_i denote an arbitrary index in A_i. Subject to the condition (5.33) and the above decomposition $[\![2p]\!]=\sqcup_{i=1}^r A_i$ (the corresponding summation being denoted by $\sum^{\star_1}$), we can bound (5.31) as follows:

$$\begin{aligned}
&\sum_{\mathbf{b}^{(1)},\dots,\mathbf{b}^{(2p)}}^{\star_1}\prod_{i=1}^{r}\left(\prod_{k\in A_i}\left(g^2\Theta_t^{(+,-)}(a_1,b_1^{(k)})\cdot\Xi_{a_2}(b_1^{(k)},b_2^{(k)})\right)\cdot\left|\mathbb{I}\mathbb{E}\mathsf{B}_{b_1^{(k)}b_2^{(k)}}\right|\right) \qquad (5.34)\\
&\prec \sum_{\mathbf{b}^{(1)},\dots,\mathbf{b}^{(2p)}}^{\star_1}\prod_{i=1}^{r}\left(\frac{1}{|a_1-b_1^{(k_i)}|^{d-2}+1}\right)^{|A_i|}\left(\frac{\ell_s}{|a_2-b_1^{(k_i)}|^{d-1}+1}\right)^{|A_i|}\prod_{k\in A_i}\left(\frac{1}{|b_1^{(k)}-b_2^{(k)}|^{d-2}+1}\right)\\
&\prec \sum_{b_1^{(k_1)},\dots,b_1^{(k_r)}}^{\star_2}\prod_{i=1}^{r}\left(\frac{1}{|a_1-b_1^{(k_i)}|^{d-2}+1}\right)^{|A_i|}\left(\frac{\ell_s}{|a_2-b_1^{(k_i)}|^{d-1}+1}\right)^{|A_i|}(\ell_s^{d+2})^{|A_i|-1}\ell_s^2\\
&\lesssim \prod_{i=1}^{r}\left(\frac{1}{|a_1-a_2|^{d-2}+1}\right)^{|A_i|}\frac{\ell_s^d}{(\ell_s^{d-2})^{|A_i|}}\cdot(\ell_s^{d+2})^{|A_i|-1}\ell_s^2=\left(\frac{\ell_s^4}{|a_1-a_2|^{d-2}+1}\right)^{2p}.
\end{aligned}$$

Here, in the first step, we use the estimates (2.66), (2.68), and (5.27), along with the conditions in (5.32) and (5.33). In the second step, we take the summation over $\mathbf{b}^{(k)}$ for $k\in A_i\setminus\{k_i\}$ and over $b_2^{(k_i)}$ under the restrictions given by (5.32) and (5.33), yielding a factor $(\ell_s^{d+2})^{|A_i|-1}\ell_s^2$ for each $i\in[\![r]\!]$, and $\sum^{\star_2}$ refers to the summation under the constraints $|b_1^{(k_i)}-a_1|\wedge|b_1^{(k_i)}-a_2|>(\log W)^4\ell_s$ for $i\in[\![r]\!]$. In the third step, we have used the following bound for any $k\ge 2$:

$$\sum_{b:|b-a_1|\wedge|b-a_2|>(\log W)^4\ell_s}\left(\frac{1}{|a_1-b|^{d-2}+1}\right)^k\left(\frac{\ell_s}{|a_2-b|^{d-1}+1}\right)^k\lesssim\left(\frac{1}{|a_1-a_2|^{d-2}+1}\right)^k\frac{\ell_s^d}{(\ell_s^{d-2})^k}. \tag{5.35}$$

To see why this holds, consider the case $|b-a_1|\ge|b-a_2|$—the case $|b-a_1|\le|b-a_2|$ can be handled similarly. We can bound the corresponding summation by

$$\left(\frac{1}{|a_1-a_2|^{d-2}+1}\right)^k\sum_{b:|b-a_1|\wedge|b-a_2|>(\log W)^4\ell_s}\left(\frac{\ell_s}{|a_2-b|^{d-1}+1}\right)^k\prec\left(\frac{1}{|a_1-a_2|^{d-2}+1}\right)^k\frac{\ell_s^d}{(\ell_s^{d-2})^k},$$

where we use that $(|a_2-b|^{d-1}+1)^{-k}$ is summable when $d\ge 3$ and $k\ge 2$. This gives (5.35).

Now, with (5.34), we obtain (5.30) under the pairing condition (5.33). It remains to control (5.31) when (5.33) does not hold. In fact, we can prove the following result: if there exists an isolated vertex $\mathbf{b}^{(i)}$ such that $\min_{j:j\neq i}|b_1^{(i)}-b_1^{(j)}|\geq 10(\log W)^3\ell_s$, then for any large constant $D>0$,

$$\mathbb{E}\bigg\{\prod_{k=1}^{p}\mathbb{I}\mathbb{E}\mathsf{B}_{b_1^{(k)}b_2^{(k)}}\cdot\prod_{k=p+1}^{2p}\mathbb{I}\mathbb{E}\overline{\mathsf{B}}_{b_1^{(k)}b_2^{(k)}}\bigg\}\leq W^{-D}. \tag{5.36}$$

Since the proof of this bound is exactly the same as that for [28, equation (7.39)] and [59, equation (A.112)] (which does not depend on the dimension d), we omit the details here. This completes the proof of (5.30), and hence concludes Lemma 5.1 together with (5.29). □

5.2. **The case $g^2/L^d\leq 1-t\leq 1-s\leq g^2/L^2$.** The $\sigma_1=\sigma_2$ case is straightforward by analyzing equation (5.1) with the estimates (2.89), (3.32), (3.35), the decomposition (5.6), and the exponential bound (2.67). We therefore omit the details and focus on the case $\sigma_1\neq\sigma_2$ in the following. In this case, we first remove the zero mode from the $(\mathcal{L}-\mathcal{K})^{(2)}$-loop using the zero-mode-removing operator introduced in Definition 4.12. Once the zero mode is removed, the rescaled propagator $(1-s)\mathring{\Theta}_t^{(+,-)}$ becomes summable by (2.70).

Applying Ward's identities (2.56) and (2.57), we can express $(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}$ as:

$$(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}-\left[Q^{(1)}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}\right]_{\mathbf{a}}=\frac{\mathrm{Im}\,(\mathcal{L}-\mathcal{K})^{(1)}_{t,+,a_2}}{N\eta_t}\prec\frac{(g^2W^d)^{-1}}{N\eta_t}, \tag{5.37}$$

where, in the second step, we apply the averaged local law (2.88). Then, we analyze the term $Q^{(1)}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}$ using equation (4.47) with $n=2$. By definition, we have $Q^{(1)}\circ\Theta_t^{(+,-)}=\mathring{\Theta}_t^{(+,-)}$. Using (2.70), we obtain the following counterparts to the equations (2.65) and (5.13):

$$\|\mathring{\Theta}_t^{(+,-)}\|_{\infty\to\infty}=\max_a\sum_b\left|\mathring{\Theta}^{(+,-)}_{t,ab}\right|\prec g^{-2}L^2, \tag{5.38}$$

$$(1-u)\sum_b\mathring{\Theta}^{(+,-)}_{t,a_1b}\cdot\widetilde{\mathcal{T}}^L_{u,D}\left(|b-a_2|\right)\prec\widetilde{\mathcal{T}}^L_{t,D}\left(|a_1-a_2|\right). \tag{5.39}$$

With these estimates in hand, and employing a similar argument to the one above (i.e., the one leading to (5.15) and (5.19)), we can derive the following estimate for $Q^{(1)}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}$:

$$\left[Q^{(1)}\circ(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}\right]_{\mathbf{a}}\prec(g^2W^d)^{-\frac{1}{5}}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}\left(|a_1-a_2|\right)+(1-s)^2\left(\mathring{\Theta}_t^{(+,-)}\cdot(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\cdot\Theta_t^{(+,-)}\right)_{\mathbf{a}}. \tag{5.40}$$

To control the second term on the RHS, we apply the bounds (2.66), (2.70), and (2.79) to get that

$$\begin{aligned}
&(1-s)^2\left(\mathring{\Theta}_t^{(+,-)}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\Theta_t^{(+,-)}\right)_{\mathbf{a}}\\
&\prec(1-s)^2\frac{(g^2W^d)^{-\frac{1}{5}}}{W^d}\sum_{b_1,b_2}\frac{g^{-2}}{|a_1-b_1|^{d-2}+1}\left(\frac{g^{-2}}{|b_1-b_2|^{d-2}+1}+\frac{1}{L^d(1-s)}\right)\left(\frac{g^{-2}}{|b_2-a_2|^{d-2}+1}+\frac{1}{L^d(1-t)}\right)\\
&\prec(1-s)\frac{(g^2W^d)^{-\frac{1}{5}}}{W^d}\sum_{b_1}\frac{g^{-2}}{|a_1-b_1|^{d-2}+1}\left(\frac{g^{-2}}{|b_1-a_2|^{d-2}+1}+\frac{1}{L^d(1-t)}\right)\\
&\prec\frac{(g^2W^d)^{-\frac{1}{5}}}{W^d}\left(\frac{(1-s)g^{-4}L^2}{|a_1-a_2|^{d-2}+1}+\frac{(1-s)g^{-2}L^2}{L^d(1-t)}\right)\lesssim(g^2W^d)^{-\frac{1}{5}}\cdot W^{-d}\widetilde{\mathcal{T}}^L_{t,D}\left(|a_1-a_2|\right),
\end{aligned}$$

where we use $B_{s,0}\asymp g^{-2}$ in the first step, and $1-s\leq g^2/L^2$ in the second and fourth steps. Plugging it into (5.40) and using (5.37), we complete the proof of (2.92) for the case $g^2/L^d\leq 1-t\leq 1-s\leq g^2/L^2$.

5.3. **The case $1-t\geq g^2$.** In this case, the length scale ℓ_u remains identically equal to 1 throughout the evolution $u\in[s,t]$ (recall (2.64)). With the estimate (2.89) established in Step 2, the proof of (2.93) is relatively straightforward and may be viewed as a special case of the argument in [69, Section 5.3], where decay estimates for 2-G-loops were derived in the context of 1D random band matrices. The key reason that the argument of [69] applies in the regime $1-t\geq g^2$ is that the prefactor $(W^{-d}B_{u,0})^2$ in the target estimate (2.93) is of order $(W^d|1-u|)^{-2}$. Unlike the regime $1-u\ll g^2$, this quantity has no polynomial dependence on $|a_1-a_2|$, precisely matching the setting in dimension 1. In fact, our proof here is much simpler, since we have already established exponential decay in (2.89) and obtained the optimal (maximum) $(\mathcal{L}-\mathcal{K})$-loop estimate (2.91).

Given $u \in [s,t]$ and a sufficiently large constant $D > 0$, we introduce the following tail function to control the tail behavior of the $(\mathcal{L}-\mathcal{K})$-loops:

$$T_{u,D}(r) := (W^d|1-u|)^{-2}\exp(-r^{1/2}) + W^{-D}, \quad \forall r \geq 0. \tag{5.41}$$

Let $\mathcal{J}^*_{u,D} \geq 1$ be a *deterministic control parameter* such that

$$\max_{\mathbf{a}=(a,b)} \max_{\boldsymbol{\sigma}} \left| (\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \right| / T_{u,D}\left(|a-b|\right) \prec \mathcal{J}^*_{u,D}. \tag{5.42}$$

With the tail function (5.41) and the control parameter $\mathcal{J}^*_{u,D}$, we can obtain the following estimates on the second to fourth terms on the RHS of (5.1).

Lemma 5.2. *In the setting of Theorem 2.24, suppose $1-t \geq g^2$ and the estimates (2.87)–(2.91) hold. Then, for any $u \in [s,t]$, $\boldsymbol{\sigma} \in \{+,-\}^2$, $\mathbf{a} = (a_1,a_2) \in (\mathbb{Z}^d_L)^2$, and large enough D (such that $W^D \geq N^{12}$), the following estimates hold:*

$$\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} / T_{u,D}(|a_1-a_2|) \prec (1-u)^{-1} \cdot \left(W^d|1-u|\right)^{-1} \left(\mathcal{J}^*_{u,D}\right)^2, \tag{5.43}$$

$$\mathcal{E}^{\mathring{G},(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} / T_{u,D}(|a_1-a_2|) \prec (1-u)^{-1} \left[\mathbf{1}\left(|a_1-a_2| \leq (\log W)^{\frac{3}{2}}\right) + \left(W^d|1-u|\right)^{-\frac{1}{2}} \left(\mathcal{J}^*_{u,D}\right)^{\frac{3}{2}}\right]. \tag{5.44}$$

Furthermore, suppose $\mathbf{a}' = (a_1',a_2') \in (\mathbb{Z}^d_L)^2$ satisfies that $\max_{i=1}^2 |a_i - a_i'| \leq (\log W)^{3/2}$. Then, we have that

$$(\mathcal{E}\otimes\mathcal{E})^{M,(2)}_{u,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}'} / T^2_{u,D}(|a_1-a_2|) \prec (1-u)^{-1} \left[\mathbf{1}\left(|a_1-a_2| \leq 4(\log W)^{\frac{3}{2}}\right) + \left(W^d|1-u|\right)^{-\frac{1}{2}} \left(\mathcal{J}^*_{u,D}\right)^3\right]. \tag{5.45}$$

Proof. The proof of this lemma is a special case of the argument for [69, Lemma 5.7], as explained above. In fact, many of the arguments can be simplified using the estimates (2.87)–(2.91) established in the previous steps. We omit the details. □

Another key ingredient in the proof is the following evolution kernel estimate for $\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}}$.

Lemma 5.3. *Suppose $1-s \geq 1-t \geq g^2$ and $\mathcal{A}$ satisfies that*

$$|\mathcal{A}_{\mathbf{a}}| \leq T_{s,D}(|a_1-a_2|), \quad \forall \mathbf{a} = (a_1,a_2) \in (\mathbb{Z}^d_L)^2,$$

for some constant $D > 0$. Then, we have

$$\left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{A}\right)_{\mathbf{a}} \lesssim T_{t,D}(|a_1-a_2|) + (|1-s|/|1-t|)^2 \cdot W^{-D}. \tag{5.46}$$

Proof. The estimate (5.46) can be proved easily with (4.51), the estimate (2.66) on Θ-propagators, and the following basic calculus fact: $\max_{a\in\mathbb{R}^d} \int_{x\in\mathbb{R}^d} \exp\left(-\sqrt{|a-x|}-\sqrt{|x|}+\sqrt{|a|}\right)\mathrm{d}x \lesssim 1$. We omit the details. □

Now, for an arbitrarily small constant $\varepsilon > 0$, we define the stopping time

$$T := \inf\left\{u \geq s : \max_{\mathbf{a}=(a,b)} \max_{\boldsymbol{\sigma}} \left| (\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \right| / T_{u,D}\left(|a-b|\right) \geq W^{\varepsilon}\right\}. \tag{5.47}$$

By the definition (5.47), for every $u \leq T$ we have $\max_{\mathbf{a},\boldsymbol{\sigma}} |(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}| \leq W^{\varepsilon}\, T_{u,D}(|a-b|)$, so all the estimates below may be applied to the stopped process $u \mapsto (\mathcal{L}-\mathcal{K})^{(2)}_{u\wedge T}$ and to the stopped hierarchy (3.13) in place of (5.1). Using Lemmas 5.2 and 5.3, we can establish the following lemma by analyzing the equation (3.13).

Lemma 5.4. *In the setting of Theorem 2.24, suppose $1-t \geq g^2$ and the estimates (2.87)–(2.91) hold. Then, for sufficiently small constant $\varepsilon > 0$, we have that $T \geq t$ with high probability.*

Proof. The proof of this lemma is analogous to, and in fact much simpler than, the proof of equation (2.76) in [69, Section 5.3], by using the estimates (5.43)–(5.45) and the evolution kernel estimate (5.46). Hence, we omit the details of the proof. □

By Lemma 5.4, we immediately obtain (2.93), since $\varepsilon > 0$ can be taken arbitrarily small. Moreover, this estimate is stronger than (2.92). This completes Step 5 in the case $1-s \geq 1-t \geq g^2$.

6. Step 6: Expected 2-loop estimates

Our proof of Step 6 relies on analyzing the expectation of the loop hierarchy in (5.1):

$$\mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} = \left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}_{u,\boldsymbol{\sigma}} + \mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathbb{E}\mathring{\mathcal{E}}^{G}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u, \tag{6.1}$$

where we omit the superscript "(2)" from some of the notation for brevity. In addition, we use the following expected averaged local law, which improves the (sharp) averaged local law (2.88) from Step 2 by an extra factor of $W^{-d}B_{u,0}$ arising from the expectation.

Lemma 6.1. *In the setting of Theorem 2.24, suppose the estimates (2.88) and (2.91) hold uniformly in $u \in [s,t]$. Then, we have*

$$\max_a |\mathbb{E}\,\mathrm{Tr}\,((G_u - M)E_a)| \prec (W^{-d}B_{u,0})^2, \quad \forall\, s \le u \le t. \tag{6.2}$$

Proof. The proof of this lemma is the same as that of [69, Lemma 5.15] by using (2.88) and (2.91). □

For the first term on the RHS of (6.1), $\mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma},\mathbf{a}}$ satisfies (2.82). For the $\mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}$ term, combining the pointwise bound (2.92) from Step 5 with the maximum bound (2.91) from Step 4 yields, for $1-u \ge g^2/L^d$ and any $\mathbf{a}=(a_1,a_2)\in(\mathbb{Z}^d_L)^2$,

$$\mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec (W^{-d}B_{u,0})^{\frac{11}{5}} \sum_b B_{u,|a_1-b|} e^{-(|a_1-b|/\ell_u)^{1/2}} \lesssim (1-u)^{-1}(W^{-d}B_{u,0})^{\frac{11}{5}}, \tag{6.3}$$

while for $1-u \le g^2/L^d$, the maximum bound (2.91) gives

$$\mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K})}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec W^d \cdot L^d (N|1-u|)^{-4} = (1-u)^{-1}(N|1-u|)^{-3}. \tag{6.4}$$

Finally, consider the light-weight term in (6.1). By the definition (2.49), we may write

$$\mathring{\mathcal{E}}^{G}_{u,\boldsymbol{\sigma},\mathbf{a}} = W^d \sum_{k=1}^{2} \sum_{\alpha,\beta\in\mathbb{Z}^d_L} \mathrm{Tr}\left(\mathring{G}_u(\sigma_k)E_\alpha\right) S^{(\mathsf{B})}_{\alpha\beta}\left(\mathcal{G}^{(\beta)}_k \circ \mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right). \tag{6.5}$$

When $1-u \le g^2/L^d$, its expectation can be bounded as

$$\begin{aligned}\mathbb{E}\mathring{\mathcal{E}}^{G}_{u,\boldsymbol{\sigma},\mathbf{a}} &= W^d \sum_{k=1}^{2}\sum_{\alpha,\beta}\left[\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}_u(\sigma_k)E_\alpha\right) S^{(\mathsf{B})}_{\alpha\beta}\left(\mathcal{G}^{(\beta)}_k \circ \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right) + \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}_u(\sigma_k)E_\alpha\right) S^{(\mathsf{B})}_{\alpha\beta}\left(\mathcal{G}^{(\beta)}_k \circ (\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right)\right] \\ &\prec W^d \cdot L^d (N|1-u|)^{-4} = (1-u)^{-1}(N|1-u|)^{-3},\end{aligned} \tag{6.6}$$

where the second step uses (6.2) and (2.58) (with $n=3$) to bound the first term, and (2.88) together with (2.91) (with $n=3$) to bound the second. On the other hand, when $1-u \ge g^2/L^d$, we control $\mathbb{E}\mathring{\mathcal{E}}^{G}_{u,\boldsymbol{\sigma},\mathbf{a}}$ as in the following lemma, which may be viewed as an analogue of Lemma 3.10 in expectation. Its proof uses the same diagrammatic techniques as in the proof of Lemma 3.10, albeit in a considerably simpler form, and is deferred to Appendix B.1.

Lemma 6.2 (Expected light-weight estimate)**.** *In the setting of Theorem 2.24, suppose $1-t \ge g^2/L^d$ and the estimates (2.87)–(2.92) hold. Then, for any $\boldsymbol{\sigma}\in\{+,-\}^2$ and $\mathbf{a}\in(\mathbb{Z}^d_L)^2$, we have*

$$\mathbb{E}\mathring{\mathcal{E}}^{G,(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \prec (1-t)^{-1}(W^{-d}B_{t,0})^{\frac{5}{2}}. \tag{6.7}$$

With the above estimates in place, we can now complete the proof of (2.94) using the tools developed in Step 3 (cf. Section 4). Here, we can obtain an additional small factor $(g^2W^d)^{-1/5} + W^{-d}B_{u,0}$ compared to (2.91), because the leading error in the integrated loop hierarchy (5.1) arises from the martingale term, which vanishes after taking the expectation.

Step 6: Proof of (2.94). In the case $1-s \ge 1-t \ge g^2$, we have $B_{u,0} \asymp |1-u|^{-1}$, while in the case $1-t \le 1-s \le g^2/L^d$, we have $B_{u,0} \asymp (L^d|1-u|)^{-1}$. In these regimes, substituting (2.82) and (6.3)–(6.7) into (6.1), applying the evolution kernel estimate (4.51) with $n=2$, and integrating over u, we obtain

$$\mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} \prec (W^{-d}B_{t,0})^2\left((g^2W^d)^{-1/5} + W^{-d}B_{t,0}\right). \tag{6.8}$$

It remains to deal with the regimes (i) $g^2/L^2 \le 1-t \le 1-s \le g^2$, and (ii) $g^2/L^d \le 1-t \le 1-s \le g^2/L^2$. In these regimes, when $\sigma_1 = \sigma_2$, substituting (2.82), (6.3), and (6.7) into (6.1), applying the evolution kernel estimate (4.54) in regime (i) and the estimate (4.57) with $A = \emptyset$ (note that $I_{\mathrm{diff}}(\boldsymbol{\sigma}) = \emptyset$) in regime (ii), and integrating over u, we obtain (6.8). (In regime (ii), (4.54) is not available since Lemma 4.16 requires $t \le 1 - g^2/L^2$, and the factor $\int_s^t (1-u)^{-1}\mathrm{d}u \le (d-2)\log L$ is absorbed by $\prec$.) The remaining and more delicate case is when $\sigma_1 \neq \sigma_2$. To handle this, we again employ the sum-zero operator from Definition 4.7 in regime (i), and the zero-mode-removing operator from Definition 4.12 in regime (ii).

Regime (i): Using Ward's identities (2.56) and (2.57), along with (6.2) and (4.27), we obtain

$$\left[\mathcal{P} \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma}}\right]_{a_1} \boldsymbol{\chi}^{(2)}_{t,\mathbf{a}} = \frac{\operatorname{Im}\mathbb{E}\operatorname{Tr}\left((G_t - M)E_{a_1}\right)}{W^d \eta_t} \boldsymbol{\chi}^{(2)}_{t,\mathbf{a}} \prec (W^{-d}B_{t,0})^3. \tag{6.9}$$

To estimate $\mathcal{Q}_t \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}}$, we take the expectation of equation (4.35) with $n=2$, yielding

$$\begin{aligned}
\mathcal{Q}_t \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} &= \left(\mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_s \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_u \circ \mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \\
&+ \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathcal{Q}_u \circ \mathbb{E}\mathring{\mathcal{E}}^{G,(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u + \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \left(\left[\mathcal{Q}_u, \Theta^{(2)}_{u,\boldsymbol{\sigma}}\right] \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right)\right)_{\mathbf{a}} \mathrm{d}u \\
&- \int_s^t \left(\mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \left\{\left[\mathcal{P} \circ \mathbb{E}\,(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right] \partial_u \boldsymbol{\chi}^{(2)}_u\right\}\right)_{\mathbf{a}} \mathrm{d}u.
\end{aligned} \tag{6.10}$$

We now estimate the RHS of (6.10). For the first three terms, we apply Claim 4.9, together with the estimates (2.82), (6.3), and (6.7), and use the improved evolution kernel estimate (4.56), which exploits the sum-zero and fast-decay properties. Integrating over u, we can bound them by $(W^{-d}B_{t,0})^2[(g^2W^d)^{-1/5} + W^{-d}B_{t,0}]$.

It remains to control the last two terms in (6.10). Arguing as in (6.9), and again using Ward's identity, (6.2), and (4.27), we find

$$\left\| \left[\mathcal{P} \circ \mathbb{E}\,(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right] \partial_u \boldsymbol{\chi}^{(2)}_u \right\|_\infty \prec (1-u)^{-1}(W^{-d}B_{u,0})^3. \tag{6.11}$$

For the commutator term, using the same argument as in (4.61) (with $n=2$), we obtain

$$\left[\mathcal{Q}_u, \Theta^{(2)}_{u,\boldsymbol{\sigma}}\right] \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}} \prec (1-u)^{-1} \left\|\boldsymbol{\chi}^{(2)}_u\right\|_\infty \cdot \left\|\mathcal{P} \circ \mathbb{E}\,(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma}}\right\|_\infty \prec (1-u)^{-1}(W^{-d}B_{u,0})^3, \tag{6.12}$$

where the second step again follows from Ward's identity and the estimates (6.2) and (4.27). Applying Claim 4.9, along with the bounds (6.11) and (6.12), and using the evolution kernel estimate (4.56) (the fast-decay property (4.52) of these two terms follows exactly as for $\mathcal{B}_5$ and $\mathcal{B}_4$ in the proof of Lemma 4.10, with $\mathcal{L}-\mathcal{K}$ replaced by $\mathbb{E}(\mathcal{L}-\mathcal{K})$), we conclude that the contribution of the last two terms in (6.10) is bounded by $(W^{-d}B_{t,0})^3$. This completes the proof of (2.94) in regime (i).

Regime (ii): We use the operator $Q^{(\{1,2\})} = Q^{(1)} \circ Q^{(2)}$ in Definition 4.12 and Ward's identity to express

$$(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} = Q^{(\{1,2\})} \circ (\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} + \frac{\operatorname{Im}\left[\operatorname{Tr}\left((G_t - M)(E_{a_1}+E_{a_2})\right) - N^{-1}\operatorname{Tr}\left(G_t - M\right)\right]}{N\eta_t} \tag{6.13}$$

for $\sigma_1 \neq \sigma_2$. Taking expectations and using (6.2), the expectation of the second term is bounded by $(W^{-d}B_{t,0})^2 \cdot (N\eta_t)^{-1} \le (W^{-d}B_{t,0})^3$. To estimate the first term, we take the expectation of equation (4.47) with $n=2$ and $A = \{1,2\}$, yielding

$$\begin{aligned}
Q^{(\{1,2\})} \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{t,\boldsymbol{\sigma},\mathbf{a}} &= \left(Q^{(\{1,2\})} \circ \mathcal{U}^{(2)}_{s,t,\boldsymbol{\sigma}} \circ \mathbb{E}(\mathcal{L}-\mathcal{K})^{(2)}_{s,\boldsymbol{\sigma}}\right)_{\mathbf{a}} + \int_s^t \left(Q^{(\{1,2\})} \circ \mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathbb{E}\mathcal{E}^{(\mathcal{L}-\mathcal{K})\times(\mathcal{L}-\mathcal{K}),(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u \\
&+ \int_s^t \left(Q^{(\{1,2\})} \circ \mathcal{U}^{(2)}_{u,t,\boldsymbol{\sigma}} \circ \mathbb{E}\mathring{\mathcal{E}}^{G,(2)}_{u,\boldsymbol{\sigma}}\right)_{\mathbf{a}} \mathrm{d}u.
\end{aligned} \tag{6.14}$$

Applying (4.43), together with the estimates (2.82), (6.3), and (6.7), and using the evolution kernel estimate (4.57), we integrate over u to obtain the desired bound: $(W^{-d}B_{t,0})^2[(g^2W^d)^{-1/5} + W^{-d}B_{t,0}]$. This completes the proof of (2.94) in regime (ii). □

7. Estimation of the light-weight term

This section is devoted to proving the key estimates for the light-weight term $\mathcal{E}^{\mathring{G},(2)}$—namely, Lemmas 3.10 and 3.11 in Step 2. The proofs build on several crucial ideas and refined graphical tools developed in earlier works [65, 66, 67]. We present the full details in the setting of the random band matrix model, while the modifications required for the block Anderson model are described separately in Appendix B.3 below. Recall the light-weight term takes the form (6.5). For $\boldsymbol{\sigma} = (\sigma', \sigma)$ and $\mathbf{a} = (a, b)$, we have

$$\begin{aligned}\mathcal{E}^{\mathring{G}}_{t,\boldsymbol{\sigma},\mathbf{a}} &= W^d \sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2} \operatorname{Tr}\left(\mathring{G}_t(\sigma)E_{a_1}\right) \operatorname{Tr}\left(G_t(\sigma)E_{a_2}G_t(\sigma)E_bG_t(\sigma')E_a\right) + ((\sigma,a)\leftrightarrow(\sigma',b)) \qquad (7.1)\\ &= W^{-2d} \sum_{x\in[a],y\in[b]} (G_t(\sigma'))_{yx} \sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2} \sum_{\alpha\in[a_2]} \operatorname{Tr}\left(\mathring{G}_t(\sigma)E_{a_1}\right)(G_t(\sigma))_{x\alpha}(G_t(\sigma))_{\alpha y} + ((\sigma,a)\leftrightarrow(\sigma',b)),\end{aligned}$$

where $((\sigma,a)\leftrightarrow(\sigma',b))$ denotes the term obtained by exchanging (σ,a) and (σ',b) in the preceding term. In the setting of Lemma 3.10, using (3.2) and (3.3), we can bound $(G_t(\sigma'))_{yx}$ as

$$|(G_t(\sigma'))_{yx}| \prec \delta_{xy} + \Psi_t(|a-b|). \qquad (7.2)$$

Note that when $1-t \le g^2/L^2$, we have $\ell_t = L$, so the exponential factor in the definition of $\mathcal{T}_t$ in (3.19) is always of order 1 for $r \lesssim L$. In this case, Lemma 3.11 is an immediate consequence of Lemma 3.10. Hence, for the proof of Lemma 3.11, we only need to focus on the $1-t > g^2/L^2$ case, where

$$W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|) \asymp [\mathsf{T}_t(|a-b|\wedge\ell)]^2 + W^{-d-D}.$$

Here, for simplicity of notation, we introduce the function $\mathsf{T}_t : [0,\infty) \to [0,\infty)$ defined as:

$$\mathsf{T}_t(r) := \frac{(g^2+|1-t|)^{-1/2}}{W^{d/2}(r+1)^{(d-2)/2}} \exp\left(-\frac{1}{2}\sqrt{r/\ell_t}\right), \quad \forall r \ge 0.$$

Then, in the setting of Lemma 3.11 and for $1-t > g^2/L^2$, we have the following estimate by (3.2) and (3.3):

$$|(G_t(\sigma'))_{yx}| \prec \delta_{xy} + \mathsf{T}_t(|a-b|\wedge\ell) + W^{-D}. \qquad (7.3)$$

With (7.2) and (7.3) in place, the proofs of Lemmas 3.10 and 3.11 reduce to estimating

$$f_{xy}(G) = W^{-d} \sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2} \sum_{\alpha\in[a_2],\beta\in[a_1]} \mathbf{1}_{\alpha\notin\{x,y\}} (\mathring{G}_t)_{\beta\beta}(G_t)_{x\alpha}(G_t)_{\alpha y} \equiv \sum_{\alpha\notin\{x,y\}} \sum_{\beta} S_{\alpha\beta}\mathring{G}_{\beta\beta}G_{x\alpha}G_{\alpha y}, \qquad (7.4)$$

where, without loss of generality, we take $\sigma = +$ and abbreviate $G \equiv G_t$. We denote by $\widetilde{f}_{xy}(G)$ the corresponding contribution with $\alpha \in \{x,y\}$. Using (7.2), (7.3), and the averaged local law (3.5), we get

$$\widetilde{f}_{xy}(G) \prec \Psi_t^2(0)\cdot(\delta_{xy} + \Psi_t(|a-b|)) \qquad (7.5)$$

in the setting of Lemma 3.10, and

$$\widetilde{f}_{xy}(G) \prec (W^{-d}B_{t,0})\cdot\left(\delta_{xy} + \mathsf{T}_t(|a-b|\wedge\ell) + W^{-D}\right) \qquad (7.6)$$

in the setting of Lemma 3.11. Combining (7.5) with (7.2), and averaging over $x\in[a]$ and $y\in[b]$, yields

$$W^{-2d} \sum_{x\in[a],y\in[b]} (G_t(\sigma'))_{yx}\widetilde{f}_{xy}(G) \prec \Psi_t^2(0)\left[W^{-d}\delta_{ab} + \Psi_t^2(|a-b|)\right] \lesssim \Psi_t^2(0)\cdot\Psi_t^2(|a-b|), \qquad (7.7)$$

which is bounded by the RHS of (3.29). Similarly, combining (7.6) with (7.3) gives

$$W^{-2d} \sum_{x\in[a],y\in[b]} (G_t(\sigma'))_{yx}\widetilde{f}_{xy}(G) \prec (W^{-d}B_{t,0})\cdot W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|), \qquad (7.8)$$

which is bounded by the RHS of (3.32). Therefore, it remains to estimate the main term $f_{xy}(G)$.

We first claim the following max-bound on $f_{xy}(G)$:

$$|f_{xy}(G)| \prec \Psi_t^2 \sum_{\alpha} |G_{x\alpha}||G_{\alpha y}| \le \Psi_t^2 \left(\operatorname{Im} G_{xx}/\eta_t\right)^{1/2}\left(\operatorname{Im} G_{yy}/\eta_t\right)^{1/2} \prec \eta_t^{-1}\Psi_t^2, \qquad (7.9)$$

where the first step uses the averaged local law (3.5), and the second applies Cauchy–Schwarz together with Ward's identity. In the setting of Lemma 3.10, we take $\Psi_t = \Psi_t(0)$, whereas in the setting of Lemma 3.11, we take $\Psi_t = \max\{W^{-d/2}, (W^{-d}B_{t,0})^{1/2}\}$. Since $B_{t,0} \ge (g^2+1)^{-1} \ge (\mathfrak{d}^{-2}+1)^{-1}$ by (2.62) and (2.10), this is comparable to $(W^{-d}B_{t,0})^{1/2}$; taking the maximum only ensures the lower bound $\Psi_t \ge W^{-d/2}$,

which is required in Lemma 3.1 and used below (7.19). For simplicity of notation, we continue to write $\Psi_t = (W^{-d}B_{t,0})^{1/2}$ in this setting. Taking Lemma 3.10 as an example, our goal is to establish the sharper bound $|f_{xy}(G)| \prec \eta_t^{-1}\Psi_t(0)\Psi_t(|a-b|)$. Thus, the estimate (7.9) lacks the long-edge factor $\Psi_t(|a-b|)$, which captures the spatial decay in $|a-b|$. This missing factor is a key technical obstacle—previously noted around (1.12) in the introduction—because all tools developed within the stochastic flow and loop hierarchy frameworks fail to extract it. We have to exploit a *global fluctuation averaging* mechanism in the sum over α in (7.4), which can yield additional powers of $L^{-d/2}$ or $\ell_t^{-d/2}$ to compensate for the missing long edge. To realize this mechanism, we estimate the high moments of $f_{xy}(G)$, adapting ideas from [65, 66, 67]. By Markov's inequality, Lemmas 3.10 and 3.11 follow from the next estimates on the high moments of $f_{xy}(G)$.

Lemma 7.1. *In the setting of Lemma 3.10, for any fixed $p \in 2\mathbb{N}$, there exists a constant $c > 0$ depending on p such that the following estimate holds:*

$$\mathbb{E}\big|f_{xy}(G)\big|^p \prec \eta_t^{-p}[\Psi_t(0)]^p \cdot [\Psi_t(c|a-b|)]^p. \tag{7.10}$$

Lemma 7.2. *In the setting of Lemma 3.11, assume that $1-t > g^2/L^2$. Then, for each fixed $p \in 2\mathbb{N}$, the following estimate holds for any large constant $D > 0$:*

$$\mathbb{E}\big|f_{xy}(G)\big|^p \prec \eta_t^{-p}(W^{-d}B_{t,0})^{p/2} \cdot [\mathsf{T}_t(|a-b| \wedge \ell)]^p + W^{-D}. \tag{7.11}$$

Proof of Lemmas 3.10 and 3.11. Applying Markov's inequality to (7.10) (with arbitrarily large p) yields

$$f_{xy}(G) \prec \eta_t^{-1}\Psi_t(0) \cdot \Psi_t(c|a-b|) \prec \eta_t^{-1}\Psi_t(0) \cdot \Psi_t(|a-b|),$$

where the second step uses the condition (3.28). Combining this bound with (7.2) and (7.7), we obtain (3.29). For the proof of Lemma 3.11, as discussed below (7.2), it suffices to assume that $1-t > g^2/L^2$. In this case, applying Markov's inequality to (7.11) yields, for any large constant $D > 0$,

$$f_{xy}(G) \prec \eta_t^{-1}(W^{-d}B_{t,0})^{1/2} \cdot [\mathsf{T}_t(|a-b| \wedge \ell)] + W^{-D}.$$

Together with (7.3) and (7.8), it gives (3.32). □

The remainder of this section is devoted to proving Lemmas 7.1 and 7.2. We begin with the proof of Lemma 7.1; the proof of Lemma 7.2 will follow the same general strategy, with several additional technical ingredients. Note that when $|a-b| \le (\log W)^{3/2}$, both (7.10) and (7.11) follow directly from (7.9). Thus, in the following proof, we restrict attention to the case

$$|a-b| > (\log W)^{3/2}. \tag{7.12}$$

Following the approach of [65, 66, 67], we represent resolvent expressions (such as $\big|f_{xy}(G)\big|^p$) as graphs, expand them using Gaussian integration by parts, and then bound the resulting diagrams according to their graphical structures. In the next subsection, we introduce the basic graphical notations, including the concepts of vertices, edges, molecules, graph values, and scaling sizes, and define the basic graph expansions that will be used in the proof.

7.1. Graphical tools and local expansions.

Our graphs are composed of matrix indices as vertices, and various types of edges representing different matrix entries. Specifically, the graphs will encode entries from the following matrices: I, S, G, $\mathring{G} = G - M$, and $S^{\pm}$, where $S^{\pm}$ are defined by

$$S^{+}_{xy}(z) := W^{-d}\big(S^{(\mathsf{B})}\Theta^{(+,+)}(z)\big)_{ab} \quad \text{for} \quad x \in [a], y \in [b], \quad \text{and} \quad S^{-}(z) := (S^{+}(z))^*, \tag{7.13}$$

with the notations $\Theta^{(+,+)}(z)$ and $S^{(\mathsf{B})}$ specified earlier in (2.21) and (2.5). The graphical notation is defined in a general setting for an arbitrary random band matrix H and its associated variance matrix S. As a special case, these definitions apply to our matrix flow H_t, where the variance matrix is given by $S_t = tS$, and the associated resolvent is denoted by $G_t \equiv (H_t - z_t)^{-1}$.

Definition 7.3 (Graphs)**.** *Given a graph with vertices and edges, we assign the following structures and call it a vertex-level graph.*

- **Vertices:** *Vertices represent matrix indices in our expressions. Every graph has some external or internal vertices: external vertices represent indices whose values are fixed, while internal vertices represent summation indices that will be summed over.*
- **Solid edges and weights:** *We will use (x, y) to denote a solid edge from vertex x to vertex y. Every solid edge represents a resolvent entry. More precisely:*

  - *A blue (resp. red) oriented solid edge from x to y represents a G_{xy} (resp. $\overline{G}_{xy}$) factor.*
  - *A blue (resp. red) oriented solid edge with a circle ($\circ$) from x to y represents a $(G-M)_{xy}$ (resp. $\overline{(G-M)}_{xy}$) factor.*
  - *A G_{xx} (resp. $\overline{G}_{xx}$) factor will be represented by a blue (resp. red) self-loop on the vertex x, while a $(G-M)_{xx}$ (resp. $\overline{(G-M)}_{xx}$) factor will be represented by a blue (resp. red) self-loop on the vertex x with a circle ($\circ$). Following the convention in [65], we will also call G_{xx} and $\overline{G}_{xx}$ as blue and red* **weights**, *and call $(G-M)_{xx}$ and $\overline{(G-M)}_{xx}$ as blue and red* **light-weights**.

  We assign a $+$ charge to each blue solid edge and a $-$ charge to each red solid edge.
- **Waved edges:**
  - *A black waved edge between x and y represents an S_{xy} factor.*
  - *A blue (resp. red) waved edge between x and y represents an S^{+}_{xy} (resp. S^{-}_{xy}) factor.*
- **Dotted edges:** *A black dotted edge between x and y represents a factor $\mathbf{1}_{x=y}$ and a $\times$-dotted edge represents a factor $\mathbf{1}_{x\neq y}$. There is at most one dotted or $\times$-dotted edge between every pair of vertices.*
- **Coefficient:** *There is a coefficient associated with every graph. Every coefficient is of order* $\mathrm{O}(1)$ *and is a polynomial in m, $\overline{m}$, m^{-1}, $\overline{m}^{-1}$, $(1-m^2)^{-1}$, and $(1-\overline{m}^2)^{-1}$.*

Edges between internal vertices are called *internal edges*, while edges with at least one end at an external vertex are called *external edges*. The orientations of non-solid edges do not matter because they all represent entries of symmetric matrices. The dotted edges are introduced solely for organizing the proof in certain steps; otherwise, we will almost always identify vertices connected by dotted edges. To each graph, we assign a *value* as follows. For simplicity, throughout this paper, we will always abuse notation by identifying a graph (a geometric object) with its value (an analytic expression).

Definition 7.4 (Values of graphs)**.** *Given a graph $\mathcal{G}$, we define its value as follows. We first take the product of all edge factors together with the coefficient associated with the graph $\mathcal{G}$. We then sum over all internal indices corresponding to internal vertices, while keeping the external indices fixed at their prescribed values. For a linear combination of graphs $\sum_i c_i\mathcal{G}_i$, we define its value in the natural way, as the linear combination of the values of the individual graphs $\mathcal{G}_i$.*

As noted in the previous works [65, 66, 64, 67], our graphs have a two-level structure: some *local structures* varying on scales of order W, called **molecules**, which are the equivalence classes of vertices connected through dotted or waved edges, along with a *global structure of blocks* varying on scales up to L. Given a graph defined in Definition 7.3, if we ignore the inner structure within each molecule, we will get a **molecular graph** with vertices being molecules. The molecular graph will be very useful in organizing the graphs and understanding their global structures. We now give its formal definition.

Definition 7.5 (Molecules and molecular graphs)**.** *We partition the set of all vertices into a union of disjoint subsets called molecules. Two vertices belong to the same molecule if and only if they are connected by a path of dotted and waved edges. Every molecule containing at least one external vertex is called an external molecule; otherwise, it is an internal molecule. An edge is said to be inside a molecule if both of its ending vertices belong to this molecule. Given a graph $\mathcal{G}$, we define its molecular graph, denoted by $\mathcal{G}_{\mathcal{M}}$, as follows:*

- *merge all vertices in the same molecule and represent them by a vertex;*
- *keep all solid edges between molecules;*
- *discard all the other components in $\mathcal{G}$ (i.e., $\times$-dotted edges, edges inside molecules, and coefficients).*

By the exponential decay of the waved edges—derived from the definitions of S and the estimate (2.67)—we see that up to an error of order $\exp(-c(\log W)^{3/2})$ for a constant $c>0$:

$$x,\ y \text{ are in the same molecule} \implies |x-y| \le W(\log W)^{3/2}. \tag{7.14}$$

We remark that molecular graphs are used solely to help with the analysis of graph structures, while all graph expansions will be applied exclusively to vertex-level graphs.

Definition 7.6 (Normal graphs)**.** *We say a graph $\mathcal{G}$ is* normal *if it satisfies the following properties:*

(i) *It contains at most* $\mathrm{O}(1)$ *many vertices and edges.*

(ii) *There are no dotted edges between vertices.*

(iii) *Every pair of vertices in the graph are connected by a $\times$-dotted edge* if and only if *they are connected by a solid edge.*

(iv) *Every weight is a light-weight.*

In a normal graph, every G edge is off-diagonal, while all the diagonal G factors are represented by weights. Given any graph with O(1) vertices and edges, we can rewrite it as a linear combination of normal graphs via the following *dotted edge partition* operation.

Definition 7.7 (Dotted edge partition)**.** *Given a graph $\mathcal{G}$, if there is at least one G-edge but no $\times$-dotted edge between a pair of vertices α and β, we write $1 = \mathbf{1}_{\alpha=\beta} + \mathbf{1}_{\alpha\neq\beta}$. If there is a $\times$-dotted edge $\mathbf{1}_{\alpha\neq\beta}$ but no G-edge between α and β, we write $\mathbf{1}_{\alpha\neq\beta} = 1 - \mathbf{1}_{\alpha=\beta}$. Expanding the product of all such identities, we can express $\mathcal{G}$ as*

$$\mathcal{G} := \sum \mathbf{Dot} \cdot \mathcal{G}, \tag{7.15}$$

where each **Dot** *is a product of dotted and $\times$-dotted edges together with a sign $\pm$. If* **Dot** *is "inconsistent"—that is, if two vertices are connected both by a $\times$-dotted edge and by a path of dotted edges—then $\mathbf{Dot} \cdot \mathcal{G} = 0$. For each consistent* **Dot***, we merge vertices connected by dotted edges in $\mathbf{Dot} \cdot \mathcal{G}$. In particular, if there is a $\times$-dotted edge between α and β, then all G edges between them are off-diagonal; otherwise, the G-edges between them become weights once α and β are merged. Finally, in each resulting graph, every diagonal factor G_{xx} (resp. $\overline{G}_{xx}$) is decomposed into a light-weight $(G_{xx} - m)$ plus a coefficient m (resp. $(\overline{G}_{xx} - \overline{m})$ plus $\overline{m}$). Taking the product over all such decompositions yields a linear combination of normal graphs.*

Given a normal graph, we can define its scaling size as in Definition 7.8.

Definition 7.8 (Scaling size)**.** *We define the scaling size of a normal graph Γ as follows:*

$$\mathsf{size}(\Gamma) := (L^d)^{n_M(\Gamma)} \cdot (\Psi_t)^{n_S(\Gamma)} \cdot W^{-d(n_W(\Gamma) - n_V(\Gamma))}. \tag{7.16}$$

where $n_S(\Gamma)$, $n_W(\Gamma)$, $n_V(\Gamma)$, and $n_M(\Gamma)$ denote the numbers of solid edges (including light-weights), waved edges, internal vertices, and internal molecules, respectively. If a graph Γ can be written as a sum of $\mathrm{O}(1)$ normal graphs Γ_k, i.e., $\Gamma = \sum_k \Gamma_k$, then we define

$$\mathsf{size}(\Gamma) := \max_k \mathsf{size}(\Gamma_k). \tag{7.17}$$

As a convention, for any (possibly non-normal) graph Γ with $\mathrm{O}(1)$ vertices and edges, we define $\mathsf{size}(\Gamma)$ by first expanding Γ into a sum of normal graphs via the dotted edge partition, and then applying (7.17).

First, by (3.2), each off-diagonal solid edge or light-weight contributes a factor of $\mathrm{O}_\prec(\Psi_t)$ under the assumption (3.4), which accounts for the $(\Psi_t)^{n_S(\Gamma)}$ factor in (7.16). Next, every molecule contains a "free vertex" that can range over the entire lattice $\mathbb{Z}^d_{WL}$, giving rise to an $N^{n_M(\Gamma)}$ factor. The remaining vertices in the molecule are confined (up to a sufficiently small error, by (7.14)) to a $\mathrm{O}(W(\log W)^{3/2})$-neighborhood of the free vertex, yielding the factor $W^{d(n_V(\Gamma)-n_M(\Gamma))}$. Finally, each waved edge contributes a W^{-d} factor, by the definition of S and the bound for $S^{\pm}$: there exist constants $c, C > 0$ such that

$$S^{\pm}_{xy} \leq CW^{-d} e^{-c|x-y|/W}, \tag{7.18}$$

which follows from the estimate (2.67). With these considerations, we obtain the following claim.

Claim 7.9. *We have the bound $\Gamma \prec \mathsf{size}(\Gamma)$ for any normal graph Γ in the setting of Lemma 3.10 (with $\Psi_t = \Psi_t(0)$) or Lemma 3.11 (with $\Psi_t = (W^{-d} B_{t,0})^{1/2}$).*

In the proofs, we will only need to keep track of the number of factors of Ψ_t and $W^{-d/2}$ appearing in the scaling size. To this end, we introduce a more convenient notion of *scaling order*.

Definition 7.10 (Scaling order)**.** *We define the* scaling order *of a normal graph Γ as*

$$\mathsf{ord}(\Gamma) := n_S(\Gamma) + 2\left(n_W(\Gamma) - n_V(\Gamma)\right). \tag{7.19}$$

Directly from (7.16) and (7.19) we have the identity $\mathsf{size}(\Gamma) = (L^d)^{n_M(\Gamma)} (\Psi_t)^{\mathsf{ord}(\Gamma)} \cdot \left(W^d \Psi_t^2\right)^{n_V(\Gamma) - n_W(\Gamma)}$, so that, since $\Psi_t \geq W^{-d/2}$, $\mathsf{size}(\Gamma) \leq (L^d)^{n_M(\Gamma)} (\Psi_t)^{\mathsf{ord}(\Gamma)}$ whenever $n_W(\Gamma) \geq n_V(\Gamma)$. For a general graph Γ that can be written as a sum of $\mathrm{O}(1)$ many normal graphs Γ_k, we define its scaling order as $\mathsf{ord}(\Gamma) = \min_k \mathsf{ord}(\Gamma_k)$.

We next state the graph expansion rules given in [65]. These expansions are stated for $G(z) = (H - z)^{-1}$, but they also apply to $G_t(z)$ with G and S replaced by G_t and $S_t = tS$, respectively.[9]

[9]Here $h_{\alpha x}$ denotes the (α, x)-th entry of H (resp. H_t), and, as in [65], $\partial_{h_{\alpha x}}$ is the partial derivative with respect to $h_{\alpha x}$, where $h_{\alpha x}$ and $h_{x\alpha} = \overline{h_{\alpha x}}$ are regarded as independent variables; in particular, $\partial_{h_{\alpha x}} G_{ij} = -G_{i\alpha} G_{xj}$.

Lemma 7.11 (Weight expansion, Lemma 3.5 of [65]). *Suppose f is a differentiable function of G. Then,*

$$\begin{aligned}\mathring{G}_{xx}f(G) =_{\mathbb{E}}\ & m\sum_{\alpha} S_{x\alpha}\mathring{G}_{xx}\mathring{G}_{\alpha\alpha}f(G) + m^3\sum_{\alpha,\beta} S^{+}_{x\alpha}S_{\alpha\beta}\mathring{G}_{\alpha\alpha}\mathring{G}_{\beta\beta}f(G)\\ & -m\sum_{\alpha} S_{x\alpha}G_{\alpha x}\partial_{h_{\alpha x}}f(G) - m^3\sum_{\alpha,\beta} S^{+}_{x\alpha}S_{\alpha\beta}G_{\beta\alpha}\partial_{h_{\beta\alpha}}f(G)\,,\end{aligned} \tag{7.20}$$

where "$=_{\mathbb{E}}$" means "equal in expectation".

Lemma 7.12 (Edge expansion, Lemma 3.10 of [65]). *Suppose f is a differentiable function of G. Consider*

$$\mathcal{G} := \prod_{i=1}^{k_1} G_{xy_i}\cdot\prod_{i=1}^{k_2}\overline{G}_{xy'_i}\cdot\prod_{i=1}^{k_3} G_{w_i x}\cdot\prod_{i=1}^{k_4}\overline{G}_{w'_i x}\cdot f(G). \tag{7.21}$$

If $k_1 \geq 1$, then we have that

$$\begin{aligned}\mathcal{G} =_{\mathbb{E}}\ & m\delta_{xy_1}\mathcal{G}/G_{xy_1} + m\sum_{\alpha} S_{x\alpha}\mathring{G}_{\alpha\alpha}\mathcal{G} \\ & + \sum_{i=1}^{k_2}|m|^2\left(\sum_{\alpha} S_{x\alpha}G_{\alpha y_1}\overline{G}_{\alpha y'_i}\right)\frac{\mathcal{G}}{G_{xy_1}\overline{G}_{xy'_i}} + \sum_{i=1}^{k_3} m^2\left(\sum_{\alpha} S_{x\alpha}G_{\alpha y_1}G_{w_i\alpha}\right)\frac{\mathcal{G}}{G_{xy_1}G_{w_i x}} \\ & + \sum_{i=1}^{k_2} m\mathring{G}^{-}_{xx}\left(\sum_{\alpha} S_{x\alpha}G_{\alpha y_1}\overline{G}_{\alpha y'_i}\right)\frac{\mathcal{G}}{G_{xy_1}\overline{G}_{xy'_i}} + \sum_{i=1}^{k_3} m\mathring{G}_{xx}\left(\sum_{\alpha} S_{x\alpha}G_{\alpha y_1}G_{w_i\alpha}\right)\frac{\mathcal{G}}{G_{xy_1}G_{w_i x}} \\ & + (k_1-1)m\sum_{\alpha} S_{x\alpha}G_{x\alpha}G_{\alpha y_1}\frac{\mathcal{G}}{G_{xy_1}} + k_4 m\sum_{\alpha} S_{x\alpha}\overline{G}_{\alpha x}G_{\alpha y_1}\frac{\mathcal{G}}{G_{xy_1}} - m\sum_{\alpha} S_{x\alpha}\frac{\mathcal{G}}{G_{xy_1}f(G)}G_{\alpha y_1}\partial_{h_{\alpha x}}f(G)\,.\end{aligned} \tag{7.22}$$

Here, the fractions are used to simplify the expression. For example, the fraction $\mathcal{G}/(G_{xy_1}\overline{G}_{xy'_i})$ is the graph obtained by removing the factor $G_{xy_1}\overline{G}_{xy'_i}$ from the product in (7.21). We refer to the above expansion as the edge expansion with respect to G_{xy_1}. *The edge expansion with respect to other G_{xy_i} can be defined in the same way. The edge expansions with respect to $\overline{G}_{xy'_i}$, $G_{w_i x}$, and $\overline{G}_{w'_i x}$ can be defined similarly by taking complex conjugates or matrix transpositions of (7.22).*

Lemma 7.13 (*GG* expansion, Lemma 3.14 of [65]). *Consider a graph $\mathcal{G} = G_{xy}G_{y'x}f(G)$, where f is a differentiable function of G. We have that*

$$\begin{aligned}\mathcal{G} =_{\mathbb{E}}\ & m\delta_{xy}G_{y'x}f(G) + m^3 S^{+}_{xy}G_{y'y}f(G) + m\sum_{\alpha} S_{x\alpha}\mathring{G}_{\alpha\alpha}\mathcal{G} + m^3\sum_{\alpha,\beta} S^{+}_{x\alpha}S_{\alpha\beta}\mathring{G}_{\beta\beta}G_{\alpha y}G_{y'\alpha}f(G) \\ & + m\mathring{G}_{xx}\sum_{\alpha} S_{x\alpha}G_{\alpha y}G_{y'\alpha}f(G) + m^3\sum_{\alpha,\beta} S^{+}_{x\alpha}S_{\alpha\beta}\mathring{G}_{\alpha\alpha}G_{\beta y}G_{y'\beta}f(G) \\ & - m\sum_{\alpha} S_{x\alpha}G_{\alpha y}G_{y'x}\partial_{h_{\alpha x}}f(G) - m^3\sum_{\alpha,\beta} S^{+}_{x\alpha}S_{\alpha\beta}G_{\beta y}G_{y'\alpha}\partial_{h_{\beta\alpha}}f(G).\end{aligned} \tag{7.23}$$

The $\overline{GG}$ expansion with respect to $\overline{G}_{xy}\overline{G}_{y'x}$ can be derived by taking the complex conjugate.

Corresponding to the three lemmas above, we can define graph operations that represent the weight, edge, and *GG* expansions. Following [65], we refer to all these operations as *local expansions at a vertex x*, meaning that they do not create new molecules: every new vertex introduced by such an expansion is connected to x by a path of dotted or waved edges. As discussed in [65, Section 3], these expansions decrease the scaling size (equivalently, increase the scaling order) of a graph. Moreover, relative to the original graph $\mathcal{G}$ prior to expansion, each new graph produced by an expansion either becomes "smaller" by a factor of Ψ_t in scaling size, or moves "closer" to being locally standard (as defined in Definition 7.14), in one of the following ways:

- It contains one fewer weight than $\mathcal{G}$.
- The solid-edge degree of a vertex is reduced by 2, a new vertex of degree 2 is introduced, and the degrees of all other vertices remain unchanged. Here, the degree refers only to the number of solid edges incident to the vertex.
- A pair of edges of the form $G_{xy}G_{yx}$ (resp. $\overline{G}_{xy}\overline{G}_{yx}$) is replaced by $m^4 S^{+}_{xy}$ (resp. $\overline{m}^4 S^{-}_{xy}$).

Consequently, by repeatedly applying the local expansions from Lemmas 7.11–7.13, any normal graph can be expanded into a sum of *locally standard graphs*, defined as follows.

Definition 7.14 (Locally standard graphs)**.** *A graph is* locally standard *if:*

(i) *It is a normal graph.*
(ii) *It has no self-loops (i.e., weights or light-weights) on vertices.*
(iii) *Each internal vertex is either standard neutral, or it is not incident to any solid (G or $\mathring{G}$) edge.*

Here, a vertex is called standard neutral *if it satisfies the following two properties:*

- *It is connected to exactly two solid edges carrying opposite charges.*
- *It has a neutral charge, where the* charge *of a vertex is defined by counting the incoming and outgoing blue solid edges (with $+$ charge) and red solid edges (with $-$ charge):*

$$\#\{\text{incoming } + \text{ or outgoing } - \text{ solid edges}\} - \#\{\text{outgoing } + \text{ or incoming } - \text{ solid edges}\}. \tag{7.24}$$

By definition, the two solid edges attached to a standard neutral vertex x take the form $G_{xy}\overline{G}_{xy'}$ or $G_{yx}\overline{G}_{y'x}$.

Simply speaking, *locally standard graphs* are those in which each G edge is paired with a unique $\overline{G}$ edge at internal vertices. As discussed in [65, Section 3.4], by repeatedly applying local expansions, any normal graph can be expanded into a sum of locally standard graphs, up to a sufficiently small error, as summarized in the following lemma. For the reader's convenience (and for later use in the proof of Lemma 7.17), we will recall the local expansion strategy from [65, Section 3] in Appendix B.2.

Lemma 7.15 (Lemma 3.22 of [65])**.** *Let Γ be an arbitrary graph with $\mathrm{O}(1)$ vertices and edges. For any large constant $D > 0$, we can expand Γ into a sum of $\mathrm{O}(1)$ many locally standard graphs Γ_μ:*

$$\Gamma =_{\mathbb{E}} \sum_{\mu} \Gamma_\mu + \mathcal{E}rr, \tag{7.25}$$

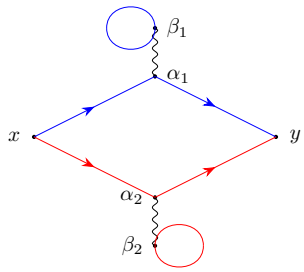
where every Γ_μ has scaling order $\geq \mathsf{ord}(\Gamma)$, and $\mathcal{E}rr$ is a sum of graphs of scaling size $\mathrm{O}(W^{-D})$.

7.2. **Examples and nested property.** To illustrate the ideas underlying the proof of Lemma 7.1, we examine the simplest case with $p = 2$.

Example 7.16. Consider the second moment of $f_{xy}(G)$:

$$\mathbb{E}|f_{xy}(G)|^2 = \mathbb{E} \sum_{\alpha_1,\alpha_2 \notin \{x,y\}} \sum_{\beta_1,\beta_2} S_{\alpha_1\beta_1} S_{\alpha_2\beta_2} \left(\mathring{G}_{\beta_1\beta_1} G_{x\alpha_1} G_{\alpha_1 y} \right) \cdot \left(\mathring{G}^*_{\beta_2\beta_2} \overline{G}_{x\alpha_2} \overline{G}_{\alpha_2 y} \right), \tag{7.26}$$

which can be represented by the following graph:



We expand (7.26) using the local expansions from Section 7.1. As shown in Lemma 7.15, these expansions decrease the scaling size of the graphs, while improving their structure for our purposes. More precisely, to obtain locally standard graphs, during the expansion process, a $-$ charged edge from the path $(x \to \alpha_2 \to y)$ or from the light-weight at β_2 must be "pulled" to the molecule containing α_1,[10] whereas a $+$ charged edge from the path $(x \to \alpha_1 \to y)$ or from the light-weight at β_1 must be pulled to the molecule containing α_2. As an illustration, consider the left panel of Figure 1, which shows a graph $\mathcal{G}$ obtained by applying two weight expansions from Lemma 7.11 ($\mathcal{G}$ is not yet locally standard, but it is already sufficient for our proof). The two short edges $G_{\beta_1\gamma_1}$ and $G_{\beta_2\gamma_2}$ contribute a $[\Psi_t(0)]^2$ factor. For the rest of the graph, we examine its molecular graph $\mathcal{G}_1$ shown in the middle panel of Figure 1, where $\mathcal{M}_1$ and $\mathcal{M}_2$ denote the molecules

[10] Pulling an edge $e = (\alpha, \beta)$ to a molecule $\mathcal{M}$ means replacing it by two edges, one connecting α to $\mathcal{M}$ and one connecting β to $\mathcal{M}$.

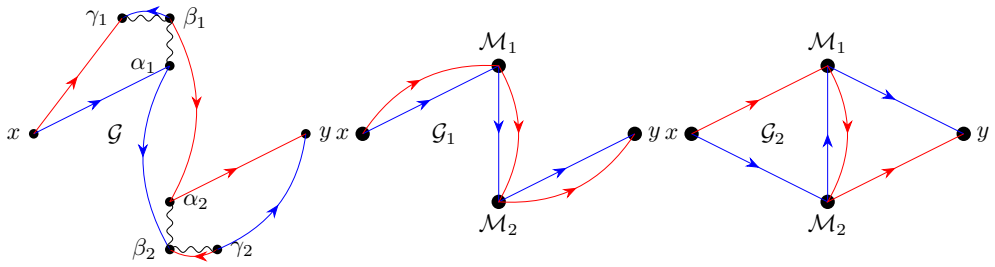


FIGURE 1. A possible expansion of (7.26) (left), its molecular graph (middle), and another possible molecular graph (right). The $\times$-dotted edges in the first graph have been omitted.

containing α_1 and α_2, respectively. Without loss of generality, suppose the two edges $(x, \mathcal{M}_1)$ provide the factor $[\Psi_t(c|a-b|)]^2$. For the remaining graph, we first sum over $\mathcal{M}_1$ and then over $\mathcal{M}_2$, applying the Cauchy–Schwarz inequality and Ward's identity to obtain a factor η_t^{-2}. Altogether, this gives the desired bound in (7.10) for $p = 2$.

In the right panel of Figure 1, we illustrate another possible molecular graph generated from the expansion of (7.26). For this graph, there are choices of long edges such that, after removing them, each summation order in the resulting graphs yields a poor bound. Nevertheless, by applying an additional AM–GM inequality, the graph still leads to the correct estimate. To demonstrate this, suppose $(y, \mathcal{M}_1)$ and $(y, \mathcal{M}_2)$ are taken as long edges. Then, in the resulting graph $\mathcal{G}_2'$ (with these edges removed), both $\mathcal{M}_1$ and $\mathcal{M}_2$ have degree 3. Summing over either one of them eliminates three incident edges and leaves a graph with only a single solid edge. The subsequent summation produces a poor bound of order $N^{1/2}\eta_t^{-1/2}$ via Cauchy-Schwarz and Ward's identity. To handle this case, we apply the AM–GM inequality $2|z_1 z_2| \le |z_1|^2 + |z_2|^2$ to the two edges $(x, \mathcal{M}_1)$ and $(x, \mathcal{M}_2)$, which yields two new graphs, one with a pair of edges $(x, \mathcal{M}_1)$ and one with a pair of edges $(x, \mathcal{M}_2)$. In each of these graphs, the summations over $\mathcal{M}_1$ and $\mathcal{M}_2$ can be carried out in a proper order, yielding the η_t^{-2} factor by applying Ward's identity twice.

To prove Lemma 7.1, we need to extend the analysis for the $p = 2$ example to arbitrarily large $p \in 2\mathbb{N}$. In the original graph $|f_{xy}(G)|^p$, there are p edge-disjoint paths from x to y, comprising a total of $2p$ solid edges. Heuristically, to obtain locally standard graphs with local expansions, each path from x to y must be pulled at least once, thereby increasing its length by 1. Now, the key graphical property required for our proof consists of the following two components: (i) We can identify at least p "long edges"—that is, edges of length $\gtrsim |x-y|$—in the graph. (ii) In the remaining graph, after removing these long edges, there exists a specific order of summation over the internal vertices such that, at each step, a sufficient number of edges (at least two per summation) remain to apply the Cauchy-Schwarz inequality and Ward's identity to control the summation. If there is no requirement of property (i), this problem was addressed in [67] using a structure called "*nested property*" of the graph. In Lemma 7.23 below, we will extend this concept to inductively select both long solid edges and a summation order to guarantee the bound (7.10).

7.3. **Proof of Lemma 7.1.** For the proof of Lemma 7.1, we begin by expanding $\big|f_{xy}(G)\big|^p = [f_{xy}(G)]^{p/2}[\overline{f_{xy}(G)}]^{p/2}$ into a sum of locally standard graphs, as in Lemma 7.15. By carefully tracking the local expansion process, we can show that each locally standard graph satisfies the graphical properties in the following lemma.

Lemma 7.17. *Given any large constant $D > 0$, we can expand $\big|f_{xy}(G)\big|^p$ into a linear combination of* $\mathrm{O}(1)$ *many* locally standard graphs $\Gamma_{\mu,xy}$ *in the sense of equal expectation:*

$$\big|f_{xy}(G)\big|^p =_{\mathbb{E}} \sum_{\mu} \Gamma_{\mu,xy} + \mathcal{E}rr, \tag{7.27}$$

where $\mathcal{E}rr$ denotes a sum of $\mathrm{O}(1)$ *many graphs of scaling size* $\mathrm{O}(W^{-D})$*. Moreover, each graph $\Gamma_{\mu,xy}$ on the RHS satisfies the following properties under the assumption* (7.12) *(recall that, up to an exponentially small error* $\exp(-c(\log W)^{3/2})$*, x and y do not belong to the same molecule by* (7.14)*):*

(1) *$\Gamma_{\mu,xy}$ is a locally standard graph with external vertices x and y, with the corresponding external molecules denoted by $\mathcal{M}_x$ and $\mathcal{M}_y$, respectively.*

(2) *$\Gamma_{\mu,xy}$ contains $0 \le q \le p$ many internal molecules, denoted by $\mathcal{M}_i$ with $i \in [\![q]\!]$. Moreover, within each (internal or external) molecule $\mathcal{M}_i$, $i \in \{1,\dots,q,x,y\}$, the number of waved edges (denoted by $n_W(\mathcal{M}_i)$) and the number of vertices (denoted by $n_V(\mathcal{M}_i)$) satisfy the relation*

$$n_V(\mathcal{M}_i) \le n_W(\mathcal{M}_i) + 1. \tag{7.28}$$

(3) *There are at least p many edge-disjoint paths $\mathfrak{P}_i$, $i \in [\![p]\!]$, in the molecular graph that connect the molecules $\mathcal{M}_x$ and $\mathcal{M}_y$ via solid edges (and possibly passing through the internal molecules).*

(4) *For each internal molecule $\mathcal{M}_i$, $i \in [\![q]\!]$, there are at least two paths $\mathfrak{P}_k \ne \mathfrak{P}_l$ passing through it on the molecular graph. As a consequence, each $\mathcal{M}_i$ has degree at least 4, i.e.,*

$$\deg(\mathcal{M}_i) \ge 4, \quad i \in [\![q]\!]. \tag{7.29}$$

(5) *For each $A \subset [\![q]\!]$, the number of paths passing through $\{\mathcal{M}_i : i \in A\}$ is at least $|A|$, that is,*

$$|\{j : \mathrm{V}(\mathfrak{P}_j) \cap \{\mathcal{M}_i : i \in A\} \ne \emptyset\}| \ge |A|,$$

where $\mathrm{V}(\mathfrak{P}_j)$ denotes the subset of vertices (i.e., molecules) in the path $\mathfrak{P}_j$ on the molecular graph.

(6) *The scaling order of $\Gamma_{\mu,xy}$ satisfies that*

$$\mathsf{ord}(\Gamma_{\mu,xy}) \ge 2p. \tag{7.30}$$

The proof of this lemma is straightforward by applying the local expansions in Lemmas 7.11 to 7.13, and we defer the details to Appendix B.2. The path properties (3)–(5) in Lemma 7.17 serve as the key structural ingredients for controlling the graphs $\Gamma_{\mu,xy}$ in (7.27). To see them, note that in the initial graph, there are p edge-disjoint paths from x to y, corresponding to the p factors $f_{xy}(G)$ or $\overline{f_{xy}(G)}$. Each path passes through a distinct internal molecule, and each internal molecule is attached to two solid edges of the same color. From the expansion rules in Lemmas 7.11 to 7.13, one can verify that these paths are preserved throughout the expansion process, which ensures properties (3) and (5). Furthermore, to obtain locally standard graphs via local expansions, each internal molecule must either (i) pull in an edge of a different color from another path or molecule, (ii) merge with another internal molecule connected by differently colored edges, or (iii) merge with an external molecule. In case (iii), the internal molecule disappears from $\Gamma_{\mu,xy}$, whereas in cases (i) and (ii), at least two paths will pass through it.

To streamline the estimation process, we now simplify the structure of the graphs $\Gamma_{\mu,xy}$ through its underlying molecular graph. Specifically, we bound them by corresponding "*auxiliary graphs*", which are obtained as quotient graphs by collapsing each molecule into a single vertex. More precisely, for each internal molecule $\mathcal{M}_i$, $i \in [\![q]\!]$, we select a representative vertex α_i within the molecule, referred to as the "center" of the molecule. Similarly, for the two external molecules, we set x and y as their respective centers. The auxiliary graph associated with $\Gamma_{\mu,xy}$ is then defined on the set of block-level vertices $[x]$, $[y]$, and $[\alpha_i]$ (for $i \in [\![q]\!]$):

Definition 7.18 (Block-level vertices)**.** *Recall that $\mathbb{Z}^d_{WL}$ is partitioned into L^d blocks as defined in (2.2). For any $x \in \mathbb{Z}^d_{WL}$, we use $[x]$ to denote the block that contains x. With slight abuse of notation, we also view these blocks as vertices in the lattice $\mathbb{Z}^d_L$. In other words, $[x]$ may be interpreted both as a vertex in $\mathbb{Z}^d_L$ and as a subset of vertices in $\mathbb{Z}^d_{WL}$.*

For clarity, throughout the following proof we use the notation $[\cdot]$ to indicate vertices in the auxiliary graph and the lattice $\mathbb{Z}^d_L$, distinguishing them from vertices in the original graphs defined in Definition 7.3 and from those in the lattice $\mathbb{Z}^d_{WL}$. To define the auxiliary graph formally, we still need to define its edges. For simplicity of notation, we will use "$\alpha \sim_{\mathcal{M}} \beta$" to mean that "vertices α and β belong to the same molecule". For any $\beta_i \sim_{\mathcal{M}} \alpha_i$, it suffices to assume that

$$|\alpha_i - \beta_i| \le W(\log W)^{1+\varepsilon_1} \tag{7.31}$$

for a small enough constant $\varepsilon_1 \in (0, 1/10)$; otherwise the graph is smaller than $\exp(-\Omega((\log W)^{1+\varepsilon_1})) \le W^{-D}$ for any constant $D > 0$. Combining Lemma 3.1 with (7.31), we can bound the solid edges between different molecules as follows for any large constant $D > 0$:

$$|(G - M)_{xy}| \prec \Psi_t(0)\delta_{xy} + \xi([x],[y]) + W^{-D}, \tag{7.32}$$

where the variables $\xi(\cdot,\cdot)$ are defined as

$$[\xi([a_1],[a_2])]^2 := \sum_{\mathbf{b}:\|\mathbf{b}-\mathbf{a}\|_\infty \le (\log W)^{1+2\varepsilon_1}} \max_{\boldsymbol{\sigma}\in\{(-,+),(+,-)\}} \mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},\mathbf{b}} + W^{-d}\mathbf{1}(|[a_1]-[a_2]| \le (\log W)^{1+2\varepsilon_1}). \quad (7.33)$$

With Ward's identity and the properties of Ψ_t given by (3.28), we readily see the following properties for the ξ variables.

Claim 7.19. *In the setting of Lemma 3.10, for any $[a_1],[a_2]\in\mathbb{Z}^d_L$, the following estimates hold:*

$$\xi([a_1],[a_2]) \prec \Psi_t(|[a_1]-[a_2]|), \quad \sum_{[a_2]}[\xi([a_1],[a_2])]^2 \prec (W^d\eta_t)^{-1}. \quad (7.34)$$

Now, we define the auxiliary graphs formed with edges representing the ξ variables.

Definition 7.20 (Auxiliary graphs)**.** *Let $\mathcal{G}_{xy}$ be an arbitrary normal graph with q internal molecules $\mathcal{M}_i$, $i\in[\![q]\!]$, and two external molecules $\mathcal{M}_x$ and $\mathcal{M}_y$ containing x and y, respectively. We fix the centers $\alpha_i\in\mathcal{M}_i$ for $i\in[\![q]\!]$. We then define the auxiliary graph $\mathcal{G}^{\mathrm{aux}}_{[x][y]}$ of $\mathcal{G}_{xy}$ as follows:*

- *The vertices are $[\alpha_i]$, $i\in\{1,\dots,q,x,y\}$, where we adopt the convention $\alpha_x\equiv x$ and $\alpha_y\equiv y$.*
- *Corresponding to each solid edge between molecules $\mathcal{M}_i$ and $\mathcal{M}_j$ in $\mathcal{G}_{xy}$, we introduce a (black) solid edge between $[\alpha_i]$ and $[\alpha_j]$ in $\mathcal{G}^{\mathrm{aux}}_{[x][y]}$, representing the factor $\xi([\alpha_i],[\alpha_j])$. (Since ξ is symmetric, these edges are not oriented.)*

In this way, we obtain the auxiliary graph $\mathcal{G}^{\mathrm{aux}}_{[x][y]}$. Its value is defined analogously to Definition 7.4, namely, as the product of all solid edges followed by summation over all internal vertices $[\alpha_i]$, $i\in[\![q]\!]$. Finally, we define the scaling order of $\mathcal{G}^{\mathrm{aux}}_{[x][y]}$ in the same manner as in (7.19):

$$\mathsf{ord}(\mathcal{G}^{\mathrm{aux}}_{[x][y]}) := \#\{\textit{solid edges in } \mathcal{G}^{\mathrm{aux}}_{[x][y]}\} - 2\#\{\textit{internal vertices in } \mathcal{G}^{\mathrm{aux}}_{[x][y]}\}. \quad (7.35)$$

Now, the estimation of $\Gamma_{\mu,xy}$ can be reduced to bounding its auxiliary graph.

Lemma 7.21. *Given a locally standard graph $\Gamma_{\mu,xy}$ with external vertices x and y that satisfies properties (1) and (2) in Lemma 7.17 (the path properties (3)–(6) are not used in this lemma, so it applies to any such graph, not only to those produced by (7.27)), we define its auxiliary graph $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$ (recall that $x\in[a]$ and $y\in[b]$) and the corresponding scaling order as in Definition 7.20. For any large constant $D>0$, we have*

$$\Gamma_{\mu,xy} \prec (\Psi_t)^{\mathsf{ord}(\Gamma_{\mu,xy})-\mathsf{ord}(\Gamma^{\mathrm{aux}}_{\mu,[a][b]})}\cdot W^{qd}\Gamma^{\mathrm{aux}}_{\mu,[a][b]} + W^{-D}. \quad (7.36)$$

Proof. As in Definition 7.20, the vertices of $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$ are denoted by $[\alpha_i]$, $i\in\{1,\dots,q,x,y\}$, corresponding to the molecule centers α_i in $\Gamma_{\mu,xy}$. First, by the definition of S together with the estimate for $S^{\pm}$ in (7.18), and recalling (7.31), we may bound a waved edge between β_i and γ_i inside molecule $\mathcal{M}_i$ by

$$\mathrm{O}_\prec\left(W^{-d}\mathbf{1}_{|\beta_i-\alpha_i|\vee|\gamma_i-\alpha_i|\le W(\log W)^{1+\varepsilon_1}} + e^{-c(\log W)^{1+\varepsilon_1}}\right).$$

Second, by Lemma 3.1, solid edges within molecules can be bounded by Ψ_t, and every solid edge between vertices $\beta_i\in\mathcal{M}_i$ and $\beta_j\in\mathcal{M}_j$ can be bounded by

$$\Big(\sum_{|[a']-[\beta_i]|\le1,|[b']-[\beta_j]|\le1}\max_{\boldsymbol{\sigma}\in\{(+,-),(-,+)\}}\mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},([a'],[b'])} + W^{-d}\mathbf{1}_{|[\beta_i]-[\beta_j]|\le1}\Big)^{1/2} \lesssim \xi([\alpha_i],[\alpha_j]),$$

where we again use (7.31) for $\beta_i\sim_{\mathcal{M}}\alpha_i$ and $\beta_j\sim_{\mathcal{M}}\alpha_j$.

With these bounds, and after summing over all *non-center* vertices inside the molecules $\mathcal{M}_i$, $i\in[\![q]\!]$, and inside the two external molecules $\mathcal{M}_x$ and $\mathcal{M}_y$, we obtain (cf. the notations in Definition 7.8)

$$\Gamma_{\mu,xy} \prec (\Psi_t)^{n_S(\Gamma_{\mu,xy})-n_S(\Gamma^{\mathrm{aux}}_{\mu,[a][b]})}W^{-d(n_W(\Gamma_{\mu,xy})-n_V(\Gamma_{\mu,xy})+q)}\cdot W^{qd}\Gamma^{\mathrm{aux}}_{\mu,[a][b]}.$$

Here, we have also used (7.28) in obtaining the factor $W^{-d(n_W(\Gamma_{\mu,xy})-n_V(\Gamma_{\mu,xy})+q)}$, while the compensating factor W^{qd} reflects the difference between summing the molecule centers α_i (in $\mathbb{Z}^d_{WL}$ for $\Gamma_{\mu,xy}$) and summing the corresponding block representatives $[\alpha_i]$ (in $\mathbb{Z}^d_L$ for $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$). Finally, using $\Psi_t\ge W^{-d/2}$ together with the definitions (7.19) and (7.35), we conclude (7.36). □

To conclude Lemma 7.1, it remains to bound the auxiliary graph $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$.

Lemma 7.22. *In the setting of Lemma 7.21, there exists a constant $c > 0$ (depending on p) such that*

$$\Gamma^{\mathrm{aux}}_{\mu,[a][b]} \prec \left(W^d\eta_t\right)^{-q} \cdot [\Psi_t(c|[a]-[b]|)]^p \cdot (\Psi_t)^{\mathsf{ord}(\Gamma^{\mathrm{aux}}_{\mu,[a][b]})-p}. \tag{7.37}$$

Before turning to the proof of Lemma 7.22, we first use it to complete the proof of Lemma 7.1.

Proof of Lemma 7.1. Using Lemmas 7.21 and 7.22, we bound the locally standard graphs in (7.27) as

$$\Gamma_{\mu,xy} \prec \eta_t^{-q} \cdot (\Psi_t)^{\mathsf{ord}(\Gamma_{\mu,xy})-p} \cdot [\Psi_t(c|[a]-[b]|)]^p + W^{-D} \le \eta_t^{-p} \cdot \Psi_t^p \cdot [\Psi_t(c|[a]-[b]|)]^p + W^{-D},$$

where we use (7.30) in the second step. Finally, we upgrade this stochastic domination bound to a bound in expectation. Since $\|G_t\| \le \eta_t^{-1}$ and $\sum_{\alpha,\beta} S_{\alpha\beta} = N$, the definition (7.4) gives the deterministic bound $|f_{xy}(G)| \lesssim N\eta_t^{-3}$. Hence, for any large constant $D' > 0$, the event on which the above $\prec$-bounds fail has probability at most $N^{-D'}$ and contributes at most $N^{-D'}(CN\eta_t^{-3})^p$ to $\mathbb{E}|f_{xy}(G)|^p$, which is negligible once D' is chosen large enough depending on p. This concludes (7.10) by taking $\Psi_t = \Psi_t(0)$. □

7.4. **Proof of Lemma 7.22.** This subsection is devoted to establishing the key technical result, Lemma 7.22. Its proof relies crucially on the path properties (3)–(5) of $\Gamma_{\mu,xy}$ stated in Lemma 7.17, which also hold for the auxiliary graph $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$. We refer to any graph satisfying these path properties as a *nested graph*. By Definition 7.20 of auxiliary graphs together with the bounds in (7.34), Lemma 7.22 follows directly from the following combinatorial result concerning nested graphs.

Lemma 7.23 (Bounding nested graphs)**.** *Let $\mathcal{G}_{\mathbf{ab}}$ be a graph with $2p$ external vertices, denoted by $\mathbf{a} = (a_1, \dots, a_p) \in (\mathbb{Z}_L^d)^p$ and $\mathbf{b} = (b_1, \dots, b_p) \in (\mathbb{Z}_L^d)^p$, together with $0 \le q \le p$ internal vertices. A solid edge between vertices α and β represents a nonnegative random variable $\xi_{\alpha\beta} = \xi_{\beta\alpha}$ (not necessarily the specific random variables appearing in (7.33)) that satisfies the bounds*

$$\xi_{\alpha\beta} \prec \Psi_t(|\alpha-\beta|), \quad \sum_\beta |\xi_{\alpha\beta}|^2 \prec (W^d\eta_t)^{-1}, \quad \forall \alpha, \beta \in \mathbb{Z}_L^d. \tag{7.38}$$

We assume that the graph contains no self-loops. Suppose $\mathcal{G}_{\mathbf{ab}}$ satisfies the following path properties:

1. *There are p edge-disjoint paths $\mathfrak{P}_i$, $i \in [\![p]\!]$, such that each path $\mathfrak{P}_i$ connects the pair a_i and b_i.*
2. *Each internal vertex α_i, for $i \in [\![q]\!]$, is traversed by at least two distinct paths $\mathfrak{P}_k \ne \mathfrak{P}_l$. As a consequence, each vertex α_i has degree at least 4:*
$$\deg(\alpha_i) \ge 4, \quad \forall i \in [\![q]\!]. \tag{7.39}$$
3. *For any subset $A \subset [\![q]\!]$, the number of paths passing through the internal vertices $\{\alpha_i : i \in A\}$ is at least $|A|$; that is, $|\{j : \mathrm{V}(\mathfrak{P}_j) \cap \{\alpha_i : i \in A\} \ne \emptyset\}| \ge |A|$, where $\mathrm{V}(\mathfrak{P}_j)$ denotes the subset of vertices in the path $\mathfrak{P}_j$.*

Then, there exists a constant $c > 0$ such that

$$\mathcal{G}_{\mathbf{ab}} \prec (W^d\eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-p} \cdot \prod_{i=1}^p \Psi_t(c|a_i-b_i|), \tag{7.40}$$

where $\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})$ is defined as in (7.35), namely $\mathsf{ord}(\mathcal{G}_{\mathbf{ab}}) := \#\{\text{solid edges in } \mathcal{G}_{\mathbf{ab}}\} - 2q$.

Proof of Lemma 7.22. By Definition 7.20 of auxiliary graphs, the bounds in (7.34), and the path properties (3)–(5) from Lemma 7.17, the auxiliary graph $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$ satisfies the assumptions of Lemma 7.23. Consequently, (7.37) becomes a special case of (7.40), with $a_1 = \cdots = a_p = [a]$ and $b_1 = \cdots = b_p = [b]$. □

In the proof of Lemma 7.23, we will select the "long edges" and determine the nested summation order of internal vertices via an inductive approach. Specifically, we assume that Lemma 7.23 holds for any graph with $(q-1)$ internal vertices. Then, given a graph with q internal vertices that satisfies the assumptions of Lemma 7.23, we provide an algorithm to identify the long edges and determine the first vertex in a valid nested order, where the long edges have been removed. We show that summing over this vertex yields the desired factors of $(W^d\eta_t)^{-1}$ and $\Psi_t(c|a_i-b_i|)$, and that the resulting graph still satisfies the assumptions of Lemma 7.23, now with $(q-1)$ internal vertices. (Note that in this reduced graph, the ending external vertices may change. This is precisely why we work with more general graphs allowing arbitrary external vertices, rather than restricting ourselves to the original external vertices $[a]$ and $[b]$.) Therefore, we can apply the inductive hypothesis to complete the argument.

Our induction argument relies critically on the path properties assumed in Lemma 7.23. However, these properties may be violated once some long edges are removed. To clarify the graphical structure and streamline the presentation of our proof, instead of deleting the chosen long edges entirely, we replace them with a new type of edges, called *ghost edges*:

- A **ghost edge** is a black dashed edge between two vertices, representing a factor of 1.

Ghost edges do not contribute to the value of a graph (and hence do not affect its scaling order); rather, they are introduced solely to preserve the connectedness of paths in our graphs. As a consequence, they will be counted when we define the degrees of vertices, i.e.,

$$\deg(\alpha) = \#\{\text{solid and ghost edges connected with } \alpha\}.$$

When we want to refer to the degree of solid edges for a vertex α, we will use the notation $\deg_s(\alpha)$. The main advantage of ghost edges is that they allow us to maintain the essential path properties—namely properties (1)–(3) in Lemma 7.23—throughout the proof. In particular, they simplify the induction: without ghost edges, we would need to distinguish between paths depending on whether long edges had been removed from them. Now, Lemma 7.23 is an easy corollary of the following more general result for graphs with ghost edges.

Lemma 7.24. *Suppose $\mathcal{G}_{\mathbf{ab}}$ is a nested graph that may contain ghost edges. Assume it satisfies the setting of Lemma 7.23, including properties (1)–(3) therein, with the convention that paths may also contain ghost edges. Furthermore, suppose that each path $\mathfrak{P}_i$, $i \in [\![p]\!]$, contains at most one ghost edge, which—if present—must appear as an ending edge of the path. Here, the ending edges of a path $\mathfrak{P}_i$ are its first and last edges, necessarily incident to an external vertex a_i or b_i. Then, there exists a constant $c > 0$ such that*

$$\mathcal{G}_{\mathbf{ab}} \prec (W^d\eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i - b_i|)]^{\mathbf{1}(\mathfrak{P}_i \text{ has no ghost edge})}, \tag{7.41}$$

where n_{ng} denotes the number of paths that have no ghost edge, i.e., $n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}}) := \sum_{i=1}^{p} \mathbf{1}(\mathfrak{P}_i \text{ has no ghost edge})$.

Proof of Lemma 7.24. We prove the estimate (7.41) by induction on q, the number of internal vertices. Suppose we aim to establish (7.41) for a target graph $\mathcal{G}_{\mathbf{ab}}$ satisfying the assumptions of Lemma 7.24. Moreover, assume that $\mathcal{G}_{\mathbf{ab}}$ contains $K \in \mathbb{N}$ many solid edges. Our induction hypothesis is the following: the estimate (7.41) holds for all nested graphs satisfying the assumptions of Lemma 7.24, with k internal vertices and at most K solid edges. (Note that we do not assume that the number of paths is smaller than p.) In the following proof, we use the notation (α, β) to denote either a solid or a ghost edge between vertices α and β. A path from α to β is denoted by $(\alpha \to\to \beta)$, where the double arrow indicates that intermediate vertices may be present along the path. More generally, a path of the form $(\alpha_1 \to\to \alpha_2 \to\to \cdots \to\to \alpha_k)$ represents a sequence of edges from α_1 to α_2, then α_2 to α_3, and so on.

First, the estimate (7.41) is trivial in the case $q = 0$, by (7.38). Now, let $1 \le q \le p$, and assume the induction hypothesis holds for all $0 \le k \le q-1$. We prove that (7.41) holds for every graph satisfying the assumptions of Lemma 7.24, with q internal vertices and at most K solid edges. Given such a graph $\mathcal{G}_{\mathbf{ab}}$, we partition the summation region over $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_q)$ into at most 2^{pq} subregions, according to whether each vertex α_i is closer to a_j or to b_j with $i \in [\![q]\!]$ and $j \in [\![p]\!]$:

$$\mathbf{D}_{\boldsymbol{\pi}} \subset (\mathbb{Z}_L^d)^q := \{\boldsymbol{\alpha} : |\alpha_i - a_j| \le |\alpha_i - b_j| \text{ if } \pi_{i,j} = 0, \text{ and } |\alpha_i - a_j| > |\alpha_i - b_j| \text{ if } \pi_{i,j} = 1\},$$

for $\boldsymbol{\pi} = (\pi_{1,1}, \dots, \pi_{1,p}, \dots, \pi_{q,1}, \dots, \pi_{q,p}) \in \{0,1\}^{pq}$. The union of these subregions covers the entire space $(\mathbb{Z}_L^d)^q$. Then, it suffices to prove that for each fixed $\boldsymbol{\pi} \in \{0,1\}^{pq}$,

$$\sum_{\boldsymbol{\alpha} \in \mathbf{D}_{\boldsymbol{\pi}}} \mathcal{G}_{\mathbf{ab}}(\boldsymbol{\alpha}) \prec (W^d\eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i - b_i|)]^{\chi(\mathfrak{P}_i)}, \tag{7.42}$$

where $\mathcal{G}_{\mathbf{ab}}(\boldsymbol{\alpha})$ denotes the graph obtained by fixing the internal vertices in $\boldsymbol{\alpha}$ as external vertices, and we abbreviate $\chi(\mathfrak{P}_i) \equiv \mathbf{1}(\mathfrak{P}_i \text{ has no ghost edge})$.

For the proof of (7.42), we look at the ending edges of the p paths. Suppose (a_j, α_i) is an ending edge of a path $\mathfrak{P}_j$ (note that $\mathfrak{P}_j$ connects a_j to b_j through α_i, but there may be no direct solid edge between b_j and α_i). On the region $\mathbf{D}_{\boldsymbol{\pi}}$, the ending edge (a_j, α_i) falls into one of the following four categories:

A1: The path $\mathfrak{P}_j = (a_j \to\to b_j)$ has no ghost edge. Moreover, for $\boldsymbol{\alpha} \in \mathbf{D}_{\boldsymbol{\pi}}$, the vertex α_i is closer to a_j than to b_j, i.e., $|\alpha_i - a_j| \le |\alpha_i - b_j|$. In particular, this implies $|\alpha_i - b_j| \ge |a_j - b_j|/2$.

A2: The path $\mathfrak{P}_j = (a_j \to\!\!\to b_j)$ has no ghost edge. Moreover, for $\boldsymbol{\alpha} \in \mathbf{D}_{\boldsymbol{\pi}}$, the vertex α_i is closer to b_j than to a_j, i.e., $|\alpha_i - a_j| > |\alpha_i - b_j|$. In particular, this implies $|\alpha_i - a_j| \geq |a_j - b_j|/2$.
B1: The path $\mathfrak{P}_j = (a_j \to\!\!\to b_j)$ contains a ghost edge, which is not (a_j, α_i).
B2: The path $\mathfrak{P}_j = (a_j \to\!\!\to b_j)$ contains a ghost edge (a_j, α_i).

By symmetry, we may also classify ending edges attached to b_j by switching the roles of a_j and b_j in the definitions above. The only difference is in the classification of type-A1 (resp. type-A2) edges: in this case, we require $|\alpha_i - b_j| < |\alpha_i - a_j|$ (resp. $|\alpha_i - b_j| \geq |\alpha_i - a_j|$) instead.

First, we notice that if an ending edge (a_j, α_i) is a type-A2 edge, then it contributes a factor $\Psi_t(|a_j - b_j|/2)$ by (7.38). In this case, we can bound the original graph $\mathcal{G}_{\mathbf{ab}}$ by the product of a new graph $\widetilde{\mathcal{G}}_{\mathbf{ab}}$, obtained by replacing the edge (a_j, α_i) with a ghost edge, and the factor $\Psi_t(|a_j - b_j|/2)$. It is easy to see that if the bound (7.42) holds for $\widetilde{\mathcal{G}}_{\mathbf{ab}}$, then it also holds for the original graph. We perform this replacement for every A2 edge that appears in the paths $\mathfrak{P}_i$, $i \in [\![p]\!]$. After performing all such replacements, it suffices to prove (7.42) for the resulting graph. With a slight abuse of notation, we continue to denote this modified graph by $\mathcal{G}_{\mathbf{ab}}$, which now satisfies the condition:

$$\text{there is no A2 edge in } \mathcal{G}_{\mathbf{ab}}. \tag{7.43}$$

We now proceed to prove the estimate (7.42), assuming (7.43) and the induction hypothesis. The proof is organized by classifying cases according to the number and type of ending edges attached to internal vertices.

(I) Two A1/B1 edges: the simple case. We first consider the case where an internal vertex is connected to two or more ending edges of type A1 or B1. That is, suppose there exists an internal vertex in the graph connected to at least two such edges—either two A1 edges, two B1 edges, or one of each. Without loss of generality, let this vertex be α_q. To illustrate the idea, we begin with a simple scenario where $\deg(\alpha_q) = 4$, so exactly two paths pass through α_q. We will address the more general case $\deg(\alpha_q) \geq 4$ in Case (II).

By definition, the two ending edges connected to α_q must belong to distinct paths. Without loss of generality, assume these edges are (a_1, α_q) and (a_2, α_q), which belong to paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$, respectively. We now define a new graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}} \equiv \mathcal{G}^{\mathrm{new}}_{\mathbf{a}'(\alpha_q),\mathbf{b}}$ as follows: it has external vertices

$$\mathbf{a}' = (a_1', a_2', a_3, \dots, a_p, a_1) \quad \text{and} \quad \mathbf{b} = (b_1, \dots, b_p, a_2), \quad \text{where} \;\; a_1' = \alpha_q, \;\; a_2' = \alpha_q,$$

and internal vertices $\boldsymbol{\alpha}' = (\alpha_1, \dots, \alpha_{q-1})$, i.e., we remove α_q from the internal vertex set. First, all edges not belonging to paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$ remain unchanged, so the paths $\mathfrak{P}_i$ for $i \in [\![3, p]\!]$ stay the same in $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}$. Second, we define new paths $\mathfrak{P}_1' = (\alpha_q \to\!\!\to b_1)$ and $\mathfrak{P}_2' = (\alpha_q \to\!\!\to b_2)$ obtained by removing the edges (a_1, α_q) and (a_2, α_q) from $\mathfrak{P}_1$ and $\mathfrak{P}_2$, respectively. Finally, we introduce an auxiliary path $(a_1 \to\!\!\to a_2)$ consisting of a single ghost edge between a_1 and a_2. The purpose of this auxiliary path is to ensure consistency with the induction hypothesis, which requires that every external vertex is an ending point of some path. (We cannot simply remove a_1 and a_2 from the graph, since some path $\mathfrak{P}_i$ may pass through them.)

The new graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}$ produced by this construction also satisfies the assumptions of Lemma 7.24, but it has one fewer internal vertex and two fewer solid edges than the original graph $\mathcal{G}_{\mathbf{ab}}$. Hence, by the induction hypothesis, we can bound it by

$$\begin{aligned}\sum_{\boldsymbol{\alpha}'} \mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}(\boldsymbol{\alpha}') &\prec \frac{\Psi_t^{\mathsf{ord}(\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}) - n_{\mathrm{ng}}(\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}})}}{(W^d \eta_t)^{q-1}} \cdot \prod_{i=1}^{2} [\Psi_t(c|\alpha_q - b_i|)]^{\chi(\mathfrak{P}_i')} \cdot \prod_{i=3}^{p} [\Psi_t(c|a_i - b_i|)]^{\chi(\mathfrak{P}_i)} \\ &\prec (W^d \eta_t)^{-(q-1)} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}}) - n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i - b_i|/2)]^{\chi(\mathfrak{P}_i)} \end{aligned} \tag{7.44}$$

for a constant $c > 0$. In the second step, we use that

$$\mathsf{ord}(\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}) = \mathsf{ord}(\mathcal{G}_{\mathbf{ab}}), \quad n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}}) = n_{\mathrm{ng}}(\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b}}), \tag{7.45}$$

since α_q is now treated as an external vertex. Moreover, we have also used that

$$\prod_{i=1}^{2} [\Psi_t(c|\alpha_q - b_i|)]^{\chi(\mathfrak{P}_i')} \leq \prod_{i=1}^{2} [\Psi_t(c|a_i - b_i|/2)]^{\chi(\mathfrak{P}_i)} \tag{7.46}$$

by the definitions of A1 and B1 edges. With (7.44), we can bound the LHS of (7.42) by

$$\sum_{\boldsymbol{\alpha}\in\mathbf{D}_{\boldsymbol{\pi}}} \mathcal{G}_{\mathbf{ab}}(\boldsymbol{\alpha}) \prec (W^d\eta_t)^{-(q-1)} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i-b_i|/2)]^{\chi(\mathfrak{P}_i)} \sum_{\alpha_q} \xi_{a_1\alpha_q}\xi_{a_2\alpha_q}$$
$$\prec (W^d\eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i-b_i|/2)]^{\chi(\mathfrak{P}_i)}, \tag{7.47}$$

where we use the Cauchy-Schwarz inequality and (7.38) in the second step. This concludes (7.42) in Case (I).

(II) Two A1/B1 edges: general case. We now consider a more general case than Case (I), where there is an internal vertex, say α_q, connected to at least two ending edges of type A1 or B1, and where $\deg(\alpha_q) \geq 4$. Without loss of generality, assume that the two ending edges connected to α_q are (a_1, α_q) and (a_2, α_q), belonging to paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$, respectively. As before, we define the new graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}} \equiv \mathcal{G}^{\mathrm{new}}_{\mathbf{a}'(\alpha_q),\mathbf{b}'(\alpha_q)}$ obtained from $\mathcal{G}_{\mathbf{ab}}$ by removing the two edges (a_1, α_q) and (a_2, α_q).

The new graph has external vertices $\alpha_q, a_1, a_2, a_3, \ldots, a_p, b_1, \ldots, b_p$ and internal vertices $\alpha_1, \ldots, \alpha_{q-1}$. To apply the induction hypothesis, we need to ensure that the structure of the paths in $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}}$ satisfies the assumptions in Lemma 7.24. For each path $\mathfrak{P}_j$, $j \in [\![3, p]\!]$, in the original graph, suppose it takes the form

$$(a_j \twoheadrightarrow \alpha_q \twoheadrightarrow \alpha_q \twoheadrightarrow \cdots \twoheadrightarrow \alpha_q \twoheadrightarrow b_j), \tag{7.48}$$

where each occurrence of α_q is explicitly listed. Here, $(a_j \twoheadrightarrow \alpha_q)$ denotes the initial segment of $\mathfrak{P}_j$ connecting a_j to the first occurrence of α_q, while $(\alpha_q \twoheadrightarrow b_j)$ denotes the final segment connecting the last occurrence of α_q to b_j. Each intermediate segment $(\alpha_q \twoheadrightarrow \alpha_q)$ represents a connection between two consecutive appearances of α_q along the path. Let k_j denote the total number of times α_q appears in the path $\mathfrak{P}_j$. Correspondingly, in $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}}$, we define the following $k_j + 1$ paths for each j: for each $1 \leq r \leq k_j + 1$, the (j, r)-th path $\mathfrak{P}'_{j,r}$ is the r-th segment of $\mathfrak{P}_j$, connecting $a_{j,r}$ to $b_{j,r}$, where

$$a_{j,r} := \begin{cases} a_j, & \text{if } r = 1 \\ \alpha_q, & \text{if } r > 1 \end{cases}, \quad b_{j,r} := \begin{cases} \alpha_q, & \text{if } r \leq k_j \\ b_j, & \text{if } r = k_j + 1 \end{cases}. \tag{7.49}$$

Next, for the first two paths $\mathfrak{P}_j$, $j \in \{1, 2\}$, assume they also take the form (7.48), with k_j appearances of α_q. After removing the edges (a_1, α_q) and (a_2, α_q), we define k_j paths in $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}}$ for each $j \in \{1, 2\}$. Specifically, for each $1 \leq r \leq k_j$, the (j, r)-th path $\mathfrak{P}'_{j,r}$ is the r-th segment of $\mathfrak{P}_j$, connecting $a_{j,r}$ to $b_{j,r}$, where

$$a_{j,r} := \alpha_q, \quad b_{j,r} := \begin{cases} \alpha_q, & \text{if } r < k_j \\ b_j, & \text{if } r = k_j \end{cases}. \tag{7.50}$$

Finally, we introduce an auxiliary path $(a_1 \twoheadrightarrow a_2)$ consisting of a single ghost edge between a_1 and a_2.

Now, we let $\mathbf{a}' := \{a_{j,r} : 1 \leq j \leq p, 1 \leq r \leq k_j + \mathbf{1}_{j\geq 3}\} \cup \{a_1\}$ and $\mathbf{b}' := \{b_{j,r} : 1 \leq j \leq p, 1 \leq r \leq k_j + \mathbf{1}_{j\geq 3}\} \cup \{a_2\}$. It is easy to see that the graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}}$ defined in this way satisfies the assumptions in Lemma 7.24, but with one fewer internal vertex and two fewer solid edges than the original graph $\mathcal{G}_{\mathbf{ab}}$. Therefore, using the induction hypothesis, we can bound it in a similar way as (7.44), that is,

$$\sum_{\boldsymbol{\alpha}'} \mathcal{G}^{\mathrm{new}}_{\mathbf{a'b'}}(\boldsymbol{\alpha}') \prec (W^d\eta_t)^{-(q-1)} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}})-n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i-b_i|/2)]^{\chi(\mathfrak{P}_i)}, \tag{7.51}$$

where $\boldsymbol{\alpha}' = (\alpha_1, \ldots, \alpha_{q-1})$, and in the derivation of this bound, we have used (7.45) and (7.46), and that

$$\prod_{r=1}^{k_j+1} [\Psi_t(c|a_{j,r}-b_{j,r}|)]^{\chi(\mathfrak{P}'_{j,r})} \leq [\Psi_t(c|a_j-b_j|/2)]^{\chi(\mathfrak{P}_j)} [\Psi_t(0)]^{\sum_{r=1}^{k_j+1}\chi(\mathfrak{P}'_{j,r})-\chi(\mathfrak{P}_j)}, \quad \forall j \in [\![3, p]\!].$$

Finally, using (7.51), we can complete the proof of (7.42) for Case (II), following a similar argument to the one used in (7.47).

(III) Two B2 edges. In this case, we assume there exists an internal vertex, say α_q, that is connected to two B2 ending edges and two solid edges, i.e., $\deg_s(\alpha_q) = 2$. By definition, these B2 edges do not belong to the same path. Without loss of generality, assume that these two B2 edges are (a_1, α_q) and (a_2, α_q), belonging to paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$, respectively. Additionally, suppose the two solid edges connected to α_q are (α_q, β_1) and (α_q, β_2), which belong to paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$, respectively. Here, β_1 and β_2 may represent external vertices in $\mathbf{a}, \mathbf{b}$ or internal vertices in $\boldsymbol{\alpha}' = (\alpha_1, \dots, \alpha_{q-1})$.

We now define a new graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}$ as follows: it still has the external vertices $\mathbf{a}$ and $\mathbf{b}$, and the internal vertices $\boldsymbol{\alpha}'$. The edges that do not belong to the paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$ remain unchanged in the new graph $\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}$. For paths $\mathfrak{P}_1$ and $\mathfrak{P}_2$, we modify them as follows:

- We create a new path $\mathfrak{P}'_1 = (a_1 \to\to b_1)$ obtained by removing the solid edge (α_q, β_1) from $\mathfrak{P}_1$ and adding a ghost edge (a_1, β_1) to the path.
- Similarly, we create a new path $\mathfrak{P}'_2 = (a_2 \to\to b_2)$ obtained by removing the solid edge (α_q, β_2) from $\mathfrak{P}_2$ and adding a ghost edge (a_2, β_2) to the path.

The reader can refer to Figure 2 for an illustration of this construction.

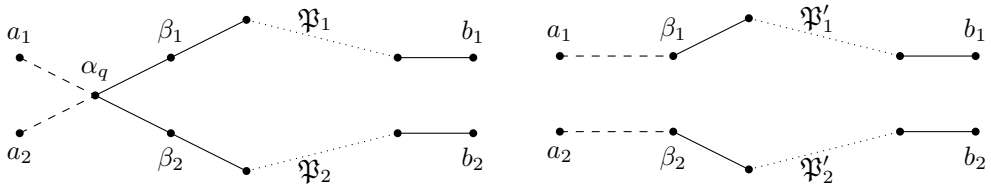


FIGURE 2. Illustration of the construction of paths $\mathfrak{P}'_1$ and $\mathfrak{P}'_2$ (right panel) from $\mathfrak{P}_1$ and $\mathfrak{P}_2$ (left panel).

Note that $\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}$ defined above is also a graph satisfying the assumptions in Lemma 7.24, but with one fewer internal vertex and two fewer solid edges compared to the original graph $\mathcal{G}_{\mathbf{ab}}$. Therefore, by the induction hypothesis, we can bound it by

$$\begin{aligned}\sum_{\boldsymbol{\alpha}'} \mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}(\boldsymbol{\alpha}') &\prec (W^d\eta_t)^{-(q-1)} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}) - n_{\mathrm{ng}}(\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}})} \cdot \prod_{i=3}^{p} [\Psi_t(c|a_i - b_i|)]^{\chi(\mathfrak{P}_i)} \\ &= (W^d\eta_t)^{-(q-1)} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}}) - n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i - b_i|)]^{\chi(\mathfrak{P}_i)},\end{aligned}$$

where, in the second step, we use that $\mathsf{ord}(\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}) = \mathsf{ord}(\mathcal{G}_{\mathbf{ab}})$, $n_{\mathrm{ng}}(\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}) = n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})$, and $\chi(\mathfrak{P}_1) = \chi(\mathfrak{P}_2) = 0$. Thus, we can bound the LHS of (7.42) as follows:

$$\begin{aligned}\sum_{\boldsymbol{\alpha} \in \mathbf{D}_{\boldsymbol{\pi}}} \mathcal{G}_{\mathbf{ab}}(\boldsymbol{\alpha}) &\prec \sum_{\boldsymbol{\alpha}'} \mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}(\boldsymbol{\alpha}') \sum_{\alpha_q} \xi_{\alpha_q\beta_1}\xi_{\alpha_q\beta_2} \prec (W^d\eta_t)^{-1} \sum_{\boldsymbol{\alpha}'} \mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}(\boldsymbol{\alpha}') \\ &\prec (W^d\eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}_{\mathbf{ab}}) - n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})} \cdot \prod_{i=1}^{p} [\Psi_t(c|a_i - b_i|/2)]^{\chi(\mathfrak{P}_i)}, \qquad (7.52)\end{aligned}$$

where we use the Cauchy-Schwarz inequality and (7.38) in the second step. This implies (7.42) in Case (III).

(IV) The remaining case. Finally, suppose that none of Cases (I)–(III) applies. In this case, we prove (7.42) directly, without using the induction hypothesis. (Note that we cannot argue by counting the ending edges attached to internal vertices: a path may pass through external vertices, so that some of its ending edges are not incident to any internal vertex, and the number q of internal vertices may be smaller than p.)

First, let K be the number of edges of $\mathcal{G}_{\mathbf{ab}}$. We further partition $\mathbf{D}_{\boldsymbol{\pi}}$ into at most K^p subregions according to which edge of each path $\mathfrak{P}_i$ without a ghost edge is the longest one; it suffices to prove (7.42) with $\mathbf{D}_{\boldsymbol{\pi}}$ replaced by any of these subregions. On such a subregion, by the triangle inequality, the chosen edge of $\mathfrak{P}_i$ has length at least $|a_i - b_i|/K$, so it contributes a factor $\Psi_t(c|a_i - b_i|)$ with $c = 1/K$ by (7.38). We bound the chosen edge of each path without a ghost edge in this way and remove it from the graph, and we also remove all ghost edges. Let $\mathcal{H}$ denote the resulting graph, which contains only solid edges, and let

$n_s(\mathcal{H}) = \mathsf{ord}(\mathcal{G}_{\mathbf{ab}}) + 2q - n_{\mathrm{ng}}(\mathcal{G}_{\mathbf{ab}})$ be the number of its solid edges. Since all graph values are nonnegative, we may then sum over all $\boldsymbol{\alpha} \in (\mathbb{Z}_L^d)^q$, so (7.42) follows from

$$\sum_{\boldsymbol{\alpha} \in (\mathbb{Z}_L^d)^q} \mathcal{H}(\boldsymbol{\alpha}) \prec (W^d \eta_t)^{-q} \cdot \Psi_t^{n_s(\mathcal{H})-2q}. \tag{7.53}$$

The key to proving (7.53) is to construct a *nested order of summation* for the q internal vertices, as in [67]: when we sum over the internal vertices in this order, each vertex being summed over must be incident to at least two solid edges whose other endpoints are external vertices or internal vertices that have not yet been summed over. Summing over such a vertex yields a factor $(W^d\eta_t)^{-1}$ by the Cauchy-Schwarz inequality and (7.38), while all other solid edges incident to it are bounded by Ψ_t. Hence, the existence of such an order implies (7.53).

We first observe that, for every nonempty subset A of the internal vertices,

$$e_{\mathcal{H}}(A, A^c) \geq |A| \quad \text{and} \quad \#\{\text{solid edges of } \mathcal{H} \text{ incident to } A\} \geq 2|A|, \tag{7.54}$$

where A^c is the set of all other (internal or external) vertices, and $e_{\mathcal{H}}(A, A^c)$ is the number of solid edges of $\mathcal{H}$ between A and A^c. Indeed, removing one edge from each path $\mathfrak{P}_i$ splits it into two edge-disjoint segments, each of which starts at an external vertex (a_i or b_i), and every vertex of $\mathfrak{P}_i$ lies on one of them. If $\mathfrak{P}_i$ passes through A, then one of its two segments enters A from A^c, which gives a solid edge between A and A^c; since the paths are edge-disjoint, the first bound in (7.54) follows from property (3) of Lemma 7.23. For the second bound, by property (2) of Lemma 7.23, every internal vertex $\alpha \in A$ lies on segments of at least two distinct paths, and each of these segments contains the edge through which it reaches α for the first time; these edges are incident to A and are distinct for distinct pairs (segment, vertex).

We now construct the order backwards, i.e., we first determine the vertex that is summed over last. Let $\mathscr{A}$ denote the set consisting of the external vertices and of the internal vertices chosen so far. As long as there is an internal vertex $\alpha \notin \mathscr{A}$ with at least two solid edges to $\mathscr{A}$, we add it to $\mathscr{A}$. Suppose that this procedure stops while the set A of the remaining internal vertices is nonempty. Then, each $\alpha \in A$ has at most one solid edge to $A^c = \mathscr{A}$, so by the first bound in (7.54) it has exactly one such edge, which we denote by (α, x_α). Let $\mathcal{C}$ be a connected component of the subgraph of $\mathcal{H}$ induced on A. Applying the second bound in (7.54) to $\mathcal{C}$, we see that $\mathcal{C}$ contains at least $|\mathcal{C}|$ edges, and hence a cycle (possibly consisting of two parallel edges). We pick two vertices $\alpha, \beta \in \mathcal{C}$ that are adjacent on this cycle and use

$$\xi_{\alpha x_\alpha} \xi_{\beta x_\beta} \leq \frac{1}{2} \left(\xi_{\alpha x_\alpha}^2 + \xi_{\beta x_\beta}^2 \right). \tag{7.55}$$

Consider the graph in which $\xi_{\alpha x_\alpha}\xi_{\beta x_\beta}$ is replaced by $\xi_{\alpha x_\alpha}^2$ (the other term is treated in the same way, with the roles of α and β exchanged). In this graph, α has two edges to $\mathscr{A}$, so it can be added to $\mathscr{A}$. Next, every vertex of $\mathcal{C} \setminus \{\beta\}$ that is adjacent to $\mathscr{A}$ has two edges to $\mathscr{A}$ (its edge to $x_\cdot$ and an edge to an added vertex), so all vertices connected to α in $\mathcal{C} \setminus \{\beta\}$ can be added one by one. These include both neighbors of β on the cycle, so β can be added as well, and then the remaining vertices of $\mathcal{C}$ can be added one by one along paths in $\mathcal{C}$ starting from β. Applying (7.55) to one such pair in each connected component of the subgraph induced on A, we obtain $\mathrm{O}(1)$ many graphs, for each of which the procedure exhausts all internal vertices; see Figure 3 for an illustration. Since (7.55) does not change the number of solid edges, summing over the internal vertices in the reverse order of their addition proves (7.53), and hence (7.42) in Case (IV). (In fact, this argument does not use the assumption that Cases (I)–(III) do not apply.)

□

7.5. **Proof of Lemma 7.2.** Note that when $|a-b| \leq (\log W)^{3/2}\ell_t$ or $\ell \leq (\log W)^{3/2}\ell_t$, $\mathsf{T}_t(|a-b| \wedge \ell)$ does not exhibit exponential decay. In this regime, Lemma 7.2 follows directly from Lemma 7.1. Therefore, for the remainder of the proof, it suffices to establish (7.11) under the conditions

$$|a-b| > (\log W)^{3/2}\ell_t, \quad \text{and} \quad (\log W)^{3/2}\ell_t \leq \ell \leq (\log W)^{10}\ell_t. \tag{7.56}$$

For this purpose, we decompose $f_{xy}(G) = f_{xy}^{>\ell}(G) + f_{xy}^{\leq \ell}(G)$ into two parts, where

$$f_{xy}^{>\ell}(G) = W^{-d} \sum_{a_1 : |a_1 - a| \wedge |a_1 - b| > \ell} \sum_{a_2} S_{a_1 a_2}^{(\mathsf{B})} \sum_{\substack{\alpha \in [a_2], \beta \in [a_1], \\ \alpha \notin \{x,y\}}} \mathring{G}_{\beta\beta} G_{x\alpha} G_{\alpha y},$$

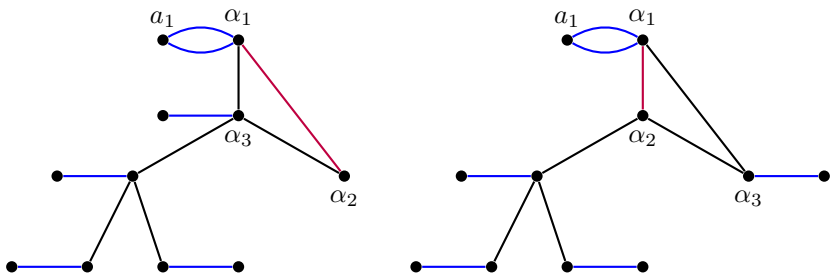


FIGURE 3. Two examples of a connected component $\mathcal{C}$ in Case (IV), after the splitting (7.55) with $(\alpha, \beta) = (\alpha_1, \alpha_2)$, in the term containing $\xi^2_{\alpha_1 a_1}$. Blue edges represent the solid edges to the set $\mathscr{A}$, here to external vertices (blue is used solely for illustration and does not indicate the + charge of G-edges as in Definition 7.3); the purple edge is the edge between α_1 and α_2 on a cycle of $\mathcal{C}$; and black edges represent the other solid edges inside $\mathcal{C}$. The vertex α_1 is added to $\mathscr{A}$ first; every other vertex except α_2 is added as soon as it is adjacent to $\mathscr{A}$, and α_2 is added once two of its neighbors belong to $\mathscr{A}$.

$$f^{\le\ell}_{xy}(G) = f^{(a)}_{xy}(G) + f^{(b)}_{xy}(G), \quad f^{(a)}_{xy}(G) := W^{-d} \sum_{a_1 : |a_1 - a| \le \ell} \sum_{a_2} S^{(\mathsf{B})}_{a_1 a_2} \sum_{\substack{\alpha \in [a_2], \beta \in [a_1], \\ \alpha \notin \{x,y\}}} \mathring{G}_{\beta\beta} G_{x\alpha} G_{\alpha y},$$

and $f^{(b)}_{xy}(G)$ is defined in the same way as $f^{(a)}_{xy}(G)$, with the summation over a_1 restricted to $|a_1 - a| > \ell \ge |a_1 - b|$. Lemma 7.2 follows immediately from the next two estimates on $f^{>\ell}_{xy}(G)$ and $f^{\le\ell}_{xy}(G)$ with $\Psi_t = (W^{-d} B_{t,0})^{1/2}$:

Lemma 7.25. *In the setting of Lemma 7.2, for any fixed $p \in 2\mathbb{N}$, the following estimate holds for any large constant $D > 0$:*

$$\mathbb{E}\big|f^{>\ell}_{xy}(G)\big|^p \prec \frac{1}{\eta_t^p} \Psi_t^p \cdot [\mathsf{T}_t(|a-b| \wedge \ell)]^p + W^{-D}. \tag{7.57}$$

Lemma 7.26. *In the setting of Lemma 7.2, for any fixed $p \in 2\mathbb{N}$, the following estimate holds for any large constant $D > 0$:*

$$\mathbb{E}\big|f^{\le\ell}_{xy}(G)\big|^p \prec \frac{1}{\eta_t^p} \Psi_t^p \cdot [\mathsf{T}_t(|a-b| \wedge \ell)]^p + W^{-D}. \tag{7.58}$$

First, the proof of Lemma 7.25 is similar to that for Lemma 7.1.

Proof of Lemma 7.25. First, similar to Lemma 7.17, we expand $\big|f^{>\ell}_{xy}(G)\big|^p$ into a sum of locally standard graphs satisfying properties (1)–(6) listed there. Next, we bound each locally standard graph using its associated auxiliary graph, constructed as in Definition 7.20. We then estimate the auxiliary graphs using a result analogous to Lemma 7.23. More precisely, suppose $\mathcal{G}^{\mathrm{aux}}_{\mathbf{ab}}$ is a nested graph that satisfies the assumptions of Lemma 7.23. In addition, all internal vertices lie in the domain $\mathbf{D}_{>\ell} := \{c \in \mathbb{Z}^d_L : |a_i - c| \wedge |b_i - c| > \ell \text{ for all } i \in [\![p]\!]\}$. Indeed, every vertex created by the local expansions is connected to an existing vertex by a path of dotted or waved edges (see the discussion below Lemma 7.13), so every internal molecule contains one of the vertices $\beta \in [a_1]$ of the p factors $f^{>\ell}_{xy}(G)$ or $\overline{f^{>\ell}_{xy}(G)}$, and we choose such a vertex as the center of the molecule in Definition 7.20. Then, we aim to show that

$$\mathcal{G}^{\mathrm{aux}}_{\mathbf{ab}} \prec (W^d \eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathcal{G}^{\mathrm{aux}}_{\mathbf{ab}}) - p} \cdot [\mathsf{T}_t(|a-b| \wedge \ell)]^p + W^{-D}. \tag{7.59}$$

In the proof of Lemma 7.24, each path contributed at least one factor of $\Psi_t(c|a_i - b_i|)$ because, roughly speaking, every path contains at least one long ending edge. In the current setting, where $a_i = [a]$ and $b_i = [b]$ for all $i \in [\![p]\!]$, we use the fact that the first edge of each path $\mathfrak{P}_i$ is "long": it joins $[a]$ either to $[b]$, in which case its length is $|a - b|$, or to an internal vertex, in which case its length is larger than ℓ, since all internal vertices lie in the domain $\mathbf{D}_{>\ell}$. Since each solid edge $\xi([\alpha], [\beta])$ is bounded by $\mathsf{T}_t(|[\alpha] - [\beta]| \wedge \ell) + W^{-D}$ and T_t is decreasing, each such long edge provides a factor of $\mathsf{T}_t(|a-b| \wedge \ell)$ (up to a negligible error W^{-D}), and we can treat it as a type-A2 edge. Replacing these A2 edges with ghost edges in all paths $\mathfrak{P}_i$, $i \in [\![p]\!]$,

yields a total factor of $[\mathsf{T}_t(|a-b|\wedge\ell)]^p$. For the remaining graph—denoted by $\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}}$—we need to establish the following bound for any constant $D>0$:

$$\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}} \prec (W^d\eta_t)^{-q}\cdot\Psi_t^{\mathsf{ord}(\mathcal{G}^{\mathrm{new}}_{\mathbf{ab}})}+W^{-D}. \tag{7.60}$$

The proof of this estimate follows exactly the same argument as in Lemma 7.24, and we omit the details. □

For the proof of Lemma 7.26, we need a new argument that makes use of the exponential decay in T_t.

Proof of Lemma 7.26. Since $|f^{\le\ell}_{xy}(G)|^p\le 2^{p-1}(|f^{(a)}_{xy}(G)|^p+|f^{(b)}_{xy}(G)|^p)$, it suffices to bound $\mathbb{E}|f^{(a)}_{xy}(G)|^p$; the term $f^{(b)}_{xy}(G)$ is handled in the same way, with a replaced by b in the definition of the domain $\mathbf{D}_{\le\ell}$ below. Similar to Lemma 7.17, we expand $\big|f^{(a)}_{xy}(G)\big|^p$ into a sum of locally standard graphs satisfying properties (1)–(6) listed there. Next, we bound each locally standard graph using its associated auxiliary graph, constructed as in Definition 7.20. This yields a class of auxiliary graphs $\mathcal{G}^{\mathrm{aux}}_{[a][b]}$ satisfying the assumptions of Lemma 7.23 with $[a_i]=[a]$ and $[b_i]=[b]$ for $i\in[\![p]\!]$. Moreover, choosing the centers of the internal molecules as in the proof of Lemma 7.25, all internal vertices of the graph lie in the domain $\mathbf{D}_{\le\ell}:=\{c\in\mathbb{Z}^d_L:|a-c|\le\ell\}$. It remains to bound such graphs as follows:

$$\mathcal{G}^{\mathrm{aux}}_{[a][b]}\prec(W^d\eta_t)^{-q}\cdot\Psi_t^{\mathsf{ord}(\mathcal{G}^{\mathrm{aux}}_{[a][b]})-p}\cdot[\mathsf{T}_t(|[a]-[b]|\wedge\ell)]^p+W^{-D}. \tag{7.61}$$

We first control $\mathcal{G}^{\mathrm{aux}}_{[a][b]}$ by bounding each solid edge, say $\xi([\alpha],[\beta])$, by its upper bound $\mathsf{T}_t(|[\alpha]-[\beta]|\wedge\ell)+W^{-D}$. Discarding the negligible error term containing W^{-D} factors, we obtain a new graph, denoted by $\mathscr{G}_{[a][b]}$, which has the same graphical structure as $\mathcal{G}^{\mathrm{aux}}_{[a][b]}$ but with each solid edge representing a T_t factor instead. Hence, to prove (7.61), it suffices to establish that

$$\mathscr{G}_{[a][b]}=\sum_{\boldsymbol{\alpha}=([\alpha_1],\dots,[\alpha_q])\in(\mathbf{D}_{\le\ell})^q}\mathscr{G}_{[a][b]}(\boldsymbol{\alpha})\prec(W^d\eta_t)^{-q}\cdot\Psi_t^{\mathsf{ord}(\mathscr{G}_{[a][b]})-p}\cdot[\mathsf{T}_t(|[a]-[b]|\wedge\ell)]^p, \tag{7.62}$$

where, recall, $\mathscr{G}_{[a][b]}(\boldsymbol{\alpha})$ denotes the graph obtained by fixing the external vertices to $\boldsymbol{\alpha}$, and $\mathsf{ord}(\mathscr{G}_{[a][b]})$ is defined in (7.35). To show (7.62), we sum over the internal vertices in $\boldsymbol{\alpha}$ one by one. Unlike in the proof of Lemma 7.23, the order of summation here is arbitrary; for definiteness, we follow the order $[\alpha_1],\dots,[\alpha_q]$. At each step, we apply the following key estimate.

Claim 7.27. *Assume $1-t\ge g^2/L^2$. Then, for any $k\ge2$ and any $0\le\ell\le(\log W)^{10}\ell_t$, the following bound holds with $\Psi_t=(W^{-d}B_{t,0})^{1/2}$:*[11]

$$\sum_{[\alpha]\in\mathbf{D}_{\le\ell}}\prod_{i=1}^k[\mathsf{T}_t(|[x_i]-[\alpha]|\wedge\ell)\cdot\mathsf{T}_t(|[y_i]-[\alpha]|\wedge\ell)]\prec(W^d\eta_t)^{-1}\cdot\Psi_t^{k-2}\cdot\prod_{i=1}^k\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell). \tag{7.63}$$

We defer the proof of Claim 7.27 to Appendix A.4, where it is derived from elementary calculus estimates. (That proof only uses that $\mathbf{D}_{\le\ell}$ is contained in a ball of radius ℓ in $\mathbb{Z}^d_L$.) The estimate (7.63) shows that when summing over an internal vertex $[\alpha]$, each path of the form $([x_i]\to[\alpha]\to[y_i])$ consisting of two solid edges through $[\alpha]$ is effectively replaced by a single solid edge $[x_i]\to[y_i]$ in the resulting graph. We refer to this as the "path-preserving phenomenon". In addition, the summation over $[\alpha]\in\mathbf{D}_{\le\ell}$ contributes a factor $(W^d\eta_t)^{-1}$ from each such pair of solid edges, along with the factor Ψ_t^{k-2} reflecting the reduction in scaling order. Note that the case $k=2$ is critical—the estimate (7.63) fails when $k=1$. Figure 4 illustrates the path-preserving phenomenon in the critical case $k=2$.

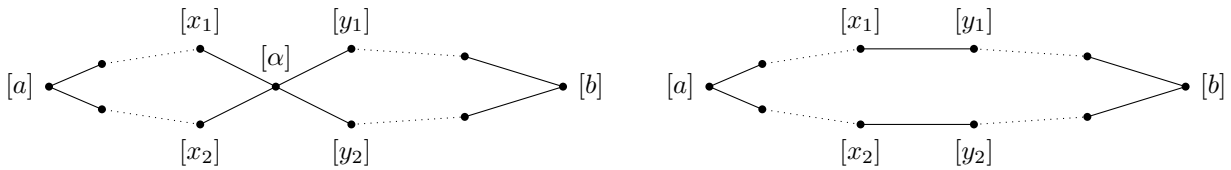


FIGURE 4. Illustration of the path-preserving phenomenon in applying (7.63).

[11] This estimate can be viewed as an extension of Lemma 3.8 in the regime $1-t\ge g^2/L^2$, except that here we obtain a "$\prec$" bound, rather than the "$\lesssim$" bound in (3.21), due to the presence of additional logarithmic factors.

We perform the summations of $\mathscr{G}_{[a][b]}$ over the internal vertices by generating a sequence of new graphs that consistently satisfy conditions (1)–(3) in Lemma 7.23 with $[a_i] \equiv [a]$ and $[b_i] \equiv [b]$. To illustrate this procedure, consider the summation over $[\alpha_1]$. Suppose there are k_1 two-edge paths passing through $[\alpha_1]$ in $\mathscr{G}_{[a][b]}$. More precisely, assume that each path $\mathfrak{P}_i$ can be decomposed as

$$([a_i] \to\to [x_{i,1}] \to [\alpha_1] \to [y_{i,1}] \to\to [x_{i,2}] \to [\alpha_1] \to [y_{i,2}] \to\to \cdots \to\to [x_{i,r_i}] \to [\alpha_1] \to [y_{i,r_i}] \to\to [b_i]),$$

where we omit intermediate vertices on the path and only list those vertices $[x_{i,j}]$ and $[y_{i,j}]$ that are directly connected to $[\alpha_1]$, and $r_i \geq 0$ is a non-negative integer. Then, we apply (7.63) to the summation

$$\sum_{[\alpha_1] \in \mathbf{D}_{\leq \ell}} \prod_{i=1}^{p} \prod_{j=1}^{r_i} \left[\mathsf{T}_t(|[x_{i,j}] - [\alpha_1]| \wedge \ell) \cdot \mathsf{T}_t(|[y_{i,j}] - [\alpha_1]| \wedge \ell) \right],$$

which yields the bound

$$\sum_{[\alpha_1] \in \mathbf{D}_{\leq \ell}} \mathscr{G}_{[a][b]}(\boldsymbol{\alpha}) \prec (W^d \eta_t)^{-1} \Psi_t^{k_1 - 2} \cdot \mathscr{G}^{(1)}_{[a][b]}([\alpha_2], \ldots, [\alpha_q]),$$

where k_1 is defined as $k_1 = \sum_{i=1}^{p} r_i$, and $\mathscr{G}^{(1)}_{[a][b]}$ is a new graph obtained by removing the vertex $[\alpha_1]$ from $\mathscr{G}_{[a][b]}$ and replacing each pair of solid edges $([x_{i,j}] \to [\alpha_1] \to [y_{i,j}])$ by $([x_{i,j}], [y_{i,j}])$. It is easy to see that the new graph $\mathscr{G}^{(1)}_{[a][b]}$ also satisfies conditions (1)–(3) in Lemma 7.23. Next, summing over the vertex $[\alpha_2]$ in $\mathscr{G}^{(1)}_{[a][b]}$ gives another graph $\mathscr{G}^{(2)}_{[a][b]}([\alpha_3], \ldots, [\alpha_q])$ that again satisfies conditions (1)–(3) in Lemma 7.23, along with a factor $(W^d \eta_t)^{-1} \Psi_t^{k_2 - 2}$ for some $k_2 \geq 2$. Continuing this procedure, after summing over all internal vertices, we can bound the LHS of (7.62) as

$$\sum_{\boldsymbol{\alpha} \in (\mathbf{D}_{\leq \ell})^q} \mathscr{G}_{[a][b]}(\boldsymbol{\alpha}) \prec (W^d \eta_t)^{-q} \cdot \Psi_t^{\mathsf{ord}(\mathscr{G}_{[a][b]}) - p} \cdot \mathscr{G}^{(q)}_{[a][b]},$$

where $\mathscr{G}^{(q)}_{[a][b]}$ is a graph consisting solely of p solid edges between $[a]$ and $[b]$, with no internal vertices. Since each solid edge is bounded by $\mathsf{T}_t(|[a] - [b]| \wedge \ell)$, this yields (7.62). Combining this with (7.36) completes the proof of Lemma 7.26. □

8. Extension to the block Anderson model

The proof of Theorem 2.7 for the block Anderson model is based on the following flow framework.

Lemma 8.1 (Lemma 3.3 of [59]). *For the block Anderson model, fix any $g > 0$ and $z \in \mathbb{C}_+$ with $\operatorname{Im} z \in (0, 1]$ and $\operatorname{Im} m(z, g) \geq \kappa$. We choose*

$$t_0 = \frac{\operatorname{Im} m(z, g)}{\operatorname{Im} m(z, g) + \operatorname{Im} z}, \quad E = \frac{t_0 \operatorname{Re} z - (1 - t_0) \operatorname{Re} m(z, g)}{\sqrt{t_0}}, \quad g_0 = \sqrt{t_0} g. \tag{8.1}$$

Then, we have that

$$\sqrt{t_0} m(E, g_0) = m(z, g), \quad z_{t_0}(E, g_0) = \sqrt{t_0} z, \quad \sqrt{t_0} M(E, g_0) = M(z, g), \quad G(z, g) \stackrel{d}{=} \sqrt{t_0} G_{t_0; E, g_0}, \tag{8.2}$$

where recall that $G(z, g) = (H - z)^{-1} = (V + g\Psi - z)^{-1}$.

Note that $t_0 \geq \kappa/(\kappa + 1)$, since $\operatorname{Im} m(z, g) \geq \kappa$ and $\operatorname{Im} z \leq 1$, and that $\operatorname{Im} m(E, g_0) = \operatorname{Im} m(z, g)/\sqrt{t_0} \geq \kappa$ by (8.2). In particular, the flow parameters (E, g_0) satisfy the bulk condition of Lemma 8.3 below.

In the following proof, we fix a target spectral parameter $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa, \varepsilon}$ for an arbitrarily small constant $\varepsilon > 0$, where the spectral domain $\mathbf{D}_{\kappa, \varepsilon}$ is now defined as

$$\mathbf{D}_{\kappa, \varepsilon} := \{z = \hat{E} + \mathrm{i}\eta \in \mathbb{C}_+ : \operatorname{Im} m(z, g) \geq \kappa, N^{-1+\varepsilon} \leq \eta \leq 1\}. \tag{8.3}$$

This domain contains the spectral parameters considered in Theorem 2.7: if $\hat{E}$ satisfies the bulk condition (2.34), i.e., $\rho_N(\hat{E}) \geq \kappa$, then $\hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa', \varepsilon}$ for all $N^{-1+\varepsilon} \leq \eta \leq 1$, where $\kappa' > 0$ is a constant depending only on d, κ, and $\mathfrak{d}$. To see this, let $f(\eta) := \operatorname{Im} m(\hat{E} + \mathrm{i}\eta, g)$ for $\eta > 0$, so that $f(0^+) = \pi \rho_N(\hat{E}) \geq \pi\kappa$. By property (4) of Lemma 8.3 below, $|f'(\eta)| \leq (2f(\eta)^2)^{-1}$, hence $f(\eta)^3 \geq (\pi\kappa)^3 - 3\eta/2 \geq (\pi\kappa)^3/2$ for $0 < \eta \leq (\pi\kappa)^3/3$. For $(\pi\kappa)^3/3 \leq \eta \leq 1$, we use that μ_N, being the free convolution of the semicircle law with a probability measure supported in $[-2dg, 2dg]$, is supported in $[-R, R]$ with $R := 2 + 2dg \leq 2 + 2d\mathfrak{d}^{-1}$, and that $|\hat{E}| \leq R$ since $\rho_N(\hat{E}) > 0$; thus, $f(\eta) = \int \eta \left((x - \hat{E})^2 + \eta^2\right)^{-1} \mu_N(\mathrm{d}x) \geq \eta/(4R^2 + 1)$. Consequently, it suffices to prove the estimates in Theorem 2.7 for $z \in \mathbf{D}_{\kappa, \varepsilon}$, for any fixed constant $\kappa > 0$. Accordingly, we choose

the parameters t_0, E, and g_0 as specified in (8.1). Again, for simplicity of presentation, unless we want to emphasize the dependence on the flow parameters E and g_0, we will omit them from various notations, such as $z_t(E, g_0)$, $E_t(E, g_0)$, $\eta_t(E, g_0)$, $m(E, g_0)$, $M(E, g_0)$, and $G_{t;E,g_0} \equiv G_t$.

Since M is not a multiple of the identity matrix, the block Anderson model also involves the following *twisted* quantities. For $c_1, c_2 \in \mathbb{Z}_L^d$, let $P_{c_1c_2}$ be the $N \times N$ matrix that maps the block $[c_2]$ to the block $[c_1]$ while preserving the position inside the block, normalized in the same way as E_a:

$$(P_{c_1c_2})_{xy} := W^{-d}\,\mathbf{1}\big(x \in [c_1],\ y \in [c_2],\ x - Wc_1 = y - Wc_2\big), \quad \text{i.e.,} \quad P_{c_1c_2} = W^{-d}\big(\mathbf{e}_{c_1}\mathbf{e}_{c_2}^\top\big) \otimes I_{W^d},$$

where $\mathbf{e}_c$ denotes the standard basis vector of $\mathbb{C}^{L^d}$ associated with $c \in \mathbb{Z}_L^d$. In particular, $P_{cc} = E_c$, $P_{c_1c_2}^* = P_{c_2c_1}$, and $\mathrm{Tr}(M(\sigma)P_{c_1c_2}) = M^{(\mathsf{B})}_{c_2c_1}(\sigma)$ by (2.33). We refer to $\mathrm{Tr}\left((G_t(\sigma) - M(\sigma))P_{c_1c_2}\right)$ as a *twisted trace*, and to

$$\widetilde{\mathcal{L}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)} := \mathrm{Tr}\left(G_t(\sigma_1)E_aG_t(\sigma_2)P_{c_1c_2}\right), \qquad \boldsymbol{\sigma} = (\sigma_1, \sigma_2) \in \{+,-\}^2, \quad a, c_1, c_2 \in \mathbb{Z}_L^d,$$

as a *twisted* 2*-loop*; when $c_1 = c_2$, it is the 2-G-loop $\mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},(a,c_1)}$. The deterministic approximation of $\widetilde{\mathcal{L}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)}$ is the *twisted tree*

$$\widetilde{\mathcal{K}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)} := W^{-d} \sum_{a' \in \mathbb{Z}_L^d} \Theta_t^{(\sigma_1,\sigma_2)}(a, a')\, M^{(\mathsf{B})}_{c_2a'}(\sigma_1)\, M^{(\mathsf{B})}_{a'c_1}(\sigma_2),$$

which coincides with $\mathcal{K}^{(2)}_{t,\boldsymbol{\sigma},(a,c_1)}$ in (2.71) when $c_1 = c_2$, and with $\widetilde{\mathcal{L}}_{0,\boldsymbol{\sigma},(a;c_1,c_2)}$ at $t = 0$, since $G_0(\sigma) = M(\sigma)$. Note also that $\mathrm{Tr}(XE_aM(\sigma)P_{c_1c_2}) = W^{-d}M^{(\mathsf{B})}_{ac_1}(\sigma)\,\mathrm{Tr}(XP_{ac_2})$ for any $N \times N$ matrix X; this relates twisted 2-loops containing $\mathring{G}_t(\sigma_2) = G_t(\sigma_2) - M(\sigma_2)$ in place of $G_t(\sigma_2)$ to $\widetilde{\mathcal{L}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)}$ and twisted traces. In the flow framework of Lemma 8.1, we can establish an analogue of Theorem 2.24 for the block Anderson model.

Theorem 8.2. *In the setting of Theorem 2.7, fix any $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$ and consider the flow framework in Lemma 8.1. Suppose the estimates (2.78)–(2.82) hold at some fixed $s \in [0, t_0]$, together with the following estimate at $u = s$:*

(e) **Twisted-loop estimate***: We have*

$$\max_{\boldsymbol{\sigma},a,c_1,c_2} \left|\widetilde{\mathcal{L}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)} - \widetilde{\mathcal{K}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)}\right| \prec \left(W^{-d}B_{u,0}\right)^{3/2}. \tag{8.4}$$

Then, there exists a constant $0 < \mathfrak{c}_d \le 10^{-2}$ such that for any $s < t < 1$ satisfying (2.83), the estimates (2.73)–(2.77) hold, and (8.4) holds at $u = t$. In addition, if $1 - t \ge g^2$, then the estimate (2.84) holds.

With Theorem 8.2 in hand, we can establish Theorem 2.7 by induction on t.

Proof of Theorem 2.7. By the discussion below (8.3), it suffices to consider $z \in \mathbf{D}_{\kappa,\varepsilon}$. Fix $z = \hat{E} + \mathrm{i}\eta \in \mathbf{D}_{\kappa,\varepsilon}$, and choose the flow as in Lemma 8.1. At $t = 0$, we have $G_0(\sigma) = M(\sigma)$, so $\mathcal{L}^{(n)}_{0,\boldsymbol{\sigma},\mathbf{a}} = \mathcal{K}^{(n)}_{0,\boldsymbol{\sigma},\mathbf{a}}$ and $\widetilde{\mathcal{L}}_{0,\boldsymbol{\sigma},(a;c_1,c_2)} = \widetilde{\mathcal{K}}_{0,\boldsymbol{\sigma},(a;c_1,c_2)}$, and the estimates (2.78)–(2.82) and (8.4) hold at $s = 0$. By Theorem 8.2, applying induction on t from $t = 0$ to $t = t_0$ (simultaneously for all target spectral parameters, see the proof of Step 1 below) yields the estimates (2.73)–(2.77) at $t = t_0$. Using (8.2), (2.63), and (2.71), we see that these estimates together imply the entrywise local law (2.14), the averaged local law (2.15), and the quantum diffusion estimates (2.22)–(2.25) for each fixed z. To extend these estimates uniformly to all $z \in \mathbf{D}_{\kappa,\varepsilon}$, we apply a standard N^{-C}-net and perturbation argument. Finally, the delocalization estimate (2.11), the QUE estimates (2.18) and (2.19), and the bulk universality (2.20) then follow as consequences, as shown in Section 2.1. For the bulk universality, two modifications are needed. First, the Green's function comparison is applied with an interpolation acting only on the random part $V = H - g\Psi$, i.e., centered at the deterministic matrix $g\Psi$, as in the flow (2.36). Second, the argument of [69, Theorem 2.6] that we follow uses the analogue of [69, Lemma 5.15] (cf. Lemma 6.1) for the matrices obtained by adding a small GUE component. For the block Anderson model, the proof of this analogue requires, in addition, a twisted averaged law for the resolvents G of these matrices, namely $\max_{c_1,c_2} |\mathrm{Tr}((G - M)P_{c_1c_2})| \prec (N\eta)^{-1}$ at the corresponding spectral parameters $z = E + \mathrm{i}\eta$, for the same reason as in the proof of Lemma 6.1 below; this bound is proved in the same way as Lemma 8.6. □

The proof of Theorem 8.2 follows the same six-step strategy outlined below Theorem 2.24. We now explain how the arguments in Section 3–7 for the random band matrix model extend to the block Anderson model. Our proof relies on the properties of the Θ-propagators stated in Lemma 2.19, together with the following properties of $m(z)$ (defined in (2.32)) and the matrix $M^{(\mathsf{B})}$ (defined in (2.33)).

Lemma 8.3. *For the block Anderson model, under the condition (2.10), $m = m(E, g)$ and $M^{(\mathsf{B})} = M^{(\mathsf{B})}(E, g)$ satisfy the following properties for any $E \in \mathbb{R}$ with $\operatorname{Im} m(E, g) \geq \kappa$ (recall that E is the spectral parameter in the flow (8.1), and that this condition holds for the flow parameters (E, g_0) of Lemma 8.1 when $z \in \mathbf{D}_{\kappa,\varepsilon}$):*

(1) **Translation invariance**: *For any $a, b, r \in \mathbb{Z}_L^d$, we have $M^{(\mathsf{B})}_{a+r,b+r} = M^{(\mathsf{B})}_{ab}$ and $M^{(\mathsf{B})}_{aa} \equiv m$.*

(2) **Ward's identity**: *We have $|m| \leq 1$, $\operatorname{Im} m \geq \kappa$, and*

$$\sum_b |M^{(\mathsf{B})}_{ab}|^2 = 1, \quad \forall a \in \mathbb{Z}_L^d . \tag{8.5}$$

(3) **Combes–Thomas bound**: *There exists a constant $C > 0$ (depending only on d and κ) such that the following estimate holds for $g < (2C)^{-1}$:*

$$C^{-1} g \mathbf{1}(a \sim b) \leq |M^{(\mathsf{B})}_{ab}| \leq (Cg)^{|a-b|}. \tag{8.6}$$

Furthermore, for $g \geq (2C)^{-1}$, there exists a constant $c > 0$ such that the following bound holds:

$$|M^{(\mathsf{B})}_{ab}| \leq c^{-1} \exp\left(-c|a-b|\right). \tag{8.7}$$

(4) **Stability**: *For every $z \in \mathbb{C}_+$ and $g > 0$, with $m = m(z, g)$ and $M^{(\mathsf{B})} = M^{(\mathsf{B})}(z)$, we have*

$$\operatorname{Re}\left(1 - L^{-d} \operatorname{Tr}\left(M^{(\mathsf{B})}\right)^2\right) \geq 2(\operatorname{Im} m)^2.$$

Consequently, $m(z, g)$ is a smooth function of $(z, g) \in \mathbb{C}_+ \times (0, \infty)$, and

$$|\partial_z m(z, g)| \leq \frac{1}{2(\operatorname{Im} m(z, g))^2}, \qquad |\partial_g m(z, g)| \leq \frac{d}{(\operatorname{Im} m(z, g))^2}.$$

Proof. Property (1) follows directly from the translation invariance of the matrix $\Psi^{(\mathsf{B})}$ in (2.31). The bound $\operatorname{Im} m \geq \kappa$ is our assumption, while Ward's identity (8.5) follows by taking the imaginary part of the equation $m = M^{(\mathsf{B})}_{aa} = (g\Psi^{(\mathsf{B})} - E - m)^{-1}_{aa}$. The identity (8.5) gives directly $|m| \leq 1$. For sufficiently small g, the estimates in (8.6) are obtained from the Taylor expansion

$$M^{(\mathsf{B})} = -\sum_{k=0}^{\infty} (E + m)^{-k-1} (g\Psi^{(\mathsf{B})})^k.$$

For g of order 1, (8.7) is given by the classical Combes–Thomas estimate (see, e.g., [7, Theorem 10.5]). For property (4), let $\psi_1, \ldots, \psi_{L^d}$ be the eigenvalues of $\Psi^{(\mathsf{B})}$ (so that $|\psi_i| \leq 2d$), and let $\mu_i := (g\psi_i - z - m)^{-1}$ be the corresponding eigenvalues of $M^{(\mathsf{B})}$. Taking the imaginary part of (2.32), i.e., of $m = L^{-d} \sum_i \mu_i$, gives $\operatorname{Im} m = \operatorname{Im}(z + m) \cdot L^{-d} \sum_i |\mu_i|^2$; hence, $L^{-d} \sum_i |\mu_i|^2 \leq 1$ and

$$\operatorname{Re}\Big(1 - L^{-d} \sum_i \mu_i^2\Big) \geq L^{-d} \sum_i \left(|\mu_i|^2 - \operatorname{Re} \mu_i^2\right) = 2L^{-d} \sum_i (\operatorname{Im} \mu_i)^2 \geq 2\Big(L^{-d} \sum_i \operatorname{Im} \mu_i\Big)^2 = 2(\operatorname{Im} m)^2.$$

Differentiating (2.32) with respect to z and g, we obtain

$$\Big(1 - L^{-d} \operatorname{Tr}\left(M^{(\mathsf{B})}\right)^2\Big) \partial_z m = L^{-d} \operatorname{Tr}\left(M^{(\mathsf{B})}\right)^2, \qquad \Big(1 - L^{-d} \operatorname{Tr}\left(M^{(\mathsf{B})}\right)^2\Big) \partial_g m = -L^{-d} \operatorname{Tr}\left(\Psi^{(\mathsf{B})} (M^{(\mathsf{B})})^2\right),$$

where the right-hand sides are bounded in absolute value by $L^{-d} \sum_i |\mu_i|^2 \leq 1$ and $2d$, respectively. Together with the implicit function theorem (the derivative of (2.32) with respect to m does not vanish by the above stability bound), this proves property (4). □

Another key ingredient in the proof of Theorem 8.2 is the following analogue of Lemma 3.1.

Lemma 8.4. *In the setting of Theorem 8.2, suppose the estimates in (3.4) hold for a deterministic control parameter $W^{-d/2} \leq \Psi_t \leq W^{-\varepsilon_0}$. Furthermore, suppose there exist deterministic control parameters $0 < \Phi_t(a, b) \leq W^{-\varepsilon_0}$ such that*

$$\mathcal{L}^{(2)}_{t,(-,+),(a,b)} \prec [\Phi_t(a, b)]^2, \quad \forall a, b \in \mathbb{Z}_L^d. \tag{8.8}$$

(a) **Local laws**: *The following entrywise and averaged local laws hold:*

$$\|G_t - M\|_{\max} \prec \Psi_t, \quad \max_a |\mathrm{Tr}\left((G_t - M)\, E_a\right)| \prec \Psi_t^2. \tag{8.9}$$

(b) **Entrywise decay estimate**: *There exists a constant $c_g > 0$ such that, for any large constant $D > 0$ and all $a, b \in \mathbb{Z}_L^d$, the following estimate holds:*

$$\max_{x\in[a],y\in[b]} |(G_t - M)_{xy}| \prec \sum_{a',b'\in\mathbb{Z}_L^d} \Phi_t(a',b') e^{-c_g(|a'-a|+|b'-b|)} + \Psi_t e^{-c_g|a-b|} + W^{-D}. \tag{8.10}$$

Proof. This lemma was proved as Lemma 6.1 in [59] under the condition $g \le W^{-\varepsilon}$ for a small constant $\varepsilon > 0$. For $W^{-\varepsilon} < g \le \mathfrak{d}^{-1}$, the argument has to be modified in two places. First, in the proof of the entrywise bounds in (a) and (b), the expansion of M in powers of g is replaced by the bounds (8.6) and (8.7), and the self-consistent equation for the diagonal entries is inverted directly in the max norm, without any expansion in g. Indeed, the matrix $1 - tM^{(+,+)}$, $t \in [0,1]$, is diagonally dominant: its diagonal entries are equal to $1 - tm^2$, while, by (8.5), its off-diagonal row sums are equal to

$$t\sum_{b\neq a} \left|M_{ab}^{(\mathsf{B})}\right|^2 = t\big(1 - |m|^2\big) = \mathrm{Re}\,\big(1 - tm^2\big) - (1-t) - 2t(\mathrm{Im}\, m)^2 \le \big|1 - tm^2\big| - \min\{1, 2\kappa^2\},$$

where we used $\kappa \le \mathrm{Im}\, m \le 1$. Hence, $\|(1 - tM^{(+,+)})^{-1}\|_{\infty\to\infty} \le (\min\{1, 2\kappa^2\})^{-1}$ uniformly in $t \in [0,1]$ and $0 < g \le \mathfrak{d}^{-1}$. Second, the averaged local law in (8.9) does not follow from the argument of [59] when $g \gg W^{-\varepsilon}$. Instead, we derive it from the entrywise local law by a fluctuation averaging argument for the block averages $\mathrm{Tr}((G_t - M)E_a)$, which uses conditional expectations with respect to the independent Gaussian blocks of V, the bounds $|m| \le 1$ and $\mathrm{Im}\, m \ge \kappa$, the above stability bound, and the row bound $\max_a \sum_b |M_{ab}^{(\mathsf{B})}| \le C$ implied by (8.6) or (8.7). This argument does not use the smallness of g and holds uniformly in $0 < g \le \mathfrak{d}^{-1}$. □

Proof of Step 1 for Theorem 8.2. The proof of Step 1 depends on the following continuity estimates in Lemma 8.5. To make the dependence on the spectral parameter z and the coupling parameter g explicit, we denote the resolvent by $G_t(z_t, g) = (V_t + g\Psi - z_t)^{-1}$ and the corresponding G-loop by $\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}(z_t, g)$, where $z_t \equiv z_t(E, g)$ is defined in (2.37).

Lemma 8.5. *Fix any $\varepsilon \le s \le t \le 1$ for a constant $\varepsilon > 0$. Given any g satisfying the condition (2.10), let $g_s := g \cdot \sqrt{s/t}$, and assume that $\mathrm{Im}\, m(E, g_s) \ge \kappa$. Then, in the flow setting given by Definition 2.8 and Lemma 8.1, we have the following continuity estimates for the G-loops and (generalized) resolvent entries.*

(1) *Assume that the bound (3.6) holds at time s for the loops $\mathcal{L}_{s,\boldsymbol{\sigma},\mathbf{a}}^{(n)}(z_s, g_s)$ for each fixed $n \in \mathbb{N}$. Then, on the event $\Omega_t := \{\|G_t(z_t, g)\|_{\max} \le C_0\}$ for a constant $C_0 > 0$, the following estimate holds for any $n \ge 2$:*

$$\mathbf{1}(\Omega_t) \cdot \max_{\boldsymbol{\sigma},\mathbf{a}} \left|\mathcal{L}_{t,\boldsymbol{\sigma},\mathbf{a}}^{(n)}(z_t, g)\right| \prec \left(\frac{\eta_s}{\eta_t} \cdot W^{-d} B_{s,0}\right)^{n-1}. \tag{8.11}$$

(2) *Given any deterministic unit vectors $\mathbf{v}, \mathbf{w} \in \mathbb{C}^N$, suppose*

$$\mathrm{Im}(G_s)_{\mathbf{vv}}(z_s, g_s) \lesssim 1, \quad \mathrm{Im}(G_s)_{\mathbf{ww}}(z_s, g_s) \lesssim 1, \quad |(G_s)_{\mathbf{vw}}(z_s, g_s)| \lesssim 1, \tag{8.12}$$

with high probability, where we adopt the simplified notation of generalized matrix entries: given a matrix $\mathcal{A}$ and any vectors $\mathbf{v}, \mathbf{w}$, we denote $\mathcal{A}_{\mathbf{vw}} := \mathbf{v}^* \mathcal{A} \mathbf{w}$. Then, the following estimates hold with high probability at time t:

$$\mathrm{Im}(G_t)_{\mathbf{vv}}(z_t, g) \lesssim \eta_s/\eta_t, \quad (G_t)_{\mathbf{vw}}(z_t, g) \lesssim \eta_s/\eta_t. \tag{8.13}$$

Proof. The proof of this lemma follows exactly the same argument as that of Lemma 7.1 in [59], except that the factor $W^{-d}B_{s,0}$ in (8.11) corresponds to $(W^d \ell_s^d \eta_s)^{-1}$ therein. The assumption $\mathrm{Im}\, m(E, g_s) \ge \kappa$ ensures that Lemma 8.3 applies to $m(E, g_s)$ and $M(E, g_s)$, which are the deterministic limits of the resolvent at time s; in particular, $|m(E, g_s)| \le 1$ and $\eta_t \le \kappa^{-1}\eta_s$. □

With Lemmas 8.4 and 8.5, Step 1 of the proof of Theorem 8.2 for the block Anderson model (i.e., the proof of (2.85) and (2.86)) is the same as that in [59, Section 7.1], except that Lemma 8.5 is applied only with source times close to 1. This is needed because the deterministic part $g_0\Psi$ of H_t does not scale with

t: Lemma 8.5 compares the flow with parameters (E, g_0) at time u with the flow with parameters (E, g'), $g' := g_0\sqrt{s'/u}$, at an earlier time s', and the latter flow must lie in the bulk. Let $c_* := \min\{1/2, \mathfrak{d}\kappa^3/(6d)\}$.

- For $u \le 1-c_*$, we have $\eta_u = (1-u)\operatorname{Im} m(E, g_0) \ge c_*\kappa$, hence $\|G_u\| \le (c_*\kappa)^{-1}$ and, deterministically, $|\mathcal{L}^{(n)}_{u,\boldsymbol{\sigma},\mathbf{a}}| \le \|G_u\|^n W^{-(n-1)d} \lesssim (W^{-d}B_{u,0})^{n-1}$, since $B_{u,0} \ge (g^2+1)^{-1}$. Thus, (2.85) holds for $u \le 1-c_*$ without using Lemma 8.5, and (2.86) then follows as in the random band matrix case.
- For $u > 1-c_*$, we apply Lemma 8.5 only with the source time $s' := \max(s, 1-c_*)$. Then, $0 \le g_0 - g' \le g_0(1-s') \le c_*/\mathfrak{d}$, so property (4) of Lemma 8.3 (applied at $E+\mathrm{i}\eta$, followed by the limit $\eta \downarrow 0$) gives $\operatorname{Im} m(E, g')^3 \ge \operatorname{Im} m(E, g_0)^3 - 3d(g_0 - g') \ge \kappa^3/2$. Hence, the source flow satisfies the assumption of Lemma 8.5 with κ replaced by $\kappa/2$. If $s' = 1-c_*$, the bound (3.6) for the source flow at time s' holds deterministically, as in the previous item.

If $s' = s$, the bound (3.6) for the source flow at time s is provided by the induction hypothesis, in the following sense. By the scaling relations (8.2), the flow with parameters (E, g') at time s coincides with the flow of Lemma 8.1 for the coupling $g'/\sqrt{s}$ and the target spectral parameter $z' := z_s(E, g')/\sqrt{s}$, stopped at its final time $t_0(z') = s$. Here, $\operatorname{Im} m(z', g'/\sqrt{s}) = \sqrt{s}\operatorname{Im} m(E, g') \ge \kappa/4$, so $z' \in \mathbf{D}_{\kappa/4,\varepsilon/2}$ for large N, and the coupling $g'/\sqrt{s}$ satisfies (2.10) up to a constant factor. Therefore, as in [59, Section 7.1], the induction on t in the proof of Theorem 2.7 is carried out simultaneously for all such target spectral parameters and couplings. With these modifications, we omit the details.

Proof of Step 2 for Theorem 8.2. In Step 2, the technical lemmas—Lemmas 3.9 to 3.11 and 3.13—remain valid in the setting of the block Anderson model, except that the estimate (3.35) is modified as follows. Let $\mathcal{J}^{\ell}_{t,D} \ge W^{-d}$ be a *deterministic* control parameter such that $\widehat{\mathcal{J}}^{\ell}_{t,D} \prec \mathcal{J}^{\ell}_{t,D}$. Then, (3.35) continues to hold with $\widehat{\mathcal{J}}^{\ell}_{t,D}$ replaced by $\mathcal{J}^{\ell}_{t,D}$; that is,

$$(\mathcal{E}\otimes\mathcal{E})^M_{t,\boldsymbol{\sigma},\mathbf{a},\mathbf{a}} \prec \frac{1}{\eta_t}\left[\left(W^{-d}B_{t,0}\right)^{1/2} + \left(\mathcal{J}^{\ell}_{t,D}\right)^3\right]\cdot\left(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|)\right)^2. \tag{8.14}$$

Assume that (2.89) holds. We combine (2.71), (8.6) (or (8.7)), and (2.66) to obtain (3.36). Together with (2.89), this yields (3.37). Then, applying Lemma 8.4 gives the desired estimates (2.87) and (2.88). Finally, we prove (2.89). Using the same argument as between (3.38) and (3.43), we obtain that for all $u \in [s,t]$,

$$\max_{\boldsymbol{\sigma},\mathbf{a}}\left|(\mathcal{L}-\mathcal{K})^{(2)}_{u,\boldsymbol{\sigma},\mathbf{a}}\right| \prec \left(\frac{1-s}{1-u}\right)^{C_0+\frac{5}{4}}(W^{-d}B_{u,0})^{\frac{5}{4}} \le (W^{-d}B_{u,0})^{\frac{1}{5}}\cdot\left(W^{-d}\widetilde{\mathcal{T}}^{\ell^{(0)}}_{u,D}(|a-b|)\right), \tag{8.15}$$

where $\ell^{(0)} = 0$ as chosen in (3.44). Next, we implement a similar inductive argument as that below (3.44): assume that (3.46) holds for length scales K_u satisfying (3.47). Moreover, suppose we have the initial estimate

$$\widehat{\mathcal{J}}^{K_u}_{u,D} \prec \mathcal{J}^{K_u}_{u,D}, \quad \forall u \in [s,t], \quad \text{where} \quad \mathcal{J}^{K_u}_{u,D} \equiv (W^{-d}B_{t,0})^{\frac{1}{50}}. \tag{8.16}$$

We then define the stopping time

$$\tau := t \wedge T, \quad \text{with} \quad T := \inf\left\{u \ge s : \widehat{\mathcal{J}}^{K_u}_{u,D} \ge (W^{-d}B_{u,0})^{\frac{1}{100}}\right\}. \tag{8.17}$$

By Lemma 3.11, the estimate (3.50) remains valid. On the other hand, using (8.14) together with Lemma 3.6, the estimate (3.51) becomes

$$\int_s^{\tau}\mathrm{d}\mathcal{E}^M_{u,\boldsymbol{\sigma},\mathbf{a}} \prec \left[\left(W^{-d}B_{\tau,0}\right)^{1/4} + \sup_{u\in[s,\tau]}\left(\mathcal{J}^{K_u}_{u,D}\right)^{3/2}\right]\cdot\left(W^{-d}\widetilde{\mathcal{T}}^{K_\tau}_{\tau,D}(|a-b|)\right). \tag{8.18}$$

Applying the same reasoning as below (3.51) and invoking Grönwall's inequality, we obtain

$$\widehat{\mathcal{J}}^{K_u}_{u,D} \prec \left(\frac{1-s}{1-u}\right)^{C_0}\left[(W^{-d}B_{t,0})^{1/5} + \left(\mathcal{J}^{K_t}_{t,D}\right)^{3/2}\right], \quad \forall u \in [s,\tau]. \tag{8.19}$$

This implies that $T \ge t$ with high probability, so (8.19) holds for all $u \in [s,t]$. We then take the RHS of (8.19) as the new control parameter $\mathcal{J}^{K_u}_{u,D}$ and repeat the above argument. Iterating this procedure O(1) times yields the improved bound

$$\widehat{\mathcal{J}}^{K_u}_{u,D} \prec (|1-s|/|1-u|)^{C_0}(W^{-d}B_{t,0})^{1/5}, \quad \forall u \in [s,t]. \tag{8.20}$$

Next, we redefine the scale K'_u as in (3.53). Under this choice, using (8.20), we can re-establish the initial estimate (8.16) with K_u replaced by K'_u, provided the constant $\mathfrak{c}_d$ in (2.83) is chosen sufficiently small.

Finally, repeating the above procedure O(1) more times yields (2.89) at $u = t$. The same argument clearly applies to all $u \in [s, t]$.

It remains to justify the validity of Lemmas 3.9 to 3.11 and 3.13 for the block Anderson model. First, the proof of Lemma 3.9 in Section 3.4 carries over with only a minor modification: in the derivation of (3.55), we additionally make use of (8.6) or (8.7). The proofs of Lemmas 3.10 and 3.11 require more substantial changes in the graphical tools, which will be detailed in Appendix B.3 below. Finally, the proof of Lemma 3.13 (with (3.35) replaced by (8.14)) follows the same argument as in Section 3.5, with the following adjustments:

Proof of Lemma 3.13 for the block Anderson model. Most of the arguments in Section 3.5 carry over directly to the block Anderson model, provided we can establish the following resolvent estimate under the assumption (3.27): for any $a, b \in \mathbb{Z}_L^d$ and $x \in [a]$, $y \in [b]$,

$$|(G_t - M)_{xy}| \prec \Psi_t(|a - b|). \tag{8.21}$$

To prove (8.21), we apply (8.10) to obtain

$$|(G_t - M)_{xy}| \prec \sum_{|a'-a|\vee|b'-b|\le(\log W)^{3/2}} \Psi_t(|a' - b'|)e^{-c_g(|a'-a|+|b'-b|)} + \Psi_t(0)e^{-c_g|a-b|} + W^{-D} \prec \Psi_t(|a - b|),$$

where $D > 0$ is an arbitrarily large constant. Here, in the second step, we have used the conditions in (3.28) to obtain that

$$\begin{aligned} &\Psi_t(|a' - b'|) \prec \Psi_t(|a - b|), \quad \Psi_t(|a - b|) \gtrsim C_1^{-1} (|a - b| + 1)^{-C_2} \Psi_t(0) \ge W^{-D}, \\ &\Psi_t(0)e^{-c_g|a-b|} \lesssim C_1^{-1}(|a - b| + 1)^{-C_2}\Psi_t(0) \lesssim \Psi_t(|a - b|). \end{aligned} \tag{8.22}$$

Under the stronger assumption (3.31), a parallel argument yields the following analogue of (8.21):

$$|(G_t - M)_{xy}| \prec \big(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a - b|)\big)^{1/2}, \quad \forall x \in [a], y \in [b]. \tag{8.23}$$

We now prove (3.33) using the estimate (8.21). As in (3.66), we need to bound

$$\mathcal{S}_1 := W^d \sum_{|c-b|>|c-a|} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\boldsymbol{\sigma})^{(1)},\mathbf{a}(c,c)}, \quad \mathcal{S}_2 := W^d \sum_{|c-b|\le|c-a|} \mathcal{L}^{(6)}_{t,(\boldsymbol{\sigma}\otimes\boldsymbol{\sigma})^{(1)},\mathbf{a}(c,c)}.$$

By symmetry, it suffices to establish (3.33) for $\mathcal{S}_1$. In this case, we apply (3.61) with $c' = c$ to get that

$$\mathcal{S}_1 \lesssim \frac{1}{\eta_t} \max_{|c-b|\ge|a-b|/2} \Big(\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c,b,c,b)}\Big)^{1/2} \cdot \max_{\sigma\in\{+,-\}} \big|\mathcal{L}^{(3)}_{t,(\sigma,\sigma_2,-\sigma_2),(a,b,a)}\big|. \tag{8.24}$$

We write the (c, b, c, b)-loop as

$$\mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c,b,c,b)} = W^{-4d} \sum_{z',z''\in[c]} \sum_{y_1,y_2\in[b]} G_{y_1z'}(\sigma_1)G_{z'y_2}(-\sigma_1)G_{y_2z''}(\sigma_1)G_{z''y_1}(-\sigma_1).$$

Expanding each resolvent entry as

$$G_{\alpha\beta}(\sigma) = M_{\alpha\beta}(\sigma) + (G - M)_{\alpha\beta}(\sigma), \quad \forall\alpha, \beta \in \{y_1, y_2, z', z''\},\ \sigma \in \{+, -\}, \tag{8.25}$$

we bound every $(G - M)_{\alpha\beta}$ using (8.21) and each $M_{\alpha\beta}$ using (8.6) or (8.7). This yields (with $\varepsilon > 0$ a small constant):

$$\begin{aligned} \mathcal{L}^{(4)}_{t,\boldsymbol{\sigma}^{(\mathrm{alt})},(c,b,c,b)} &\prec \Psi_t^4(|b - c|) + \big[W^{-d}\Psi_t^3(|b - c|) + W^{-2d}\Psi_t^2(|b - c|) + W^{-3d}\Psi_t(|b - c|) + W^{-3d}\big]\, e^{-\varepsilon|b-c|} \\ &\lesssim \Psi_t^4(|b - c|) \lesssim \Psi_t^4(|a - b|), \end{aligned}$$

where the second step uses a bound analogous to (8.22), and the last step follows from $|c - b| \ge |a - b|/2$ together with the condition (3.28). Inserting this bound into (8.24), we obtain the estimate (3.68) again. To handle the 3-loop on the RHS of (3.68), we again expand each resolvent entry as in (8.25), bound every $(G - M)_{\alpha\beta}$ with (8.21), and each $M_{\alpha\beta}$ with (8.6) or (8.7). This gives the same bound as in (3.69), where the factor $\Psi_t^2(|a - b|)$ comes from the entries $G_{x'y}$ and G_{yx} between blocks a and b, while the short leg $G_{xx'}$ contributes a factor $\Psi_t(0) + W^{-d} \lesssim \Psi_t(0)$ by (8.9). (Specifically, $(G - M)_{xx'}$ contributes $\Psi_t(0)$, while the deterministic part $M_{xx'}$ yields a factor W^{-d}.) Substituting (3.69) into (3.68) concludes (3.33).

Next, the exponential decay bound (8.14) can be established by an argument analogous to that used for (3.35) below (3.70). First, it follows directly from the polynomial decay bound (3.33) whenever $|a - b| \le \ell_t^\dagger$. It therefore remains to consider the case (3.71), where we must control the three terms $\widetilde{\mathcal{S}}_i$, $i \in \{1, 2, 3\}$,

appearing in (3.72) with $c' = c$. The terms $\widetilde{\mathcal{S}}_1$ and $\widetilde{\mathcal{S}}_2$ can be bounded by the same argument used between (3.73) and (3.75), except that here we apply (8.23) in place of the estimates (3.3) and (3.31) employed there. For the term $\widetilde{\mathcal{S}}_3$, we bound all legs of the original 6-G-loop directly using (8.10) together with the control parameter $\mathcal{J}^{\ell}_{t,D}$. In this setting, (3.76) remains valid with $\widehat{\mathcal{J}}_{t,D}$ replaced by $\mathcal{J}^{\ell}_{t,D}$, where the 2-$\mathcal{K}$-loop $\mathcal{K}^{(2)}_{t,(-,+),(a,b)}$ defined in (2.71) is of order $\mathrm{O}(W^{-D})$ thanks to the exponential decay of the Θ-propagators in (2.66) and the decay of the 2-M-loop due to (8.6) or (8.7). Using (3.76) (with $\widehat{\mathcal{J}}^{\ell}_{t,D}$ replaced by $\mathcal{J}^{\ell}_{t,D}$) and an argument analogous to that below (8.21), we obtain that for any $x \in [a]$ and $y \in [b]$ with $|a-b| \gtrsim \ell_t^*$,

$$|(G_t - M)_{xy}| \prec \big(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|)\big)^{1/2} \implies |(G_t)_{xy}| \prec \big(W^{-d}\widetilde{\mathcal{T}}^{\ell}_{t,D}(|a-b|)\big)^{1/2}, \tag{8.26}$$

where the implication uses the bound (8.6) or (8.7). Finally, combining (8.26) with the same reasoning as below (3.76), we conclude the exponential decay bound (8.14). □

Proof of Steps 3–6 for Theorem 8.2. The proofs in Sections 4 to 6 extend verbatim to the block Anderson model, except for the arguments in Section 5.3, the proof of Lemma 6.1, and the proof of Lemma 6.2. In Section 5.3, the random band matrix case relies on Lemma 3.1 and follows the approach of [69, Section 5.3], whereas the corresponding arguments for the block Anderson model are based on Lemma 8.4 and parallel those in [59, Section 7.3]. We therefore omit the details. The proof of Lemma 6.2 is deferred to Appendix B.1 below. In Step 6, i.e., in the proofs of Lemmas 6.1 and 6.2, we use in addition the following twisted averaged law, which is a consequence of Steps 2 and 3, and the twisted-loop estimate (8.4) for $u \in [s,t]$, which we derive after Step 4 below.

Lemma 8.6 (Twisted averaged law)**.** *In the setting of Theorem 8.2, suppose that the estimates (2.87), (2.88), and (2.90) with $n \le 4$ hold at some $u \in [s,t]$. Then, we have*

$$\max_{\sigma\in\{+,-\}} \max_{c_1,c_2\in\mathbb{Z}^d_L} \Big|\operatorname{Tr}\Big(\big(G_u(\sigma) - M(\sigma)\big)P_{c_1c_2}\Big)\Big| \prec W^{-d}B_{u,0}.$$

In particular, this bound holds uniformly in $u \in [s,t]$ after Steps 2 and 3.

Proof. Since $\operatorname{Tr}((G_u^* - M^*)P_{c_1c_2}) = \overline{\operatorname{Tr}((G_u - M)P_{c_2c_1})}$, it suffices to consider $\sigma = +$. Denote $\delta_b := \operatorname{Tr}((G_u - M)E_b)$ and $\Lambda := \max_{c_1,c_2} |\operatorname{Tr}((G_u - M)P_{c_1c_2})|$. By (2.33) and (2.37), we have $H_u - z_u = V_u + M^{-1} + u\,m$, where $V_u := H_u - H_0$. Together with $\mathcal{S}[M] = mI_N$, this gives the identity

$$G_u - M = -M\,\underline{V_uG_u} + u\,M\,\mathcal{S}[G_u - M]\,G_u, \qquad \underline{V_uG_u} := V_uG_u + u\,\mathcal{S}[G_u]\,G_u, \tag{8.27}$$

where $\mathcal{S}[G_u - M] = W^d\sum_b \delta_b E_b$ by (2.41) and (2.29). Multiplying (8.27) by $P_{c_1c_2}$ and taking the trace, we obtain from $M = M^{(\mathsf{B})} \otimes I_{W^d}$ that

$$\operatorname{Tr}\big((G_u - M)P_{c_1c_2}\big) = -\operatorname{Tr}\big(M\underline{V_uG_u}P_{c_1c_2}\big) + u\sum_b \delta_b M^{(\mathsf{B})}_{c_2b}M^{(\mathsf{B})}_{bc_1} + u\sum_b \delta_b M^{(\mathsf{B})}_{c_2b}\operatorname{Tr}\big((G_u - M)P_{c_1b}\big).$$

By (2.88), the Cauchy–Schwarz inequality, and (8.5), the second term on the right-hand side is $\prec W^{-d}B_{u,0}$. By (2.88) and the row bound $\max_c \sum_b |M^{(\mathsf{B})}_{cb}| \le C$, which follows from (8.6) or (8.7), the third term is $\prec W^{-d}B_{u,0}\cdot\Lambda$. Since $W^{-d}B_{u,0} \le W^{-c}$ for a constant $c > 0$ by (2.10), this term can be absorbed into the left-hand side after taking the maximum over c_1, c_2.

It remains to show that $X := \operatorname{Tr}(M\underline{V_uG_u}P_{c_1c_2}) \prec W^{-d}B_{u,0}$, which follows from a high-moment estimate. Applying Gaussian integration by parts to the entries of V_u in one factor X of $\mathbb{E}|X|^{2p}$, $p \in \mathbb{N}$, the term in which the derivative acts on the resolvent in the same factor is cancelled by the renormalization in $\underline{V_uG_u}$. In the remaining terms, the derivative acts on one of the other $2p-1$ factors, and, using (8.27) again (i.e., $\underline{V_uG_u} = -M^{-1}(G_u - M) + u\,\mathcal{S}[G_u - M]G_u$), it produces sums of products of resolvent entries with deterministic weights. The leading one is $u\sum_{x,w} S_{xw}(G_uP_{c_1c_2}M)_{wx}(G_uP_{c_1c_2}M)_{xw}$ (or its analogue with one factor replaced by its complex conjugate). Since $(G_uP_{c_1c_2}M)_{wx} = W^{-d}M^{(\mathsf{B})}_{c_2[x]}(G_u)_{w\tilde{x}}$, where $\tilde{x} \in [c_1]$ has the same position inside its block as x, the Cauchy–Schwarz inequality and (8.5) bound it by

$$u\,W^{-d}\sum_b \big|M^{(\mathsf{B})}_{c_2b}\big|^2\,\mathcal{L}^{(2)}_{u,(+,-),(b,c_1)} \prec (W^{-d}B_{u,0})^2,$$

where we used (2.90) with $n = 2$. The other terms are bounded in the same way, using (2.87) and (2.90) with $n \le 4$, by $(W^{-d}B_{u,0})^2$ times a product of $2p-2$ factors X or $\overline{X}$. This yields $\mathbb{E}|X|^{2p} \prec (W^{-d}B_{u,0})^2\,\mathbb{E}|X|^{2p-2}$, and hence $X \prec W^{-d}B_{u,0}$ by Hölder's and Markov's inequalities, since p is arbitrary. □

Proof of the twisted-loop estimate (8.4) for $u \in [s,t]$. Fix $\boldsymbol{\sigma} = (\sigma_1, \sigma_2)$ and $c_1, c_2 \in \mathbb{Z}_L^d$, and regard $Z_u(a) := \widetilde{\mathcal{L}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)} - \widetilde{\mathcal{K}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)}$ as a vector indexed by $a \in \mathbb{Z}_L^d$. The derivation of the loop hierarchy (2.47) by Itô's formula does not use the special form of the matrices E_{a_i}, so it also applies to $\widetilde{\mathcal{L}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)}$, viewed as a 2-loop in which E_{a_2} is replaced by $P_{c_1c_2}$. Since $S^{(\mathsf{B})} = I_{L^d}$, the cut-and-glue term in (2.47) then becomes $W^d \sum_b \mathcal{L}^{(2)}_{u,\boldsymbol{\sigma},(a,b)} \widetilde{\mathcal{L}}_{u,\boldsymbol{\sigma},(b;c_1,c_2)}$. On the other hand, since $\partial_u \Theta_u^{(\sigma_1,\sigma_2)} = \Theta_u^{(\sigma_1,\sigma_2)} M^{(\sigma_1,\sigma_2)} \Theta_u^{(\sigma_1,\sigma_2)}$, we have $\partial_u \widetilde{\mathcal{K}}_{u,\boldsymbol{\sigma},(a;c_1,c_2)} = W^d \sum_b \mathcal{K}^{(2)}_{u,\boldsymbol{\sigma},(a,b)} \widetilde{\mathcal{K}}_{u,\boldsymbol{\sigma},(b;c_1,c_2)}$ by (2.71). Hence, with the evolution kernel $\widetilde{\mathcal{U}}_{v,u} := \Theta_u^{(\sigma_1,\sigma_2)} (\Theta_v^{(\sigma_1,\sigma_2)})^{-1}$ acting on the index a, we have

$$Z_u = \widetilde{\mathcal{U}}_{s,u} Z_s + \int_s^u \widetilde{\mathcal{U}}_{v,u} \Big(\widetilde{\mathcal{B}}_{1,v} + \widetilde{\mathcal{B}}_{2,v} + \widetilde{\mathcal{E}}_v^{\mathring{G}} \Big) \mathrm{d}v + \int_s^u \widetilde{\mathcal{U}}_{v,u} \, \mathrm{d}\widetilde{\mathcal{E}}_v^{M}, \tag{8.28}$$

where $\widetilde{\mathcal{B}}_{1,v}(a) := W^d \sum_b (\mathcal{L} - \mathcal{K})^{(2)}_{v,\boldsymbol{\sigma},(a,b)} \widetilde{\mathcal{K}}_{v,\boldsymbol{\sigma},(b;c_1,c_2)}$ and $\widetilde{\mathcal{B}}_{2,v}(a) := W^d \sum_b (\mathcal{L} - \mathcal{K})^{(2)}_{v,\boldsymbol{\sigma},(a,b)} Z_v(b)$, while $\widetilde{\mathcal{E}}_v^{\mathring{G}}$ and $\mathrm{d}\widetilde{\mathcal{E}}_v^{M}$ are the light-weight and martingale terms defined as in (2.49) and (2.48), with E_{a_2} replaced by $P_{c_1c_2}$. In particular, $\widetilde{\mathcal{E}}_v^{\mathring{G}}(a)$ is a sum of twisted 3-loops $\mathrm{Tr}(G_v(\sigma_1) E_b G_v(\sigma_1) E_a G_v(\sigma_2) P_{c_1c_2})$ and $\mathrm{Tr}(G_v(\sigma_1) E_a G_v(\sigma_2) E_b G_v(\sigma_2) P_{c_1c_2})$, multiplied by $W^d \mathrm{Tr}(\mathring{G}_v(\sigma_i) E_b)$ and summed over b.

We reduce twisted loops to G-loops using the following two facts. First, by the Cauchy–Schwarz inequality, for any $N \times N$ matrices X_1, X_2,

$$\big| \mathrm{Tr}\,(X_1 E_a X_2 P_{c_1c_2}) \big|^2 \le \mathrm{Tr}\,(E_{c_2} X_1 E_a X_1^*)\ \mathrm{Tr}\,(E_a X_2 E_{c_1} X_2^*);$$

for example, $|\widetilde{\mathcal{L}}_{v,\boldsymbol{\sigma},(a;c_1,c_2)}|^2 \le \mathcal{L}^{(2)}_{v,(\sigma_1,-\sigma_1),(a,c_2)} \mathcal{L}^{(2)}_{v,(\sigma_2,-\sigma_2),(a,c_1)}$. Second, since $G - G^* = 2\mathrm{i}\eta G G^* = 2\mathrm{i}\eta G^* G$ and $\sum_a E_a = W^{-d} I_N$, we have the Ward identity

$$\sum_a \widetilde{\mathcal{L}}_{v,(\sigma,-\sigma),(a;c_1,c_2)} = \frac{1}{2\mathrm{i} W^d \eta_v} \Big(\mathrm{Tr}\,(G_v P_{c_1c_2}) - \mathrm{Tr}\,(G_v^* P_{c_1c_2}) \Big), \quad \sigma \in \{+,-\},$$

and the same identity holds for $\widetilde{\mathcal{K}}$, with $\mathrm{Tr}(G_v P_{c_1c_2})$ and $\mathrm{Tr}(G_v^* P_{c_1c_2})$ replaced by $M^{(\mathsf{B})}_{c_2c_1}$ and $(M^{(\mathsf{B})})^*_{c_2c_1}$, respectively. Using these two facts together with the estimates of Steps 1–4, we obtain the following bounds, uniformly in $v \in [s,u]$, $\boldsymbol{\sigma}$, a, c_1, and c_2:

- $|\widetilde{\mathcal{B}}_{1,v}(a)| + |\widetilde{\mathcal{E}}_v^{\mathring{G}}(a)| \prec \eta_v^{-1} (W^{-d} B_{v,0})^2$. For $\widetilde{\mathcal{B}}_{1,v}$, this follows from (2.91) with $n = 2$ and the bound $\sum_b |\widetilde{\mathcal{K}}_{v,\boldsymbol{\sigma},(b;c_1,c_2)}| \le (W^d(1-v))^{-1}$, which is a consequence of (2.65), the symmetry of $\Theta_v^{(\sigma_1,\sigma_2)}$, and (8.5). For $\widetilde{\mathcal{E}}_v^{\mathring{G}}$, we use (2.88), and we bound the sum over b of the twisted 3-loops by $(W^d\eta_v)^{-1} W^{-d} B_{v,0}$, using the above Cauchy–Schwarz inequality with the vertex E_b in place of E_a, the Cauchy–Schwarz inequality in b, Ward's identity (2.56) for the resulting 2- and 4-loops at the vertex b, and (2.90).
- The quadratic variation of the martingale term is bounded in the same way, by $\eta_v^{-1} (W^{-d} B_{v,0})^{4-\delta}$ for any constant $\delta > 0$ (in the sense of $\prec$).
- If $|Z_v(b)| \le \Xi (W^{-d} B_{v,0})^{\gamma}$ for all $v \in [s,u]$ and $b \in \mathbb{Z}_L^d$, for some $\gamma \in [1,2]$ and $\Xi \ge 1$, then $|\widetilde{\mathcal{B}}_{2,v}(a)| \prec \Xi\, \eta_v^{-1} (W^{-d} B_{v,0})^{\gamma + 1/6}$, by (2.89) and (2.83) (provided $\mathfrak{c}_d$ is small enough).
- All these sources, as well as $Z_v(a)$, are bounded by W^{-D} for any constant $D > 0$ when $|a - c_1| \ge W^{\varepsilon} \ell_v$, by the decay of the Θ-propagators in (2.66), of $M^{(\mathsf{B})}$ in (8.6) or (8.7), and of the 2-G-loops in (2.89).

With these bounds, (8.4) at time u follows from (8.28) in the same way as (2.91) for $n = 2$ in Step 4: we use the one-index analogues of the evolution kernel estimates in Lemmas 4.15 to 4.17 for $\widetilde{\mathcal{U}}_{v,u}$ (which are proved in the same way, with c_1 playing the role of the second index), the Burkholder–Davis–Gundy inequality for the martingale term, and a bootstrap in γ starting from $\gamma = 1$; the initial bound $|Z_v(a)| \le |\widetilde{\mathcal{L}}_{v,\boldsymbol{\sigma},(a;c_1,c_2)}| + |\widetilde{\mathcal{K}}_{v,\boldsymbol{\sigma},(a;c_1,c_2)}| \prec W^{-d} B_{v,0}$ follows from the above Cauchy–Schwarz inequality, (2.90), (2.66), and (8.5). Finally, in the regime $1 - u \le g^2/L^2$ and for $\sigma_1 \ne \sigma_2$, the zero mode of Z_u is controlled directly by the two Ward identities above and Lemma 8.6, which give $|\sum_a Z_u(a)| \prec (W^d \eta_u)^{-1} W^{-d} B_{u,0}$.

Modification in the proof of Lemma 6.1. Since $M = M^{(\mathsf{B})} \otimes I_{W^d}$ is not a multiple of the identity matrix, the argument of [69, Lemma 5.15] has to be modified as follows. Taking $P_{c_1c_2} = E_a$ in the identity

derived in the proof of Lemma 8.6 and using $\mathbb{E}\,\underline{V_u G_u} = 0$, which holds by Gaussian integration by parts, we obtain that $\delta_b = \mathrm{Tr}((G_u - M)E_b)$ satisfies

$$\mathbb{E}\delta_a - u\sum_b M_{ab}^{(+,+)}\,\mathbb{E}\delta_b = u\sum_b M_{ab}^{(\mathsf{B})}\,\mathbb{E}\Big[\delta_b\ \mathrm{Tr}\left((G_u - M)P_{ab}\right)\Big], \qquad a \in \mathbb{Z}_L^d, \tag{8.29}$$

where $M_{ab}^{(+,+)} = (M_{ab}^{(\mathsf{B})})^2$. For $b \neq a$, the right-hand side contains the twisted traces $\mathrm{Tr}((G_u - M)P_{ab})$, which are controlled neither by (2.88) nor by (2.91). By Lemma 8.6, (2.88), and the row bound $\max_a \sum_b |M_{ab}^{(\mathsf{B})}| \leq C$, the right-hand side of (8.29) is $\prec (W^{-d}B_{u,0})^2$ (to pass from $\prec$ to expectations, we use the deterministic bound $\|G_u\| \leq \eta_u^{-1}$ on the exceptional event). Inverting $1 - uM^{(+,+)}$, whose inverse $\Theta_u^{(+,+)}$ is bounded in the $(\infty \to \infty)$-norm by (2.67), yields (6.2) for the block Anderson model.

Appendix A. Proofs of some deterministic estimates

This appendix is devoted to proving several deterministic estimates used in the main proof.

A.1. **Proof of Lemma 2.19.** Property 1 follows from the underlying symmetry of $S^{(\mathsf{B})}$ in the random band matrix model, where $M^{(\sigma_1,\sigma_2)}$ is a scalar matrix, or from the symmetry of $M^{(\sigma_1,\sigma_2)}$ in the block Anderson model, where $S^{(\mathsf{B})}$ is the identity matrix. Property 2 is a consequence of the translation invariance of the matrix $M^{(\sigma_1,\sigma_2)}S^{(\mathsf{B})}$. Property 3 follows from the fact that $\Theta_t^{(\sigma_1,\sigma_2)}$ is a rational function of $S^{(\mathsf{B})}$ in the random band matrix model, and from the fact that $S^{(\mathsf{B})} = I_{L^d}$ in the block Anderson model. To prove property 4, we expand $\Theta_t^{(\sigma_1,\sigma_2)}$ using the Taylor series

$$\Theta_t^{(\sigma_1,\sigma_2)} = \sum_{k=0}^{\infty} t^k \big(M^{(\sigma_1,\sigma_2)}S^{(\mathsf{B})}\big)^k. \tag{A.1}$$

Since $|M_{ab}^{(\sigma_1,\sigma_2)}| \leq M_{ab}^{(+,-)}$ for both models, it follows that for any $\sigma_1, \sigma_2 \in \{+,-\}$,

$$|\Theta_{t,ab}^{(\sigma_1,\sigma_2)}| \leq \sum_{k=0}^{\infty} t^k \big(M^{(+,-)}S^{(\mathsf{B})}\big)_{ab}^k = \Theta_{t,ab}^{(+,-)}.$$

The estimate (2.65) then follows directly from this inequality and the identity $\sum_b \Theta_{t,ab}^{(+,-)} \equiv (1-t)^{-1}$, which holds because $M^{(+,-)}S^{(\mathsf{B})}$ is doubly stochastic. This uses $|m| = 1$ in the random band matrix model, and Ward's identity (8.5) in the block Anderson model.

To prove (2.67) for the random band matrix model, we use the following shifted Taylor expansion:

$$\Theta_t^{(+,+)} = \frac{1}{1+\varepsilon}\sum_{k=0}^{\infty}\left(\frac{tm^2S^{(\mathsf{B})} + \varepsilon}{1+\varepsilon}\right)^k, \tag{A.2}$$

where $\varepsilon > 0$ is a positive constant. As shown in [14, Lemma 4.2], one has $\|(tm^2S^{(\mathsf{B})}+\varepsilon)/(1+\varepsilon)\|_{\infty\to\infty} \leq 1-c$ for some constant $c > 0$ depending on ε and κ. Applying this estimate to the expansion (A.2) and noticing that the off-diagonal entries of $\left(tm^2S^{(\mathsf{B})} + \varepsilon\right)/(1+\varepsilon)$ contain a g^2 factor (by the definition (2.5)), we derive the bound (2.67). For the block Anderson model, we instead use the Taylor expansion

$$\left(1 - tM^{(+,+)}\right)_{0a}^{-1} = \left[\left(1 - tm^2\right)I - tM'\right]_{0a}^{-1} = \sum_{k=0}^{\infty}\left(1 - tm^2\right)^{-(k+1)}(tM')_{0a}^k, \tag{A.3}$$

where $M' := M^{(+,+)} - m^2 I$ is obtained from $M^{(+,+)}$ by setting its diagonal entries to zero. By property (2) of Lemma 8.3, there exists a constant $\varepsilon > 0$ such that $|1 - tm^2| \geq \varepsilon$ and

$$\|M'\|_{\infty\to\infty} = \sum_{a\neq 0}|M'_{0a}| = 1 - |m|^2 \leq (1-\varepsilon)|1 - tm^2|, \quad \forall t \in [0,1]. \tag{A.4}$$

Applying this bound to (A.3), we deduce the following estimate: there exists a constant $C > 0$ such that for any $\delta > 0$,

$$\left|\left(1 - tM^{(+,+)}\right)_{0a}^{-1}\right| \leq \sum_{0\leq k\leq \delta|a|}\left(1 - tm^2\right)^{-(k+1)}(tM')_{0a}^k + \varepsilon^{-1}\sum_{k>\delta|a|}(1-\varepsilon)^k$$

$$\leq \sum_{0\leq k\leq \delta|a|} C^k \exp(-c|a|) + \varepsilon^{-2}(1-\varepsilon)^{\delta|a|} \leq C' e^{-c'|a|}. \tag{A.5}$$

In the second inequality, we use the exponential decay of M' by (8.6) or (8.7), together with the convolution bound $\sum_b e^{-c|a_1-b|-c|a_2-b|} \lesssim e^{-c|a_1-a_2|}$ for all $a_1, a_2 \in \mathbb{Z}_L^d$, and in the third inequality we choose $\delta > 0$ sufficiently small so that $C^\delta \leq e^{c/2}$. The constants c', C' depend only on C, c, ε, and δ. The bound (A.5) yields (2.67) in the regime $g \gtrsim 1$. When $g \ll 1$, we return to (A.3) and note that $\|M'\|_{\infty\to\infty} \lesssim g^2$ by (8.6). In this case, for $a \neq 0$, the same argument as in (A.5) applies, leading to the bound (2.67).

The bounds (2.66) and (2.68)–(2.70) follow directly from (2.67) in the case $\sigma_1 = \sigma_2$. It therefore remains to consider the case $\sigma_1 \neq \sigma_2$. In this case, the estimates (2.70) and (2.69) were proved in [68] by analyzing the Fourier series of $\mathring{\Theta}_t^{(+,-)}$ through a standard summation-by-parts argument.[12] Specifically, (2.70) is proved in Lemma 3.1 of [68], while (2.69) appears as equation (E.19) therein. Although the bound (2.68) is not stated explicitly in [68], its proof proceeds analogously to those of (2.70) and (2.69), by applying the same summation-by-parts technique to the corresponding Fourier expansion. We therefore omit the details. These bounds (2.68)–(2.70) have also been derived for dimension $d = 2$ in [59, Lemma 3.10], where the summation-by-parts argument is explained in Appendix B. The same reasoning extends directly to dimensions $d \geq 3$.

Finally, it remains to prove the bound (2.66) for the case $\sigma_1 \neq \sigma_2$. Its proof is similar to that of [28, Lemma 2.14], using the Taylor expansion, along with the random walk representation of $(S^{(\mathsf{B})})^k$ (for the random band matrix model), or $(M^{(+,-)})^k$ (for the block Anderson model). More precisely, for the random band matrix model, we consider the random walk $\{X_k : k \geq 0\}$ on $\mathbb{Z}^d$, with transition probabilities $\{p(0,a) = S_{0a}^{(\mathsf{B})} : a \in \mathbb{Z}^d\}$. First, given any $a \in \mathbb{Z}^d \setminus \{0\}$, by applying the Bernstein inequality to $X_k \cdot \widehat{a}$ with $\widehat{a}$ denoting the unit vector $a/\|a\|_2$, we can derive the following large deviation estimate: there exist constants $c, C > 0$ (which do not depend on a) such that

$$\mathbb{P}_0(X_k = a) \leq C \exp\left(-c\left(\frac{|a|^2}{g^2 k} \wedge |a|\right)\right), \quad \forall a \in \mathbb{Z}^d,\ k \geq 1.$$

On the other hand, using the local CLT for X_k (see e.g., [48, Section 2]), we obtain that

$$\mathbb{P}_0(X_k = a) \leq 1 \wedge \left(C(g^2 k)^{-d/2}\right), \quad \forall a \in \mathbb{Z}^d,\ k \geq 1.$$

Combining the above two bounds and using the argument below equation (8.3) of [28], we obtain the large deviation bound

$$\mathbb{P}_0(X_k = a) \leq C\left((g^2 k)^{-d/2} \wedge 1\right) \exp\left(-c\left(\frac{|a|^2}{g^2 k} \wedge |a|\right)\right), \quad \forall a \in \mathbb{Z}^d,\ k \geq 1,$$

for some constants $c, C > 0$. Projecting the random walk onto the torus $\mathbb{Z}_L^d$, i.e., summing this bound over the periodic images of a, we obtain that

$$\left(M^{(+,-)} S^{(\mathsf{B})}\right)_{0a}^k \leq C\left((g^2 k)^{-d/2} \wedge 1 + L^{-d}\right) \exp\left(-c\left(\frac{|a|^2}{g^2 k} \wedge |a|\right)\right), \quad \forall a \in \mathbb{Z}_L^d,\ k \geq 1, \tag{$\star$}$$

for some constants $c, C > 0$ that do not depend on L.

For the block Anderson model, the matrix $P := M^{(+,-)} S^{(\mathsf{B})} = M^{(+,-)}$, $P_{ab} = |M_{ab}^{(\mathsf{B})}|^2$, is symmetric, translation invariant, and doubly stochastic by properties (1) and (2) of Lemma 8.3; in particular, $P_{0,-a} = P_{a0} = P_{0a}$. However, P depends on L and m, and it is not the periodization of a fixed transition kernel on $\mathbb{Z}^d$, so the above projection argument and the local CLT do not apply. Instead, we prove ($\star$) directly on the torus, with constants $c, C > 0$ depending only on d, κ, and $\mathfrak{d}$. For $x \in \mathbb{Z}_L^d$ and $j \in [\![d]\!]$, let $|x_j|_L$ denote the periodic distance between the j-th coordinate of x and 0, so that $|x| = \max_j |x_j|_L$.

Step 1: Fourier symbol. The eigenvalues of $\Psi^{(\mathsf{B})}$ are $\psi_q := 2\sum_{j=1}^d \cos q_j$, $q \in (\mathbb{Z}_L^d)^* := (2\pi/L)\mathbb{Z}_L^d \subset (-\pi, \pi]^d$, with eigenvectors $(L^{-d/2} e^{\mathrm{i}q\cdot a})_{a\in\mathbb{Z}_L^d}$. Hence, setting $f_q := (g\psi_q - E - m)^{-1}$, we have $M_{0a}^{(\mathsf{B})} = L^{-d}\sum_{q\in(\mathbb{Z}_L^d)^*} f_q e^{-\mathrm{i}q\cdot a}$, and Ward's identity (8.5) reads $L^{-d}\sum_q |f_q|^2 = 1$. In particular, $|f_q| \geq 1$ for some q, which gives $|E| \leq 2dg + 2$ (recall $|m| \leq 1$), and therefore $|f_q| \geq R^{-1}$ for all q, where $R := 4dg + 3 \lesssim 1$. The

[12] In [68], the assumption $g \ll 1$ was imposed; however, the same argument (in fact, slightly simpler) applies when $g \gtrsim 1$.

Fourier transform of $P_{0\cdot}$ is

$$\widehat{P}(p) := \sum_{a\in\mathbb{Z}_L^d} P_{0a} e^{-\mathrm{i}p\cdot a} = \frac{1}{L^d} \sum_{q\in(\mathbb{Z}_L^d)^*} f_q \overline{f_{q+p}}, \qquad p \in (\mathbb{Z}_L^d)^*,$$

which is real since $P_{0,-a} = P_{0a}$. Using $L^{-d}\sum_q |f_q|^2 = 1$, $f_{q+p} - f_q = g(\psi_q - \psi_{q+p}) f_q f_{q+p}$, and $L \geq 3$, we obtain that

$$1 - \widehat{P}(p) = \frac{1}{2L^d} \sum_q |f_{q+p} - f_q|^2 \geq \frac{g^2}{2R^4} \cdot \frac{1}{L^d} \sum_q (\psi_{q+p} - \psi_q)^2 = \frac{2g^2}{R^4} \sum_{j=1}^d (1 - \cos p_j).$$

On the other hand, $1 + \widehat{P}(p) = \sum_a P_{0a}(1 + \cos(p\cdot a)) \geq 2P_{00} = 2|m|^2 \geq 2(\operatorname{Im} m)^2 \gtrsim 1$ by property (2) of Lemma 8.3. Since $g \leq \mathfrak{d}^{-1}$, these two bounds yield that, for a constant $c_1 > 0$,

$$|\widehat{P}(p)| \leq 1 - c_1 g^2 |p|^2, \quad \forall p \in (\mathbb{Z}_L^d)^*.$$

Step 2: on-diagonal bound. By Fourier inversion and the bound in Step 1, $P_{00}^{2k} = L^{-d}\sum_p \widehat{P}(p)^{2k} \leq L^{-d}\sum_p e^{-2c_1 k g^2 |p|^2} \leq C\big((g^2 k)^{-d/2} + L^{-d}\big)$, where the term L^{-d} comes from $p = 0$. Moreover, by the Cauchy–Schwarz inequality, the symmetry and the translation invariance of P, we have $P_{0a}^{2k} = \sum_b P_{0b}^k P_{ba}^k \leq (P_{00}^{2k} P_{aa}^{2k})^{1/2} = P_{00}^{2k}$ and $P_{0a}^{2k+1} = \sum_b P_{0b} P_{ba}^{2k} \leq P_{00}^{2k}$. Together with $P_{0a}^k \leq 1$, this gives

$$\max_a P_{0a}^k \leq C\left((g^2 k)^{-d/2} \wedge 1 + L^{-d}\right), \quad \forall k \geq 1.$$

Step 3: tail bound. Fix $j \in [\![d]\!]$ and $\mu \geq 0$, and let $F_\mu(x) := \cosh(\mu |x_j|_L)$. For $x, y \in \mathbb{Z}_L^d$, choose integer representatives $\widetilde{x}_j, \widetilde{y}_j$ of x_j, y_j with $|\widetilde{x}_j| = |x_j|_L$ and $|\widetilde{y}_j| = |y_j|_L$. Since $|x_j \pm y_j|_L \leq |\widetilde{x}_j \pm \widetilde{y}_j|$, we have $F_\mu(x+y) + F_\mu(x-y) \leq 2F_\mu(x)\cosh(\mu |y_j|_L)$. Together with $P_{0,-y} = P_{0y}$, this implies

$$(PF_\mu)(x) = \frac{1}{2}\sum_y P_{0y}\big(F_\mu(x+y) + F_\mu(x-y)\big) \leq F_\mu(x) \sum_y P_{0y} \cosh(\mu|y|) \leq \big(1 + Cg^2\mu^2\big) F_\mu(x), \quad \forall\, 0 \leq \mu \leq c,$$

where in the last step we used $\sum_{y\neq 0} P_{0y}|y|^2 e^{2c|y|} \leq Cg^2$, which follows from (8.6) when $g < (2C)^{-1}$ and from (8.7) when $g \geq (2C)^{-1}$. Iterating this bound and applying the exponential Chebyshev inequality, we get $\sum_{x:|x_j|_L \geq r} P_{0x}^k \leq e^{Cg^2\mu^2 k}/\cosh(\mu r)$ for all $r \geq 0$. Optimizing over $\mu \in [0, c]$ and taking a union bound over $j \in [\![d]\!]$, we obtain that

$$\sum_{x:|x|\geq r} P_{0x}^k \leq 2d \exp\left(-c\left(\frac{r^2}{g^2 k} \wedge r\right)\right), \quad \forall r \geq 0,\ k \geq 1.$$

Step 4: conclusion. For $k \geq 2$, write $k = k_1 + k_2$ with $k_1 = \lceil k/2 \rceil$ and $k_2 = \lfloor k/2 \rfloor$. Since $|b| \geq |a|/2$ or $|a - b| \geq |a|/2$ for every $b \in \mathbb{Z}_L^d$, we have

$$P_{0a}^k = \sum_b P_{0b}^{k_1} P_{ba}^{k_2} \leq \max_x P_{0x}^{k_2} \sum_{b:|b|\geq|a|/2} P_{0b}^{k_1} + \max_x P_{0x}^{k_1} \sum_{b:|b-a|\geq|a|/2} P_{ab}^{k_2}.$$

Applying the bounds of Steps 2 and 3 to the right-hand side yields ($\star$) for $k \geq 2$, while the case $k = 1$ follows directly from (8.6) and (8.7).

Finally, for both models, applying ($\star$) to the expansion (A.1) and summing over k, we can derive the desired bound (2.66). Since the argument closely follows that in [28, Section 8], we omit the details. In fact, the proof here is somewhat simpler than in [28], because $d = 2$ is the critical dimension, where logarithmic corrections (e.g., $\log L$) appear. In contrast, for $d \geq 3$, all relevant series (or integrals) are summable, resulting in a dimension-dependent constant C_d in (2.66).

A.2. **Proofs of evolution kernel estimates.** In this subsection, we present the proofs of Lemmas 4.15 to 4.17. Parts of the proofs parallel those in [69, 28, 59] for random band matrices and for the block Anderson model in dimensions 1 and 2. However, certain key arguments must be adapted to handle the higher-dimensional setting $d \geq 3$. We therefore outline the proofs of the evolution kernel estimates, emphasizing the modifications needed compared to the arguments in [69, 28, 59].

Proof of Lemma 4.15. Notice the following decomposition:

$$\frac{1-s\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}=1+\Xi^{(i)},\quad \text{where}\quad \Xi^{(i)}:=(t-s)\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}\Theta_t^{(\sigma_i,\sigma_{i+1})}. \tag{A.6}$$

Using (2.65), along with (8.5) in the case of block Anderson model, we obtain that

$$\|\Xi^{(i)}\|_{\infty\to\infty}=\max_a\sum_b|\Xi^{(i)}_{ab}|\le(t-s)\,\|\Theta_t^{(\sigma_i,\sigma_{i+1})}\|_{\infty\to\infty}\le\frac{t-s}{1-t}. \tag{A.7}$$

Together with (A.6), this implies that

$$\left\|\frac{1-s\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}\right\|_{\infty\to\infty}\le\frac{1-s}{1-t}.$$

With this estimate, we conclude (4.51) immediately using the definition (3.12). □

Proof of Lemma 4.16. With the decomposition (A.6), we can express $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}\circ\mathcal{A}$ as

$$\left(\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}\circ\mathcal{A}\right)_{\mathbf{a}}=\sum_{\mathbf{b}\in(\mathbb{Z}_L^d)^n}\prod_{i=1}^n\left(\delta_{a_ib_i}+\Xi^{(i)}_{a_ib_i}\right)\cdot\mathcal{A}_{\mathbf{b}}=\sum_{\mathbf{b}\in(\mathbb{Z}_L^d)^n}\sum_{A\subset[\![n]\!]}\prod_{i\in A}\delta_{a_ib_i}\cdot\prod_{i\in A^c}\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\mathbf{b}}\,. \tag{A.8}$$

By the estimate (2.66) (along with (8.6) or (8.7) in the case of block Anderson model), we have that

$$\Xi^{(i)}_{a_ib_i}\lesssim(1-s)\frac{(g^2+|1-t|)^{-1}}{(|a_i-b_i|+1)^{d-2}}e^{-c|a_i-b_i|/\ell_t} \tag{A.9}$$

for a constant $c>0$. We claim that for any subset A with $|A|=k\in[\![1,n]\!]$,

$$\sum_{\mathbf{b}\in(\mathbb{Z}_L^d)^n}\prod_{i\in A}\delta_{a_ib_i}\cdot\prod_{i\in A^c}\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\boldsymbol{b}}\le W^{C\varepsilon}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^{n-k}\|\mathcal{A}\|_\infty+W^{-D+C}, \tag{A.10}$$

for a constant C that does not depend on ε or D, while for $A=\emptyset$, we claim that

$$\sum_{\mathbf{b}\in(\mathbb{Z}_L^d)^n}\prod_{i=1}^n\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\boldsymbol{b}}\le W^{C\varepsilon}\frac{\ell_t^2}{\ell_s^2}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^n\|\mathcal{A}\|_\infty+W^{-D+C}. \tag{A.11}$$

Note that combining (A.10) and (A.11) concludes the proof of (4.53).

To show the estimate (A.10), we assume that $A=[\![1,k]\!]$ without loss of generality. With the notations $\mathbf{a}'=(a_1,\dots,a_k)$ and $\mathbf{b}'=(b_{k+1},\dots,b_n)$, we can bound the LHS of (A.10) as

$$\begin{aligned}\sum_{\mathbf{b}'\in(\mathbb{Z}_L^d)^{n-k}}\prod_{i=k+1}^n\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\mathbf{a}',\mathbf{b}'}&\lesssim\frac{(1-s)^{n-k}}{(g^2+|1-t|)^{n-k}}\|\mathcal{A}\|_\infty\sum_{\mathbf{b}'}\prod_{i=k+1}^n\frac{\mathbf{1}\left(|b_i-a_1|\le W^\varepsilon\ell_s\right)}{|a_i-b_i|^{d-2}+1}+W^{-D+(n-k)}\\&\lesssim(W^{2\varepsilon})^{n-k}\left(\frac{\ell_s^2|1-s|}{g^2+|1-t|}\right)^{n-k}\|\mathcal{A}\|_\infty+W^{-D+(n-k)}\\&\lesssim\left(W^{2\varepsilon}\frac{g^2+|1-s|}{g^2+|1-t|}\right)^{n-k}\|\mathcal{A}\|_\infty+W^{-D+(n-k)},\end{aligned}$$

where we have used (A.9), the decay property (4.52) for $\mathcal{A}_{\mathbf{b}}$, and

$$(1-s)\ell_s^2\asymp g^2+|1-s|,\quad\text{for}\quad s\le1-g^2/L^2, \tag{A.12}$$

along with the condition $(1-t)/(1-s)\ge W^{-1}$ and the bound (A.7) in getting the $W^{-D+(n-k)}$ term. This concludes (A.10) for any constant $C>2(n-k)$. For (A.11), with the notation $\mathbf{b}'=(b_2,\dots,b_n)$, we get that

$$\begin{aligned}\sum_{\mathbf{b}}\prod_{i=1}^n\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\mathbf{b}}&=\sum_{\mathbf{b}}\prod_{i=1}^n\Xi^{(i)}_{a_ib_i}\cdot\mathcal{A}_{\mathbf{b}}\cdot\mathbf{1}\left(\max_{i\ne j}|b_i-b_j|\le W^\varepsilon\ell_s\right)+\mathrm{O}\left(W^{-D+n}\right)\\&\lesssim\frac{(1-s)^{n-1}}{(g^2+|1-t|)^{n-1}}\|\mathcal{A}\|_\infty\sum_{b_1}\left|\Xi^{(1)}_{a_1b_1}\right|\sum_{\mathbf{b}'}\prod_{i=2}^n\frac{\mathbf{1}\left(|b_i-b_1|\le W^\varepsilon\ell_s\right)}{|a_i-b_i|^{d-2}+1}+W^{-D+n}\end{aligned}$$

$$\lesssim W^{2(n-1)\varepsilon}\frac{1-s}{1-t}\left(\frac{\ell_s^2|1-s|}{g^2+|1-t|}\right)^{n-1}\|\mathcal{A}\|_\infty + W^{-D+n}$$
$$\lesssim W^{2(n-1)\varepsilon}\frac{\ell_t^2}{\ell_s^2}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^{n}\|\mathcal{A}\|_\infty + W^{-D+n}, \tag{A.13}$$

where in the first step, we use the decay property (4.52) for $\mathcal{A}_{\mathbf{b}}$ and the bound (A.7) (in getting the W^{-D+n} term); in the second step, we apply (A.9) for $i \in [\![2,n]\!]$; in the third step, we take the summation over $\mathbf{b}'$ and apply (A.7); in the last step, we use (A.12) again. This gives (A.11) for any constant $C > 2(n-1)$.

For the estimates (4.54) and (4.56), we notice that when $|A| \geq 1$, (A.10) already gives a good enough bound. Hence, we only need to focus on the case where $A = \emptyset$. First, for Case I, due to (2.67), we can use the better bound $\sum_{b_1} |\Xi^{(1)}_{a_1 b_1}| = \mathrm{O}(1)$ in the third step of (A.13), which leads to (4.54).

Next, for the estimate (4.56) in Case II, we need to show that

$$\sum_{\mathbf{b}\in(\mathbb{Z}_L^d)^n}\prod_{i=1}^n \Xi^{(i)}_{a_i b_i}\cdot\mathcal{A}_{\boldsymbol{b}} \leq W^{C_n\varepsilon}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^n\|\mathcal{A}\|_\infty + W^{-D+C_n}. \tag{A.14}$$

It suffices to assume that $\sigma_i \neq \sigma_{i+1}$ for all $i \in [\![n]\!]$. We decompose $\mathbb{Z}_L^d$ into the following two regions:

$$S_{\mathrm{far}} := \Big\{b\in\mathbb{Z}_L^d : \min_{i\in[\![n]\!]}|b-a_i| > W^{2\varepsilon}\ell_s\Big\}, \quad S_{\mathrm{near}} := \Big\{b\in\mathbb{Z}_L^d : \min_{i\in[\![n]\!]}|b-a_i| \leq W^{2\varepsilon}\ell_s\Big\}.$$

Following a similar argument as in (A.13), and using that $\sum_{b_1\in S_{\mathrm{near}}}|\Xi^{(1)}_{a_1b_1}| \lesssim W^{4\varepsilon}(g^2+|1-s|)/(g^2+|1-t|)$ by (A.9) and (A.12), we obtain

$$\sum_{b_1\in S_{\mathrm{near}}}\sum_{\mathbf{b}'}\prod_{i=1}^n \Xi^{(i)}_{a_i b_i}\cdot\mathcal{A}_{\mathbf{b}} \lesssim W^{2(n+1)\varepsilon}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^n\|\mathcal{A}\|_\infty + W^{-D+n}. \tag{A.15}$$

It remains to control the sum over $b_1 \in S_{\mathrm{far}}$. In this case, we decompose $\Xi^{(i)}$ as

$$\Xi^{(i)}_{a_i b_i} = \Xi^{(i)}_{a_i b_1} + \Delta\Xi^{(i)}_{a_i;b_1b_i}, \quad \text{with} \quad \Delta\Xi^{(i)}_{a_i;b_1b_i} := \Xi^{(i)}_{a_ib_i} - \Xi^{(i)}_{a_ib_1}.$$

Then, we expand the LHS of (A.14) as

$$\sum_{A:A\subset[\![2,n]\!]} f(A), \quad \text{with} \quad f(A) := \sum_{b_1\in S_{\mathrm{far}}}\sum_{\mathbf{b}'\in(\mathbb{Z}_L^d)^{n-1}}\prod_{i\in A^c}\Xi^{(i)}_{a_ib_1}\cdot\prod_{i\in A}\Delta\Xi^{(i)}_{a_i;b_1b_i}\cdot\mathcal{A}_{\mathbf{b}}. \tag{A.16}$$

By the sum-zero property (4.55), the leading term with $A = \emptyset$ vanishes. For the remaining terms, we will use (A.9) to control the factors $\Xi^{(i)}_{a_ib_1}$, and apply (A.9) and (2.68) (along with (8.6) or (8.7) in the case of block Anderson model) to control the factors $\Delta\Xi^{(i)}_{a_i;b_1b_i}$ for $|a_i-b_1| \geq W^{2\varepsilon}\ell_s$ and $|b_i-b_1| \leq W^{\varepsilon}\ell_s$:

$$\Xi^{(i)}_{a_ib_1} \lesssim \frac{W^{-(d-1)\varepsilon}}{\ell_s^d}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right), \ \Delta\Xi^{(i)}_{a_i;b_1b_i} \lesssim \frac{1-s}{g^2+|1-t|}\frac{|b_i-b_1|}{|a_i-b_1|^{d-1}} \prec \frac{W^{-d\varepsilon}}{\ell_s^d}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right), \tag{A.17}$$

where we have also used $2(d-2) \geq d-1$ and (A.12) in the derivation. Without loss of generality, suppose $2 \in A$. Then, using the estimates in (A.9) and (A.17), along with the decay property (4.52) for $\mathcal{A}$, by a similar argument as in (A.13), we can bound $f(A)$ by

$$\begin{aligned}
f(A) &\lesssim W^{\varepsilon(n-1)}\frac{(1-s)(g^2+|1-s|)^{n-2}}{(g^2+|1-t|)^{n-1}}\|\mathcal{A}\|_\infty\sum_{b_1\in S_{\mathrm{far}}}\left|\Xi^{(1)}_{a_1b_1}\right|\frac{(W^\varepsilon\ell_s)^{d+1}}{|a_2-b_1|^{d-1}} + W^{-D+n}L^{d(n-1)}\\
&\lesssim W^{(n+d+1)\varepsilon}\frac{(1-s)^2\ell_s^{d+1}(g^2+|1-s|)^{n-2}}{(g^2+|1-t|)^n}\|\mathcal{A}\|_\infty\sum_{b_1\in S_{\mathrm{far}}}\frac{\exp(-c|a_1-b_1|/\ell_t)}{|a_1-b_1|^{d-2}\cdot|a_2-b_1|^{d-1}} + W^{-D+n}L^{d(n-1)}\\
&\lesssim W^{(n+d+1)\varepsilon}\|\mathcal{A}\|_\infty\cdot\frac{(1-s)^2\ell_s^{d+1}}{(W^{2\varepsilon}\ell_s)^{d-3}}\log L\cdot\frac{(g^2+|1-s|)^{n-2}}{(g^2+|1-t|)^n} + W^{-D+n}L^{d(n-1)}\\
&\lesssim W^{(n+5)\varepsilon}\left(\frac{g^2+|1-s|}{g^2+|1-t|}\right)^n\|\mathcal{A}\|_\infty + W^{-D+n}L^{d(n-1)}.
\end{aligned} \tag{A.18}$$

In the third step, we used the lattice sum estimate

$$\sum_{b:\,|a_1-b|\wedge|a_2-b|>R}\frac{\exp(-c|a_1-b|/\ell_t)}{|a_1-b|^{d-2}\cdot|a_2-b|^{d-1}} \lesssim \frac{\log L}{R^{d-3}}, \quad \forall\, 1\leq R\leq L, \tag{A.19}$$

which holds for every $d \geq 3$; the logarithm is needed only in the borderline case $d = 3$, where the summand decays exactly like $|b|^{-d}$ and the sum over $R \leq |b| \lesssim \ell_t$ is of order $\log(\ell_t/R)$. In the last step we absorbed it using $\log L \leq W^{\varepsilon}$, together with (A.12) in the form $(1-s)^2\ell_s^4 \asymp (g^2+|1-s|)^2$. The error term $W^{-D+n}L^{d(n-1)}$ comes from the region where $|\mathcal{A}_{\mathbf{b}}| \leq W^{-D}$ by (4.52): there, since $\Xi^{(i)}_{a_ib_1}$ does not depend on b_i, we have $\sum_{b_i}\left(|\Xi^{(i)}_{a_ib_1}|+|\Delta\Xi^{(i)}_{a_i;b_1b_i}|\right) \leq (2L^d+1)\|\Xi^{(i)}\|_{\infty\to\infty}$ for $i \in [\![2,n]\!]$, so the free summations over $b_i \in \mathbb{Z}^d_L$ produce factors L^d; the remaining factors are controlled by (A.7) and the condition $(1-t)/(1-s) \geq W^{-1}$. Since $L^d \leq N \leq W^{1/\mathfrak{c}}$ by (2.9), this term is bounded by $W^{-D+n+(n-1)/\mathfrak{c}}$. This concludes the estimate (A.14) with a constant C_n that depends only on n, d, and $\mathfrak{c}$ (and not on ε or D), which further concludes (4.56). □

Proof of Lemma 4.17. Let $\mathbf{e} \in \mathbb{C}^{L^d}$ denote the unit vector with $\mathbf{e}(a) \equiv L^{-d/2}$, and define the projection matrix $\mathsf{Proj}_{\mathbf{e}^\perp}$ onto the orthogonal complement of $\mathbf{e}$: $\mathsf{Proj}_{\mathbf{e}^\perp} = I - \mathbf{e}\mathbf{e}^\top$. When $\sigma_i = \sigma_{i+1}$, we have

$$\|\Xi^{(i)}\|_{\infty\to\infty} \lesssim t-s, \quad \text{and} \quad \|\mathsf{Proj}_{\mathbf{e}^\perp}\cdot\Xi^{(i)}\|_{\infty\to\infty} \lesssim t-s, \tag{A.20}$$

by using the definition of $\Xi^{(i)}$ in (A.6) and the estimate (2.67). On the other hand, when $\sigma_i \neq \sigma_{i+1}$, we have

$$\mathsf{Proj}_{\mathbf{e}^\perp}\cdot\Xi^{(i)} = (t-s)\mathsf{Proj}_{\mathbf{e}^\perp}\cdot M^{(+,-)}S^{(\mathsf{B})}\Theta_t^{(+,-)} = (t-s)M^{(+,-)}S^{(\mathsf{B})}\mathring{\Theta}_t^{(+,-)}.$$

Then, using (2.70) (along with (8.5) in the case of block Anderson model), we get

$$\|\mathsf{Proj}_{\mathbf{e}^\perp}\cdot\Xi^{(i)}\|_{\infty\to\infty} \lesssim (t-s)\max_a\sum_b\left|\mathring{\Theta}^{(+,-)}_{t,ab}\right| \prec (1-s)\cdot g^{-2}L^2 \leq 1, \tag{A.21}$$

where we use $1-s \leq g^2/L^2$ in the last step. Plugging (A.20) and (A.21) into (A.6), we obtain that

$$\left\|\mathsf{Proj}_{\mathbf{e}^\perp}\cdot\frac{1-s\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}\right\|_{\infty\to\infty} \lesssim 1$$

for all $i \in [\![n]\!]$. Finally, we deduce (4.57) as follows. By (3.12), the kernel $\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ acts on each of the n indices separately, and by Definition 4.12 the operator $Q^{(A)} = \prod_{i\in A}Q^{(i)}$ acts on the indices in A only, with $Q^{(i)}$ acting on the i-th index as $\mathsf{Proj}_{\mathbf{e}^\perp}$. Since $\mathsf{Proj}_{\mathbf{e}^\perp}$ commutes with the translation-invariant matrix $M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}$, we may therefore write $Q^{(A)}\circ\mathcal{U}^{(n)}_{s,t,\boldsymbol{\sigma}}$ as the tensor product over $i \in [\![n]\!]$ of the one-index operators

$$\mathsf{Proj}_{\mathbf{e}^\perp}\cdot\frac{1-s\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}} \quad (i\in A), \qquad \frac{1-s\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}}{1-t\cdot M^{(\sigma_i,\sigma_{i+1})}S^{(\mathsf{B})}} \quad (i\notin A).$$

The operators of the first kind are bounded in $(\infty\to\infty)$-norm by the display above. For those of the second kind, the assumption $A \supset I_{\mathrm{diff}}(\boldsymbol{\sigma})$ forces $\sigma_i = \sigma_{i+1}$, so (A.6) and the first bound in (A.20) give $\|1+\Xi^{(i)}\|_{\infty\to\infty} \lesssim 1$. Multiplying the n one-index bounds yields (4.57). □

A.3. **Proof of Lemma 3.8.** First, we consider the case $1-t \leq 1-u \leq g^2/L^2$, where we have $\ell_t = \ell_u = L$. In this case, the exponential factor is of order 1, and we have

$$\begin{aligned}\sum_c \mathcal{T}_u(|a-c|)\cdot\mathcal{T}_t(|c-b|) &\lesssim \sum_c\left(\frac{g^{-2}}{|a-c|^{d-2}+1}+\frac{1}{L^d|1-u|}\right)\left(\frac{g^{-2}}{|c-b|^{d-2}+1}+\frac{1}{L^d|1-t|}\right)\\ &\lesssim \frac{g^{-4}L^2}{|a-b|^{d-2}+1}+\frac{g^{-2}}{L^{d-2}|1-t|}+\frac{1}{1-u}\frac{1}{L^d|1-t|}\\ &\lesssim \frac{1}{1-u}\left(\frac{g^{-2}}{|a-b|^{d-2}+1}+\frac{1}{L^d|1-t|}\right) \lesssim \frac{1}{1-u}\mathcal{T}_t(|a-b|),\end{aligned} \tag{A.22}$$

where, in the third and fourth steps, we use that $1-u \leq g^2/L^2$. Next, we consider the case $1-u \geq 1-t \geq g^2/L^2$, where we have $\ell_u \leq \ell_t = \max(g(1-t)^{-1/2},1) \leq L$. In this case, the term $(L^d|1-t|)^{-1}$ can always be neglected in the function $\mathcal{T}_t$. Then, we get that

$$\begin{aligned}&\mathcal{T}_u(|a-c|)\cdot\mathcal{T}_t(|c-b|)/\mathcal{T}_t(|a-b|)\\ &\lesssim \frac{1}{g^2+|1-u|}\left\{1+\sum_{c\notin\{a,b\}}\frac{(|a-b|+1)^{d-2}}{|a-c|^{d-2}|c-b|^{d-2}}\exp\left(-\frac{\sqrt{|a-c|}+(\ell_u/\ell_t)^{1/2}(\sqrt{|c-b|}-\sqrt{|a-b|})}{\ell_u^{1/2}}\right)\right\}\\ &\lesssim \ell_u^2/\left(g^2+|1-u|\right) \lesssim |1-u|^{-1},\end{aligned} \tag{A.23}$$

where, in the second step, we use the following basic calculus fact for $d \geq 3$ and any $0 \leq \varepsilon \leq 1$:

$$\max_{a\in\mathbb{R}^d} \int_{x\in\mathbb{R}^d} \frac{|a|^{d-2}}{|a-x|^{d-2}\cdot|x|^{d-2}} \exp\left(-\sqrt{|a-x|} - \varepsilon\left(\sqrt{|x|}-\sqrt{|a|}\right)\right) \mathrm{d}x \leq C_d$$

for a constant $C_d > 0$. Combining the two cases (A.22) and (A.23) completes the proof of (3.21).

A.4. **Proof of Claim 7.27.** To prove the estimate (7.63), we begin by partitioning the summation region over $[\alpha]$ into 2^{2k} subregions according to whether each $|[x_i]-[\alpha]|$ or $|[y_i]-[\alpha]|$ is larger than ℓ or not. Namely, we define

$$\mathbf{D}_{\leq \ell,\boldsymbol{\sigma}} := \{[\alpha] : |[w_i]-[\alpha]| \leq \ell \text{ if } \sigma_i = 0, \text{ and } |[w_i]-[\alpha]| > \ell \text{ if } \sigma_i = 1, \ \forall i \in [\![2k]\!]\},$$

where $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_{2k}) \in \{0,1\}^{2k}$, and $[w_{2i-1}] = [x_i]$ and $[w_{2i}] = [y_i]$ for $i \in [\![k]\!]$. It therefore suffices to show that, for each fixed $\boldsymbol{\sigma} \in \{0,1\}^{2k}$,

$$\sum_{[\alpha]\in\mathbf{D}_{\leq\ell,\boldsymbol{\sigma}}} \prod_{i=1}^{k} [\mathsf{T}_t(|[x_i]-[\alpha]|\wedge\ell)\cdot\mathsf{T}_t(|[y_i]-[\alpha]|\wedge\ell)] \prec (W^d\eta_t)^{-1}\Psi_t^{k-2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell). \tag{A.24}$$

We now analyze three cases, depending on how many paths consist entirely of "short edges", i.e., edges of length $\leq \ell$.

1. Assume that there are at least two indices i such that $|[x_i]-[\alpha]| \vee |[y_i]-[\alpha]| \leq \ell$. Without loss of generality, suppose this condition holds for $1 \leq i \leq r$, and $|[x_i]-[\alpha]| \vee |[y_i]-[\alpha]| > \ell$ for $r+1 \leq i \leq k$, where $2 \leq r \leq k$. For $1 \leq i \leq r$, we have

$$\mathsf{T}_t(|[x_i]-[\alpha]|\wedge\ell)\,\mathsf{T}_t(|[y_i]-[\alpha]|\wedge\ell) \lesssim \mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell)\cdot\frac{(W^{-d}B_{t,0})^{1/2}}{(|[x_i]-[\alpha]|\wedge|[y_i]-[\alpha]|+1)^{(d-2)/2}}, \tag{A.25}$$

and whenever $|[y_i]-[\alpha]| > \ell$ (in particular for $r+1 \leq i \leq k$, after relabelling $[x_i]$ and $[y_i]$ if necessary), we have

$$\mathsf{T}_t(|[x_i]-[\alpha]|\wedge\ell)\,\mathsf{T}_t(|[y_i]-[\alpha]|\wedge\ell) \lesssim \mathsf{T}_t(\ell)\cdot(W^{-d}B_{t,0})^{1/2}\cdot(|[x_i]-[\alpha]|\wedge\ell+1)^{-\frac{d-2}{2}}. \tag{A.26}$$

Both bounds follow from the definition of T_t, the monotonicity of T_t, and the two elementary facts $\mathsf{T}_t(0) \leq (W^{-d}B_{t,0})^{1/2}$ and $\mathsf{T}_t(r\wedge\ell) \leq \mathsf{T}_t(0)\,(r\wedge\ell+1)^{-(d-2)/2}$. In (A.26) we will bound the last factor by 1 whenever it is not needed. Using these two estimates, we can bound the LHS of (A.24) by

$$\begin{aligned}
&(W^{-d}B_{t,0})^{k/2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell)\cdot\sum_{[\alpha]\in\mathbf{D}_{\leq\ell,\boldsymbol{\sigma}}}\prod_{i=1}^{r}(|[x_i]-[\alpha]|\wedge|[y_i]-[\alpha]|+1)^{-\frac{d-2}{2}}\\
&\lesssim \ell^2(W^{-d}B_{t,0})^{k/2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell) \prec (W^d\eta_t)^{-1}\Psi_t^{k-2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell),
\end{aligned}$$

where in the first step, we use

$$\sum_{[\alpha]\in\mathbf{D}_{\leq\ell,\boldsymbol{\sigma}}}\prod_{i=1}^{r}(|[x_i]-[\alpha]|\wedge|[y_i]-[\alpha]|+1)^{-\frac{d-2}{2}} \lesssim \sum_{i=1}^{r}\sum_{[\alpha]\in\mathbf{D}_{\leq\ell}}(|[x_i]-[\alpha]|\wedge|[y_i]-[\alpha]|+1)^{-(d-2)} \lesssim \ell^2,$$

and in the second step, we use the facts that $\ell \leq (\log W)^{10}\ell_t$ and $\ell_t^2 B_{t,0} \lesssim |1-t|^{-1} \lesssim \eta_t^{-1}$ when $1-t \geq g^2/L^2$.

2. Suppose there is exactly one index i such that $|[x_i]-[\alpha]| \vee |[y_i]-[\alpha]| \leq \ell$. Without loss of generality, assume that $|[x_1]-[\alpha]| \vee |[y_1]-[\alpha]| \leq \ell$, and $|[y_2]-[\alpha]| \geq \ell$. Then, applying (A.25) for $i=1$ and (A.26) for $2 \leq i \leq k$—keeping the last factor of (A.26) for $i=2$ and bounding it by 1 for $i \geq 3$—we can bound the LHS of (A.24) as

$$\begin{aligned}
&(W^{-d}B_{t,0})^{k/2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell)\cdot\sum_{[\alpha]\in\mathbf{D}_{\leq\ell,\boldsymbol{\sigma}}}(|[x_1]-[\alpha]|\wedge|[y_1]-[\alpha]|+1)^{-\frac{d-2}{2}}(|[x_2]-[\alpha]|\wedge\ell+1)^{-\frac{d-2}{2}}\\
&\lesssim \ell^2(W^{-d}B_{t,0})^{k/2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell) \prec (W^d\eta_t)^{-1}\Psi_t^{k-2}\prod_{i=1}^{k}\mathsf{T}_t(|[x_i]-[y_i]|\wedge\ell).
\end{aligned}$$

3. Finally, assume that for all $i \in [\![k]\!]$, we have $|[x_i]-[\alpha]| \vee |[y_i]-[\alpha]| > \ell$. Without loss of generality, suppose $|[y_1]-[\alpha]| \geq \ell$ and $|[y_2]-[\alpha]| \geq \ell$. Then, applying (A.26) for $1 \leq i \leq k$—keeping the last factor for $i \in \{1,2\}$ and bounding it by 1 for $i \geq 3$—we can bound the LHS of (A.24) by

$$(W^{-d}B_{t,0})^{k/2} \prod_{i=1}^{k} \mathsf{T}_t(|[x_i]-[y_i]| \wedge \ell) \cdot \sum_{[\alpha] \in \mathbf{D}_{\leq \ell, \boldsymbol{\sigma}}} (|[x_1]-[\alpha]| \wedge \ell + 1)^{-\frac{d-2}{2}} (|[x_2]-[\alpha]| \wedge \ell + 1)^{-\frac{d-2}{2}}$$

$$\lesssim \ell^2 (W^{-d}B_{t,0})^{k/2} \prod_{i=1}^{k} \mathsf{T}_t(|[x_i]-[y_i]| \wedge \ell) \prec (W^d \eta_t)^{-1} \Psi_t^{k-2} \prod_{i=1}^{k} \mathsf{T}_t(|[x_i]-[y_i]| \wedge \ell).$$

By combining all three cases, we conclude that (A.24) holds, which, in turn, implies (7.63).

A.5. **Basic properties of $\mathcal{K}$-loops.** In this subsection, we collect several basic properties of the $\mathcal{K}$-loops used in the analysis of the loop hierarchy and apply them to establish the $\mathcal{K}$-loop bounds stated in Lemmas 2.16 and 4.2. The results presented here are higher-dimensional analogues (in dimensions $d \geq 3$) of those in [69, Section 3] and [59, Section 4]. We begin by introducing a dimension-independent *tree representation formula* for $\mathcal{K}$-loops, first discovered in [69] for random band matrices and later extended to the block Anderson model in [59]. This tree representation is constructed using the notion of *canonical partitions of polygons*. Roughly speaking, a canonical partition of an oriented polygon $\mathcal{P}_{\mathbf{a}}$ is a partition in which each edge of the polygon is in one-to-one correspondence with each region in the partition.

Definition A.1 (Canonical partitions)**.** *Fix $n \geq 3$ and let $\mathcal{P}_{\mathbf{a}}$ be an oriented polygon with vertices $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ arranged in a (counterclockwise) cyclic order, where we adopt the cyclic convention that $a_i = a_j$ if and only if $i = j \mod n$. Let (a_{k-1}, a_k) denote the k-th side of $\mathcal{P}_{\mathbf{a}}$. A planar partition of the polygonal domain enclosed by $\mathcal{P}_{\mathbf{a}}$ is called* **canonical** *if the following properties hold:*

- *Every sub-region in the partition is also a polygonal domain.*
- *There is a one-to-one correspondence between the sides of the polygon and the sub-regions, where every side (a_{k-1}, a_k) belongs to exactly one sub-region, denoted by R_k, and each sub-region contains exactly one side of $\mathcal{P}_{\mathbf{a}}$.*
- *Every vertex a_k of $\mathcal{P}_{\mathbf{a}}$ belongs to exactly two regions, R_k and R_{k+1} (with the convention $R_{n+1} = R_1$).*

Note that given a canonical partition, by removing the n sides of the polygon $\mathcal{P}_{\mathbf{a}}$, the remaining interior edges form a tree, with the leaves being the vertices of $\mathcal{P}_{\mathbf{a}}$. Following the definitions in [69], we define the equivalence classes of all such trees under graph isomorphism, and denote the collection of equivalence classes by $\mathbf{T}(\mathcal{P}_{\mathbf{a}})$. We will consider each element of $\mathbf{T}(\mathcal{P}_{\mathbf{a}})$ as an abstract tree structure rather than as an equivalence class, and call it a canonical tree partition*.*

In a canonical tree partition $\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})$, we call an edge that contains exactly one external vertex a_k an *external edge*, and an edge connecting two internal vertices an *internal edge*. Two regions R_k and R_l are said to be *neighbors* if they share a common side, which may be either an external or an internal edge. In the case of an external edge, we necessarily have $k - l = \pm 1 \pmod n$, and we refer to R_k and R_l as *trivial neighbors*; otherwise, they are called *nontrivial neighbors*. Given $\boldsymbol{\sigma} \in \{+,-\}^n$, we assign charges to the subregions as follows: each subregion R_k carries the charge of the edge (a_{k-1}, a_k), which is given by σ_k. An illustration is provided in the left panel of Figure 6, which shows a canonical tree partition $\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})$ of a polygon with six vertices, where R_4 and R_6 form a pair of nontrivial neighbors. We note that the figures in this section are all reproduced from [59].

In the context of random band matrices, we assign a value to Γ according to the following rule.

Definition A.2. *Given any $t \in [0,1)$ and $\boldsymbol{\sigma} \in \{+,-\}^n$, we define the values of the edges in Γ as follows:*

(1) *If $e = (a_k, b)$ is an external edge lying between regions R_k and R_{k+1}, then we define*

$$f_{t,\boldsymbol{\sigma}}(e) := \Theta_t^{(\sigma_k, \sigma_{k+1})}(a_k, b). \tag{A.27}$$

(2) *If $e = (b_1, b_2)$ is an internal edge lying between regions R_k and R_l, then*

$$f_{t,\boldsymbol{\sigma}}(e) := \left(\Theta_t^{(\sigma_k,\sigma_l)} - I\right)(b_1, b_2) = m(\sigma_k)m(\sigma_l) \cdot \left(tS^{(\mathsf{B})}\Theta_t^{(\sigma_k,\sigma_l)}\right)(b_1, b_2). \tag{A.28}$$

Then, we assign a value $\Gamma^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ *to* Γ *as:*

$$\Gamma^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} := \left(\prod_{i=1}^{n} m(\sigma_i)\right) \cdot \sum_{\mathbf{b}} \prod_{e} f_{t,\boldsymbol{\sigma}}(e) \ , \tag{A.29}$$

where $\mathbf{b} = (b_1, \dots, b_r)$ *denotes the internal vertices in* Γ *and* e *denotes all the edges in* Γ.

With these definitions, we recall the tree representation formula of the $\mathcal{K}$-loops in Lemma 3.4 of [69]. Recall that the formulas for the $\mathcal{K}$-loops of length 2 and 3 have been given in (2.71) and (2.72).

Lemma A.3 (Lemma 3.4 of [69])**.** *In the setting of random band matrices, for any* $n \geq 4$, $t \in [0,1)$, $\boldsymbol{\sigma} \in \{+,-\}^n$, *and* $\mathbf{a} \in (\mathbb{Z}_L^d)^n$, *we have the following representation formula for* $\mathcal{K}$*-loops:*

$$\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = W^{-d(n-1)} \sum_{\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})} \Gamma^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}. \tag{A.30}$$

When extending (A.30) to the block Anderson model, certain factors of m must be replaced by entries of the matrix M. A canonical procedure for this replacement is described in [59, Section 4]. To formulate it, we first extend Definition A.1 to include loops that contain M*-edges*. As the name suggests, these edges correspond to entries of M, while the remaining edges in our graphs are left *unlabeled* (i.e., without label M): external edges represent entries of Θ_t, and internal edges represent entries of $tS^{(\mathsf{B})}\Theta_t$.

Definition A.4 (Canonical partitions with M-loops)**.** *Let* $\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})$ *be a canonical tree partition of the oriented polygon* $\mathcal{P}_{\mathbf{a}}$, *and denote the internal vertices of* Γ *by* $\mathbf{b} = (b_1, \dots, b_r)$. *We define the graph* Γ_M *by replacing each* b_i *with an* M*-loop in the following way.*

(1) *Consider the subgraph* $(V^{(b_i)}, E^{(b_i)})$ *of all vertices in* Γ *connected to* b_i. *More precisely, we let*

$$V^{(b_i)} := \{b_i, c_1, \dots, c_{k_i}\}, \quad \text{and} \quad E^{(b_i)} = \{(c_1, b_i), \dots, (c_{k_i}, b_i)\} \tag{A.31}$$

denote the subsets of all vertices (including b_i*) and edges connected to* b_i. *Reordering the* c_k*'s if necessary, we can ensure that* $(c_1, \dots, c_{k_i})$ *form a loop without any crossing edges and with vertices arranged in counterclockwise order.*

(2) *Next, we construct a new graph* $(\widetilde{V}^{(b_i)}, \widetilde{E}^{(b_i)})$ *with vertices and edges*

$$\widetilde{V}^{(b_i)} = \{b_{i,1}, \dots, b_{i,k_i}, c_1, \dots, c_{k_i}\}, \quad \widetilde{E}^{(b_i)} = \{(c_j, b_{i,j}) : j \in [\![k_i]\!]\} \cup \{(b_{i,j}, b_{i,j+1}; M) : j \in [\![k_i]\!]\}, \tag{A.32}$$

where $e = (e_i, e_f; M)$ *refers to an edge labeled with* M, *and we adopt the cyclic convention that* $b_{i,k_i+1} = b_{i,1}$. *We will call* $(e_i, e_f; M)$ *an* M-edge.

(3) *Lastly, we replace the subgraph* $(V^{(b_i)}, E^{(b_i)})$ *with* $(\widetilde{V}^{(b_i)}, \widetilde{E}^{(b_i)})$.

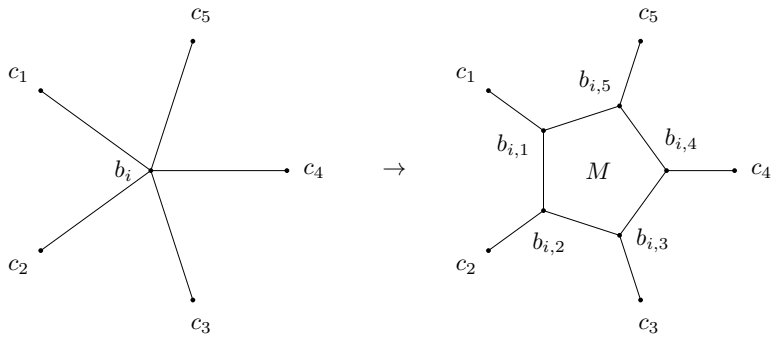


FIGURE 5. Replacing b_i with an M-loop with 5 sides.

In Figure 5, we illustrate the above procedure for an example with $k_i = 5$. *Repeating these steps for each internal vertex* b_i, $i \in [\![r]\!]$, *we get a graph* Γ_M, *which we will refer to as the* M-graph corresponding to Γ. *Note that the order in which we replace* b_i*'s by* M*-loops does not matter, and every boundary edge* (a_{k-1}, a_k) *still belongs to exactly one polygonal region in the* M*-graph* Γ_M. *With a slight abuse of notation, we still use* R_k *to denote the sub-region containing* (a_{k-1}, a_k) *in* Γ_M, *and assign the charge* σ_k *of* (a_{k-1}, a_k) *to* R_k.

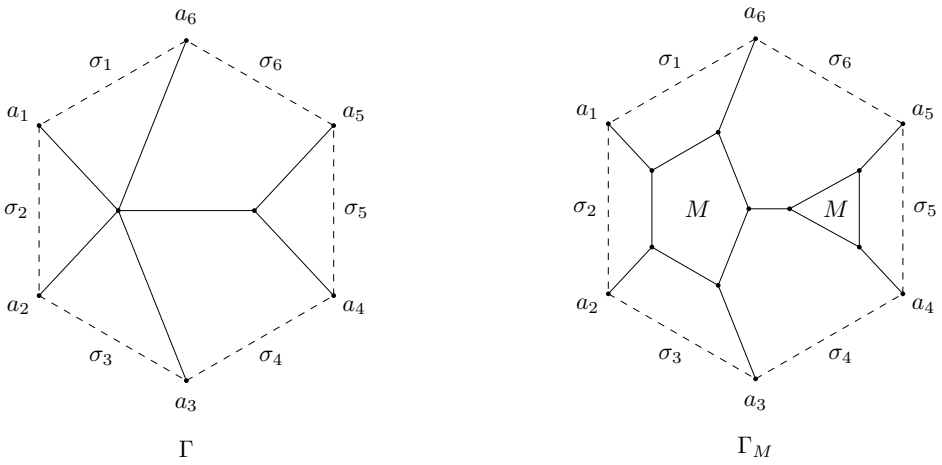


FIGURE 6. Example of $\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})$ and its corresponding M-graph Γ_M.

We refer readers to Figure 6 for an example of a canonical tree partition Γ and its M-graph Γ_M.

Definition A.5. *Given an M-graph Γ_M, we assign a value to Γ according to the following rule. For any $t \in [0,1)$ and $\boldsymbol{\sigma} \in \{+,-\}^n$, we define the values of the edges in Γ_M as follows.*

(1) *If $e = (a_k, b)$ is an external edge lying between regions R_k and R_{k+1}, then we define*

$$g_{t,\boldsymbol{\sigma}}(e) := \Theta_t^{(\sigma_k,\sigma_{k+1})}(a_k, b). \tag{A.33}$$

(2) *If $e = (b_1, b_2)$ is an internal unlabeled edge lying between regions R_k and R_l, then we define*

$$g_{t,\boldsymbol{\sigma}}(e) := \left(tS^{(\mathsf{B})}\Theta_t^{(\sigma_k,\sigma_l)}\right)(b_1, b_2). \tag{A.34}$$

(3) *Corresponding to each loop of M-edges as in (A.32), we define its value as*

$$\mathcal{F}(b_i) \equiv W^{(k_i-1)d}\mathcal{M}^{(k_i)}_{\boldsymbol{\sigma}(b_i),\mathbf{a}(b_i)}, \tag{A.35}$$

where we recall the M-loop defined in (2.53). Here, the charges $\sigma_j(b_i)$ and vertices $a_j(b_i)$ are fixed according to the following rule: for each $j \in [\![k_i]\!]$, the edge $(b_{i,j-1} \to b_{i,j})$ (with the cyclic convention that $b_{i,0} = b_{i,k_i}$) belongs to exactly one region R_{k_j}. Then, we set $\sigma_j(b_i) = \sigma_{k_j}$ and $a_j(b_i) = b_{i,j}$.

Now, we assign a value $\Gamma^{(n)}_{M;t,\boldsymbol{\sigma},\mathbf{a}}$ to Γ_M as:

$$\Gamma^{(n)}_{M;t,\boldsymbol{\sigma},\mathbf{a}} := \sum_{b_{i,j}\in\mathbb{Z}_L^d,\ i\in[\![r]\!],\ j\in[\![k_i]\!]} \prod_{\text{unlabeled } e} g_{t,\boldsymbol{\sigma}}(e) \cdot \prod_{i=1}^{r} \mathcal{F}(b_i), \tag{A.36}$$

where $b_{i,j}$'s denote the internal vertices in Γ_M and e denotes all unlabeled (i.e., non-M) edges.

It is straightforward to verify that the definition in (A.36) is consistent with (A.29) in the case of random band matrices, where $M(\sigma) = m(\sigma)I$. Furthermore, in the block Anderson model, the tree representation formula (A.30) extends as follows.

Lemma A.6 (Lemma 4.16 of [59])**.** *In the setting of the block Anderson model, for any $n \geq 4$, $t \in [0,1)$, $\boldsymbol{\sigma} \in \{+,-\}^n$, and $\mathbf{a} \in (\mathbb{Z}_L^d)^n$, we have the following representation formula for $\mathcal{K}$-loops:*

$$\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = W^{-d(n-1)} \sum_{\Gamma\in\mathbf{T}(\mathcal{P}_{\mathbf{a}})} \Gamma^{(n)}_{M;t,\boldsymbol{\sigma},\mathbf{a}}. \tag{A.37}$$

Next, we recall the molecule structure of $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ given in Section 3.3 of [69]. We split the M-graphs Γ_M with $\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}})$ according to which edges of the tree are "long"—we refer to a $\Theta_t^{(\sigma_1,\sigma_2)}$-edge as a *long edge* if $\sigma_1 \neq \sigma_2$, and a *short edge* otherwise. We adopt this terminology because, according to (2.66) and (2.67), the short edge decays on a scale of order 1, which is much shorter than the decay scale ℓ_t for a long edge.

Given any $\boldsymbol{\sigma} \in \{+,-\}^n$, define the subset of long internal edges (i.e., the boundary between two nontrivial neighbors of different charges) as

$$\mathcal{F}_{\text{long}}(\Gamma, \boldsymbol{\sigma}) := \left\{ \{k,l\} \in \mathbb{Z}_n^{\text{off}} : R_k \cap R_l \neq \emptyset,\ \sigma_k \neq \sigma_l \right\},$$

where we define the subset

$$\mathbb{Z}_n^{\text{off}} := \{\{k,l\} : 1 \leq k < l \leq n,\ k-l \pmod n \notin \{1,-1\}\}.$$

Given any subset $\pi \subset \mathbb{Z}_n^{\text{off}}$, we use $\mathbf{T}(\mathcal{P}_{\mathbf{a}}, \boldsymbol{\sigma}, \pi) := \left\{ \Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}}) : \mathcal{F}_{\text{long}}(\Gamma, \boldsymbol{\sigma}) = \pi \right\}$ to denote the set of all Γ such that π labels the pairs of all non-trivial neighbors in Γ (note π can be $\emptyset$). Then, we define

$$\mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) := \sum_{\Gamma \in \mathbf{T}(\mathcal{P}_{\mathbf{a}}, \boldsymbol{\sigma}, \pi)} \Gamma^{(n)}_{M;t,\boldsymbol{\sigma},\mathbf{a}}. \tag{A.38}$$

Note that $\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}}$ can be decomposed as

$$\mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} = W^{-d(n-1)} \sum_{\pi \subset \mathbb{Z}_n^{\text{off}}} \mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}). \tag{A.39}$$

Moreover, we have the following molecule decomposition of $\mathcal{K}^{(\pi)}$ given in equation (3.53) of [69]:

$$\mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{b},\mathbf{c}} \prod_{k=1}^{n} \Theta^{(\sigma_k,\sigma_{k+1})}_{t,a_k b_k} \cdot \prod_{k=1}^{r-1} \left(tS^{(\mathsf{B})} \Theta^{(+,-)}_t \right)_{c_{2k-1} c_{2k}} \cdot \prod_{k=1}^{r} \Sigma^{(\pi)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{b}^{(k)}). \tag{A.40}$$

Here, each term $\Sigma^{(\pi)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{b}^{(k)})$ represents a *molecule*, defined as a maximal subgraph consisting solely of M-loops and *short (unlabeled) internal edges*.[13] In other words, if each molecule is collapsed into a single vertex, the resulting quotient graph contains only long internal edges connecting different molecules, along with external edges, which may be either short or long. In (A.40), we assume there are r molecules in total. The terms $tS^{(\mathsf{B})}\Theta^{(+,-)}_t$ correspond to the long internal edges (c_{2k-1}, c_{2k}) between molecules, while the terms $\Theta^{(\sigma_k,\sigma_{k+1})}_{t,a_k b_k}$ correspond to the external edges (a_k, b_k). The vectors $\mathbf{b} = (b_1, \ldots, b_n)$ and $\mathbf{c} = (c_1, c_2, \ldots, c_{2r-3}, c_{2r-2})$ denote the internal vertices that serve as endpoints of these external and long internal edges, respectively. Furthermore, $\boldsymbol{\sigma}^{(k)}$ and $\mathbf{b}^{(k)}$ specify the charges and distinguished vertices associated with the k-th molecule. All other internal vertices within the molecules are implicitly summed over.

Using the estimate (2.67), we can derive the following exponential decay estimate on "pure loops" where all charges are identical.

Lemma A.7. *For $\boldsymbol{\sigma} \in \{+,-\}^n$ with $\sigma_1 = \sigma_2 = \cdots = \sigma_n$, there exist constants $c_n, C_n > 0$ such that*

$$\left| \mathcal{K}^{(n)}_{t,\boldsymbol{\sigma},\mathbf{a}} \right| \leq C_n W^{-d(n-1)} \exp\left(-c_n \max_{i,j} |a_i - a_j| \right). \tag{A.41}$$

Proof. In the tree representation (A.37), each term $\Gamma^{(n)}_{M;t,\boldsymbol{\sigma},\mathbf{a}}$ on the RHS contains only M-edges and short unlabeled edges. Using the exponential decay from (2.67), along with the bound (8.6) or (8.7) for M-edges in the setting of the block Anderson model, we obtain the desired estimate. □

We are now ready to prove the key bounds on $\mathcal{K}$-loops, stated in Lemma 2.16. The proof is analogous to that of Lemma 3.11 in [69], but requires additional modifications to handle the higher-dimensional setting $d \geq 3$. For the reader's convenience, we provide the proof below.

Proof of Lemma 2.16. For $n = 1$ we have $\mathcal{K}^{(1)}_{t,\boldsymbol{\sigma},\mathbf{a}} = m(\sigma_1) = \mathrm{O}(1)$, and for $n \in \{2,3\}$ the bound follows from the explicit formulas (2.71) and (2.72) together with (2.66) and, for $n = 3$, (2.67) (and (8.6) or (8.7) for the M-loop in the block Anderson model). Indeed, the three charge pairs (σ_1, σ_2), (σ_2, σ_3), and (σ_3, σ_1) in (2.72) cannot all consist of different charges, so at least one of the three Θ-propagators in (2.72) is short-range by (2.67), and its sum over the free index is $\mathrm{O}(1)$ rather than $(1-t)^{-1}$; the other two propagators are bounded pointwise by $\mathrm{O}(B_{t,0})$ using (2.66). This gives $|\mathcal{K}^{(3)}_{t,\boldsymbol{\sigma},\mathbf{a}}| \lesssim (W^{-d} B_{t,0})^2$. We may therefore assume

[13] With a slight abuse of notation, we again use the term "molecule" here. This definition is in the same spirit as that in Definition 7.5, although the graph considered here is different from the one in Definition 7.3.

$n \geq 4$, which is the range in which the tree representation (A.30) is available. By (A.39), it suffices to show that for any $\pi \subset \mathbb{Z}_n^{\mathrm{off}}$, we have

$$\left|\mathcal{K}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a})\right| \prec B_{t,0}^{n-1}. \tag{A.42}$$

We prove the above bound by induction on the number of molecules r in π. In order to carry out the induction step, we will prove (A.42) for a slightly generalized quantity $\tilde{\mathcal{K}}^{(\pi)}$, defined by

$$\tilde{\mathcal{K}}^{(\pi)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{b},\mathbf{c}} \prod_{k=1}^{n} \tilde{\Theta}_{t,a_k b_k} \cdot \prod_{k=1}^{r-1} \left(tS^{(\mathsf{B})}\Theta_t^{(+,-)}\right)_{c_{2k-1}c_{2k}} \cdot \prod_{k=1}^{r} \Sigma^{(\pi)}(t, \boldsymbol{\sigma}^{(k)}, \mathbf{b}^{(k)}), \tag{A.43}$$

where each $\tilde{\Theta}_t$ represents either $\Theta_t^{(\sigma,\sigma')}$ for some $\sigma, \sigma' \in \{+,-\}$ or $tS^{(\mathsf{B})}\Theta_t^{(+,-)}$.

We start with the single molecule case with $\pi = \emptyset$. In this case, we can write

$$\tilde{\mathcal{K}}^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{\mathbf{b}} \prod_{i=1}^{n} \tilde{\Theta}_{t,a_i b_i} \cdot \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{b}).$$

First, the M-edges and short unlabeled edges in $\tilde{\mathcal{K}}$, including all edges contained within the molecule, have constant size and fast exponential decay outside their constant range by (2.67), together with (8.6) or (8.7) in the block Anderson model. Thus, the molecule $\Sigma^{(\emptyset)}$ has constant size and fast exponential decay as well:

$$\left|\Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{b})\right| \leq C \exp\left\{-c \max_{i,j} |b_i - b_j|\right\} \tag{A.44}$$

for some constants $c, C > 0$. Additionally, if we have a pure loop (i.e., $\boldsymbol{\sigma}$ consists entirely of the same charge), then every edge in $\tilde{\mathcal{K}}^{(\emptyset)}$ is short, so we have

$$\left|\tilde{\mathcal{K}}^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{a})\right| \leq C \exp\left\{-c \max_{i,j} |a_i - a_j|\right\} \lesssim B_{t,0}^{n-1}.$$

If $\boldsymbol{\sigma}$ is not a pure loop, then without loss of generality, we can assume that $\sigma_1 \neq \sigma_2$. In this case, we have

$$\tilde{\mathcal{K}}^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{a}) = \sum_{b_1} \tilde{\Theta}_{t,a_1 b_1} \sum_{\mathbf{b}\setminus\{b_1\}} \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{b}) \prod_{i=2}^{n} \tilde{\Theta}_{t,a_i b_i}.$$

We now claim that

$$\sum_{b_1} \left| \sum_{\mathbf{b}\setminus\{b_1\}} \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{b}) \prod_{i=2}^{n} \tilde{\Theta}_{t,a_i b_i} \right| \prec B_{t,0}^{n-2}. \tag{A.45}$$

Combining this estimate with the bound (2.66) for the Θ-propagators, we obtain that

$$\left|\tilde{\mathcal{K}}^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{a})\right| \lesssim B_{t,0} \sum_{b_1} \left| \sum_{\mathbf{b}\setminus\{b_1\}} \Sigma^{(\emptyset)}(t, \boldsymbol{\sigma}, \mathbf{b}) \prod_{i=2}^{n} \tilde{\Theta}_{t,a_i b_i} \right| \prec B_{t,0}^{n-1}.$$

This concludes (A.42) for the case $\pi = \emptyset$.

To prove the bound (A.45), we consider two cases depending on whether (i) $\boldsymbol{\sigma}$ is non-alternating, or (ii) $\boldsymbol{\sigma}$ is alternating, where $\sigma_j \neq \sigma_{j+1}$ for all $j \in [\![n]\!]$.

(i) If $\boldsymbol{\sigma}$ is non-alternating, then there exists $j \in [\![2, n]\!]$ such that $\sigma_j = \sigma_{j+1}$. The corresponding short edge has fast exponential decay on the constant scale by (2.67):

$$\left|\tilde{\Theta}_{t,a_j b_j}\right| \leq C \exp\{-c|a_j - b_j|\}. \tag{A.46}$$

For the remaining $(n-2)$ factors $\tilde{\Theta}_{t,a_i b_i}$, we can bound them pointwise using (2.66) as

$$\left|\tilde{\Theta}_{t,a_i b_i}\right| \lesssim B_{t,0}, \quad \forall i \in [\![2, n]\!] \setminus \{j\}. \tag{A.47}$$

The above two bounds, together with the exponential decay for the molecule weight in (A.44), allow us to bound the summation on the LHS of (A.45) by $B_{t,0}^{n-2}$.

(ii) It remains to handle the challenging case of alternating $\boldsymbol{\sigma}$. In this setting, in addition to the constant-range exponential decay, we must also use the sum-zero property of molecule weights:

$$\sum_{\mathbf{b}\setminus\{b_1\}} \Sigma^{(\emptyset)}(t,\boldsymbol{\sigma},\mathbf{b}) = \mathrm{O}(|1-t|), \qquad \sum_{\mathbf{b}\setminus\{b_1\}} \big|\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma},\mathbf{b})\big| = \mathrm{O}(g^2+|1-t|). \tag{A.48}$$

The first estimate was established in [69, Lemma 3.10] for 1D random band matrices and in [59, Lemma 4.29] for 1D and 2D block Anderson models, while the second estimate was proved in [59, Claim 4.30]. Importantly, these proofs are dimension-independent: they rely only on the pure loop estimate (Lemma A.7), the short-edge bound (2.67), the M-edge bound (8.6) or (8.7), and Ward's identity (2.57) for $\mathcal{K}$-loops.

We now decompose the long external edges into three parts as follows:

$$\tilde{\Theta}^{(+,-)}_{t,a_jb_j} = f_0(a_j,s_j)+f_1(a_j,s_j)+f_2(a_j,s_j), \quad \tilde{\Theta}^{(+,-)}_t \in \big\{\Theta^{(+,-)}_t,\ tS^{(\mathsf{B})}\Theta^{(+,-)}_t\big\},$$

where we denote $f(a_j,s_j) = \tilde{\Theta}^{(+,-)}_t(a_j,b_1+s_j)$ with $s_j = b_j - b_1$, and

$$f_0(a_j,s_j) = f(a_j,0), \quad f_1(a_j,s_j) = \frac{1}{2}f(a_j,s_j) - \frac{1}{2}f(a_j,-s_j),$$
$$f_2(a_j,s_j) = \frac{1}{2}f(a_j,s_j) + \frac{1}{2}f(a_j,-s_j) - f(a_j,0).$$

By Lemma 2.19, we have the following bounds under the condition $|s_j| \prec 1$ for $j \in [\![2,n]\!]$ (if $|a_j-b_1| \geq 4|s_j|$, the bounds on f_1 and f_2 follow from (2.68) and (2.69) with $c_0 = 1/2$, recalling that $S^{(\mathsf{B})}$ has range 1; otherwise, $|a_j-b_1| < 4|s_j| \prec 1$, and they follow from (2.70), since the zero mode cancels in f_1 and f_2):

$$|f_0(a_j,s_j)| \lesssim B_{t,|a_j-b_1|} \leq B_{t,0}, \quad |f_1(a_j,s_j)| \prec \frac{(g^2+|1-t|)^{-1}}{|a_j-b_1|^{d-1}+1}, \quad |f_2(a_j,s_j)| \prec \frac{(g^2+|1-t|)^{-1}}{|a_j-b_1|^{d}+1}. \tag{A.49}$$

We view $\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma},\mathbf{b}) = g(\mathbf{s})$ as a function of the shifts $\mathbf{s} = (s_2,\dots,s_n)$. Then, we can write

$$\sum_{\mathbf{b}\setminus\{b_1\}} \Sigma^{(\emptyset)}(t,\boldsymbol{\sigma},\mathbf{b}) \prod_{j=2}^n \tilde{\Theta}_{t,a_jb_j} = \sum_{\mathbf{s}} g(\mathbf{s}) \prod_{j=2}^n \sum_{\xi_j\in\{0,1,2\}} f_{\xi_j}(a_j,s_j).$$

We split the sum above into several parts.

(1) Consider the terms where $\xi_j = 0$ for all $j \in [\![2,n]\!]$. Then, we use the sum-zero property (A.48) to get

$$\sum_{b_1}\Big|\sum_{\mathbf{s}} g(\mathbf{s})\prod_{j=2}^n f_0(a_j,s_j)\Big| \lesssim (1-t)\sum_{b_1}\prod_{j=2}^n\big|\tilde{\Theta}_{t,a_jb_1}\big| \lesssim (1-t)B_{t,0}^{n-2}\sum_{b_1}\big|\tilde{\Theta}_{t,a_2b_1}\big| \lesssim B_{t,0}^{n-2},$$

where in the second step, we use (A.47) for all but one $\tilde{\Theta}_t$ factor, and in the last step, we use (2.65).

(2) Consider the case where exactly one ξ_j equals 1 while all other ξ_j's are zero. In this situation, the sum vanishes by the skew symmetry of f_1 (namely, $f_1(a_j,s_j) = -f_1(a_j,-s_j)$) together with the symmetry of $\Sigma^{(\emptyset)}$ (i.e., $g(\mathbf{s}) = g(-\mathbf{s})$). The latter symmetry follows from the translation invariance and symmetry of the Θ-propagators (Lemma 2.19) and of the M-loops (Lemma 8.3).

(3) Suppose there is at least one $i \in [\![2,n]\!]$ such that $\xi_i = 2$. Then, using (A.49) and (A.48), we get

$$\sum_{b_1}\Big|\sum_{\mathbf{s}} g(\mathbf{s})f_2(a_i,s_i)\prod_{j\notin\{1,i\}} f_{\xi_j}(a_j,s_j)\Big| \prec \frac{B_{t,0}^{n-2}}{g^2+|1-t|}\sum_{b_1}\frac{1}{|a_i-b_1|^d+1}\sum_{\mathbf{b}\setminus\{b_1\}}\big|\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma},\mathbf{b})\big| \prec B_{t,0}^{n-2}.$$

(4) Suppose there are at least two $i,k \in [\![2,n]\!]$ such that $i \neq k$ and $\xi_i = \xi_k = 1$. In this case, using (A.49) and (A.48), we get

$$\sum_{b_1}\Big|\sum_{\mathbf{s}} g(\mathbf{s})f_1(a_i,s_i)f_1(a_k,s_k)\prod_{j\notin\{1,i,k\}} f_{\xi_j}(a_j,s_j)\Big| \prec \frac{B_{t,0}^{n-3}}{g^2+|1-t|}\sum_{b_1}\frac{1}{|a_i-b_1|^{d-1}+1}\frac{1}{|a_k-b_1|^{d-1}+1}$$
$$\lesssim B_{t,0}^{n-2}\sum_{b_1}\left[\frac{1}{|a_i-b_1|^d+1}\frac{1}{|a_k-b_1|^{d-2}+1}+\frac{1}{|a_i-b_1|^{d-2}+1}\frac{1}{|a_k-b_1|^d+1}\right] \lesssim B_{t,0}^{n-2},$$

where, in the second step, we also use $(g^2+|1-t|)^{-1} \lesssim B_{t,0}$ and the elementary inequality

$$\frac{1}{(x^{d-1}+1)(y^{d-1}+1)} \leq 2\left[\frac{1}{(x^d+1)(y^{d-2}+1)}+\frac{1}{(x^{d-2}+1)(y^d+1)}\right], \qquad \forall\, x,y \geq 0.$$

This asymmetric splitting is the modification of the argument in [69, Lemma 3.11] needed for $d \geq 3$: since $d-2 \geq 1$, each of the two resulting lattice sums over b_1 in the second line is $\mathrm{O}(1)$, uniformly in a_i, a_k, and L.

This concludes the proof of the claim (A.45).

Finally, we establish (A.42) by induction on the number of molecules r. Assume that (A.42) holds for every π with at most n external vertices and at most $(r-1)$ molecules. Now, consider a configuration π consisting of r molecules. In the molecule-level quotient graph—where each molecule is represented as a single vertex—the molecules, together with the long internal edges, form a tree. Therefore, there exists a leaf in this tree. Without loss of generality, assume that the molecule indexed by $k=1$ in (A.40) is such a leaf, and that it is connected to $a_1, \dots, a_l$ by external edges. Then, we can decompose $\tilde{\mathcal{K}}^{(\pi)}$ in (A.43) as

$$\begin{aligned}\left|\tilde{\mathcal{K}}^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{a})\right| &= \left|\sum_{\mathbf{b}^{(1)}}\prod_{i=1}^{l}\tilde{\Theta}_{t,a_ib_i}\cdot\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma}^{(1)},\mathbf{b}^{(1)})\cdot\tilde{\mathcal{K}}^{(\pi')}(t,\boldsymbol{\sigma}',\mathbf{a}')\right| \\ &\leq \sum_{c_1}\left|\sum_{\mathbf{b}^{(1)}\setminus\{c_1\}}\prod_{i=1}^{l}\tilde{\Theta}_{t,a_ib_i}\cdot\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma}^{(1)},\mathbf{b}^{(1)})\right|\cdot\left|\tilde{\mathcal{K}}^{(\pi')}(t,\boldsymbol{\sigma}',\mathbf{a}')\right|, \end{aligned} \tag{A.50}$$

where π' consists of the remaining $(r-1)$ molecules, $\mathbf{b}^{(1)}=(b_1,\dots,b_l,c_1)$ with c_1 denoting the vertex in the first molecule that connects to other molecules through a long internal edge, $\boldsymbol{\sigma}^{(1)}=(\sigma_1,\dots,\sigma_l,\sigma_{l+1})$, $\mathbf{a}'=(c_1,a_{l+1},\dots,a_n)$, and $\boldsymbol{\sigma}'=(\sigma_{l+1},\dots,\sigma_n,\sigma_1)$. Applying the induction hypothesis to bound $\tilde{\mathcal{K}}^{(\pi')}$, and using (A.45) to estimate the remaining part of the product (noting that (A.45) is applicable because $\sigma_1\neq\sigma_{l+1}$ by the definition of a molecule), we obtain that

$$\left|\tilde{\mathcal{K}}^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{a})\right| \prec B_{t,0}^{l-1}\cdot B_{t,0}^{n-l} = B_{t,0}^{n-1}.$$

This concludes the proof of Lemma 2.16. □

Finally, Lemma 4.2 follows from the bound (A.45), together with an induction argument analogous to the one used above in the proof of Lemma 2.16.

Proof of Lemma 4.2. By (A.39), it suffices to prove that for any $\pi\subset\mathbb{Z}_n^{\mathrm{off}}$,

$$\sum_{a_n}\left|\mathcal{K}^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{a})\right| \prec \eta_t^{-1}B_{t,0}^{n-2}. \tag{A.51}$$

In the case $\pi=\emptyset$, using (2.65) and (A.45), we can bound the LHS of (A.51) by

$$\sum_{b_n}\sum_{a_n}\Theta_{t,a_nb_n}^{(+,-)}\left|\sum_{\mathbf{b}\setminus\{b_n\}}\Sigma^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{b})\prod_{i=1}^{n-1}\tilde{\Theta}_{t,a_ib_i}\right| \leq \frac{1}{1-t}\sum_{b_n}\left|\sum_{\mathbf{b}\setminus\{b_n\}}\Sigma^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{b})\prod_{i=1}^{n-1}\tilde{\Theta}_{t,a_ib_i}\right| \prec \eta_t^{-1}B_{t,0}^{n-2}.$$

Here, in the last step, we have used that the external edge at a_n is $\Theta_t^{(\sigma_n,\sigma_1)}$ with $\sigma_n\neq\sigma_1$, so that (A.45) applies with b_n playing the role of b_1. If $\sigma_n=\sigma_1$, this edge is short-range, and (2.67) gives $\sum_{a_n}|\Theta_{t,a_nb_n}^{(\sigma_n,\sigma_1)}| = \mathrm{O}(1)$. Then, bounding all but one of the remaining external edges pointwise as in (A.47), summing the last one using (2.65), and summing over the other vertices of the molecule using (A.44), we obtain the bound $\mathrm{O}\left((1-t)^{-1}B_{t,0}^{n-2}\right) \lesssim \eta_t^{-1}B_{t,0}^{n-2}$ in this case as well. For $\pi\neq\emptyset$, we argue by induction as in the proof of Lemma 2.16. Assume that (A.51) holds for any π with at most n external vertices and at most $(r-1)$ molecules. Using the notation of (A.50), and applying (A.45) together with the induction hypothesis, we obtain

$$\begin{aligned}\sum_{a_n}\left|\tilde{\mathcal{K}}^{(\pi)}(t,\boldsymbol{\sigma},\mathbf{a})\right| &\leq \sum_{c_1}\left|\sum_{\mathbf{b}^{(1)}\setminus\{c_1\}}\prod_{i=1}^{l}\tilde{\Theta}_{t,a_ib_i}\cdot\Sigma^{(\emptyset)}(t,\boldsymbol{\sigma}^{(1)},\mathbf{b}^{(1)})\right|\cdot\sum_{a_n}\left|\tilde{\mathcal{K}}^{(\pi')}(t,\boldsymbol{\sigma}',\mathbf{a}')\right| \\ &\prec B_{t,0}^{l-1}\cdot\eta_t^{-1}B_{t,0}^{n-l-1} = \eta_t^{-1}B_{t,0}^{n-2}.\end{aligned}$$

This concludes the proof of (A.51). □

Appendix B. Proofs of auxiliary graphical lemmas

B.1. **Proof of Lemma 6.2.** Recall the light-weight term in (7.1). To prove Lemma 6.2, it suffices to establish

$$W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]} \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}_t E_{a_1}\right)(G_t(\sigma))_{yx}(G_t)_{x\alpha}(G_t)_{\alpha y} \prec \eta_t^{-1}(W^{-d}B_{t,0})^{5/2} \tag{B.1}$$

for any $\sigma\in\{+,-\}$. Here $[a]$ and $[b]$ denote the two external blocks appearing in (7.1), whereas a_1 and a_2 are summation indices. For brevity, write $G_t\equiv G$, and assume $\sigma=-$ in the following proof. The case $\sigma=+$ is analogous. We first focus on the random band matrix model. Performing the GG expansion in (7.23) with respect to $G_{x\alpha}G_{\alpha y}$, we can expand the LHS of (B.1) as

$$\begin{aligned}
&mW^{-2d}\sum_{a_1} S^{(\mathsf{B})}_{a_1b}\sum_{x\in[a],y\in[b]} \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)|G_{xy}|^2 && (I_1)\\
&+ m^3W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]} S^{+}_{\alpha y}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)|G_{xy}|^2 && (J_1)\\
&+ mW^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta} \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)(S_{\alpha\beta}\mathring{G}_{\beta\beta})G_{x\alpha}G_{\alpha y}\overline{G}_{xy} && (I_2)\\
&+ m^3W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta,\gamma} S^{+}_{\alpha\beta}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)(S_{\beta\gamma}\mathring{G}_{\gamma\gamma})G_{x\beta}G_{\beta y}\overline{G}_{xy} && (J_2)\\
&+ mW^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta} \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)(\mathring{G}_{\alpha\alpha}S_{\alpha\beta})G_{x\beta}G_{\beta y}\overline{G}_{xy} && (I_3)\\
&+ m^3W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta,\gamma} S^{+}_{\alpha\beta}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)(\mathring{G}_{\beta\beta}S_{\beta\gamma})G_{x\gamma}G_{\gamma y}\overline{G}_{xy} && (J_3)\\
&- mW^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta} S_{\alpha\beta}\mathbb{E}G_{x\alpha}G_{\beta y}\partial_{\beta\alpha}\left(\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)\overline{G}_{xy}\right) && (I_4)\\
&- m^3W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]}\sum_{\beta,\gamma} S^{+}_{\alpha\beta}S_{\beta\gamma}\mathbb{E}G_{x\beta}G_{\gamma y}\partial_{\gamma\beta}\left(\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)\overline{G}_{xy}\right), && (J_4)
\end{aligned}$$

where we recall that $\partial_{\beta\alpha}\equiv\partial_{(H_t)_{\beta\alpha}}$. We will only estimate the terms I_i for $i\in[\![4]\!]$; the terms J_i can be treated in exactly the same way by using (7.13), together with the bound (2.67).

For the term I_1, we have

$$\begin{aligned}
I_1 &= m\sum_{a_1} S^{(\mathsf{B})}_{a_1b}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)\mathcal{L}^{(2)}_{t,(-,+),(a,b)}\\
&= m\sum_{a_1} S^{(\mathsf{B})}_{a_1b}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)\mathcal{K}^{(2)}_{t,(-,+),(a,b)} + m\sum_{a_1} S^{(\mathsf{B})}_{a_1b}\mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)(\mathcal{L}-\mathcal{K})^{(2)}_{t,(-,+),(a,b)} \prec (W^{-d}B_{t,0})^3,
\end{aligned} \tag{B.2}$$

where in the last step, we use (2.58) and (6.2) to bound the first term, and (2.88) together with (2.91) to bound the second. For I_2, applying the averaged local law (2.88) to both $\mathrm{Tr}(\mathring{G}E_{a_1})$ and $\sum_\beta S_{\alpha\beta}\mathring{G}_{\beta\beta}$ yields

$$\begin{aligned}
I_2 &\prec (W^{-d}B_{t,0})^2\cdot W^{-2d}\sum_{a_1,a_2} S^{(\mathsf{B})}_{a_1a_2}\sum_{x\in[a],y\in[b],\alpha\in[a_2]} \mathbb{E}|G_{x\alpha}||G_{\alpha y}||G_{xy}|\\
&\lesssim (W^{-d}B_{t,0})^2\cdot W^{-2d}\sum_{x\in[a],y\in[b]}\sum_{\alpha}\mathbb{E}|G_{x\alpha}||G_{\alpha y}||G_{xy}|\\
&\prec \eta_t^{-1}(W^{-d}B_{t,0})^2\cdot W^{-2d}\sum_{x\in[a],y\in[b]}\mathbb{E}|G_{xy}| \prec \eta_t^{-1}(W^{-d}B_{t,0})^{5/2},
\end{aligned} \tag{B.3}$$

where in the third step we use Cauchy–Schwarz together with Ward's identity, and in the last step the local law (2.87) to control the averages over $x\in[a]$ and $y\in[b]$. The term I_3 can be handled in exactly the same way. It remains to estimate I_4. Splitting according to which factor the partial derivative acts on, we obtain

$$I_4 = mW^{-3d}\sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2}S^{(\mathsf{B})}_{a_2a_3}\sum_{x\in[a],y\in[b],\alpha\in[a_2],\beta\in[a_3]} \mathbb{E}\,\mathrm{Tr}\left(\mathring{G}E_{a_1}\right)|G_{x\alpha}|^2|G_{\beta y}|^2$$

$$
\begin{aligned}
&+ mW^{-4d} \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \sum_{x\in[a],y\in[b],\gamma\in[a_1],\alpha\in[a_2],\beta\in[a_3]} \mathbb{E} G_{x\alpha} G_{\alpha\gamma} G_{\gamma\beta} G_{\beta y} \overline{G}_{xy} \\
&= mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)} \mathcal{L}^{(2)}_{t,(-,+),(a_3,b)} + mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \mathcal{L}^{(5)}_{t,\boldsymbol{\sigma}_5,\mathbf{a}_5} \\
&=: I_{41} + I_{42},
\end{aligned}
$$

where $\boldsymbol{\sigma}_5 = (+,+,+,+,-)$ and $\mathbf{a}_5 = (a_2, a_1, a_3, b, a)$. For I_{41}, we decompose each 2-G-loop as $(\mathcal{L}-\mathcal{K})^{(2)} + \mathcal{K}^{(2)}$, giving

$$
\begin{aligned}
I_{41} &= mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)} \\
&\quad + mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)} (\mathcal{L}-\mathcal{K})^{(2)}_{t,(-,+),(a_3,b)} \\
&= mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)} + \mathrm{O}_{\prec}\left((W^{-d}B_{t,0})^3\right) \cdot W^d \sum_{a_2} \mathbb{E} \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)} \\
&= mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) \mathcal{K}^{(2)}_{t,(-,+),(a,a_2)} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)} \\
&\quad + mW^d \sum_{a_1,a_2,a_3} S^{(\mathsf{B})}_{a_1a_2} S^{(\mathsf{B})}_{a_2a_3} \mathbb{E} \operatorname{Tr}\left(\mathring{G} E_{a_1}\right) (\mathcal{L}-\mathcal{K})^{(2)}_{t,(-,+),(a,a_2)} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)} + \mathrm{O}_{\prec}\left(\eta_t^{-1}(W^{-d}B_{t,0})^3\right) \\
&\prec (W^{-d}B_{t,0})^3 \cdot W^d \sum_{a_3} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)} + \eta_t^{-1}(W^{-d}B_{t,0})^3 \prec \eta_t^{-1}(W^{-d}B_{t,0})^3.
\end{aligned} \tag{B.4}
$$

Here, we use (2.91) and (2.88) in the second step; Ward's identity (2.56) for $\sum_{a_2} \mathcal{L}^{(2)}_{t,(-,+),(a,a_2)}$ and the local law (2.87) in the third; (2.91), (2.58), and (6.2) in the fourth; and Ward's identity (2.57) for $\sum_{a_3} \mathcal{K}^{(2)}_{t,(-,+),(a_3,b)}$ in the last. Finally, for the term I_{42}, we rewrite it as an average over the graphs $\mathcal{G}_{xy}$:

$$
I_{42} = mW^{-2d} \sum_{x\in[a],y\in[b]} \mathbb{E}\mathcal{G}_{xy}, \quad \text{where} \quad \mathcal{G}_{xy} := \sum_{\gamma,\alpha,\beta} S_{\gamma\alpha} S_{\alpha\beta} \left(G_{x\alpha} G_{\alpha\gamma} G_{\gamma\beta} G_{\beta y} \overline{G}_{xy}\right). \tag{B.5}
$$

We claim the following bound for $\mathbb{E}\mathcal{G}_{xy}$:

$$
\mathbb{E}\mathcal{G}_{xy} \prec \mathbf{1}_{x=y} \cdot \eta_t^{-1}(W^{-d}B_{t,0})^2 + \mathbf{1}_{x\neq y} \cdot \eta_t^{-1}(W^{-d}B_{t,0})^{5/2}. \tag{B.6}
$$

Substituting this estimate into (B.5) gives

$$
I_{42} \prec \eta_t^{-1}(W^{-d}B_{t,0})^{5/2}.
$$

Together with (B.2)–(B.4), this completes the proof of (B.1).

Finally, we prove the estimate (B.6). To this end, we apply the GG-expansion from Lemma 7.13 at the non-standard neutral vertices α, γ, β, which expands $\mathcal{G}_{xy}$ into a collection of new graphs:

$$
\mathcal{G}_{xy} =_{\mathbb{E}} \sum_{\mu} \Gamma_{\mu,xy} \,.
$$

By Lemma 7.21, each $\Gamma_{\mu,xy}$ can be bounded through its auxiliary graph $\Gamma^{\mathrm{aux}}_{\mu,[a][b]}$ as in (7.36), with $q \in \{0,1\}$ internal molecules and $\Psi_t = (W^{-d}B_{t,0})^{1/2}$. If an auxiliary graph contains an internal molecule, then that molecule is attached to at least two solid edges corresponding to the ξ-variables defined in (7.33). Indeed, every vertex of $\mathcal{G}_{xy}$ is incident to an even number of solid half-edges (a light-weight counting twice), and $\mathcal{G}_{xy}$ is connected through its solid and waved edges; both properties are preserved by the local expansions and by the identification of vertices. Since different molecules are joined only by solid edges, an internal molecule of $\Gamma_{\mu,xy}$ is therefore joined to the other molecules by a positive even number of solid edges. Applying Cauchy–Schwarz together with (7.34) therefore gives

$$
\Gamma^{\mathrm{aux}}_{\mu,[a][b]} \prec (W^d\eta_t)^{-q} (\Psi_t)^{\mathsf{ord}(\Gamma^{\mathrm{aux}}_{\mu,[a][b]})}.
$$

Combining this with (7.36) implies that

$$
\Gamma_{\mu,xy} \prec \eta_t^{-1}(W^{-d}B_{t,0})^{\frac{1}{2}\mathsf{ord}(\Gamma_{\mu,xy})}. \tag{B.7}
$$

With this estimate in hand, to conclude (B.6) it remains to show that the scaling order of each $\Gamma_{\mu,xy}$ satisfies

$$\mathsf{ord}(\Gamma_{\mu,xy}) \geq 4 \cdot \mathbf{1}_{x=y} + 5 \cdot \mathbf{1}_{x\neq y}, \tag{B.8}$$

where x and y are regarded as external vertices.

The three remaining cases below all rest on the following fact about the graphs $\mathcal{G}^{(21)}_{xy}$ and $\mathcal{G}^{(4)}_{xy}$ defined below (and the analogue of $\mathcal{G}^{(21)}_{xy}$ in case (3)), which is verified by inspecting the terms on the right-hand side of (7.23):

applying the GG expansion (7.23) at a non-standard neutral vertex strictly increases the scaling order. (B.9)

This fact does not hold for arbitrary graphs. For instance, if the two solid edges at the expansion vertex form a 2-cycle, i.e., $\mathcal{G} = G_{xy}G_{yx}f(G)$ in the notation of (7.23), then the parts of the first two terms on the right-hand side of (7.23) in which the resulting diagonal entry G_{yy} is replaced by m have the same scaling order as $\mathcal{G}$. To obtain (B.8), we decompose $\mathcal{G}_{xy}$ into four parts $\mathcal{G}^{(i)}_{xy}$, $i \in \{1,2,3,4\}$, corresponding to the cases: (1) $\alpha = \gamma = \beta$, (2) $\alpha = \gamma \neq \beta$, (3) $\alpha \neq \gamma = \beta$, and (4) $\alpha \neq \gamma$ and $\gamma \neq \beta$. It is easy to see that $\mathsf{ord}(\mathcal{G}^{(1)}_{xy})$ already satisfies the lower bound in (B.8). Next, in case (2), we write

$$\mathcal{G}^{(2)}_{xy} = m\sum_{\alpha\neq\beta} S_{\alpha\alpha}S_{\alpha\beta}\left(G_{x\alpha}G_{\alpha\beta}G_{\beta y}\overline{G}_{xy}\right) + \sum_{\alpha\neq\beta} S_{\alpha\alpha}S_{\alpha\beta}\mathring{G}_{\alpha\alpha}\left(G_{x\alpha}G_{\alpha\beta}G_{\beta y}\overline{G}_{xy}\right) =: \mathcal{G}^{(21)}_{xy} + \mathcal{G}^{(22)}_{xy}.$$

The second graph $\mathcal{G}^{(22)}_{xy}$ already meets the lower bound in (B.8), while the first graph $\mathcal{G}^{(21)}_{xy}$ has scaling order $3 \cdot \mathbf{1}_{x=y} + 4 \cdot \mathbf{1}_{x\neq y}$. To raise its scaling order, we expand $\mathcal{G}^{(21)}_{xy}$ using the GG expansion (7.23) at the vertex α. By (B.9), every graph produced by this expansion has scaling order strictly larger than that of $\mathcal{G}^{(21)}_{xy}$, and thus satisfies (B.8). Case (3) is handled in exactly the same manner as case (2). Finally, in case (4), the graph $\mathcal{G}^{(4)}_{xy}$ has scaling order $2\cdot\mathbf{1}_{x=y}+3\cdot\mathbf{1}_{x\neq y}$, which has to be raised by 2. We first apply the GG expansion (7.23) at the vertex α; by (B.9), every resulting graph has scaling order at least $3\cdot\mathbf{1}_{x=y}+4\cdot\mathbf{1}_{x\neq y}$. A second GG expansion need not raise the scaling order further, because the first one may create 2-cycles: for instance, the term in which the derivative $\partial_{h_{\alpha'\alpha}}$ in (7.23) acts on $G_{\gamma\beta}$ contains the factor $G_{\alpha'\gamma}G_{\gamma\alpha'}$. We therefore proceed as follows: in every graph that does not yet satisfy (B.8), we apply (7.23) at one of its remaining non-standard neutral vertices, and we repeat. A direct inspection of the resulting finite expansion tree shows that, for any choice of the expansion vertices, every branch ends after at most three GG expansions in a graph satisfying (B.8), for some branches with equality. This yields (B.8) and thus completes the proof of (B.6).

For the block Anderson model, the proof of Lemma 6.2 follows the same outline, in the graphical notation of Appendix B.3 below: we have $S^{(\mathsf{B})} = I_{L^d}$ (so that $a_2 = a_1$ in (B.1)), every solid edge is split as $G = \mathring{G}+M$, and the GG expansion (B.20) (and its complex conjugate) is used in place of (7.23). However, the argument above does not carry over verbatim, and the following three modifications are needed.

(i) *Twisted loops.* Since $M_{uv} = 0$ unless $u - W[u] = v - W[v]$ (see (2.31) and (2.33)), summing over the two ends of an M-dotted edge produces, besides G-loops, the twisted traces $\mathrm{Tr}(\mathring{G}_t(\sigma)P_{c_1c_2})$ and the twisted 2-loops $\widetilde{\mathcal{L}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)} = \mathrm{Tr}(G_t(\sigma_1)E_aG_t(\sigma_2)P_{c_1c_2})$ introduced in Section 8. In coordinates,

$$\mathrm{Tr}\left(\mathring{G}_tP_{c_1c_2}\right) = W^{-d}\sum_{u\in[c_1]}(\mathring{G}_t)_{u'u}, \qquad \widetilde{\mathcal{L}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)} = W^{-2d}\sum_{x\in[a]}\sum_{u\in[c_1]}\big(G_t(\sigma_1)\big)_{u'x}\big(G_t(\sigma_2)\big)_{xu},$$

where, for $u \in [c_1]$, u' denotes the vertex of $[c_2]$ with $u' - W[u'] = u - W[u]$. For $c_1 = c_2$ these are the light-weight $\mathrm{Tr}(\mathring{G}_tE_{c_1})$ and the 2-loop $\mathcal{L}^{(2)}_{t,\boldsymbol{\sigma},(a,c_1)}$. For $c_1 \neq c_2$ they are not G-loops in the sense of Definition 2.11, and bounding them entrywise by (2.87) loses a factor $(W^{-d}B_{t,0})^{-1/2}$. Twisted traces replace the light-weights at the expansion vertex in the analogues of the terms I_3 and J_3, and twisted 2-loops replace the 2-loop $\mathcal{L}^{(2)}_{t,(-,+),(a,a_2)}$ in the analogue of I_{41}. We bound the twisted traces by Lemma 8.6, which gives the same bound $W^{-d}B_{t,0}$ as (2.88) gives for light-weights. We write each twisted 2-loop as the twisted tree $\widetilde{\mathcal{K}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)}$ plus a fluctuation, which is bounded by (8.4); if a leg is circled, we write $\mathring{G}_t(\sigma) = G_t(\sigma) - M(\sigma)$, and the contribution of $M(\sigma)$ is W^{-d} times an entry of $M^{(\mathsf{B})}(\sigma)$ times a twisted trace. By (2.66), (2.65) and (8.5), the twisted tree satisfies $|\widetilde{\mathcal{K}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)}| \lesssim W^{-d}B_{t,0}$ and $\sum_a|\widetilde{\mathcal{K}}_{t,\boldsymbol{\sigma},(a;c_1,c_2)}| \lesssim (W^d\eta_t)^{-1}$, which are the properties of the $\mathcal{K}$-loops used in (B.4). With these replacements, the arguments leading to (B.2)–(B.4) carry over.

Since (8.4) gives the power $3/2$ of $W^{-d}B_{t,0}$ instead of the power 2 in (2.91), the analogue of (B.4) is $\eta_t^{-1}(W^{-d}B_{t,0})^{5/2}$, which still suffices for (B.1).

(ii) *Scaling orders at the level of atoms.* Scaling orders are now defined by (B.17), and the analogue of (B.9) for (B.20) holds only under additional conditions. Before the new solid edges are split as $G = \mathring{G} + M$, the first sum on the right-hand side of (B.20) removes two solid edges, adds one waved edge, and merges the atoms of y and y'; hence it raises the scaling order by 2ν, where $\nu \in \{0, 1\}$ is the number of internal atoms lost in this merger. In particular, it does not raise the scaling order if y and y' lie in the same atom, or if both lie in external atoms. The other terms raise the scaling order by exactly 1. Accordingly, we apply (B.20) to $\mathring{G}_{y'v}\mathring{G}_{vy}f(G)$ at a vertex v only if the following conditions hold:

(H1) the atom of v is internal, and $\mathring{G}_{y'v}$ and $\mathring{G}_{vy}$ are the only solid edges incident to it;
(H2) the atoms of v, y, and y' are pairwise distinct;
(H3) at least one of the atoms of y and y' is internal.

Under these conditions, every graph produced by (B.20) at v has scaling order at least that of the input graph plus 1. Indeed, $\nu = 1$ by (H2) and (H3). Moreover, if j of the new solid edges of a term are replaced by M-dotted edges, the scaling order of the term changes by $2\nu' - j$, where ν' is the number of internal atoms lost in the resulting mergers. Each such M-dotted edge joins two different atoms, at least one of which is internal (the atom of v, which is internal by (H1), an atom created by the expansion, or one of the atoms of y and y' as in (H3)), and checking the terms of (B.20) one by one shows that $\nu' \geq \lceil j/2 \rceil$. The target (B.8) is replaced by the atom-level bound

$$\mathsf{ord}(\Gamma_{\mu,xy}) \geq 5 - 2 \cdot \mathbf{1}\,(x \text{ and } y \text{ lie in the same atom})\,.$$

The version with $\mathbf{1}_{x=y}$ fails for the block Anderson model: for instance, the part of case (1) in which all solid edges are replaced by M-dotted edges has scaling order 4, while x and y lie in the same atom. By (B.15), $x \in [a]$ and $y \in [b]$ can lie in the same atom only if $x - W[x] = y - W[y]$, which holds for at most W^d of the W^{2d} pairs (x, y). Since $B_{t,0} \geq (\mathfrak{d}^{-2} + 1)^{-1}$ by (2.10), the resulting factor $W^{-d} \lesssim W^{-d}B_{t,0}$, obtained after averaging over $x \in [a]$ and $y \in [b]$, compensates for the weaker bound. Finally, the analogue of (B.7), which follows from (B.25) and (B.22), requires that each internal molecule be joined to the other molecules by at least two solid edges. This follows from two properties that hold for $\mathcal{G}_{xy}$ and are preserved by every expansion step and by the splitting $G = \mathring{G} + M$: each atom is incident to an even number of solid half-edges (a light-weight counting twice), and the graph is connected through its solid, waved and dotted edges. Since different molecules are joined only by solid edges, an internal molecule is then joined to the other molecules by a positive even number of solid edges.

(iii) *Folding.* The case analysis below (B.9) is replaced by the following procedure. We expand the block Anderson analogue of the graph $\mathcal{G}_{xy}$ in (B.5) by (B.20), only at vertices satisfying (H1)–(H3), and we stop as soon as a graph satisfies the bound in (ii). Along each branch, at most two such expansions are needed, and every graph at which the procedure stops either satisfies the bound in (ii) and the attachment property, or is a *folded graph*: a graph of scaling order 4, with x and y in different atoms, whose solid edges form the cycle $\mathring{G}_{xu_1}\mathring{G}_{u_1'u_2}\mathring{G}_{u_2'y}(\mathring{G}_t(\sigma))_{yx}$, where u_i and u_i' are joined by M-dotted edges, $i = 1, 2$. An example is the graph $\mathring{G}_{xw}M_{w\gamma}\mathring{G}_{\gamma v}M_{v\delta}\mathring{G}_{\delta y}(\mathring{G}_t(\sigma))_{yx}$ (with the waved edges of $\mathcal{G}_{xy}$), obtained from $\mathcal{G}_{xy}$ by the splitting $G = \mathring{G} + M$. In a folded graph, no vertex is incident to two solid edges of the same charge, so (B.20) cannot be applied, although each of the two internal atoms carries two blue solid edges. For these graphs, we do not prove the pointwise bound (B.6), but only its average over $x \in [a]$ and $y \in [b]$, which is all that is used in (B.5). Since $M_{uu'} \neq 0$ only if $u - W[u] = u' - W[u']$, every folded graph, after summing over all vertices other than x, y, u_1, u_2 and averaging over $x \in [a]$ and $y \in [b]$, takes the form

$$\mathfrak{F}_\rho := W^{-4d} \sum_{\mathbf{c} \in (\mathbb{Z}_L^d)^4} \rho(\mathbf{c}) \sum_{x \in [a],\, y \in [b]} \sum_{u \in [c_1],\, v \in [c_3]} (\mathring{G}_t)_{xu}\, (\mathring{G}_t)_{u'v}\, (\mathring{G}_t)_{v'y}\, \big(\mathring{G}_t(\sigma)\big)_{yx},$$

where $\mathbf{c} = (c_1, c_2, c_3, c_4)$, the vertices $u' \in [c_2]$ and $v' \in [c_4]$ are determined by $u' - W[u'] = u - W[u]$ and $v' - W[v'] = v - W[v]$, and the coefficients $\rho(\mathbf{c})$ collect the entries of $M^{(\mathsf{B})}$, the waved edges and the remaining factors, which do not depend on the vertices of the cycle. In all cases, one checks

using (8.5) and (8.6)–(8.7) that

$$r_1 := \max_{c_1} \sum_{c_2,c_3,c_4} |\rho(\mathbf{c})| \prec B_{t,0}, \qquad r_2 := \max_{c_4} \sum_{c_1,c_2,c_3} |\rho(\mathbf{c})| \prec B_{t,0}.$$

We now show that $|\mathfrak{F}_\rho| \prec \eta_t^{-1}(W^{-d}B_{t,0})^{5/2}$ whenever $r_1 r_2 \prec B_{t,0}^2$. Writing $R := \mathring{G}_t E_b \mathring{G}_t(\sigma)$, we have $\mathfrak{F}_\rho = W^{-3d}\sum_{\mathbf{c},x,u,v} \rho(\mathbf{c})(\mathring{G}_t)_{xu}(\mathring{G}_t)_{u'v}R_{v'x}$. We apply the Cauchy–Schwarz inequality to this sum, with weight $|\rho(\mathbf{c})|$, putting $(\mathring{G}_t)_{xu}(\mathring{G}_t)_{u'v}$ into the first factor and $R_{v'x}$ into the second. Since $R_{v'x}$ does not depend on $u \in [c_1]$, we obtain

$$|\mathfrak{F}_\rho|^2 \le W^{-4d} r_1 r_2 \, \|\mathring{G}_t\|_{\max}^2 \, S_A S_C, \qquad S_A := \sum_{x\in[a]}\sum_{w} |(\mathring{G}_t)_{xw}|^2, \quad S_C := \sum_{x\in[a]} \big\|R\mathbf{e}_x\big\|^2,$$

where $\mathbf{e}_x$ denotes the standard basis vector. By (2.87), $\|\mathring{G}_t\|_{\max}^2 \prec W^{-d}B_{t,0}$. Since $\sum_w |(G_t)_{xw}|^2 = \operatorname{Im}(G_t)_{xx}/\eta_t$ by Ward's identity, and $\sum_w |M_{xw}|^2 = \sum_c |M^{(\mathsf{B})}_{[x]c}|^2 = 1$ by (8.5), we have $S_A \le 2\eta_t^{-1}W^d \operatorname{Im}\operatorname{Tr}(G_t E_a) + 2W^d \prec \eta_t^{-1}W^d$. For S_C, let $\phi_x := E_b\mathring{G}_t(\sigma)\mathbf{e}_x$, a vector supported in $[b]$. We have $\|\mathring{G}_t\phi\|^2 \le 2\|G_t\phi\|^2 + 2\|M\phi\|^2$, where $\|G_t\phi\|^2 = \eta_t^{-1}\operatorname{Im}\langle\phi, G_t\phi\rangle$ by Ward's identity, and $\|M\phi\| = \|\phi\|$ for every vector ϕ supported in a single block by (8.5). Splitting $\phi_x = E_bG_t(\sigma)\mathbf{e}_x - E_bM(\sigma)\mathbf{e}_x$ and using that $\phi \mapsto \operatorname{Im}\langle\phi, G_t\phi\rangle$ is a nonnegative quadratic form, we obtain

$$S_C \le 4\eta_t^{-1}W^d \big|\mathcal{L}^{(3)}_{t,(-\sigma,+,\sigma),(b,b,a)}\big| + 4\eta_t^{-1}W^{-d}\big|\operatorname{Tr}(G_tE_b)\big| + 2\|\mathring{G}_t\|_{\max}^2 \prec \eta_t^{-1}W^d(W^{-d}B_{t,0})^2.$$

Here, the second term comes from $E_bM(\sigma)\mathbf{e}_x$, which is supported on the single vertex of $[b]$ with the same offset as x, and in the last step we used (2.90) with $n = 3$, $|\operatorname{Tr}(G_tE_b)| \prec 1$, $\eta_t \le 1$, and $B_{t,0} \gtrsim 1$. Combining these bounds yields $|\mathfrak{F}_\rho|^2 \prec r_1r_2\,\eta_t^{-2}W^{-2d}(W^{-d}B_{t,0})^3 \prec \eta_t^{-2}(W^{-d}B_{t,0})^5$. Since all quantities involved are deterministically bounded by W^C, the same bound holds for the expectations. Together with (ii), we conclude

$$W^{-2d}\sum_{x\in[a],\,y\in[b]} \mathbb{E}\,\mathcal{G}_{xy} \prec \eta_t^{-1}(W^{-d}B_{t,0})^{5/2},$$

which is the form in which (B.6) is used in (B.5).

With these modifications, the remaining steps are the same as for the random band matrix model.

B.2. **Proof of Lemma 7.17.** Our local expansion strategy leading to the proof of Lemma 7.17 proceeds as follows. Given a graph Γ that is not locally standard, we first find all weights in it, and apply the weight expansion (7.20) to remove them one by one. Once the graph is free of weights, we find all vertices connected to more than two solid edges, and apply the edge expansion (7.22) iteratively to reduce their degrees. After removing weights and ensuring each vertex is connected to at most two solid edges, we identify all vertices that are not standard neutral—that is, vertices connected to two G edges or two $\overline{G}$ edges—and apply the GG expansion (7.23) to correct them. It is important to note that after each expansion, the resulting graphs may require earlier steps to be re-applied. For instance, an edge expansion might introduce new weights, which must be removed via weight expansions before proceeding with further edge or GG expansions. We now formally state the local expansion rules. Throughout the following proof, when we refer to the degree of a vertex α, we mean *the number of solid edges connected to it*, excluding any solid self-loops (i.e., weights).

Strategy B.1 (Local expansion strategy). Given an arbitrarily large constant $D > 0$, we apply the following local expansion strategy.

Step 1: Given an input graph, we apply the *weight expansion* $\mathcal{O}_{\text{weight}}$ as follows. If the graph contains no weights (neither regular nor light), then $\mathcal{O}_{\text{weight}}$ is a null operation, and the graph is passed to the next step. Otherwise, we select one of its weights, say on vertex α. If this is a light-weight, we apply the expansion (7.20) directly. If it is a regular weight, i.e., $G_{\alpha\alpha}$ or $\overline{G}_{\alpha\alpha}$, we decompose it as a sum of a light-weight and a factor of m or $\overline{m}$: $(G_{\alpha\alpha} - m) + m$ or $(\overline{G}_{\alpha\alpha} - \overline{m}) + \overline{m}$, and then apply the expansion (7.20) to the resulting light-weight term. For each new graph generated by the weight expansion, we apply the operations in Definition 7.7 to write it as a sum of normal graphs. For each normal graph $\mathcal{G}$, if its scaling size is sufficiently small:

$$\mathsf{size}(\mathcal{G}) \le W^{-D}, \tag{B.10}$$

then we send it directly to the output; if it contains no weights, we pass it to Step 2; if it still contains weights, we restart Step 1.

Step 2: At this stage, the input graph has no weights. We apply the edge expansion $\mathcal{O}_{\text{edge}}$ as follows. If there exists a vertex α whose degree is $\notin \{0,2\}$, or whose charge is not neutral (recall the definition in (7.24)),[14] then we apply the edge expansion (7.22) at α. For each resulting graph, we again apply the operations in Definition 7.7 to express it as a sum of normal graphs. Those satisfying the small-size condition (B.10) are sent to the output, while the rest are sent back to Step 1. If all internal vertices have degree $\in \{0,2\}$ and are neutral in charge, then $\mathcal{O}_{\text{edge}}$ is a null operation, and the graph is passed to Step 3.

Step 3: Now, the input graph has no weights, and all internal vertices have degree $\in \{0,2\}$ and neutral charge. We apply the GG expansion $\mathcal{O}_{GG}$ as follows. If there exists a vertex α connected to two solid edges of the same charge (i.e., two G edges or two $\overline{G}$ edges), we apply the expansion (7.23) at α. Each resulting graph is expanded into a sum of normal graphs using the operations in Definition 7.7. Those satisfying (B.10) are sent to the output, while the rest are sent back to Step 1. If all internal vertices are standard neutral, then $\mathcal{O}_{GG}$ is a null operation, and the graph is sent to the output.

We now apply Strategy B.1 to complete the proof of Lemma 7.17.

Proof of Lemma 7.17. By Lemma 7.15, we obtain the expansion of $\big|f_{xy}(G)\big|^p$ as in (7.27), where the graphs $\Gamma_{\mu,xy}$ trivially satisfy property (1). For property (2), note that the original graph

$$\Gamma_{xy} = \big|f_{xy}(G)\big|^p = \sum_{\boldsymbol{\alpha},\boldsymbol{\beta}} \prod_{i=1}^{p} \Big[S_{\alpha_i\beta_i} \mathbf{1}_{x\neq\alpha_i} \mathbf{1}_{y\neq\alpha_i} \cdot \mathring{G}^{(i)}_{\beta_i\beta_i} G^{(i)}_{x\alpha_i} G^{(i)}_{\alpha_i y} \Big] \tag{B.11}$$

where $G^{(i)} := G$ for $1 \le i \le p/2$ and $G^{(i)} := \overline{G}$ for $p/2 < i \le p$ (recall $p \in 2\mathbb{N}$), contains p internal molecules, corresponding to the vertices $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_p)$ and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$. During the expansions, the number of internal molecules never increases; it may decrease when two molecules—internal or external—are merged due to new dotted or waved edges created in the process. Consequently, $\Gamma_{\mu,xy}$ may contain strictly fewer molecules than Γ_{xy}. In particular, if all internal molecules merge with external ones, then $\Gamma_{\mu,xy}$ contains no internal molecules, in which case $q = 0$. Finally, (7.28) follows directly from the definition of a molecule, since all vertices within a molecule are connected by paths of waved edges.

For properties (3)–(5), note that the original graph Γ_{xy} contains p edge-disjoint paths between $\mathcal{M}_x$ and $\mathcal{M}_y$, each passing through an internal molecule $\mathcal{M}_i$ that contains the vertex α_i, $i \in [\![p]\!]$. We denote the path through $\mathcal{M}_i$ by $\mathfrak{P}_i$. During the expansions, molecules may merge, but for notational convenience, we retain their original labels: if molecules $\mathcal{M}_i$ and $\mathcal{M}_j$ merge, we continue to refer to the resulting molecule as both $\mathcal{M}_i$ and $\mathcal{M}_j$. Similarly, although the paths $\mathfrak{P}_i$ may change during the expansion, we keep their names and regard each $\mathfrak{P}_i$ as the path associated with $\mathcal{M}_i$. Checking Lemmas 7.11 to 7.13, we observe that these paths never disappear under local expansions. More precisely, if e is a solid edge between two distinct molecules $\mathcal{M}$ and $\mathcal{M}'$, then in any local expansion either:

- the edge e remains unaffected on the molecular graph; or
- e is replaced by two new edges e_1 and e_2, which still form a connected path between $\mathcal{M}$ and $\mathcal{M}'$ (though this connectedness may fail at the vertex level); or
- the molecules $\mathcal{M}$ and $\mathcal{M}'$ merge, in which case e disappears on the molecular graph, but the connectedness between $\mathcal{M}$ and $\mathcal{M}'$ holds trivially.

From this observation, properties (3) and (5) follow.

For property (4), let $\mathcal{M}$ be an internal molecule in $\Gamma_{\mu,xy}$. If $\mathcal{M}$ arises from merging at least two internal molecules, say $\mathcal{M}_i$ and $\mathcal{M}_j$, then both associated paths $\mathfrak{P}_i$ and $\mathfrak{P}_j$ pass through $\mathcal{M}$. If the previous scenario does not occur, then we are in the case $\mathcal{M} = \mathcal{M}_i$, which corresponds to the following path in the original graph Γ_{xy} (assuming, without loss of generality, that the path carries a + charge):

$$\sum_{\alpha_i,\beta_i} S_{\alpha_i\beta_i} \mathbf{1}_{x\neq\alpha_i} \mathbf{1}_{y\neq\alpha_i} \cdot \mathring{G}_{\beta_i\beta_i} G_{x\alpha_i} G_{\alpha_i y}.$$

[14] If α is attached to two + solid edges and has non-neutral charge, then the two edges incident to it must be of the form $G_{\alpha y} G_{\alpha y'}$ or $G_{y\alpha} G_{y'\alpha}$. Such configurations cannot be expanded using the GG expansion (7.23), and therefore must first be removed via the edge expansion (7.22).

To ensure that $\Gamma_{\mu,xy}$ is locally standard, the molecule $\mathcal{M}_i$ must pull in some $\overline{G}$ edges from other paths or molecules during the expansions. If a $\overline{G}$-edge from path $\mathfrak{P}_j$ is pulled to $\mathcal{M}_i$, then $\mathfrak{P}_j$ also passes through $\mathcal{M}_i$. Similarly, if a $\overline{G}$-edge inside $\mathcal{M}_j$ is pulled to $\mathcal{M}_i$, then the two new edges between $\mathcal{M}_i$ and $\mathcal{M}_j$ can be incorporated into path $\mathfrak{P}_j$, so that $\mathfrak{P}_j$ again passes through $\mathcal{M}_i$. In either case, property (4) holds.

Finally, we examine the scaling order of $\Gamma_{\mu,xy}$. For the original graph Γ_{xy}, the scaling order is

$$\mathsf{ord}(\Gamma_{xy}) = p, \tag{B.12}$$

which follows from the presence of $2p$ off-diagonal solid edges, p light-weights, p waved edges, and $2p$ internal vertices. The scaling order itself is not monotone along the expansions; for instance, the contraction of a 2-cycle discussed below (B.9) does not increase it. We therefore track a quantity that also accounts for the vertices that still have to be expanded, and that is minimized over all identifications of vertices that may occur later.

Let $\mathcal{G}$ be a normal graph. A *partition* π of $\mathcal{G}$ is a partition of its vertex set into classes; a class is *external* if it contains an external vertex, and *internal* otherwise. A solid edge is *kept* by π if it is a light-weight or joins two different classes, and $\mathcal{G}/\pi$ denotes the graph obtained from $\mathcal{G}$ by identifying the vertices in each class and removing the solid edges that are not kept.[15] An internal class of π is called *elementary* if, as a vertex of $\mathcal{G}/\pi$, it either carries exactly one light-weight and no other solid edge, or carries no light-weight and exactly two solid edges, which have the same color (two G edges or two $\overline{G}$ edges). We define the cost of π and the local cost of $\mathcal{G}$ by

$$c_\pi(\mathcal{G}) := \mathsf{ord}(\mathcal{G}/\pi) + \#\{\text{elementary classes of } \pi\}, \qquad \Phi(\mathcal{G}) := \min_\pi c_\pi(\mathcal{G}), \tag{B.13}$$

where $\mathsf{ord}(\mathcal{G}/\pi) = \#\{\text{solid edges kept by } \pi\} + 2n_W(\mathcal{G}) - 2\#\{\text{internal classes of } \pi\}$ as in (7.19), and we denote by $\Phi^{\mathrm{far}}(\mathcal{G})$ the minimum of $c_\pi(\mathcal{G})$ over the partitions π in which x and y lie in different classes. These quantities have the following three properties.

(a) Let π_0 be the partition of $\mathcal{G}$ into single vertices. Then $\mathcal{G}/\pi_0 = \mathcal{G}$, and the elementary classes of π_0 are the internal vertices that carry either exactly one light-weight and no other solid edge, or exactly two solid edges of the same color and no light-weight. A locally standard graph has no such vertices. Hence, if $\mathcal{G}$ is locally standard and $x \neq y$, then $\mathsf{ord}(\mathcal{G}) = c_{\pi_0}(\mathcal{G}) \geq \Phi^{\mathrm{far}}(\mathcal{G})$.

(b) We have $\Phi(\Gamma_{xy}) = 2p$ and $\Phi^{\mathrm{far}}(\Gamma_{xy}) = 3p$. Indeed, all internal vertices α_i and β_i of Γ_{xy} are elementary, so every internal class of a partition π that consists of a single vertex is elementary. Hence $-2\#\{\text{internal classes}\}+\#\{\text{elementary classes}\}$ is at least minus the number of internal vertices lying in internal classes, and each β_i contributes at least -1 to $c_\pi(\Gamma_{xy})$. Each α_i contributes at least 1, unless its class contains both x and y, in which case it contributes at least 0: if α_i lies in an internal class, its two edges $G^{(i)}_{x\alpha_i}$ and $G^{(i)}_{\alpha_i y}$ are kept, and if it lies in an external class not containing both x and y, at least one of them is kept. Together with the p light-weights and the contribution $2p$ of the waved edges, we obtain

$$c_\pi(\Gamma_{xy}) \geq 2p + \#\{i \in [\![p]\!] : \text{the class of } \alpha_i \text{ does not contain both } x \text{ and } y\},$$

which gives $\Phi(\Gamma_{xy}) \geq 2p$ and $\Phi^{\mathrm{far}}(\Gamma_{xy}) \geq 3p$. Equality is attained by putting x, y, and all α_i in one class, and by π_0, respectively.

(c) Φ and Φ^{far} do not decrease along the expansions: if $\mathcal{G}_1$ is a normal graph obtained from $\mathcal{G}_0$ by one step of Strategy B.1, then $\Phi(\mathcal{G}_1) \geq \Phi(\mathcal{G}_0)$ and $\Phi^{\mathrm{far}}(\mathcal{G}_1) \geq \Phi^{\mathrm{far}}(\mathcal{G}_0)$.

To prove (c), given a partition π_1 of $\mathcal{G}_1$, we construct a partition π_0 of $\mathcal{G}_0$ with $c_{\pi_0}(\mathcal{G}_0) \leq c_{\pi_1}(\mathcal{G}_1)$, such that x and y lie in different classes of π_0 whenever they do in π_1. Since the dotted edge partition only identifies vertices, and an identification of vertices amounts to passing to a coarser partition (with a cost that is not smaller), it suffices to consider the graphs on the right-hand sides of (7.20), (7.22), and (7.23) before the dotted edge partition. Each of these graphs is obtained from $\mathcal{G}_0$ by a bounded number of local operations at the expansion vertex: adding a new vertex joined to it by a waved edge, adding or removing a light-weight, moving an end of a solid edge or of a light-weight (this includes the action of the partial derivatives), and contracting a solid edge, i.e., identifying its two ends (this comes from the factors δ in (7.22) and (7.23)). Placing a new vertex in a class that contains at most one of its neighbors (counted with multiplicity) never lowers the cost, so it suffices to consider partitions π_1 in which each new vertex either forms a class of its own

[15] Intuitively, π describes an identification of vertices that may occur in later expansions: an off-diagonal solid edge whose two ends are identified may be replaced by its deterministic part m or $\overline{m}$, and then loses its factor Ψ_t.

or lies in a class containing at least two of its neighbors. For each of the local operations, one checks directly that the partition π_0 induced by π_1 on the vertices of $\mathcal{G}_0$ (in which identified vertices are put into the same class) is a suitable choice, with one exception: when the two solid edges of a 2-cycle $G_{uv}G_{vu}$ (or $\overline{G}_{uv}\overline{G}_{vu}$) at an internal vertex u are removed without identifying u and v, as in the second term of the example below (B.9), the cost of the partition into single vertices may decrease by 1. In this case, we put in addition u into the class of v in π_0. Since u is internal, this never merges two external classes, so x and y remain in different classes. This finite case-by-case verification over the terms of (7.20), (7.22), and (7.23) replaces a bookkeeping of individual light-weights and vertices, which is not monotone along the expansions.

Applying (c) along the expansions in Lemma 7.15 and using (a) and (b), we obtain that every locally standard graph $\Gamma_{\mu,xy}$ satisfies

$$\mathsf{ord}(\Gamma_{\mu,xy}) \geq \Phi^{\mathrm{far}}(\Gamma_{\mu,xy}) \geq \Phi^{\mathrm{far}}(\Gamma_{xy}) = 3p > 2p, \tag{B.14}$$

since $x \neq y$ under (7.12). (Without (7.12), the same argument with Φ in place of Φ^{far} gives $\mathsf{ord}(\Gamma_{\mu,xy}) \geq 2p$, also when $x = y$.) This concludes (7.30). □

Remark B.2. The bound (B.14) is in fact the stronger bound $\mathsf{ord}(\Gamma_{\mu,xy}) \geq 3p$, which in turn improves (7.10) by replacing the factor $[\Psi_t(0)]^p$ with $[\Psi_t(0)]^{2p}$. We do not use this refinement, however, since the bottleneck of the main proof lies instead in the martingale estimate of Lemma 3.13 (see Remark 3.14).

B.3. **Proof of Lemmas 3.10 and 3.11 for the block Anderson model.** Analogously to the random band matrix model, the proofs of Lemmas 3.10 and 3.11 for the block Anderson model reduce to Lemmas 7.1 and 7.2. In this reduction (see the beginning of Section 7), Lemma 8.4 gives, in place of (7.2) and (7.3),

$$|(G_t(\sigma'))_{yx}| \prec |(M(\sigma'))_{yx}| + \Psi_t(|a-b|) \quad \text{and} \quad |(G_t(\sigma'))_{yx}| \prec |(M(\sigma'))_{yx}| + \mathsf{T}_t(|a-b| \wedge \ell) + W^{-D},$$

respectively, where $M(\sigma')$ is defined in (2.45). Unlike δ_{xy}, the term $|(M(\sigma'))_{yx}|$ need not vanish for $x \neq y$; it is of order g for some pairs in neighboring blocks. However, it is harmless after averaging over $x \in [a]$ and $y \in [b]$: by (2.31) and (2.33), $M_{yx} \neq 0$ only if $x - W[x] = y - W[y]$, so at most W^d of the W^{2d} pairs (x, y) contribute, each with $|M_{yx}| \leq Ce^{-c|a-b|}$ by (8.6) or (8.7). Hence the additional contributions carry a factor $\mathrm{O}(W^{-d}e^{-c|a-b|})$, which is bounded by $\Psi_t(|a-b|)$ by (3.28), and by $\mathsf{T}_t(|a-b| \wedge \ell)$ since $\mathsf{T}_t(0) \gtrsim W^{-d/2}$ and $\mathsf{T}_t(0)e^{-cr} \lesssim \mathsf{T}_t(r \wedge \ell)$. To establish Lemmas 7.1 and 7.2, we introduce new graphical notations and corresponding local expansion rules. In the setting of the block Anderson model, we continue to define the matrices $S^{\pm}$ as in (7.13), with $\Theta^{(+,+)}(z)$ defined in (2.35) and $S^{(\mathsf{B})} = I_{L^d}$. In addition to the components introduced in Definition 7.3, we introduce new types of edges that represent the matrix entries of Ψ and M, specific to this model:

- **Ψ-dotted edge**: A black dotted edge labeled Ψ between vertices x and y represents a factor $g\Psi_{xy}$.[16]
- **M-dotted edge**: A blue (resp. red) dotted edge labeled M between vertices x and y represents a factor M_{xy} (resp. $\overline{M}_{xy}$).

With these new types of dotted edges, the definitions of molecules and molecular graphs remain the same as in Definition 7.5. However, graphs in the block Anderson model carry an additional level of structure, which we refer to as *atoms*. More precisely, our graphs exhibit microscopic structures, called atoms, which are equivalence classes of vertices connected via dotted edges. These atoms form mesoscopic structures within each molecule, and the global (macroscopic) structure is represented by the molecular graph.

Definition B.3 (Atoms and atomic graphs)**.** *We partition the set of all vertices in a graph into disjoint subsets called atoms. Two vertices belong to the same atom if and only if they are connected by a path consisting entirely of dotted edges—that is, any combination of regular dotted edges, Ψ-dotted edges, or M-dotted edges. An atom is called external if it contains at least one external vertex; otherwise, it is called internal.*

Given a graph $\mathcal{G}$, we define its atomic graph as follows:

- *Merge all vertices belonging to the same atom into a single vertex.*
- *Retain all solid and waved edges that connect different atoms.*
- *Discard all other components within $\mathcal{G}$, including $\times$-dotted edges, edges between vertices within the same atom, and coefficients.*

[16] Here Ψ denotes the block-hopping matrix of the model, cf. (2.31); it should not be confused with the control parameter Ψ_t of (3.28).

By the definition of (regular) dotted edges, the form of the Ψ-dotted edges in (2.31), and the bounds for the M-dotted edges in (8.6) or (8.7), we deduce that, up to an error of order $e^{-c(\log W)^{1+\varepsilon_1}}$, where $\varepsilon_1 \in (0, 1/10)$ is the constant fixed in (7.31),

$$x,\, y \text{ belong to the same atom } \implies x - W[x] = y - W[y], \text{ and } |[x] - [y]| \le (\log W)^{1+\varepsilon_1}. \tag{B.15}$$

Here, $[x], [y] \in \mathbb{Z}_L^d$ are the block-level vertices as defined in Definition 7.18. The concept of atoms is introduced for two main purposes: first, to define the scaling size and scaling order of our graphs; and second, to enable a structural comparison between the block Anderson model and the random band matrix model. In particular, if every solid edge between two vertices of the same atom is retained as a solid self-loop of the merged vertex, then the atomic graphs in the block Anderson model have the same form as the vertex-level graphs in the random band matrix setting, where the M matrices reduce to scalars. This correspondence transfers the *statements* concerning molecular graphs from the random band matrix model to the block Anderson model, but not the *arguments* step by step. A local expansion in the block Anderson model acts at a vertex rather than at an atom (for example, (B.20) below requires two solid edges at the same vertex, not merely in the same atom), and in general it is not the image of a single expansion step of the random band matrix model; the correspondence holds only for the outputs of the expansions, after grouping consecutive steps. The structural properties used below are therefore verified directly for the block Anderson expansions; see the end of Appendix B.1 and the proof of Lemmas 7.1 and 7.2 below.

Definition B.4 (Normal graphs)**.** *We say a graph is* normal *if it satisfies the following properties:*

(i) *It contains at most* $\mathrm{O}(1)$ *many vertices and edges.*
(ii) *There are no regular dotted edges between vertices (note that* Ψ*- or* M*-dotted edges are allowed).*
(iii) *Every solid edge carries a* $\circ$ *(i.e., it represents an entry of* $\mathring{G}$ *or* $\mathring{G}^*$*). In particular, all weights are light-weights.*

Given an arbitrary graph with $\mathrm{O}(1)$ many vertices and edges, we can decompose it into a linear combination of normal graphs by expanding each G_{xy} (resp. G^*_{xy}) edge into a $\mathring{G}_{xy}$ edge plus an M_{xy} edge (resp. a $\mathring{G}^*_{xy}$ edge plus an M^*_{xy} edge), and by merging any vertices connected by regular dotted edges. For a normal graph, we define its scaling size in the same way as in Definition 7.8, except that we replace the number of internal vertices with the number of internal atoms.

Definition B.5 (Scaling size and scaling order)**.** *We define the scaling size of a normal graph* Γ *as*

$$\mathsf{size}(\Gamma) := (L^d)^{n_M(\Gamma)} \cdot (\Psi_t)^{n_S(\Gamma)} \cdot W^{-d(n_W(\Gamma) - n_A(\Gamma))}, \tag{B.16}$$

where $n_S(\Gamma)$*,* $n_W(\Gamma)$*,* $n_A(\Gamma)$*, and* $n_M(\Gamma)$ *denote the numbers of solid edges (including light-weights), waved edges, internal atoms, and internal molecules, respectively. The scaling order of* Γ *is then defined as*

$$\mathsf{ord}(\Gamma) := n_S(\Gamma) + 2\,(n_W(\Gamma) - n_A(\Gamma))\,. \tag{B.17}$$

If Γ *can be expressed as a sum of* $\mathrm{O}(1)$ *many normal graphs* Γ_k*, i.e.,* $\Gamma = \sum_k \Gamma_k$*, we define its scaling size as in* (7.17)*, and its scaling order by* $\mathsf{ord}(\Gamma) = \min_k \mathsf{ord}(\Gamma_k)$*.*

Next, we state the local expansion rules for the block Anderson model, as given in [68].

Lemma B.6 (Basic expansion, Lemma B.9 of [68])**.** *In the setting of the block Anderson model, let* f *be a differentiable function of* G*. Then, we have the expansion*

$$\mathring{G}_{xy} f(G) =_{\mathbb{E}} \sum_{\alpha,\beta} M_{x\alpha} S_{\alpha\beta} \mathring{G}_{\beta\beta} G_{\alpha y} f(G) - \sum_{\alpha,\beta} M_{x\alpha} S_{\alpha\beta} G_{\beta y} \partial_{h_{\beta\alpha}} f(G). \tag{B.18}$$

The purpose of the basic expansion is to expand any graph as a sum of graphs in which every vertex has a solid-edge degree $\in \{0, 2\}$. If a vertex still carries self-loops (i.e., weights), we then apply the following weight expansion.

Lemma B.7 (Weight expansion, Lemma B.10 of [68])**.** *In the setting of the block Anderson model, let* f *be a differentiable function of* G*. Then, we have the expansion*

$$\mathring{G}_{xx} f(G) =_{\mathbb{E}} \sum_{y} (1 + M^+ S^+)_{xy} \Big(\sum_{\alpha,\beta} M_{y\alpha} S_{\alpha\beta} \mathring{G}_{\alpha y} \mathring{G}_{\beta\beta} f(G) - \sum_{\alpha,\beta} M_{y\alpha} S_{\alpha\beta} G_{\beta y} \partial_{h_{\beta\alpha}} f(G) \Big), \tag{B.19}$$

where M^+ *is the* $N \times N$ *matrix with entries* $M^+_{xy} := M_{xy} M_{yx}$*.*

If there are vertices incident to two solid edges of the same charge, we use the following GG expansion.

Lemma B.8 (GG expansion, Lemma B.11 of [68]). *In the setting of the block Anderson model, let f be a differentiable function of G. Then, we have the expansion*

$$\mathring{G}_{y'x}\mathring{G}_{xy}f(G) =_{\mathbb{E}} \sum_{\beta}(M^+S^+)_{x\beta}M_{\beta y}M_{y'\beta}f(G) + \sum_{\beta}(M^+S^+)_{x\beta}\left(\mathring{G}_{\beta y}M_{y'\beta} + M_{\beta y}\mathring{G}_{y'\beta}\right)f(G) \tag{B.20}$$
$$+ \sum_{w,\alpha,\beta}\left(1+M^+S^+\right)_{xw}M_{w\alpha}S_{\alpha\beta}\left(\mathring{G}_{\beta\beta}G_{\alpha y}\mathring{G}_{y'w}f(G) + \mathring{G}_{\alpha w}G_{\beta y}G_{y'\beta}f(G) - G_{\beta y}\mathring{G}_{y'w}\partial_{h_{\beta\alpha}}f(G)\right).$$

The expansion of $\overline{\mathring{G}_{y'x}}\,\overline{\mathring{G}_{xy}}f(G)$ (two red solid edges) can be derived by taking the complex conjugate of (B.20).

As explained in [68, Appendix B], repeated applications of the above local expansions to an arbitrary normal graph yield a sum of O(1) *locally standard graphs* (see Definition 7.14), subject to the additional requirement that all solid edges are $\mathring{G}$ edges. This allows us to establish an analogue of Lemma 7.15 for the block Anderson model. With the above preparations, we can now proceed to the proof of Lemmas 7.1 and 7.2 for the block Anderson model. Since the argument is very similar to that in Section 7, we only sketch the proof and omit the repetitive details.

Proof of Lemmas 7.1 and 7.2 for the block Anderson model. We begin by applying the local expansions (B.18), (B.19), and (B.20) (and their complex conjugates) to the graph $|f_{xy}(G)|^p$, splitting every solid edge as $G = \mathring{G} + M$. This yields the analogue of Lemma 7.17 for the block Anderson model, with properties (1)–(6) read on the atomic graphs. In property (2), $n_V(\mathcal{M}_i)$ denotes the number of atoms in $\mathcal{M}_i$, in accordance with Definition B.5; counted with vertices, (7.28) fails already for the initial graph, since M-dotted edges join vertices of a molecule without waved edges. With atoms in place of vertices, (2) follows as in Appendix B.2, since the atoms of a molecule are connected by paths of waved edges.

These properties are not inherited from the random band matrix model step by step (see the discussion below (B.15)), and we verify them directly. For the path properties (3)–(5), one checks term by term that every expansion step, including the splitting $G = \mathring{G} + M$, acts on the molecular graph in one of the three ways listed in Appendix B.2: a solid edge between two molecules is unaffected, or it is replaced by two edges forming a path between the same molecules, or it disappears because the two molecules merge. Hence the family of p edge-disjoint paths from $\mathcal{M}_x$ to $\mathcal{M}_y$ (which may revisit molecules), labeled as in Appendix B.2, is preserved. This includes the terms of (B.20) that have no single counterpart in (7.23), namely its first sum and the terms in which both expanded solid edges become edges inside one molecule.

Property (6) is used, and proved, when x and y lie in different atoms, which holds under (7.12) up to a negligible error by (B.15). For (6), we use the local cost Φ^{far} of Appendix B.2, computed on the atomic graphs, in which solid edges between different atoms are treated as off-diagonal solid edges and solid edges inside an atom as light-weights. A locally standard graph has no elementary atoms, since each of its vertices carries either no solid edge or one blue and one red solid edge. The splitting $G = \mathring{G} + M$ of a solid edge between two atoms replaces it by an M-dotted edge and merges the two atoms; this does not decrease Φ^{far}, because every partition of the new atomic graph comes from a partition of the old one in which the two atoms lie in the same class, the removed edge is not kept by the latter, and the two quotient graphs coincide. Most of the remaining terms act on the atomic graphs as compositions of the local operations considered in Appendix B.2; for the terms that have no counterpart in (7.20), (7.22), and (7.23) (for instance, the first sum in (B.20)), the same comparison of partitions is carried out directly. Hence $\mathsf{ord}(\Gamma_{\mu,xy}) \geq 3p > 2p$ whenever x and y lie in different atoms.

Next, we define the auxiliary graph in a similar manner as in Definition 7.20, with minor modifications that we now describe. Let $\mathcal{G}_{xy}$ be a locally standard graph obtained from the local expansions, with q internal molecules $\mathcal{M}_i$, $i \in [\![q]\!]$, and two external molecules $\mathcal{M}_x$ and $\mathcal{M}_y$, as in Definition 7.20. For each $i \in \{1,\dots,q,x,y\}$, we fix a center $\alpha_i \in \mathcal{M}_i$, and we also choose centers $x_{i,j}$ for the atoms $\mathcal{A}_{i,j} \subset \mathcal{M}_i$, where $j \in [\![k_i]\!]$ and k_i denotes the number of atoms in $\mathcal{M}_i$. Without loss of generality, we set $\alpha_i \equiv x_{i,1}$ and assume that α_i is connected to other molecules via solid edges. (By the definition of locally standard graphs, the solid-edge degree of each α_i is at least 2.) If $(\beta \to \beta')$ is a solid edge connecting the atoms $\mathcal{A}_{i,j}$ and $\mathcal{A}_{i',j'}$, then by (B.15), the form of the Ψ-dotted edges in (2.31), and the exponential decay estimates (8.6) or (8.7)

for the M-dotted edges, we obtain

$$\left|(\mathring{G}_t)_{\beta\beta'}\right| \prec \zeta(x_{i,j}, x_{i',j'}) := \sum_{|[a]|+|[b]|\le(\log W)^{1+\varepsilon_1}} \left|(\mathring{G}_t)_{x_{i,j}+W[a],x_{i',j'}+W[b]}\right| + W^{-D} \tag{B.21}$$

for any large constant $D > 0$. This quantity satisfies bounds analogous to (7.34):

$$\zeta(\alpha,\beta) \prec \Psi_t(|[\alpha]-[\beta]|), \qquad \sum_{\alpha\in\mathbb{Z}^d_{WL}} (|\zeta(\alpha,\beta)|^2 + |\zeta(\beta,\alpha)|^2) \prec \eta_t^{-1}, \tag{B.22}$$

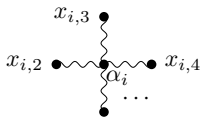
where the first estimate follows from Lemma 8.4, and the second from Ward's identity for G together with (8.6) or (8.7) for M. If (β,β') is a waved edge connecting the atoms $\mathcal{A}_{i,j}$ and $\mathcal{A}_{i',j'}$, then by (B.15), the definition of S in (2.29), and the estimate (2.67), we obtain

$$\left|S_{\beta\beta'}\right| + \left|S^+_{\beta\beta'}\right| \prec W^{-d}\mathbf{1}\left(|x_{i,j}-x_{i',j'}| \le W(\log W)^{1+\varepsilon_1}\right) + W^{-D}. \tag{B.23}$$

We first define a graph $\mathcal{G}^a_{xy}$ as an "atomic reduction" of $\mathcal{G}_{xy}$:

- Its vertices are the atom centers $x_{i,j}$ for $i \in \{1,\dots,q,x,y\}$ and $j \in [\![k_i]\!]$.
- For each solid edge in $\mathcal{G}_{xy}$ from the atom $\mathcal{A}_{i,j}$ to the atom $\mathcal{A}_{i',j'}$, we introduce an oriented solid edge from $x_{i,j}$ to $x_{i',j'}$ in $\mathcal{G}^a_{xy}$, representing the factor $\zeta(x_{i,j}, x_{i',j'})$.
- For each waved edge in $\mathcal{G}_{xy}$ connecting the atoms $\mathcal{A}_{i,j}$ and $\mathcal{A}_{i',j'}$, we introduce a waved edge between $x_{i,j}$ and $x_{i',j'}$ in $\mathcal{G}^a_{xy}$, representing the factor $W^{-d}\mathbf{1}\left(|x_{i,j}-x_{i',j'}| \le W(\log W)^{1+\varepsilon_1}\right)$.

Next, we further simplify the structure of $\mathcal{G}^a_{xy}$ and define the auxiliary graph $\mathcal{G}^{\mathrm{aux}}_{xy}$ as follows. First, we remove all solid and waved edges that lie entirely within individual molecules of $\mathcal{G}^a_{xy}$. Second, for each molecule $\mathcal{M}_i$, we retain only those vertices (including the center α_i) that connect to other molecules through solid edges; all remaining vertices inside $\mathcal{M}_i$ are discarded. Third, for each retained vertex $x_{i,j} \neq \alpha_i$ inside $\mathcal{M}_i$, we add a new waved edge connecting $x_{i,j}$ to α_i, representing the factor $W^{-d}\mathbf{1}(|x_{i,j}-\alpha_i| \le W(\log W)^{1+2\varepsilon_1})$. In other words, the molecular structure is reduced to the following simplified form:



For the auxiliary graph $\mathcal{G}^{\mathrm{aux}}_{xy}$ constructed above, we define its scaling order by

$$\mathsf{ord}(\mathcal{G}^{\mathrm{aux}}_{xy}) := \#\{\text{solid edges in } \mathcal{G}^{\mathrm{aux}}_{xy}\} + 2\#\{\text{waved edges in } \mathcal{G}^{\mathrm{aux}}_{xy}\} - 2\#\{\text{internal vertices in } \mathcal{G}^{\mathrm{aux}}_{xy}\}. \tag{B.24}$$

From the construction, together with (B.22) and by repeating the argument used in the proof of Lemma 7.21, we can control $\mathcal{G}_{xy}$ via its auxiliary graph as

$$\mathcal{G}_{xy} \prec (\Psi_t)^{\mathsf{ord}(\mathcal{G}_{xy})-\mathsf{ord}(\mathcal{G}^{\mathrm{aux}}_{xy})} \cdot \mathcal{G}^{\mathrm{aux}}_{xy} + W^{-D}. \tag{B.25}$$

Finally, using (B.22), the auxiliary graph $\mathcal{G}^{\mathrm{aux}}_{xy}$ can be bounded by an argument parallel to that in Section 7.4. In particular, the argument there, when applied to the molecular graph of $\mathcal{G}^{\mathrm{aux}}_{xy}$, provides a systematic procedure for selecting the long solid edges between molecules and for establishing a nested summation order over the internal molecules. The only difference from the random band matrix case is that the "summation over a molecule" here also includes summing over certain non-center vertices within the molecule. However, only two solid edges incident to $\mathcal{M}_i$ are used in the summation step, while every other solid edge contributes either a long-edge factor or a Ψ_t-factor. Applying the Cauchy-Schwarz inequality together with the bound (B.22) to these two solid edges, the summation over all vertices in the molecule $\mathcal{M}_i$ produces the desired factor of η_t^{-1}. With this modification, the argument of Section 7.4 carries over verbatim, completing the proof of Lemma 7.1 for the block Anderson model.

For the proof of Lemma 7.2, the bounds in (B.22) remain valid with $\Psi_t(|[\alpha]-[\beta]|) = \mathsf{T}_t(|[\alpha]-[\beta]| \wedge \ell) + W^{-D}$. Then, by following the same reasoning as in Section 7.5, we complete the proof of Lemma 7.2 for the block Anderson model. □

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